

Selberg Integrals and Particle–Hole Duality in the Richardson–Gaudin Pairing Model

Exact Periods, Self-Dual States,
and Finite-Size Crossover

LLM-assisted working manuscript

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Authorship, LLM contribution, and verification status

The scientific problem, original method, and research direction belong to W. V. Pogosov. This project builds on his published and unpublished work developed before the involvement of an advanced large language model (LLM). During the subsequent LLM-assisted phase, the LLM produced detailed derivations, calculations, and verification programs through an extended collaboration with the author. The author has not exhaustively verified every calculation and proof.

The research developed through many iterations and branching lines of enquiry over a substantial period of time. The author posed and refined questions, assessed proposed directions, and guided the revision, rejection, and combination of intermediate approaches; the LLM supplied calculations and arguments that prompted further questions and revisions. The resulting manuscript is a record of this reciprocal, sustained research process.

The historical sequence is explicit: the author’s two papers published in 2012, doctoral dissertation of 2014, the joint paper on particle–hole duality published in 2015, and the author’s 2015 analysis of BCS applicability belong to the pre-LLM research programme. Further author-developed results and working materials remained unpublished. Only after this corpus had accumulated was the LLM brought into the project. The earlier published and unpublished research must not be attributed retrospectively to the LLM. Chapter 1 describes the provenance in more detail.

This document is an openly inspectable working record of an ongoing attempt to complete that programme. Statements presented as identities, lemmas, or theorems record the arguments developed in the manuscript; their formatting is not evidence of exhaustive human verification. Several known mathematical gaps remain and are identified explicitly as assumptions, conjectures, or open proof obligations; undetected errors may also remain.

Supporting programs provide finite exact-arithmetic tests and numerical comparisons. Here, “separately implemented” or “independent check” describes independence of implementations within the LLM-assisted workflow, not independent human authorship, peer review, or certification. Agreement in these tests does not establish the unresolved contour-selection, Gaussian-remainder, or physical finite-defect assumptions.

Status of this deposit version. Version 1.3 is the fourth prepared edition. It retains the exact period, Pfaffian, general even-exponent, and one-pair saddle developments of the earlier editions. It adds three chapters on duality and self-dual states, using finite polynomial identities, a real boundary formulation, controlled leading BCS limits, finite defect and gap estimates, and a scalar-response scheme for the first self-dual excited branch. The consolidated checkpoint v0.1 and its recovered finite spectral dependency are included. New finite checks and reviewed arguments are identified separately from historical computations. The universal excited crossover limit and full pair-preserving spectral ordering remain open; short-iteration accuracy is numerical evidence. The new route does not remove the earlier period bridges outside its own stated sector.

Version 1.3: scope and proof status

This fourth deposit preparation edition extends version 1.2. The opening authorship and verification declaration applies to both retained and new material. The release contains twenty-five chapters and fourteen appendices. It does not claim a completed independent solution of the full physical excitation spectrum.

Duality and self-dual states. Chapters 23–25 incorporate the consolidated duality checkpoint v0.1 and its finite spectral dependencies. The starting point is a finite polynomial identity for the Richardson roots at half filling in a reflection-symmetric spectrum. Ground-branch localization and a real boundary equation lead to an energy-area identity and controlled leading BCS limits without postulating a continuum arc of roots. A finite defect response, the first self-dual gap, and a scalar equation coupled to a smooth real response provide a second route to physical quantities. The distinction between the condensation-energy criterion and the spectral-gap criterion is known physics recovered by this method.

The first self-dual excitation is distinguished throughout from the minimum over the full pair-preserving spectrum. The universal excited crossover limit and uniform accuracy of a single Newton correction are not proved; the latter is a numerical observation on the stated test set. Exact finite identities, inherited finite spectral proofs, asymptotic bounds, and numerical evidence are separately identified in the new chapters. This route does not establish the older contour-to-spectrum assumptions outside its own half-filled self-dual setting.

Arbitrary even auxiliary inverse temperature, Case A. Chapter 21 gives the common all-pole period in terms of a confluent hypergeometric function of Jack type for every positive integer multiplier p , with Vandermonde exponent $\beta = 2p$. The complete normalization and the logarithmic derivative are specified. An exact vector integral obeys an imaginary-time equation in $u = 1/V$; its positive branch and scalar projection establish $E_p^{\text{per}} = E_0^{\text{RG}} + O(p^{-1})$ for a fixed finite system as $p \rightarrow \infty$. This supplies a physical link for that limit without assuming dominance of a complex scalar saddle. It does not identify a finite- p period energy with an eigenenergy, or supply estimates uniform in the thermodynamic limit.

The one-pair saddle route at $\beta = 2$, or $p = 1$. Chapter 22 proves the cancellation of every higher extensive coefficient in the formal Gaussian expansion by orthogonal polynomial recurrence identities. The statement is coefficientwise; it is not a convergence theorem for the full integral. A separate exact finite-spacing calculation provides a comparison. The chapter also collects the blocked and marked spectral scales, the Case B ground cascade estimate, and the distinction between established period statements and conditional excitation transfers. In particular, control of the unblocked ground cascade is not silently reused as a proof for a Christoffel-deformed blocked ensemble.

Retained results and unresolved thermodynamic bridges. Chapter 20 retains the fourth-power Pfaffian and positive compact integral introduced in version 1.1. No new higher- p mixed-contour Case B solution is included. The earlier twenty chapters retain their definitions and arguments; their forward-looking scope statements are qualified by the new fixed-system result in Chapter 21.

The repaired source-index interpolation in Appendix J continues to use a saddle-localized lattice sum, not the invalid unrestricted gamma continuation. The ground contour-to-thimble estimate B_0 , the independent Hessian-to-spectral-shift identification G , and the stated finite-defect estimates remain separate requirements for the physical thermodynamic and finite-size claims in the earlier chapters. The fixed-system $p \rightarrow \infty$ theorem does not settle these questions by an interchange of limits. The Lambert law concerns a rate-defined period endpoint; its identification with the physical real-root count remains conjectural.

Verification. The new duality checks and retained source packages are in `code/duality_selfdual/`; the earlier additions remain in `code/even_beta_caseA/` and `code/one_pair_saddle/`. The release report identifies which checks were run again and which reports are retained historical outputs. The older programs and their reports remain available for inspection. Finite tests support the formulas tested; they do not replace the proof or establish its unresolved uniformity and physical matching conditions.

Abstract

Computational provenance. This manuscript continues the author’s published and unpublished work through prolonged author–LLM collaboration. The author has not exhaustively checked every calculation and proof; limitations are stated explicitly.

We develop two complementary approaches to the zero-temperature Richardson–Gaudin pairing problem: exact Selberg-type contour periods and direct equations for dual and self-dual states. The main quantitative thermodynamic and crossover results use equally spaced levels; the finite algebra extends beyond them where specified.

The period approach yields a fully normalized Selberg–Nørlund–Rice factorization, a complementary Krawtchouk-type kernel, and finite determinant generators for marked, blocked, and nested contours. A repaired lattice argument controls confluent minors. The Case B cascade has a small effect on the chosen ground-period energy in compact interior regimes; its Lambert- W endpoint law is a statement about periods, whose identification with a physical real-root count remains conjectural. Period logarithmic derivatives are not automatically Hamiltonian eigenenergies. The older thermodynamic and finite-size interpretations retain explicit contour, Gaussian, and finite-defect conditions.

For the common Case A cycle, the fourth-power Vandermonde gives an exact moment Pfaffian and a positive quaternionic compact integral. Every even auxiliary exponent $\beta = 2p$ has a normalized confluent Jack-hypergeometric representation. An off-shell vector period obeys an inverse-coupling imaginary-time equation; positivity and its scalar projection prove $E_p^{\text{per}} = E_0^{\text{RG}} + O(p^{-1})$ at fixed finite size as $p \rightarrow \infty$. This does not exchange that limit with the thermodynamic limit. At $\beta = 2$, an alternative expansion about the one-pair saddle has coefficientwise cancellation of all higher extensive terms. Analytic summation and some blocked Case B transfers remain separate questions.

The fourth edition adds the direct duality method. Starting from the particle–hole duality equations, a finite self-dual polynomial identity gives access to the Richardson roots at half filling in a reflection-symmetric spectrum. The physical ground branch is uniquely characterized by its roots lying in the left half-plane, and its coefficient Jacobian remains nonsingular at finite coupling, including rapidity collisions. A real boundary formulation gives an exact energy-area identity and a positive expression for the correlation energy. Comparison estimates recover the leading BCS energy without postulating a continuum arc or density of roots. A finite spectral representation and blocked-defect response give bounds on the first self-dual pair-preserving gap.

The finite-size analysis distinguishes the known criterion for the bulk condensation energy from the criterion for the low spectral gap. Their weak-coupling exponential scales are associated, respectively, with the Cooper-pair binding energy and the collective BCS gap; the explicit condensation-energy comparison also contains a coupling prefactor. Selected joint limits and their error bounds are stated without turning a logarithmic loss in a bound into a new physical scale.

For the first self-dual excited branch, the positive rapidity is extracted analytically. The remaining roots are encoded by a smooth real response, coupled to one scalar spectral condition and an exact energy functional. Small-system root reconstruction, direct diagonalization, and finite crossover calculations check this construction. The observed accuracy of one Newton correction is numerical evidence. A universal limiting excited-state system is proposed, while the required root selection, stability, and energy-limit estimates remain open. The first self-dual gap is not asserted to be the minimum of the full pair-preserving spectrum.

These routes retain complementary exact information. Their agreement on familiar physical scales does not close their separate open estimates; the finite self-dual construction provides a basis for further extensions.

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Reader's guide and hierarchy of claims

The following distinctions are operative throughout this working revision.

Exact identity. A finite algebraic equality between specified sums, determinants, or periods, including the Pfaffian and Jack-hypergeometric identities in the Case A chapters.

Formal coefficient theorem. An identity at every order of a formal series. It does not assert convergence or a uniform remainder after summation.

Period asymptotics. A large-size estimate for a specified period. It is not automatically an eigenenergy estimate.

Branch identification. A label and an existence/continuation statement for a Hamiltonian eigenbranch. Numerical continuation is a diagnostic, not a proof of contour dominance.

Finite self-dual theorem. A statement about solutions of the self-dual polynomial identity and their Hamiltonian interpretation, with finite branch-selection arguments. It does not use a contour-dominance hypothesis.

Conditional consequence. A physical formula obtained after explicitly imposing the contour and response conditions. A valid conditional calculation is not an independent proof of those conditions.

The remaining bridges

B_0 . Control of the full ground period relative to its selected thimble, including possible cancellations, and a uniform local Gaussian energy remainder. See Assumption [14.6](#).

G. Identification of the Gaudin logarithmic derivative with the auxiliary spectral-shift functional. See Assumption [15.2](#).

Finite defects. The coalescent blocked-period prefactor estimate and the physical determinant comparison are distinct pending inputs. The latter is recorded in Assumption [14.8](#); the former is Assumption [16.4](#).

These assumptions do not enter the exact finite residue algebra or the repaired Case B lattice-minor argument. They are not established by matching a weak-coupling subset. The ground and excitation formulas in Chapters 15–18 must be read with their stated conditions, including any inherited ones.

Energy conventions

The spectral parameter Δ , odd threshold Δ_{qp} , and neutral fixed-particle-number threshold $2\Delta_{\text{qp}}$ are different objects. The coordinate scale $d/\log M$ of an edge root is not a new gapless excitation. There are $M+1$ active levels; first finite-size coefficients use the actual $\nu_M = N/M$ and the exact dual count $M+1-N$.

Reading this iteration

For the new fourth-power Case A construction, read Chapter 20. Its positive representation proves nonvanishing of the full period, while physical saddle selection remains open. For the retained completed repair, read Chapter 12 and Appendix J. For the remaining obstruction, read Chapter 14 and the first sections of Chapter 15. The conclusion states the next tasks without treating them as already solved.

The third edition

Chapter 21 gives the arbitrary-even- β exact period and its fixed-system large- p physical ground-state limit. This last theorem uses a positive vector evolution, and does not assume B_0 or G . It also does not prove these assumptions at fixed p and growing system size. Chapter 22 returns to the original $\beta = 2$ integral, proves a formal all-order cancellation, and separates the Case A, Case B, marked, and blocked conclusions. The scope table in that chapter identifies the remaining excitation-sector hypotheses.

The fourth edition: a direct route through duality

Start with Chapter 23 for the finite polynomial problem, root reconstruction, ground-branch regularity, and the energy-area formula. Chapter 24 gives the finite defect response, self-dual gap, and the two observable-dependent criteria for bulk BCS accuracy. Chapter 25 treats the first self-dual excited branch through one real rapidity and a smooth real response. These results apply at half filling in the stated symmetric spectrum. The finite algebra permits more general symmetric levels where explicitly stated; the quantitative BCS and crossover estimates use equal spacing. Removing equal spacing from the title does not remove this hypothesis.

No assumed continuum root arc enters this route. The relevant finite root localization is derived, and the maximum-principle estimates are applied before taking a large-system limit. This avoids specific root-distribution assumptions of the continuum construction, not every hypothesis of asymptotic analysis. In particular, it does not prove the universal excited crossover function or identify the first self-dual level with the first level of every pair-preserving sector.

Stable notation and conventions

Symbol	Definition	Convention and warning
M	highest one-particle lattice index	There are $L = M + 1$ levels, indexed by $n = 0, \dots, M$.
L	$M + 1$	Used when the actual number of levels, rather than the last index, matters.
N	number of Cooper pairs in the seniority-zero reference sector	The thermodynamic filling is defined as $\nu = N/M$, not N/L .
$\bar{N}$	$M + 1 - N$, the reflected hole-pair number	The exact finite transformation is $N \leftrightarrow \bar{N}$.
δ_N	$N - \bar{N} = 2N - M - 1$	Controls the additive electron-hole energy shifts.
ρ	one-spin density of states in the Debye window	The level spacing is $d = \rho^{-1}$.
Ω	width of the one-particle interaction window	The scaling convention is $M/\rho \rightarrow \Omega$.
d	$\rho^{-1} = \Omega/M$ in the chosen finite-size sequence	Differential symbols are written as dx , never as plain dx , to avoid ambiguity.
V	reduced pairing matrix element	$V = O(M^{-1})$ in the thermodynamic limit.
u	$1/V$	Natural logarithmic-derivative variable for all periods.
v	ρV	Fixed dimensionless coupling.
σ	$\exp[-2/(\rho V)] = \exp(-2/v)$	Primary exponential coupling variable.
x	$\sigma/(1 - \sigma)$	The positive variable used in residue polynomials.
ε_n	$\varepsilon_{F0} + n/\rho$	One-particle energies; pair poles are $2\varepsilon_n$.
E_n	$2\varepsilon_n$, the pair-pole energy	Used in the eigenvalue-based equations and Baxter reconstruction.
λ_n	$V \sum_j (E_n - R_j)^{-1}$	Regular eigenvalue-based branch coordinate; at $V = 0$ it is the binary subset indicator.
$\mathcal{Q}(z)$	$\prod_j (z - R_j)$, the Baxter polynomial	Reconstructed from $\mathcal{Q}'(E_n) = (\lambda_n/V)\mathcal{Q}(E_n)$.

Symbol	Definition	Convention and warning
$\mathcal{B}_S$	maximal continuation of the local branch with binary label S	Total-energy crossings do not exchange this complete eigenvalue-based label.
ν	N/M	Thermodynamic filling. Finite electron–hole duality sends $N \mapsto M + 1 - N$, hence $\nu \mapsto 1 - \nu + M^{-1}$.
$\bar{\nu}_M$	$\bar{N}/M = 1 - \nu + M^{-1}$	Exact finite hole filling; its M^{-1} shift contributes to upper-chart coefficients of order d .
$\nu^\sharp$	$(N - \frac{1}{2})/M$	Exactly symmetric finite-size filling: $\bar{\nu}^\sharp = 1 - \nu^\sharp$.
$\tilde{\mu}$	$\Omega[\nu(1 + 2x) - x]$	Chemical potential measured from the lower Debye edge.
Δ	$2\Omega\sqrt{x(1+x)\nu(1-\nu)}$	Square-root spectral parameter; it is not always the minimum quasiparticle energy.
ε_c	$2\Omega x = 2\Omega\sigma/(1-\sigma)$, the isolated-pair binding energy	In the dilute limit $\Delta_{\text{qp}} \rightarrow \varepsilon_c/2$ and $\Delta_{\text{even}} \rightarrow \varepsilon_c$.
$\mathcal{E}(\xi)$	$\sqrt{(\xi - \tilde{\mu})^2 + \Delta^2}$	One-quasiparticle dispersion in the Debye window.
Δ_{qp}	$\min_{0 \leq \xi \leq \Omega} \mathcal{E}(\xi)$	Odd/parity threshold after chemical-work subtraction.
Δ_{even}	$2\Delta_{\text{qp}}$	Leading neutral even-sector threshold.
Δ_R	real-root, seniority-zero even excitation	One Richardson root remains trapped between levels.
Δ_{B_2}	two-blocked-level, seniority-two excitation	The physically broken-pair sector.
Z_Γ	meromorphic period in contour class Γ	Defined before any saddle or branch identification.
$\mathcal{Z}_\Gamma$	reduced pole-supported residue period	Defined by stripping the common factor $\mathcal{C}_{M,N,\rho} e^{-2N\varepsilon_{F0}/V}$; ratios in a fixed N -pair sector are unchanged.
E_Γ^{per} E^{RG}	$-\partial_{1/V} \log Z_\Gamma$ energy of a named Richardson eigenbranch	Exact period barycenter. Equals the period result only after the branch bridge and fluctuation correction are included.
S	residue subset of $\{0, \dots, M\}$	Usually $ S = N$.
$\mathcal{R}(A)$	$\{M - a : a \in A\}$, reflection of a lattice subset	Used in the exact finite electron–hole map.
$\mathcal{D}(S)$	$\mathcal{R}(\mathcal{L}_M \setminus S)$, reflected complement	Bijection between N - and $\bar{N}$ -element residue sets.
$W(S)$	signed discrete Vandermonde weight	It is not a positive probability weight.
$\chi_S(z)$	$\prod_{n \in S} (z - n)$, the characteristic polynomial of a residue subset	Blocking a level inserts $\chi_S(p)$; this is not the Boolean excluded-pole insertion ξ_p .

Symbol	Definition	Convention and warning
$\pi_s^{(M)}(z; x)$	$s!x^s \mathbf{K}_s^{(M)}(z; \sigma)$, the monic blocked polynomial	One-characteristic-polynomial averages equal π_s ; finite-seniority averages are Heine–Christoffel determinants.
$\mathfrak{h}_s$	signed squared norm of the monic polynomial π_s	Alternates in sign; Christoffel norms are defined algebraically.
$Q_n^{[\mathbf{p}]}$	complementary kernel of the Christoffel-deformed blocked ensemble	Generates hole insertions over a finite-seniority background; it is distinct from the pure excluded-pole kernel Q_n .
Z_0	unlabelled residue sum $\sum_{ S =N} W(S)$	The ordered all-pole residue sum is $N!Z_0$.
$z_{M,N}$	ordered all-pole Selberg–Nørlund–Rice sum	Exactly $N!Z_0$; factorizes in Result R1.
$A_{M,N}$	coupling-independent factor in $z_{M,N}$	Includes the ordered factor $N!$; superfactorial convention is fixed in Appendix A.
$\mathbf{K}_s(n)$	Krawtchouk-type polynomial defined by $\sum_s \mathbf{K}_s(n) z^s = (1+z)^{M-n} (1+z/\sigma)^n$	Orthogonal only in the signed bilinear form generated by $w(n)$.
K_N^{occ}	finite occupation projector	Written with the weight on the second index; it need not be Hermitian in the Euclidean inner product.
Q_N	$I - K_N^{\text{occ}}$, the complementary hole kernel	Principal minors generate hole insertions algebraically; they are not ordinary probabilities.
ξ_A	$\prod_{a \in A} \mathbf{1}_{a \notin S}$	Canonical pure excluded-pole insertion; $\xi_A \xi_B = \xi_{A \cup B}$.
J	diagonal sign matrix, $J_{jj} = (-1)^{N-j}$	Removes the columnwise sign of the lower restriction of Q_N .
$\mathbf{K}^{(k)}$	sign-gauged lower restriction $Q_N^{(k)} J$	Factorizes as $FD_1CD_2BD_3$.
$P_a^\downarrow$	$(-1)^{N+a} Q_{N,aa}$, lower marked factor for $a < N$	A sign-corrected excluded-pole period.
$P_a^\uparrow$	$(-1)^{N+a+1} K_{N,aa}^{\text{occ}}$, upper marked factor for $a \geq N$	Exactly $P_{M-a}^\downarrow$ in the $\overline{N}$ -pair hole sector.
$\mathfrak{P}(A)$	sign-corrected principal minor	Has a positive Cauchy–Binet expansion.
$X_r(S)$	number of occupied residue nodes below cutoff r	$X_r = \#\{n \in S : n < r\}$.
$H_r(S)$	number of holes below cutoff r	$H_r = r - X_r = \sum_{j < r} \xi_j$.
$C_r(A)$	canonical Boolean–Möbius coefficient for nested cutoffs	Expands the raw multiplicity polynomial into pure insertions ξ_A .

Symbol	Definition	Convention and warning
$\widehat{C}_{k,c}(A)$	normalized Boolean coefficient of the consecutive cascade	Defined by $\prod_{r=1}^k (c + H_r) = c^k \sum_A \widehat{C}_{k,c}(A) \xi_A$; it is nonnegative.
$\mathfrak{Z}_{N,k}^\downarrow$	reduced physical lower consecutive-cascade period	The contour label k is held fixed under $-\partial_{1/V}$.
$\mathcal{R}_{N,k}^\downarrow[E]$	normalized complete support sum for an excluded set E over the cascade	$\mathfrak{Z}_{N,k}^\downarrow[E] = (N-k)!(N-k)^k Z_0 \mathcal{R}_{N,k}^\downarrow[E]$.
k, κ	number and density of trapped contours	$k \sim \kappa M$ in macroscopic Case B.
d_κ	$\partial_\alpha \Phi(\kappa) > 0$, the endpoint rate slope	Controls geometric localization of endpoint supports in compact Case B.
$u(\alpha), v(\gamma), \mathcal{G}(\alpha, \gamma)$	separated off-diagonal kernel rates and full prefactor	$\mathbf{K}_{ij} = M^{-1} \mathcal{G} \exp\{M[u + v]\} [1 + O(M^{-1})]$.
$\mathcal{M}_{N,k}^\downarrow(a)$	normalized complete marked-background factor	Defined by $\mathcal{R}_{N,k}^\downarrow[\{a\}]/Q_{N,aa}$; equals $1 + O(M^{-2})$ in compact Case B for separated a .
$\Phi(\alpha)$	one-point large-deviation rate	The Case B endpoint satisfies $\Phi(\kappa) = 0$.
$q(\alpha)$	$\tau_\alpha - \beta_\alpha = \sqrt{D_\alpha}/(1+x)$	Leading degree density in the positive one-point sum.
$C(\alpha, \nu, x)$	complete interior prefactor of P_a	Defined in Chapter 7; diverges as $\alpha^{-1/2}$ toward the lower edge.
Φ_0, Λ	lower-edge rate and fixed-index scale	Control the discrete layer $a = O(1)$ and its matching to the interior saddle.
ν_B	lower Case A–B transition filling	Defined by $\Phi_0(\nu_B, \sigma) = 0$; its upper dual is $1 - \nu_B$.
δ_Φ	positive edge-rate deficit $-\Phi_0$ on the Case B side	The birth law is $\kappa \sim \delta_\Phi / \log(1/\delta_\Phi)$.
W_{-1}	negative real Lambert branch	Solves the implicit small-cascade endpoint equation.
$\mathcal{G}$	Gaudin Hessian of the certified Richardson saddle	Its continuous determinant branch gives the Gaussian period-to-branch correction.
$\tilde{\mu}_M, \Delta_M$	finite spectral-curve parameters solving the discrete gap and number equations	They approach $\tilde{\mu}, \Delta$ with $O(M^{-1})$ corrections.
$\mathcal{F}_M(z)$	$\sum_n [(z - \xi_n) \mathcal{E}_{n,M}]^{-1}$, the finite auxiliary Gaudin spectral function	It has one zero y_ℓ in each one-particle interval.
$a(\epsilon)$	$-\pi^{-1} \arg \mathcal{F}(\epsilon + i0)$, the continuum zero displacement measured from the upper pole	$a(0+) = 1$ and $a(\Omega-) = 0$.

Symbol	Definition	Convention and warning
u_-, u_+	$\operatorname{arsinh}(-\tilde{\mu}/\Delta)$ and $\operatorname{arsinh}[(\Omega - \tilde{\mu})/\Delta]$	Their separation is $u_+ - u_- = 2/v$.
δE_{fl}	$\lim(E_0^{\text{RG}} - E_0^{\text{per}})$, the full Gaussian $O(M^0)$ ground-energy correction	Smooth across $\tilde{\mu} = 0$, $\tilde{\mu} = \Omega$, and the A–B topology line.
$\dot{\mu}_\xi, \dot{\Delta}_\xi$	First spectral-curve response per removed-level energy scale d to two blocked levels coalescing at ξ	Defined by $\tilde{\mu}_{[\xi]} = \tilde{\mu} + d\dot{\mu}_\xi + O(d^2)$ and $\Delta_{[\xi]} = \Delta + d\dot{\Delta}_\xi + O(d^2)$.
$\mathcal{H}(\xi)$	Finite-defect Gaudin response of the collective sea	Gives the common physical one-loop correction $d\mathcal{H}(\xi)$; at an interior minimum it is shared by real-root and broken-pair sectors through order d .
y_ℓ	zeros of the auxiliary Gaudin spectral function	They are not Richardson roots.
$y_1^{(F)}$	first zero of the auxiliary function F_M	Has the sharpened logarithmic expansion.
$r_1^{\text{RG}}, s_1^{\text{RG}}$	lower physical trapped-root coordinate and upper distance from the edge	Each obeys $d/[\log M + \gamma + \mathcal{C}_\mp] + O(d/\log^2 M)$; they are distinct from the auxiliary zeros.
$\mathcal{C}_\mp, L_\mp$	edge constants and $L_\mp = \log M + \gamma + \mathcal{C}_\mp$	Control lower/upper logarithmic soft-root expansions and the $d/\log M$ approach of the edge gap splitting.

Permanent factor-of-two convention

The marked-root period is an even-sector object:

$$E_{\text{mark}}(\xi) - E_0 = 2\mathcal{E}(\xi) + o(1).$$

The odd blocked-level sector gives $\mathcal{E}(\xi)$ after subtracting the appropriate chemical work.

Notation in the Case A Pfaffian extension

Chapter 20 uses $L = M + 1$ for the number of active levels, $a_n = 2\varepsilon_n$, $u = 1/V$, and the multiplier p of the original auxiliary inverse temperature. The new calculation is at $p = 2$, hence Vandermonde power four and $t = 2u$. The quantities E_1, E_2 are common-cycle period derivatives; $\hat{E} = 2E_2 - E_1$ is an extrapolation, not an exact physical eigenenergy. Its matrix and projector notation is local to that chapter.

Auxiliary inverse temperature in the third edition

Throughout Chapters 21–22, $\beta = 2p$ denotes the Vandermonde exponent. Thus $\beta = 2$ is the original determinant integral ($p = 1$), and $\beta = 4$ is the Pfaffian extension ($p = 2$). The pole order and the coefficient pu in the exponential must change together with p . The derivative is

$E_p^{\text{per}} = -p^{-1} \partial_u \log Z_p$. This auxiliary parameter is unrelated to a physical Gibbs temperature. The Jack parameter $\alpha_J = 1/p$ is local to Chapter 21; it is distinct from the scaled marked position $\alpha = a/M$ used elsewhere. The Gaussian coordinate x_j in Chapter 22 is also distinct from the coupling variable $x = \sigma/(1 - \sigma)$.

Dictionary for the self-dual chapters

Chapters 23–25 use a centered, half-filled spectrum. Local dimensionless variables are explicitly reintroduced there. The correspondence with the earlier Hamiltonian is

Quantity	Self-dual chapters	Earlier period chapters
Active orbitals	$2N$, with N pairs	$L = M + 1 = 2N$
Single-particle levels	$\varepsilon = \pm(i - \frac{1}{2})d$	$\varepsilon_{F0} = -(N - \frac{1}{2})d$
Spacing and coupling	$d, G = \lambda d$	$\rho = 1/d, V = G, v = \lambda$
Bandwidth convention	$\omega_D = Nd$	$\Omega = (2N - 1)d$
Positive pair energies	$da_i, a_i = 2i - 1$	Pair energies $2\varepsilon_j$
Dimensionless coupling	$g = \lambda, q = 1/\lambda$	$q(\alpha)$ elsewhere is unrelated

Thus $2\omega_D$ is the width associated with the midpoint cutoff, whereas Ω is the distance between the actual endpoint levels. They differ by d at finite size and must not be interchanged in subleading terms. The diagonal interaction is included throughout. The new correlation energy is $C = E_{\text{FS}} - E_0$, with the Fermi-sea expectation evaluated for the same interacting Hamiltonian. The bulk single-quasiparticle parameter is $\Delta_0 = \omega_D / \sinh(1/\lambda)$; the compared two-particle gap is $2\Delta_0$. Local r is a dimensionless Richardson rapidity, not a gap.

Registry of principal results

ID	Type	Compact statement	Final location
R1	Exact identity	The all-pole Selberg–Nørlund–Rice sum factorizes exactly, yielding an exact finite- M period energy.	Chapter 4
R2	Exact identity	The complementary kernel admits both a Krawtchouk-tail representation and the finite factorization $Q^{(k)}J = FD_1CD_2BD_3$; sign-corrected principal minors have positive Cauchy–Binet expansions.	Chapter 5
R3	Exact identity	The reduced excluded-pole period is $\mathcal{Z}_{\text{all}}[A] = N!Z_0 \det Q_{N,A}$; the projected local marked period is $(N-1)!Z_0P_a$, and arbitrary nested raw periods have a canonical pure-sector generator.	Chapter 6
R4	Period asymptotics	In Case A, the marked period obeys $-\partial_{1/V} \log P_a = 2\mathcal{E}(\xi) + O(M^{-1})$, with a matched discrete edge layer.	Chapter 7
R5	Exact identity	Reflected complementation gives $N \mapsto M+1-N$: excluded sectors map to occupation sectors, lower and upper marked charts are dual, period energies shift by $\delta_N(2\varepsilon_{F0} + \Omega)$, and physical paired energies by $\delta_N(2\varepsilon_{F0} + \Omega - V)$.	Chapter 8
R6	Exact identity / period asymptotics	A blocked level inserts a characteristic polynomial whose average is the monic Krawtchouk polynomial; the odd particle/hole periods obey $E_{\text{odd}} - E_{\text{even}} \mp \mu = \mathcal{E}(\xi) + O(M^{-1})$.	Chapter 9
R7	Exact identity / period asymptotics	Finite-seniority periods are Heine–Christoffel determinants, and the marked factor is an exact particle–hole convolution; hence the real-root and broken-pair sectors have the common leading threshold $2\Delta_{\text{qp}}$.	Chapter 10

ID	Type	Compact statement	Final location
R8	Exact identity	The physical consecutive cascade has exact finite ground, complete marked, and arbitrary finite-seniority periods; all supports are resummed either as complementary-kernel minor sums or by one determinant differential generator.	Chapter 11
R9	Period asymptotics	In compact Case B, complete support resummation gives $\mathcal{R}_{N,k} = 1 + O(M^{-2})$ and the same estimate for a separated marked background; hence the cascade changes the ground period barycenter only by $O(M^{-2})$ and leaves the marked period gap unchanged through order M^{-1} .	Chapter 12
R10	Period asymptotics / conjecture	The cascade is born as $\kappa = -\delta_\Phi/W_{-1}[-\delta_\Phi/(e\Lambda)] \sim \delta_\Phi/\log(1/\delta_\Phi)$, where $\delta_\Phi = -\Phi_0$; the finite transition width is $(\log M)/M$, and representative continuation data show finite real-root counts tracking $M\kappa$ up to bounded parity/edge corrections; this root-count statement is numerical evidence, not part of the period theorem.	Chapter 13
R11	Branch identification	Weak-coupling residue labels coincide with unique local eigenvalue-based branches; the paired ground branch is globally unique, and the marked and blocked paths used in the manuscript are numerically checked by continuation, Hamiltonian-spectrum, and Baxter diagnostics.	Chapter 14
R12	Physical consequence under $\mathbf{B}_0, \mathbf{G}$	The physical ground energy is $E^{\text{RG}} = E^{\text{per}} + \delta E_{\text{fl}} + o(1)$, where δE_{fl} is a one-dimensional Gaudin spectral-shift integral, smooth across both crossovers.	Chapter 15
R13	Physical consequence under $\mathbf{B}_0, \mathbf{G}, \mathbf{B}_{\text{def}}, \mathbf{C}_{\text{per}}$	If the quasiparticle minimum is interior, both Δ_R and Δ_{B_2} equal $2\Delta + d[-(1+2x) + \Delta/(\Omega(1-\nu)) + \mathcal{H}(\tilde{\mu})] + O(d^2)$ under the exact finite convention $\nu = N/M$; hence their difference is $O(d^2)$. The confluent blocked-period prefactor estimate is additionally required. The contour and real-axis definitions of $\mathcal{H}$ are equivalent, and the result holds in compact Cases A and B.	Chapter 16

ID	Type	Compact statement	Final location
R14	Physical consequence under B_0, G, B_{def}	At a Debye edge, the auxiliary Gaudin zero has a full harmonic expansion, the physical Richardson root obeys $r_1^{\text{RG}} = d/[\log M + \gamma + \mathcal{C}_-] + O(d/\log^2 M)$, and $\Delta_{B_2} - \Delta_R$ has a leading slope-times-spacing term approached through a $d/\log M$ correction.	Chapter 17
R15	Physical synthesis under the bridge conditions	The full ground energy through order M^0 , the odd/parity threshold, the two even sectors, Cases A/B, the Debye-edge and topology crossovers, and the hierarchy $d \gg d/\log M \gg d^2$ form one consistent physical picture; no additional leading anomaly occurs at the A–B line.	Chapter 18
R16	Exact identity	For the Case A common cycle, the fourth-power Vandermonde period is an explicit $2N \times 2N$ moment Pfaffian; double-pole residues determine every entry and a trace gives its energy derivative.	Chapter 20
R17	Exact identity / positivity	The full fourth-power period is $(-1)^N C t^{N[2(L-N)+1]}$ times a positive normalized quaternionic Grassmann integral. It has no zeros for real levels and $t > 0$.	Chapter 20
R18	Fixed-spectrum coupling limits	The extrapolation $2E_2 - E_1$ reproduces the first two weak-coupling corrections but has an explicit cubic error; its strong-coupling error is also known. These are fixed-size limits of Case A periods, not a general eigenenergy theorem.	Chapter 20

Results introduced in the third edition

- R19. Exact normalized Jack-hypergeometric common-cycle period and its energy derivative at every even auxiliary exponent $\beta = 2p$; Chapter 21.
- R20. Exact vector inverse-coupling evolution and scalar projection. Positivity of the regular branch implies $E_p^{\text{per}} \rightarrow E_0^{\text{RG}}$ with error $O(p^{-1})$ on compact positive- u intervals for a fixed finite system; Chapter 21. This is not a uniform thermodynamic estimate.
- R21. Coefficientwise cancellation of all higher extensive terms in the one-pair Gaussian expansion at $\beta = 2$, through the exact formal recurrence symbols $B(s) = i(1 + \sigma)s$ and $A(s) = s(1 - \sigma s)$; Chapter 22. No convergence claim is part of this result.
- R22. One-pair-route synthesis of blocked and marked spectral scales and a scalar ground-

cascade estimate in compact Case B. The blocked Case B and physical transfers retain the specific conditions displayed in Chapter 22.

Results introduced in the fourth edition

- R23. Finite self-dual polynomial identity, reconstruction of Richardson roots, unique Hurwitz ground branch, and nonsingular coefficient Jacobian; Chapter 23.
- R24. Exact real boundary and energy-area formulations, positive correlation-energy representation, and controlled leading BCS energy with finite-size bounds; Chapter 23.
- R25. Finite blocked response and first self-dual gap, coalescence comparison, and distinct energy and spectral criteria in specified limits; Chapter 24. The full pair-preserving minimum is a separate spectral-ordering question.
- R26. Exact separation of a positive excited rapidity: nonlinear real response, scalar spectral condition, and energy functional; Chapter 25. The conditional limiting crossover system and observed accuracy of short Newton iteration are not part of an unconditional asymptotic theorem.

Logical dependency map

Finite algebra. R1 and R2 are independent base results. Together they imply R3, and provide the algebraic input for all later residue sectors.

Case A. R3 plus positive lattice Laplace asymptotics imply R4. R5 extends the lower chart to the upper chart.

Finite seniority. The same moment algebra plus the monic Krawtchouk normalization gives R6; the multi-Heine determinant and exact particle-hole convolution then give R7.

Case B. The pure-sector algebra gives R8; off-diagonal kernel asymptotics and Boolean-coefficient bounds give R9; the lower-edge expansion of the one-point rate gives R10.

Physical branches. Result R11 supplies permanent Hamiltonian branch labels for the sectors of R4, R6, R7, and R9. Their physical subleading energies additionally require bridge conditions B_0 , G , and B_{def} , specified in Chapters 14 and 15, before the Gaudin calculations of R12–R14 are applied.

Subleading physics. Under B_0, G , the ground comparison gives R12. Finite-defect responses additionally require B_{def} for R13 and R14; the common interior gap in R13 also requires C_{per} , Assumption 16.4.

Pfaffian Case A. R16–R18 form a separate common-cycle extension at doubled auxiliary inverse temperature. They do not assume the physical bridges and do not establish a higher-temperature Case B cascade.

Synthesis. R15 is a conditional assembly of R1–R14. Its statement about the absence of an $O(1)$ ground-energy anomaly uses R12, while its finite-size qualifications use R13–R14.

Minimal reading paths

1. **Leading thermodynamic energy and gap only:** Chapters 2, 4, 7, 11, 12, 13, and 18.
2. **Exact residue algebra:** Chapters 3–6 and Appendices A–E.
3. **Spectral identification and finite-size physics:** Chapters 14–17 and Appendices K–N.
4. **New Case A Pfaffian results:** Chapter 20, with the stand-alone programs in `code/pfaffian_caseA/`.

New logical routes in version 1.2

The exact full-period Jack formula and the finite-dimensional vector evolution in Chapter 21 imply a fixed-system $p \rightarrow \infty$ ground-energy limit. They do not require the older thermodynamic bridge assumptions. Their error constants are not asserted to be uniform in L . The coefficientwise theorem in Chapter 22 uses formal orthogonal-polynomial identities. Transfer to the Case B ground period uses an additional exact cascade identity and its scalar bounds. Transfer to a blocked cascade requires the corresponding deformed-kernel bounds; the unblocked estimate alone is insufficient.

The direct duality route in version 1.3

The particle-hole duality equations lead to a finite self-dual polynomial identity. Its Hurwitz ground branch, uniqueness, and nonsingular coefficient Jacobian give the input for the real boundary formulation. The energy-area identity and comparison bounds then yield the leading BCS ground energy. Finite spectral ordering and coalescence, together with a blocked-background response, give the first self-dual gap and its finite error bound. The latest excited-state factorization supplies an exact response equation, a scalar condition, and a gap functional. These dependencies are detailed in Chapters 23–25. They do not pass through B_0 , G , or the older period finite-defect bridges. Nor do they prove those bridges in general.

The limiting excited crossover equation is a candidate obtained from finite identities. Existence and selection of its physical scalar root, and the energy-limit interchange, remain additional obligations. Reproduced finite computations test the stated identities and selected branches; they are not proofs of these limiting steps.

Part I

Problem, method, and results

Chapter 1

Introduction

1.1 Origin of the method and history of the project

The present project continues an analytical approach proposed by the author in “‘Probabilistic’ approach to Richardson equations,” published in *Journal of Physics: Condensed Matter* **24**, 075701 (2012), and first circulated as arXiv:1111.2907 [1]. That work reformulated the Richardson problem through a meromorphic weight inspired by the classical electrostatic analogy and a Selberg-type integral. The application was restricted to the ground state. It extracted the dependence on the pairing interaction needed for the ground-state energy, but did not carry out the full direct evaluation of the determinants, including their normalization. In particular, the coupling-independent factor $A(N, N_\Omega)$ remained undetermined in its Eq. (32). This distinction matters: recovering the coupling dependence of a period is less than evaluating every factor in the finite determinant.

Within this line of work, the 2012 paper was also the first to formulate explicitly the Hamiltonian origin of electron–hole symmetry in the analysis of the exact Richardson solution. Appendix A introduces hole operators and derives the relation between the electron and hole descriptions.¹ Electron–hole symmetry consequently became an explicit analytical constraint on the exact solution. Its finite-size consequences are developed systematically in Chapter 8 below.

The subsequent paper, “Excited states in Richardson pairing model: ‘probabilistic’ approach,” *Progress of Theoretical Physics* **128**, 1–14 (2012), arXiv:1208.1088, extended the method to excited states [2]. It analyzed the effect of blocked levels and the energy gap at arbitrary filling, identifying dilute-pair, BCS, and dilute-hole regimes. The determinant structure was again not fully evaluated by a direct calculation. In Sec. 3.4, Eqs. (20)–(26), symmetry determines the relative coefficients required for the coupling dependence, while a common normalization factor remains unspecified.

A further step was taken in the author’s 2014 doctoral dissertation, *Superconductors and dilute superfluid Bose systems: From the microscopic to the macroscopic scale*, prepared at the Institute for Theoretical and Applied Electrodynamics of the Russian Academy of Sciences [3]. The thesis is listed in the Lebedev Physical Institute dissertation-council record; its complete Russian text is also available through ResearchGate. It evaluates the determinants themselves using a comparatively systematic procedure. Section 2.2.5, especially Eqs. (2.34)–(2.35), obtains the ground-period normalization through superfactorials. Section 2.2.8, Eqs. (2.49)–(2.56), brings sums inside the determinant and uses successive row differences and a triangular form to evaluate the excited-state normalization. Thus the direct determinant calculation

¹This priority statement concerns the explicit Hamiltonian formulation in the development of this approach. Appendix A of Ref. [1] itself acknowledges earlier observations of spectral electron–hole duality. It is not a claim that electron–hole symmetry as a general physical concept originated in that paper.

has an antecedent in the dissertation and is not first introduced in the present manuscript.

From the perspective of the broader program pursued here, those developments provided a partial solution: they treated particular sectors and calculations, rather than a complete, controlled treatment of the contour sectors, physical branch selection, and finite-size asymptotics. The present project attempts to carry the program through by developing the finite determinant algebra, marked and blocked sectors, nested contours, and their asymptotic analysis in a unified text. This is an objective, not a claim of completion. The remaining contour-to-spectrum and fluctuation estimates are stated explicitly in Chapters 14–17 and in the open-proof-obligations section.

After the two papers and the dissertation, the author continued the programme and obtained additional results and working materials that remained unpublished. This intervening stage preceded the use of an LLM. Only later was an advanced LLM brought into the project to work with the accumulated corpus of publications, unpublished results, and preliminary developments. The present calculations therefore build on a body of author-led research that already existed at that point.

The LLM-assisted stage was a prolonged collaboration involving many iterations and branching investigations. The author formulated and reformulated questions, selected and assessed directions, and revisited earlier steps in response to the calculations; the LLM developed derivations, computational checks, and alternative arguments that were then revised, rejected, or combined through further exchanges. Considerable time was spent on this reciprocal process. The text records its accumulated outcome rather than the response to one prompt.

The opening authorship statement distinguishes these stages and records the incomplete extent of human verification. The earlier publications and dissertation should be cited for their original contributions; the subsequent unpublished author-developed material is also part of the project’s antecedents. Version 1.1 adds the higher-inverse-temperature Pfaffian analysis in Chapter 20, restricted here to the common all-pole Case A construction.

1.2 Origin and scope of the duality method

The duality construction developed here continues a specific line of work on the exact Richardson problem. The earlier Selberg-integral treatment explicitly rewrote the pairing Hamiltonian in the hole representation and used the electron–hole relation to connect complementary fillings [1, Appendix A]. This is the relevant starting point in the genealogy of the present project; it is not a claim that particle–hole symmetry itself was first recognized in 2012. That appendix acknowledges earlier observations of the symmetry in the energy spectrum.

Particle–hole identities for the Russian doll BCS model and their rational Richardson limit were subsequently derived in Ref. [4]. The half-filled construction classifies states of a reflection-symmetric spectrum as self-dual or mutually dual and gives alternative equations directly for the rapidities. To display these equations without a convention ambiguity, let L be the number of active levels, $N = L/2$, $g > 0$, and $\epsilon_m = \epsilon_0 + md$, $m = 0, \dots, L - 1$. Write the pair rapidities as $E_j = 2\epsilon_0 + e_j$. The usual Richardson equations are

$$\frac{1}{g} = \sum_{m=0}^{L-1} \frac{1}{2md - e_j} + 2 \sum_{k \neq j} \frac{1}{e_j - e_k}, \quad j = 1, \dots, N. \quad (1.1)$$

The alternative half-filled relations are

$$\sum_{j=1}^N \frac{1}{2md - e_j^\alpha} + \sum_{j=1}^N \frac{1}{2(L-1-m)d - e_j^\beta} = \frac{1}{g}, \quad (1.2)$$

$$\sum_{j=1}^N e_j^\alpha = \sum_{j=1}^N e_j^\beta. \quad (1.3)$$

These are Eqs. (30)–(31) of the published article [4, Sec. 4.3]; they are numbered (32)–(33) in arXiv:1412.1016v3. For a self-dual state the two sets coincide, leaving one equation for each reflected pair of levels, $m = 0, \dots, N-1$. The formulas extend to other reflection-symmetric spectra by replacing the equally spaced level energies with the actual energies.

Equation (1.2) has no explicit pairwise collision denominator $1/(e_j - e_k)$. Its level-rapidity denominators remain, and this algebraic observation alone establishes neither global numerical regularity nor admissibility of every algebraic solution. The 2015 paper tested its classification using solutions of the original Richardson equations and proposed direct continuation with the alternative relations. The present work develops that direct route and its polynomial consequences; no exhaustive priority claim about all subsequent uses of these equations is needed.

A related development uses variables encoding the eigenvalues of the commuting Gaudin charges. This approach already predates the 2015 half-filled classification [5, 6] and was extended to the XXZ family [7]. For the convention above, its basic variables are

$$\Lambda_m = \sum_{j=1}^N \frac{1}{2\epsilon_m - E_j} = \frac{Q'(2\epsilon_m)}{Q(2\epsilon_m)}, \quad Q(z) = \prod_{j=1}^N (z - E_j). \quad (1.4)$$

Thus a self-dual solution obeys $\Lambda_m + \Lambda_{L-1-m} = 1/g$. Given admissible Λ_m , the equations $Q'(2\epsilon_m) = \Lambda_m Q(2\epsilon_m)$ are linear in the coefficients of the monic polynomial Q ; extracting its zeros is a further numerical step. Polynomial reconstruction is therefore a bridge between representations, not evidence that the direct self-dual rapidity problem has already been resolved [5, 8].

Dual representations also underlie determinant formulas for scalar products. Faribault and Schuricht obtained expressions in the charge variables by combining opposite reference vacua [6]. Claeys, Van Neck, and De Baerdemacker subsequently established their relation to the rapidity-based Slavnov determinant through Cauchy-matrix identities [8, Sec. 5]. These results concern the evaluation of overlaps and form factors once the relevant states are characterized. They should be distinguished from the nonlinear task of selecting and determining the physical rapidities.

Retaining access to the rapidities also keeps the resulting states compatible with the extensive Bethe-state machinery, including Slavnov’s determinant for an on-shell/off-shell scalar product [9]. This is a practical reason to solve the alternative rapidity equations themselves, while using charge variables as a complementary representation and numerical check.

1.3 The reduced pairing problem

The reduced pairing Hamiltonian is one of the simplest interacting many-fermion models that retains a nontrivial collective spectrum. In its conventional form, fermions occupying time-reversed one-particle states interact through an attractive matrix element independent of the level labels. Cooper’s one-pair problem first exposed the bound-state scale generated by an arbitrarily weak attraction above a filled Fermi sea [10]. The Bardeen–Cooper–Schrieffer construction captures the thermodynamic paired state through a quasiparticle vacuum and an energy gap [11]. Richardson subsequently found exact, particle-number-conserving eigenstates whose energies are sums of nonlinear spectral parameters satisfying coupled algebraic equations [12, 13]. The solution became the rational member of the broader Richardson–Gaudin family;

its integrability, Bethe-ansatz structure, and applications have been developed from several complementary viewpoints [14–17].

The exact solution is particularly valuable when level discreteness, number parity, or the topology of individual Bethe roots matters. These questions arose prominently in the theory of ultrasmall metallic grains, where the one-particle level spacing can compete with the bulk gap and canonical even–odd effects become spectroscopically relevant [18–21]. Large- N continuum analyses recover the usual BCS energy scales from the Richardson equations [22]; controlled inverse-size expansions can also be constructed directly from the exact solution [23]. The present work develops complementary routes through meromorphic contour periods and through particle–hole duality. The first organizes exact residue algebra and contour sectors; the second returns directly to finite Richardson polynomials in a self-dual sector. Both address thermodynamic energies, excitation scales, and finite-size structure, with the different hypotheses stated separately.

The equally spaced model that provides the main setting of this manuscript has $M + 1$ equally spaced, doubly degenerate one-particle levels,

$$\varepsilon_n = \varepsilon_{F0} + \frac{n}{\rho}, \quad n = 0, \dots, M, \quad (1.5)$$

and N Cooper pairs in the active window. The equally spaced spectrum is restrictive, but it is precisely what allows the residue sums to close into finite determinant and Selberg-type structures. The thermodynamic sequence keeps

$$\frac{N}{M} \rightarrow \nu \in (0, 1), \quad \frac{M}{\rho} \rightarrow \Omega, \quad v = \rho V > 0 \quad (1.6)$$

fixed. The central observables are the ground-state energy, the odd or parity threshold, and the two neutral even-sector thresholds associated with a real Richardson root and with a broken pair.

1.4 Why replace root solving by meromorphic periods?

Let $R_1, \dots, R_N$ be the Richardson spectral parameters. The same nonlinear equations that determine them arise as the stationary conditions of a meromorphic weight

$$S_N(\mathbf{R}) = \frac{\prod_{i < j} (R_i - R_j)^2}{\prod_{i=1}^N \prod_{n=0}^M (2\varepsilon_n - R_i)} \exp \left(-\frac{1}{V} \sum_{i=1}^N R_i \right). \quad (1.7)$$

Instead of solving for every stationary point, we integrate this object over specified product cycles:

$$Z_\Gamma = \frac{1}{(2\pi i)^N} \int_\Gamma S_N(\mathbf{R}) d^N R. \quad (1.8)$$

Its exact logarithmic derivative is the period barycenter

$$E_\Gamma^{\text{per}} = -\frac{\partial}{\partial(1/V)} \log Z_\Gamma. \quad (1.9)$$

Because the integrand is meromorphic, the periods reduce to finite residue sums. For the all-pole cycle the result is an exactly factorized Selberg–Nørlund–Rice sum. More selective cycles lead to determinant insertions, complementary kernels, and nested-contour multiplicities.

This replacement changes the technical problem. The original Richardson equations couple $N = O(M)$ moving roots. The period formulation first produces exact finite- M sums and determinants, and only then applies large- M asymptotics to positive, sign-corrected

components. It therefore does not assume in advance that the roots lie on one or several continuum arcs. Nor does it introduce a collective Hubbard–Stratonovich field at the Hamiltonian level. The method is not claimed to supersede those approaches. Its value is an independent exact finite algebra and an explicit description of specified contour sectors. The finite algebra is not itself a proof of the physical spectrum. Contour selection and Gaussian control remain assumptions in [Section 14.6](#), and the additional Hessian-to-spectral-shift identification is stated in [Assumption 15.2](#).

The earlier stages reviewed in [Section 1.1](#) used a probabilistic language for the same meromorphic object [\[1–3\]](#). The present construction retains their useful contour idea but replaces that interpretation by exact signed residue algebra. The residue weights are signed. Consequently, the finite sums are not probability distributions, and derivatives of their logarithms are not ordinary cumulants of a positive measure. In the original squared-Vandermonde construction, positivity is used after an explicit sign correction of complementary-kernel minors. This distinction preserves the exact integral construction without assigning probabilities to signed residues. The doubled-inverse-temperature construction of [Chapter 20](#) supplies an additional, separately proved positive compact representation of the full fourth-power period. That representation does not make its individual residues positive and does not itself identify the physical saddle.

1.5 Two contour regimes

The ground-state Richardson roots do not have one universal finite-size topology. At sufficiently strong coupling for a given filling, the roots relevant to the ground branch are detached from the one-particle axis and form complex-conjugate configurations. The corresponding residue description uses the all-pole homology class. We call this *Case A*.

At weaker coupling in a central range of fillings, a macroscopic set of roots remains real: one root lies below the active window and the others are trapped between consecutive one-particle levels. The proposed residue description then uses a nested cascade of contours with lower cutoffs

$$1, 2, \dots, k, \quad k = \kappa M + o(M). \quad (1.10)$$

This is *Case B*. The exact finite- M cascade period is not postulated from a continuum root density. It follows from a counting identity for the admissible residue assignments, and it can be expanded or resummed through the same complementary kernel that controls the marked-root sector.

Both contour families have the same weak-coupling ground label. The physical ground branch is unique, but the equality of labels does not prove that either contour localizes on it. Complete endpoint resummation proves the small effect of the cascade on period barycenters. The corresponding physical interpretation requires the separate contour bridge, and the equality of the rate-defined endpoint with a physical real-root density remains to be established.

1.6 Three distinct crossovers

Three notions that can appear similar in a root plot must be separated.

1.6.1 The constrained-minimum crossover

The quasiparticle dispersion derived from the periods is

$$\mathcal{E}(\xi) = \sqrt{(\xi - \tilde{\mu})^2 + \Delta^2}, \quad 0 \leq \xi \leq \Omega. \quad (1.11)$$

When $\tilde{\mu}$ crosses the lower edge, the constrained minimum moves from $\xi = 0$ into the interior. The crossing occurs at

$$\nu_0 = \frac{\sigma}{1 + \sigma}, \quad \sigma = e^{-2/v}. \quad (1.12)$$

The minimum is continuous through its first derivative but not its second derivative. This is a kinematic change in the location of a minimum, not a change in the ground-root homology class.

1.6.2 The root-topology crossover

The Case A–B transition is controlled by a different rate function. At the lower edge,

$$\Phi_0(\nu, \sigma) = \nu \log \sigma - \log(1 - \sigma) - \nu \log \nu - (1 - \nu) \log(1 - \nu). \quad (1.13)$$

The rate-defined cascade endpoint is zero for $\Phi_0 \geq 0$ and positive for $\Phi_0 < 0$. This is an analytic statement about the period rate, not yet a theorem about the number of physical real roots. At the chemical-potential crossing,

$$\Phi_0(\nu_0, \sigma) = \log \frac{1 + \sigma}{1 - \sigma} > 0, \quad (1.14)$$

so the two crossovers are parametrically and conceptually distinct.

1.6.3 The nonuniform mesoscopic weak-coupling limit

All main thermodynamic theorems keep $v > 0$ fixed. Then $M\sigma \rightarrow \infty$, and the binding scale of an isolated pair is resolved by increasingly many levels. A different problem arises if

$$M\sigma = O(1). \quad (1.15)$$

In this joint limit the isolated-pair binding energy competes directly with the level spacing. Several estimates that are uniform at fixed v cease to be uniform, and the hierarchy between d , $d/\log M$, and higher powers of d must be reconsidered. The earlier period chapters identify this regime without a uniform solution. Chapters 23–25 now give separate half-filled self-dual energy bounds, selected joint-limit results, and a finite excited-state response formulation. The boundary scaling curves and the full-spectrum ordering problem are not thereby completely solved.

1.7 Excitation sectors and gap conventions

The word “gap” is potentially ambiguous in a number-conserving Richardson problem. We distinguish four quantities.

First,

$$\Delta = 2\Omega \frac{\sqrt{\sigma\nu(1-\nu)}}{1-\sigma} \quad (1.16)$$

is the parameter entering the square-root dispersion. It equals the minimum quasiparticle energy only when $\tilde{\mu}$ lies inside the active window.

Second, the odd or parity threshold is

$$\Delta_{\text{qp}} = \min_{0 \leq \xi \leq \Omega} \mathcal{E}(\xi). \quad (1.17)$$

It is obtained from a blocked-level sector after the chemical work has been subtracted.

Third, a neutral seniority-zero excitation can contain one real Richardson root trapped between levels. Its threshold is denoted Δ_R .

Fourth, a neutral seniority-two state contains two blocked levels and represents a broken pair. Its threshold is denoted Δ_{B_2} .

In the thermodynamic limit,

$$\Delta_R = \Delta_{B_2} = 2\Delta_{\text{qp}} + o(1). \quad (1.18)$$

Under the finite-defect and prefactor estimates stated later, the equality persists through order M^{-1} if the minimum is interior. At a Debye edge the two sectors split at order $d = \rho^{-1}$, with a logarithmically slow $d/\log M$ approach to the limiting coefficient.

1.8 Questions and current answers

1. The exact all-pole period and the complete nested cascade can be evaluated without solving a continuum root problem. Their physical interpretation is a separate step.
2. Marked and blocked periods produce the square-root spectral scale and an exact particle-hole convolution. Turning the corresponding logarithmic derivatives into named excitation energies requires control of the selected contours.
3. The constrained minimum of the candidate dispersion is C^1 , but not C^2 , at an active-window edge. This kinematic fact does not establish smoothness of every response function.
4. The period cascade has a rate-defined Lambert birth law. Its full resummation has a small effect on the ground and marked period barycenters. Identification with a physical real-root cascade remains an open bridge.
5. The proposed finite-size physical formulas are consistent with known limits, but their new Gaussian and finite-defect derivation is conditional. Revision 01 repairs the separate confluent lattice-minor argument in Appendix J.

1.9 Relation to standard methods

The physical scales obtained here are familiar: the square-root quasiparticle dispersion, the chemical potential, the paired gap parameter, and the dilute-pair binding energy coincide with the corresponding large-system results. The novelty is not a new thermodynamic equation of state for the rational reduced pairing model. The intended contribution is a derivation along the following chain, whose contour-to-physical links remain to be completed:

$$\begin{aligned} \text{meromorphic period} &\longrightarrow \text{finite residue determinant} \longrightarrow \text{positive sign-corrected asymptotics} \\ &\longrightarrow \text{branch identification} \longrightarrow \text{physical observable.} \end{aligned} \quad (1.19)$$

This chain exposes information that a leading mean-field calculation does not organize by itself: the exact pure contour sectors, the Case B cascade, the Lambert- W birth law, the distinction between real-root and broken-pair excitations, and the logarithmically soft edge coordinate.

The dilute-pair interpretation and the distinction between the single-pair binding scale and the many-body spectral gap have also been emphasized in canonical analyses of Cooper pairs [24–26]. The manuscript also makes direct contact with known inverse-size results. In particular, the half-filled $O(1)$ ground-energy correction and the first excitation correction agree with the finite-size expansion obtained from the traditional Richardson analysis [23].

Here the known spectral-shift expressions are retained for comparison with a proposed Gaudin determinant bridge. Appendix K explicitly distinguishes the known Richardson formulas from the missing new determinant identification. Macroscopic exact expectation values and multiple-gap Richardson states provide a broader context for why retaining the exact branch structure can matter even when the leading equilibrium energy is mean-field-like [27].

1.9.1 Which continuum assumptions the duality route avoids

The usual electrostatic continuum treatment introduces a limiting support and density of rapidities, and solves the resulting field problem with an analytic ansatz. For example, the derivation reviewed in Ref. [22, Secs. V–VI] explicitly assumes a single arc in obtaining the standard gap and number equations. Control of that support and of the passage from finite roots to its continuum field is an additional issue, especially when finite-size errors or changing root configurations matter.

The new self-dual route avoids these particular inputs. It derives the finite half-plane localization of the ground roots, rewrites their polynomial identity on an auxiliary real boundary, and establishes comparison bounds before taking a thermodynamic limit. Its profile is not an assumed rapidity density. This gives a way to recover the leading BCS results without relying on an unproved root-support or resolvent ansatz in that derivation. It does not challenge the established BCS limit or erase the controlled results of the traditional finite-size expansion [23]. Its own limiting and spectral-ordering obligations are stated in the new chapters.

1.10 Organization of the manuscript

Chapter 2 fixes the Hamiltonian, scaling conventions, energy scales, and result map. Chapters 3–6 develop the exact finite- M period algebra. Chapters 7–10 treat Case A, electron–hole duality, blocked levels, and finite seniority. Chapters 11–13 construct and resum Case B and analyze the birth of the cascade. Chapter 14 assigns physical branch labels and states the unresolved contour-selection conditions. Chapters 15–17 add the Gaudin one-loop bridge and the finite-size excitation spectrum. Chapter 18 collects the conditional physical formulas, and Chapter 19 summarizes the methodological outcome and open extensions.

A reader interested only in the leading energy and thresholds may proceed from Chapter 2 directly to Chapter 18. A reader checking the new contour algebra should follow Chapters 3–13 and Appendices A–J. The subextensive and finite-size results depend additionally on Chapters 14–17 and Appendices K–N.

The addition in Chapter 20 develops the doubled-inverse-temperature extension for the common all-pole Case A period, including the moment Pfaffian, its positive compact representation, and fixed-spectrum coupling comparisons. Its finite identities permit arbitrary distinct real levels; they do not extend the mixed-contour Case B analysis at the higher inverse temperature.

1.11 Additions in the third edition

Chapter 21 gives a normalized Jack-hypergeometric formula at arbitrary even auxiliary β and an exact vector route to the physical ground state when the multiplier tends to infinity at fixed system size. Chapter 22 develops the alternative one-pair expansion at $\beta = 2$. Its new all-order coefficient theorem explains the cancellations in the earlier integral approach. The familiar thermodynamic energy and square-root spectral scales are retained as previously

known results; the new contribution is the stated derivation and the control or explicit qualification of each transfer.

1.12 Additions in the fourth edition

The title now makes explicit the second principal method: particle–hole duality. Chapter 23 develops the finite self-dual polynomial identity, its physical ground branch, and a real boundary representation with an exact energy functional. The leading BCS energy is obtained by comparison estimates, without a guessed continuum support for the complex rapidities. Chapter 24 then treats a finite blocked defect and the first self-dual pair-preserving excitation, and separates the energy and spectral criteria for bulk BCS accuracy. The Cooper-pair binding interpretation follows Ref. [26]; the earlier mesoscopic scale of Ref. [20] and the finite-size spectrum of Ref. [23] provide essential comparisons.

Chapter 25 gives the latest finite reduction: one positive Richardson rapidity, a smooth nonlinear real response of the remaining roots, a scalar spectral condition, and an energy integral. It includes numerical checks in the middle of the crossover and a candidate universal limiting system. Exact finite relations are not identified with a theorem on that entire limiting system. The first self-dual excitation is not silently promoted to the minimum of the full pair-preserving spectrum. Further development can address this ordering, controlled crossover limits, and more general spectra or dual sectors.

Chapter 2

Model, scaling limit, and result map

Remark 2.1 (Revision 01 interpretation). The exact formulas below define periods, while the stated physical energy and finite-size gap formulas inherit the contour-selection, Gaussian spectral-shift, and finite-defect conditions of Chapters 14–17. The blocked-period confluent prefactor is also a pending estimate. A branch label alone does not supply any of these estimates. Case A/B and κ are first defined through period rates; their physical root-count interpretation is conjectural.

2.1 Reduced pairing Hamiltonian

Let $c_{n\sigma}^\dagger$ create a fermion in the doubly degenerate level n with time-reversal label $\sigma = \uparrow, \downarrow$. The reduced pairing Hamiltonian is

$$H = \sum_{n=0}^M \sum_{\sigma} \varepsilon_n c_{n\sigma}^\dagger c_{n\sigma} - V \sum_{n,m=0}^M c_{n\uparrow}^\dagger c_{n\downarrow}^\dagger c_{m\downarrow} c_{m\uparrow}, \quad V > 0. \quad (2.1)$$

Introduce hard-core pair operators

$$b_n^\dagger = c_{n\uparrow}^\dagger c_{n\downarrow}^\dagger, \quad b_n = c_{n\downarrow} c_{n\uparrow}. \quad (2.2)$$

In the seniority-zero N -pair sector,

$$H_N = \sum_{n=0}^M 2\varepsilon_n b_n^\dagger b_n - V \sum_{n,m=0}^M b_n^\dagger b_m. \quad (2.3)$$

Blocked singly occupied levels are removed from the active pair sums and contribute their one-particle energies separately.

The one-particle spectrum is

$$\boxed{\varepsilon_n = \varepsilon_{F0} + \frac{n}{\rho}, \quad n = 0, \dots, M.} \quad (2.4)$$

There are $M + 1$ active levels. Throughout the manuscript M denotes the last index and $\Omega = M/\rho$ the width between the lowest and highest one-particle energies. This convention is responsible for several genuine $O(M^{-1})$ shifts and must not be replaced by an informal identification of M with the number of levels.

2.2 Richardson equations and sector energies

For a seniority-zero eigenstate with N pairs, Richardson's ansatz introduces N spectral parameters R_j . In the normalization used here they satisfy

$$\boxed{\frac{1}{V} = \sum_{n=0}^M \frac{1}{2\varepsilon_n - R_j} + 2 \sum_{\ell \neq j} \frac{1}{R_j - R_\ell}, \quad j = 1, \dots, N.} \quad (2.5)$$

The energy is

$$\boxed{E^{\text{RG}} = \sum_{j=1}^N R_j} \quad (2.6)$$

in the unblocked sector.

If s levels $p_1, \dots, p_s$ are singly occupied and n active pairs remain, the same equations hold with those levels omitted from the one-body sum, and

$$E_{n;p}^{\text{RG}} = \sum_{j=1}^n R_j + \sum_{i=1}^s \varepsilon_{p_i}. \quad (2.7)$$

The conserved seniority s therefore separates the odd, broken-pair, and pair-preserving excitation sectors.

2.3 Thermodynamic sequence and dimensionless variables

The principal limit is

$$M, N, \rho \longrightarrow \infty, \quad \frac{N}{M} \longrightarrow \nu \in (0, 1), \quad \frac{M}{\rho} \longrightarrow \Omega \in (0, \infty), \quad (2.8)$$

with

$$v = \rho V \quad (2.9)$$

fixed and positive. We use

$$\boxed{\sigma = \exp(-2/v), \quad x = \frac{\sigma}{1 - \sigma}, \quad d = \frac{1}{\rho} = \frac{\Omega}{M}.} \quad (2.10)$$

The exact finite electron-hole transformation is

$$\boxed{\bar{N} = M + 1 - N.} \quad (2.11)$$

Consequently,

$$\frac{\bar{N}}{M} = 1 - \frac{N}{M} + \frac{1}{M}. \quad (2.12)$$

The M^{-1} term must be retained whenever a first finite-size coefficient is transformed from one edge to the other.

The derivative conversions used throughout are

$$-\frac{\partial}{\partial(1/V)} = \frac{2\sigma}{\rho} \frac{\partial}{\partial\sigma} = \frac{2x}{\rho(1 - \sigma)} \frac{\partial}{\partial x}. \quad (2.13)$$

2.4 Thermodynamic spectral curve

The period asymptotics and branch continuation lead to the two parameters

$$\tilde{\mu} = \Omega \frac{\nu(1+\sigma) - \sigma}{1-\sigma} = \Omega[\nu(1+2x) - x], \quad (2.14)$$

$$\Delta = 2\Omega \frac{\sqrt{\sigma\nu(1-\nu)}}{1-\sigma} = 2\Omega \sqrt{x(1+x)\nu(1-\nu)}. \quad (2.15)$$

The one-quasiparticle dispersion is

$$\mathcal{E}(\xi) = \sqrt{(\xi - \tilde{\mu})^2 + \Delta^2}, \quad 0 \leq \xi \leq \Omega. \quad (2.16)$$

At the edges,

$$\mathcal{E}(0) = \Omega(\nu + x), \quad \mathcal{E}(\Omega) = \Omega(1 - \nu + x). \quad (2.17)$$

Three gap quantities are kept distinct:

$$\Delta, \quad \Delta_{\text{qp}} = \min_{[0, \Omega]} \mathcal{E}, \quad \Delta_{\text{even}} = 2\Delta_{\text{qp}}. \quad (2.18)$$

The sector-resolved neutral thresholds are Δ_R for a real-root seniority-zero excitation and Δ_{B_2} for a two-blocked-level seniority-two excitation.

2.5 Case A, Case B, and their endpoint rate

Case A is the all-pole contour regime of the ground branch. Case B contains a consecutive nested cascade with $k = \kappa M + o(M)$ trapped contours. The lower edge rate is

$$\Phi_0(\nu, \sigma) = \nu \log \sigma - \log(1 - \sigma) - \nu \log \nu - (1 - \nu) \log(1 - \nu). \quad (2.19)$$

For the lower half of the filling interval,

$$\Phi_0 \geq 0 \implies \text{Case A}, \quad (2.20)$$

whereas

$$\Phi_0 < 0 \implies \text{Case B with } \Phi(\kappa) = 0. \quad (2.21)$$

The upper case is obtained by the exact reflected-complement transformation.

At half filling the topology crossover disappears above

$$\sigma_c = 3 - 2\sqrt{2}, \quad v_c = \frac{1}{\log(1 + \sqrt{2})}. \quad (2.22)$$

Near its birth on the Case B side,

$$\kappa = -\frac{\delta_\Phi}{W_{-1}[-\delta_\Phi/(e\Lambda)]} \sim \frac{\delta_\Phi}{\log(1/\delta_\Phi)}, \quad \delta_\Phi = -\Phi_0 > 0. \quad (2.23)$$

2.6 Principal physical statements

The detailed proofs are distributed through the manuscript. This section records the final formulas and their domains.

2.6.1 Ground-state energy

The exact all-pole period barycenter is

$$E_{\text{all}}^{\text{per}} = 2N\varepsilon_{F0} + \frac{N(N-1)}{\rho} - \frac{2N(M-N+1)}{\rho} \frac{\sigma}{1-\sigma}. \quad (2.24)$$

The physical ground-state energy through order M^0 is

$$E_0^{\text{RG}} = E_{\text{all}}^{\text{per}} + \delta E_{\text{fl}}(\nu, v) + o(1), \quad (2.25)$$

where δE_{fl} is the one-dimensional Gaudin spectral-shift integral of Chapter 15. The same formula holds on both sides of the Case A–B line.

2.6.2 Odd and even thresholds

The leading odd threshold is

$$\Delta_{\text{qp}} = \min_{0 \leq \xi \leq \Omega} \sqrt{(\xi - \tilde{\mu})^2 + \Delta^2}. \quad (2.26)$$

Explicitly,

$$\Delta_{\text{qp}} = \begin{cases} \Omega(\nu + x), & 0 < \nu \leq \frac{\sigma}{1+\sigma}, \\ \Delta, & \frac{\sigma}{1+\sigma} \leq \nu \leq \frac{1}{1+\sigma}, \\ \Omega(1 - \nu + x), & \frac{1}{1+\sigma} \leq \nu < 1. \end{cases} \quad (2.27)$$

The leading neutral threshold is

$$\Delta_{\text{even}} = 2\Delta_{\text{qp}}. \quad (2.28)$$

2.6.3 First finite-size corrections

If $0 < \tilde{\mu} < \Omega$, the two neutral sectors coincide through order d :

$$\Delta_R = \Delta_{B_2} = 2\Delta + d \left[-(1+2x) + \frac{\Delta}{\Omega(1-\nu)} + \mathcal{H}(\tilde{\mu}) \right] + O(d^2). \quad (2.29)$$

The asymmetric coefficient reflects the exact finite convention $\nu = N/M$; its reflected expression must use $\bar{N} = M + 1 - N$.

If $\tilde{\mu} < 0$, define

$$v_- = -\frac{\tilde{\mu}}{\mathcal{E}(0)}, \quad L_-(M) = \log M + \gamma + \mathcal{C}_-. \quad (2.30)$$

Then

$$\Delta_{B_2} - \Delta_R = dv_- \left[1 - \frac{2}{L_-(M)} \right] + O\left(\frac{d}{L_-(M)^2} + d^2\right). \quad (2.31)$$

The upper-edge formula is the exact electron–hole image.

2.7 Result map R1–R15

The results form five logical blocks, but each permanent identifier is listed separately so that the map remains useful if theorem numbering changes.

ID	Level	Compact content	Location
R1	Exact identity	Exact all-pole Selberg–Nørlund–Rice factorization and exact finite- M period energy.	4
R2	Exact identity	Krawtchouk-tail complementary kernel, factorization $Q^{(k)}J = FD_1CD_2BD_3$, and positive sign-corrected minors.	5
R3	Exact identity	Pure excluded-pole periods, local marked sector, arbitrary nested cutoffs, Möbius expansion, and determinant generator.	6
R4	Period asymptotics	Case A marked-root square root, full interior prefactor, discrete edge layer, matching, and finite staircase.	7
R5	Exact symmetry	Finite- M reflected-complement duality of weights, kernels, pure sectors, marked charts, and energy shifts.	8
R6	Exact identity / asymptotics	Blocked-level Heine polynomial and odd particle/hole quasiparticle energy.	9
R7	Exact identity / asymptotics	Finite-seniority Heine–Christoffel determinants and exact particle–hole convolution of the real-root period.	10
R8	Exact identity	Complete finite ground, marked, and finite-seniority periods over the physical consecutive Case B cascade.	11
R9	Period asymptotics	Complete endpoint resummation: the cascade changes the ground and separated marked periods only beyond the orders controlling the energy and gap.	12
R10	Topology asymptotics	Lambert- W_{-1} cascade birth, $(\log M)/M$ onset window, mesoscopic regimes, and root-count scaling.	13
R11	Branch identification	Weak-coupling labels, regular eigenvalue-based continuation, global ground uniqueness, and Baxter diagnostics.	14
R12	Physical consequence	Gaudin spectral-shift bridge and complete $O(M^0)$ correction to the physical ground energy, under B_0 .	15
R13	Physical consequence	Common interior $O(M^{-1})$ correction to the real-root and broken-pair thresholds through $\mathcal{H}$, under B_0, B_{def} .	16
R14	Physical consequence	Edge soft root, $d/\log M$ scale, and first finite-size splitting of the two neutral sectors, under the finite-defect bridge.	17
R15	Physical synthesis	Unified ground energy, odd and even thresholds, Cases A/B, two crossovers, and finite-size hierarchy; subleading physical terms use the bridge conditions.	18

The permanent registry in the front matter gives a more granular statement of each result and remains the authoritative cross-reference if theorem numbering changes.

2.8 Hierarchy of errors and uniformity

Five orders must be kept separate:

$$O(M), \quad O(1), \quad O(d), \quad O\left(\frac{d}{\log M}\right), \quad O(d^2). \quad (2.32)$$

They enter different observables:

Order	Principal origin	Observable
$O(M)$	Exact all-pole exponent	Ground-state energy
$O(1)$	Gaudin one-loop determinant	Physical minus period ground energy
$O(d)$	Marked/blocked prefactors and finite-defect response	First excitation correction
$O(d/\log M)$	Soft real root in the first edge interval	Approach to edge-sector splitting
$O(d^2)$	Interior difference between real-root and broken-pair sectors; cascade corrections	Next finite-size structure

The principal asymptotic statements assume fixed $v > 0$, fixed $\nu \in (0, 1)$, and compact separation from any boundary explicitly excluded in the relevant theorem. The Case B estimates are uniform in compact interiors and are matched separately to the birth region $\kappa \downarrow 0$. Edge formulas are one-sided and use their own logarithmic scaling. These period estimates do not supply a uniform statement in the joint limit $M\sigma = O(1)$. The separate half-filled self-dual joint-limit results are stated in Chapter 24.

2.9 Suggested reading routes

Physical results only. Read this chapter, Chapter 18, and the conclusions.

Exact period algebra. Read Chapters 3–6, followed by Chapters 11–13.

Odd and even excitations. Read Chapters 7, 9, and 10, then the finite-size Chapters 16–17.

Physical branch identification and subleading terms. Read Chapters 14–17 and Appendices K–N.

Supporting computations. Appendix N lists the supporting programs. Their commands and scope are documented in `code/README.md`. They provide exact-arithmetic tests, asymptotic comparisons, continuation diagnostics, and spectral-shift calculations. The opening authorship statement specifies their LLM provenance and the limits of human verification.

2.10 Case A Pfaffian extension in version 1.1

The added Chapter 20 forms a separate exact-period extension at doubled auxiliary inverse temperature. Its finite moment Pfaffian, explicit residue entries, positive quaternionic compact integral, and fixed-spectrum coupling limits are recorded as R16–R18. They do not inherit or

establish a higher-inverse-temperature mixed Case B cascade. The physical bridge conditions above are retained.

2.11 Arbitrary even exponent and the one-pair route in version 1.2

The additions are logically distinct. Chapter 21 provides an exact finite-system special-function formula and a physical large- p limit through a vector identity. Chapter 22 proves a coefficientwise all-order cancellation at $p = 1$, and records the period-level thermodynamic conclusions for ground, blocked, and marked sectors. The latter chapter explicitly states what the Case B ground estimate proves and what requires a blocked-sector extension. These results do not assert that the finite- p thermodynamic physical bridges listed above have all been completed.

2.12 Direct self-dual results in version 1.3

Results R23–R26 start with the finite polynomial identity rather than a period barycenter. Their half-filled convention is specified in the front-matter dictionary and in Chapter 23. The finite branch arguments lead to a real boundary formulation, controlled leading energy and self-dual gap limits, and a direct scalar-response scheme for the first excited branch. The physical interpretation is established within that specified sector; it neither requires nor completes the general period bridges listed above. For this independent reading path, proceed through Chapters 23–25.

Part II

Exact finite- M residue algebra

Chapter 3

Meromorphic Richardson periods

3.1 The meromorphic differential form

The one-particle levels in the interaction window are

$$\varepsilon_n = \varepsilon_{F0} + \frac{n}{\rho}, \quad n = 0, 1, \dots, M, \quad (3.1)$$

so that the poles relevant to a pair energy are $2\varepsilon_n$. For N pairs define

$$\mathcal{S}_N(R_1, \dots, R_N; V) = \frac{\prod_{1 \leq i < j \leq N} (R_i - R_j)^2}{\prod_{j=1}^N \prod_{n=0}^M (2\varepsilon_n - R_j)} \exp \left[-\frac{1}{V} \sum_{j=1}^N R_j \right]. \quad (3.2)$$

The only finite singular hyperplanes are $R_j = 2\varepsilon_n$. The diagonals $R_i = R_j$ are zeros, not singularities.

Let γ_j be a positively oriented finite union of closed contours in $\mathbb{C} \setminus \{2\varepsilon_0, \dots, 2\varepsilon_M\}$ and put

$$\Gamma = \gamma_1 \times \dots \times \gamma_N. \quad (3.3)$$

The normalized period is

$$Z_\Gamma(V) = \int_\Gamma \mathcal{S}_N(R; V) \prod_{j=1}^N \frac{dR_j}{2\pi i}. \quad (3.4)$$

Because the contours are compact, differentiation with respect to $1/V$ may be performed under the integral sign. Contour deformation is allowed as long as no pole is crossed. Later chapters will use all-pole contours, excluded-pole contours, and nested systems of contours; all are special cases of (3.3).

Remark 3.1 (Overall contour normalization). The factors $(2\pi i)^{-N}$ are included so that residue formulas are literal equalities. Multiplication of Z_Γ by a nonzero V -independent constant never affects its logarithmic derivative, but such constants will nevertheless be displayed when the all-pole period is evaluated in [Chapter 4](#).

3.2 Exact logarithmic-derivative identities

Write

$$u = \frac{1}{V}. \quad (3.5)$$

Whenever $Z_\Gamma \neq 0$, define the period average of a symmetric meromorphic insertion $A(R)$ by

$$\langle A \rangle_\Gamma = \frac{1}{Z_\Gamma} \int_\Gamma A(R) \mathcal{S}_N(R; V) \prod_{j=1}^N \frac{dR_j}{2\pi i}. \quad (3.6)$$

This notation is algebraic; it does not imply that the integrand is a probability density.

Exact identity 3.2 (Period barycenter). *For every admissible contour class and every value of V for which $Z_\Gamma \neq 0$,*

$$E_\Gamma^{\text{per}} := -\frac{\partial}{\partial u} \log Z_\Gamma = \left\langle \sum_{j=1}^N R_j \right\rangle_\Gamma. \quad (3.7)$$

Equivalently, $-\partial_u \log Z_\Gamma$ means $-(\partial_u Z_\Gamma)/Z_\Gamma$ and therefore requires no choice of a branch of the logarithm.

Proof. The u dependence of (3.2) is entirely contained in $\exp[-u \sum_j R_j]$. Hence

$$\partial_u Z_\Gamma = - \int_\Gamma \left(\sum_j R_j \right) \mathcal{S}_N(R; V) \prod_j \frac{dR_j}{2\pi i},$$

which gives (3.7) after division by Z_Γ . □

Higher logarithmic derivatives generate algebraic connected moments. In particular,

$$\frac{\partial^2}{\partial u^2} \log Z_\Gamma = \left\langle \left(\sum_j R_j \right)^2 \right\rangle_\Gamma - \left\langle \sum_j R_j \right\rangle_\Gamma^2. \quad (3.8)$$

For a positive real measure the right-hand side would be a nonnegative variance. In the present meromorphic problem it need not be nonnegative; an explicit negative value for the all-pole period will be obtained in [Corollary 4.8](#).

3.3 Stationary points and the Richardson equations

The logarithmic derivative of (3.2) with respect to R_j is

$$\frac{\partial}{\partial R_j} \log \mathcal{S}_N = -\frac{1}{V} + \sum_{n=0}^M \frac{1}{2\varepsilon_n - R_j} + 2 \sum_{\substack{\ell=1 \\ \ell \neq j}}^N \frac{1}{R_j - R_\ell}. \quad (3.9)$$

Therefore the stationary points satisfy

$$\frac{1}{V} = \sum_{n=0}^M \frac{1}{2\varepsilon_n - R_j} + 2 \sum_{\ell \neq j} \frac{1}{R_j - R_\ell}, \quad j = 1, \dots, N. \quad (3.10)$$

These are the Richardson equations in the normalization used throughout this manuscript [[12–14](#), [17](#), [28](#)]. An equivalent form is

$$\frac{1}{2V} + \frac{1}{2} \sum_{n=0}^M \frac{1}{R_j - 2\varepsilon_n} - \sum_{\ell \neq j} \frac{1}{R_j - R_\ell} = 0. \quad (3.11)$$

Proposition 3.3 (Common meromorphic origin). *The residue periods and the Richardson saddles are generated by the same finite-dimensional meromorphic form: residues of (3.2) determine Z_Γ , while its stationary points determine solutions of (3.10).*

The proposition does *not* by itself identify a particular period with a particular saddle. A contour period can be a linear combination of steepest-descent contributions, whereas a Richardson eigenstate is associated with a named stationary branch. We therefore maintain the distinction

$$E_{\Gamma}^{\text{per}} \quad \text{versus} \quad E^{\text{RG}} = \sum_{j=1}^N R_j^{(\text{branch})}. \quad (3.12)$$

The branch identification and the Gaussian correction relating these quantities are developed only in [Chapters 14](#) and [15](#). No such identification is used in the exact residue algebra of [Chapters 4](#) to [6](#).

3.4 Coupling variables and derivative conversion

The two coupling variables used below are

$$\sigma = \exp\left(-\frac{2}{\rho V}\right) = e^{-2u/\rho}, \quad x = \frac{\sigma}{1-\sigma}. \quad (3.13)$$

For attractive coupling $V > 0$ one has $0 < \sigma < 1$ and $x > 0$. Differentiation gives

$$\frac{\partial \sigma}{\partial u} = -\frac{2\sigma}{\rho}, \quad \frac{\partial x}{\partial u} = -\frac{2x}{\rho(1-\sigma)}. \quad (3.14)$$

Consequently,

Exact identity 3.4 (Derivative conversion). *For every differentiable function of the coupling,*

$$-\frac{\partial}{\partial(1/V)} = \frac{2\sigma}{\rho} \frac{\partial}{\partial \sigma} = \frac{2x}{\rho(1-\sigma)} \frac{\partial}{\partial x}. \quad (3.15)$$

The σ form is most convenient for the exact all-pole factorization. The x form will be used for the positive polynomial representations of marked and blocked sectors.

3.5 Why the residue representation is signed

Already the one-variable residue at a pole has an alternating sign. Let

$$P_M(R) = \prod_{m=0}^M (2\varepsilon_m - R). \quad (3.16)$$

Using [\(3.1\)](#), one finds

$$\text{Res}_{R=2\varepsilon_n} \frac{e^{-uR}}{P_M(R)} = (-1)^{n+1} \left(\frac{\rho}{2}\right)^M \frac{e^{-2u\varepsilon_n}}{n!(M-n)!}. \quad (3.17)$$

Equivalently, after a common n -independent normalization, the lattice weight is

$$w(n) = (-1)^n \sigma^n \binom{M}{n}. \quad (3.18)$$

Thus the residue sum alternates with n . The Vandermonde square is nonnegative on the real lattice, but it does not remove this sign alternation.

Warning 3.5 (No probability interpretation). The notation (3.6) is a normalized meromorphic average, not a probability expectation. Positivity, variance inequalities, and probabilistic concentration arguments cannot be invoked before a separate sign-corrected positive representation has been constructed. Such a representation will exist for specific complementary-kernel minors, not for the original all-pole residue weight itself.

This distinction does not weaken the residue method. Exact contour identities and positive Laplace representations obtained after finite algebraic transformations remain valid. It only removes the former argument that attempted to infer saddle localization directly from a putative variance of (3.2).

3.6 Logical output of this chapter

The remainder of the exact finite- M analysis uses only the following inputs from this chapter:

1. the normalized period (3.4);
2. the exact barycenter identity (3.7);
3. the Richardson stationary equations (3.10);
4. the derivative conversion (3.15);
5. the signed lattice residue structure (3.18).

No assertion about a dominant saddle is made at this stage.

Chapter 4

The all-pole Selberg–Nørlund–Rice period

4.1 The all-pole contour and its lattice reduction

Let γ_{all} be a positively oriented contour enclosing all pair poles

$$2\varepsilon_n = 2\varepsilon_{F0} + \frac{2n}{\rho}, \quad n = 0, \dots, M, \quad (4.1)$$

and no other singularity. The all-pole product cycle is

$$\Gamma_{\text{all}} = \gamma_{\text{all}}^{\times N}. \quad (4.2)$$

We first evaluate the corresponding exact period, without assigning it to a physical eigenbranch. The terminology reflects the relation of the resulting determinant to Selberg-type products and to finite Nørlund–Rice sums [29–31].

Introduce scaled variables

$$R_j = 2\varepsilon_{F0} + \frac{2z_j}{\rho}. \quad (4.3)$$

The Vandermonde, pole denominator, and measure transform as

$$\prod_{i < j} (R_i - R_j)^2 = \left(\frac{2}{\rho}\right)^{N(N-1)} \prod_{i < j} (z_i - z_j)^2, \quad (4.4)$$

$$\prod_{j=1}^N \prod_{n=0}^M (2\varepsilon_n - R_j) = \left(\frac{2}{\rho}\right)^{N(M+1)} \prod_{j=1}^N \prod_{n=0}^M (n - z_j), \quad (4.5)$$

$$\prod_{j=1}^N \frac{dR_j}{2\pi i} = \left(\frac{2}{\rho}\right)^N \prod_{j=1}^N \frac{dz_j}{2\pi i}. \quad (4.6)$$

Moreover,

$$\exp \left[-\frac{1}{V} \sum_j R_j \right] = \exp \left(-\frac{2N\varepsilon_{F0}}{V} \right) \exp \left[-\frac{2}{\rho V} \sum_j z_j \right]. \quad (4.7)$$

At the pole $z = n$,

$$\text{Res}_{z=n} \frac{\exp[-2z/(\rho V)]}{\prod_{m=0}^M (m - z)} = -\frac{1}{M!} (-1)^n \sigma^n \binom{M}{n}. \quad (4.8)$$

The exact geometric prefactor is therefore

$$\mathcal{C}_{M,N,\rho} = (-1)^N (M!)^{-N} \left(\frac{2}{\rho}\right)^{N(N-M-1)}. \quad (4.9)$$

Exact identity 4.1 (Exact all-pole residue reduction). *For $0 \leq N \leq M + 1$,*

$$Z_{\text{all}} = \mathcal{C}_{M,N,\rho} \exp\left(-\frac{2N\varepsilon_{F0}}{V}\right) z_{M,N}(\sigma), \quad (4.10)$$

where the ordered Selberg–Nørlund–Rice sum is

$$z_{M,N}(\sigma) = \sum_{n_1, \dots, n_N=0}^M \left[\prod_{j=1}^N (-1)^{n_j} \sigma^{n_j} \binom{M}{n_j} \right] \prod_{1 \leq i < j \leq N} (n_j - n_i)^2. \quad (4.11)$$

Proof. Substitution of (4.3) into the normalized period (3.4) gives the common scale factor

$$\left(\frac{2}{\rho}\right)^{N(N-1)-N(M+1)+N} = \left(\frac{2}{\rho}\right)^{N(N-M-1)}.$$

Taking the residue (4.8) independently in every variable produces $(-1)^N (M!)^{-N}$ and the lattice weights in (4.11). Repeated lattice indices do not contribute because the Vandermonde square vanishes. $\square$

The sum is ordered because the integration variables are labelled. Equivalently, define

$$W(S) = \prod_{n \in S} \left[(-1)^n \sigma^n \binom{M}{n} \right] \Delta(S)^2, \quad \Delta(S) = \prod_{\substack{m, n \in S \\ m < n}} (n - m), \quad (4.12)$$

and

$$Z_0 = \sum_{\substack{S \subset \{0, \dots, M\} \\ |S|=N}} W(S). \quad (4.13)$$

Then

$$\boxed{z_{M,N} = N! Z_0.} \quad (4.14)$$

This factor will be important when excluded-pole periods and kernel minors are introduced in Chapters 5 and 6.

4.2 Mixed falling-factorial moments

Write

$$(y)_{\underline{a}} = y(y-1) \cdots (y-a+1), \quad (y)_{\underline{0}} = 1. \quad (4.15)$$

The basic one-variable moment is

$$\mu_{ab} = \sum_{n=0}^M (-1)^n \sigma^n \binom{M}{n} (n)_{\underline{a}} (M-n)_{\underline{b}}. \quad (4.16)$$

We adopt the finite-lattice convention that $(q)_{\underline{r}} = 0$ for nonnegative integers $q < r$.

Lemma 4.2 (Mixed binomial moment). *For nonnegative integers a, b ,*

$$\boxed{\mu_{ab} = (-1)^a \sigma^a (1-\sigma)^{M-a-b} (M)_{\underline{a+b}}.} \quad (4.17)$$

In particular, the right-hand side vanishes when $a+b > M$.

Proof. Consider

$$\begin{aligned}\mathcal{B}(u, v) &:= \sum_{n=0}^M (-1)^n \sigma^n \binom{M}{n} (1+u)^n (1+v)^{M-n} \\ &= [(1+v) - \sigma(1+u)]^M = [(1-\sigma) + v - \sigma u]^M.\end{aligned}\quad (4.18)$$

Since

$$(1+u)^n = \sum_{a \geq 0} \frac{(n)_a}{a!} u^a, \quad (1+v)^{M-n} = \sum_{b \geq 0} \frac{(M-n)_b}{b!} v^b,$$

$a!b!$ times the coefficient of $u^a v^b$ in (4.18) is precisely μ_{ab} . Coefficient extraction on the final polynomial gives (4.17). $\square$

4.3 Vandermonde splitting and discrete Andréief reduction

Let

$$s_N = \binom{N}{2} = \frac{N(N-1)}{2}. \quad (4.19)$$

Because falling factorials are monic polynomials of their degree,

$$\det[(n_i)_{\underline{a}}]_{\substack{i=1,\dots,N \\ a=0,\dots,N-1}} = \prod_{i < j} (n_j - n_i). \quad (4.20)$$

Likewise,

$$\det[(M - n_i)_{\underline{b}}]_{\substack{i=1,\dots,N \\ b=0,\dots,N-1}} = (-1)^{s_N} \prod_{i < j} (n_j - n_i). \quad (4.21)$$

Hence

$$\prod_{i < j} (n_j - n_i)^2 = (-1)^{s_N} \det[(n_i)_{\underline{a}}] \det[(M - n_i)_{\underline{b}}]. \quad (4.22)$$

The finite discrete Andréief identity now gives

$$z_{M,N} = (-1)^{s_N} N! \det[\mu_{ab}]_{a,b=0}^{N-1}. \quad (4.23)$$

A direct permutation proof of this identity is included in [Appendix A](#).

Using (4.17), factor the powers of σ and $1 - \sigma$ by rows and columns:

$$\det[\mu_{ab}] = (-1)^{s_N} \sigma^{s_N} (1 - \sigma)^{NM-2s_N} \det[(M)_{\underline{a+b}}]_{a,b=0}^{N-1}. \quad (4.24)$$

The remaining determinant is elementary.

Lemma 4.3 (Falling-factorial Hankel determinant). *For $0 \leq N \leq M + 1$,*

$$\boxed{\det[(M)_{\underline{a+b}}]_{a,b=0}^{N-1} = (-1)^{s_N} \prod_{a=0}^{N-1} a! (M)_{\underline{a}}.} \quad (4.25)$$

Proof. Use

$$(M)_{\underline{a+b}} = (M)_{\underline{a}} (M - a)_{\underline{b}}. \quad (4.26)$$

Factoring $(M)_{\underline{a}}$ from row a leaves the falling-factorial Vandermonde matrix evaluated at the descending nodes $M, M-1, \dots, M-N+1$. Its determinant is

$$\prod_{0 \leq i < j \leq N-1} [(M-j) - (M-i)] = (-1)^{s_N} \prod_{j=1}^{N-1} j!,$$

which proves (4.25). $\square$

The two signs in (4.24) and (4.25) cancel, so

$$\det[\mu_{ab}] = \sigma^{s_N} (1 - \sigma)^{NM - 2s_N} \prod_{a=0}^{N-1} a!(M)_{\underline{a}}. \quad (4.27)$$

4.4 Exact all-pole factorization

Exact identity 4.4 (All-pole factorization, R1). *For $0 \leq N \leq M + 1$,*

$$z_{M,N}(\sigma) = A_{M,N} \sigma^{N(N-1)/2} (1 - \sigma)^{NM - N(N-1)}, \quad (4.28)$$

where

$$A_{M,N} = (-1)^{N(N-1)/2} N! \prod_{a=0}^{N-1} a!(M)_{\underline{a}}. \quad (4.29)$$

Equivalently, with

$$\text{sf}(K) = \prod_{j=0}^{K-1} j!, \quad \text{sf}(0) = 1, \quad (4.30)$$

one has

$$A_{M,N} = (-1)^{N(N-1)/2} (M!)^N \frac{\text{sf}(N+1) \text{sf}(M-N+1)}{\text{sf}(M+1)}. \quad (4.31)$$

Proof. Substitute (4.27) into (4.23). Since $NM - 2s_N = NM - N(N-1)$, the powers of the coupling variables agree with (4.28), and the remaining constant is (4.29). Finally,

$$N! \prod_{a=0}^{N-1} a! = \text{sf}(N+1)$$

and

$$\prod_{a=0}^{N-1} (M)_{\underline{a}} = (M!)^N \frac{\text{sf}(M-N+1)}{\text{sf}(M+1)},$$

which gives (4.31). □

Remark 4.5 (One-step superfactorial shift). With convention (4.30), the numerator is $\text{sf}(N+1)$, not $\text{sf}(N)$. This is the ordered factor $N!$ in (4.14). Omitting it produces the common one-step shift in the normalization constant.

4.5 Exact all-pole period energy

The factor $\mathcal{C}_{M,N,\rho} A_{M,N}$ in

$$Z_{\text{all}} = \mathcal{C}_{M,N,\rho} A_{M,N} \exp\left(-\frac{2N\varepsilon_{F0}}{V}\right) \sigma^{N(N-1)/2} (1 - \sigma)^{N(M-N+1)} \quad (4.32)$$

is independent of V . Therefore only the three displayed coupling-dependent factors contribute to the logarithmic derivative.

Exact identity 4.6 (Exact all-pole period energy). *The all-pole period barycenter is*

$$E_{\text{all}}^{\text{per}} = 2N\varepsilon_{F0} + \frac{N(N-1)}{\rho} - \frac{2N(M-N+1)}{\rho} \frac{\sigma}{1-\sigma}. \quad (4.33)$$

Equivalently, using $\Omega = M/\rho$,

$$E_{\text{all}}^{\text{per}} = 2N\varepsilon_{F0} + \frac{N(N-1)}{\rho} - 2N \left(\Omega - \frac{N-1}{\rho} \right) \frac{\sigma}{1-\sigma}. \quad (4.34)$$

Proof. From (4.32),

$$\log Z_{\text{all}} = \text{const} - \frac{2N\varepsilon_{F0}}{V} + \frac{N(N-1)}{2} \log \sigma + N(M-N+1) \log(1-\sigma).$$

Using

$$\frac{\partial \log \sigma}{\partial(1/V)} = -\frac{2}{\rho}, \quad \frac{\partial \log(1-\sigma)}{\partial(1/V)} = \frac{2\sigma}{\rho(1-\sigma)},$$

and the barycenter identity (3.7) gives (4.33). $\square$

4.6 Thermodynamic density and the explicit period-level constant

Choose the standard finite-size sequence

$$\frac{M}{\rho} = \Omega \quad (4.35)$$

exactly, and define the finite filling

$$\nu_M = \frac{N}{M}. \quad (4.36)$$

Then (4.33) can be rewritten without approximation as

$$E_{\text{all}}^{\text{per}} = M e_{\text{all}}^{\text{per}}(\nu_M) - \Omega \nu_M (1 + 2x), \quad (4.37)$$

where

$$e_{\text{all}}^{\text{per}}(\nu) = 2\nu\varepsilon_{F0} + \Omega\nu^2 - 2\Omega\nu(1-\nu)x. \quad (4.38)$$

Thus, for $\nu_M \rightarrow \nu \in (0, 1)$,

$$\frac{E_{\text{all}}^{\text{per}}}{M} \longrightarrow 2\nu\varepsilon_{F0} + \Omega\nu^2 - 2\Omega\nu(1-\nu) \frac{\sigma}{1-\sigma}. \quad (4.39)$$

Remark 4.7 (Meaning of the $O(1)$ term). The explicit constant $-\Omega\nu_M(1+2x)$ in (4.37) belongs to the exact period barycenter. It is not yet the complete $O(1)$ term of a named physical Richardson branch. The latter also contains the Gaussian Gaudin-determinant correction derived in [Chapter 15](#).

4.7 A direct diagnostic of the signed measure

The exact factorization also provides a concise check of the warning in [Warning 3.5](#).

Corollary 4.8 (Negative second connected moment). *For $1 \leq N \leq M$,*

$$\boxed{\frac{\partial^2}{\partial(1/V)^2} \log Z_{\text{all}} = -\frac{4N(M-N+1)}{\rho^2} \frac{\sigma}{(1-\sigma)^2} < 0.} \quad (4.40)$$

Proof. Only the factor $(1-\sigma)^{N(M-N+1)}$ has a nonzero second derivative with respect to $1/V$. Since

$$\frac{\partial^2}{\partial(1/V)^2} \log(1-\sigma) = -\frac{4\sigma}{\rho^2(1-\sigma)^2},$$

formula (4.40) follows. $\square$

By (3.8), the left-hand side is the algebraic second connected moment of $\sum_j R_j$. Its strict negativity rules out an interpretation as the variance of a positive real probability distribution. No conclusion about the validity of the exact period algebra follows from this sign; it only fixes the permissible type of asymptotic argument.

4.8 Boundary checks and conventions

Several limiting cases are useful for auditing conventions.

One pair. For $N = 1$,

$$z_{M,1} = \sum_{n=0}^M (-1)^n \sigma^n \binom{M}{n} = (1-\sigma)^M, \quad (4.41)$$

and

$$E_{\text{all}}^{\text{per}} = 2\varepsilon_{F0} - 2\Omega \frac{\sigma}{1-\sigma}. \quad (4.42)$$

Full lattice occupancy. For $N = M + 1$, the exponent $N(M - N + 1)$ vanishes, so the all-pole period barycenter is independent of V :

$$E_{\text{all}}^{\text{per}} = 2N\varepsilon_{F0} + \frac{N(N-1)}{\rho}. \quad (4.43)$$

This is a statement about the period barycenter, not an identification with the finite- M saddle energy.

Ordered versus unordered sums. The determinant factorization naturally evaluates the ordered sum (4.11). The subset sum (4.13) differs by exactly $N!$, not by an asymptotic factor.

Zeros and logarithms. For $0 < \sigma < 1$, the closed form (4.28) is nonzero. Its overall sign is $(-1)^{N(N-1)/2}$ times the fixed contour sign. All energy formulas use the logarithmic derivative Z'/Z , so this fixed sign creates no ambiguity.

4.9 Verification record for R1

The accompanying script

`code/verify_all_pole.py`

uses exact SymPy arithmetic. In the archived default run it verified:

- 852 mixed-moment identities;
- 456 falling-factorial Vandermonde splits;
- 30 Hankel determinant evaluations;
- 90 closed all-pole sums at three independent rational values of σ ;
- 66 direct checks of the original- R contour prefactor at rational ρ ;
- the ordered/unordered factor $N!$;
- the symbolic logarithmic-derivative energy formula;
- the negative second connected moment (4.40);
- the exact finite-size decomposition (4.37).

The tested range was $0 \leq M \leq 7$ and $1 \leq N \leq \min(5, M + 1)$. All comparisons were exact; no floating-point tolerance was used. The detailed derivations and the generated report are collected in [Appendix A](#) and the project code archive.

Chapter 5

Moment matrix, complementary kernel, and corrected positivity

5.1 The signed polynomial ensemble and its moment matrix

Recall the unordered all-pole sum from [Equation \(4.13\)](#):

$$Z_0 = \sum_{\substack{S \subset \{0, \dots, M\} \\ |S|=N}} W(S), \quad W(S) = \prod_{n \in S} w(n) \Delta(S)^2, \quad (5.1)$$

with

$$w(n) = (-\sigma)^n \binom{M}{n}, \quad \Delta(S) = \prod_{\substack{i < j \\ i, j \in S}} (j - i). \quad (5.2)$$

For $0 < \sigma < 1$, the closed formula of [Exact identity 4.4](#) implies $Z_0 \neq 0$ whenever $0 \leq N \leq M + 1$. We use the normalized signed average

$$\langle F \rangle_0 = \frac{1}{Z_0} \sum_{|S|=N} W(S) F(S). \quad (5.3)$$

As in [Warning 3.5](#), this is an algebraic normalization and not a probability expectation.

Let

$$p_a(n) = (n)_{\underline{a}}, \quad a = 0, \dots, N - 1, \quad (5.4)$$

and define the $N \times N$ moment matrix

$$G_{ab} = \sum_{n=0}^M w(n) p_a(n) p_b(n). \quad (5.5)$$

The falling-factorial Vandermonde identity gives

$$\det[p_a(n_i)]_{i=1, \dots, N}^{a=0, \dots, N-1} = \Delta(n_1, \dots, n_N). \quad (5.6)$$

Therefore finite Cauchy–Binet gives the first exact representation of the ensemble normalization.

Exact identity 5.1 (Moment determinant). *For $0 \leq N \leq M + 1$,*

$$\boxed{Z_0 = \det G.} \quad (5.7)$$

For $N = 0$ both sides are understood as 1.

Proof. Let Φ be the $(M+1) \times N$ evaluation matrix

$$\Phi_{na} = p_a(n), \quad (5.8)$$

and let $W = \text{diag}(w(0), \dots, w(M))$. Then $G = \Phi^T W \Phi$. Cauchy–Binet gives

$$\det G = \sum_{|S|=N} \det(\Phi_S)^2 \prod_{n \in S} w(n),$$

which is exactly (5.1) by (5.6). $\square$

The matrix G can be evaluated directly from the one-dimensional moments

$$\mu_r := \sum_{n=0}^M w(n) (n)_{\underline{r}} = (-\sigma)^r (M)_{\underline{r}} (1-\sigma)^{M-r}. \quad (5.9)$$

Together with

$$(n)_{\underline{a}} (n)_{\underline{b}} = \sum_{r=0}^{\min(a,b)} \binom{a}{r} \binom{b}{r} r! (n)_{\underline{a+b-r}}, \quad (5.10)$$

this yields an explicit finite expression for every entry of G . For the kernel construction, however, a diagonal basis is substantially more useful.

5.2 Krawtchouk-type diagonalization

The polynomial system below is a signed-parameter specialization of the classical Krawtchouk family; standard conventions and neighboring families are reviewed in [32, 33].

Define polynomials $K_s(n) = K_s^{(M)}(n; \sigma)$ by the generating function

$$\boxed{\sum_{s=0}^M K_s(n) z^s = (1+z)^{M-n} \left(1 + \frac{z}{\sigma}\right)^n.} \quad (5.11)$$

The parameter

$$x = \frac{\sigma}{1-\sigma} \quad (5.12)$$

will be used to display their coefficients. Expanding (5.11) gives

$$\boxed{K_s(n) = \sum_{b=0}^s \frac{x^{-b}}{b!} \binom{M-b}{s-b} (n)_{\underline{b}}.} \quad (5.13)$$

Terms with $b > n$ vanish automatically. In particular,

$$K_s(n) = \frac{x^{-s}}{s!} (n)_{\underline{s}} + \text{a polynomial of degree smaller than } s. \quad (5.14)$$

Thus $K_0, \dots, K_M$ form a basis of functions on the $(M+1)$ -point lattice.

Exact identity 5.2 (Signed orthogonality). *The polynomials (5.11) obey*

$$\boxed{\sum_{n=0}^M w(n) K_r(n) K_s(n) = h_s \delta_{rs},} \quad (5.15)$$

where

$$h_s = (1 - \sigma)^M \left(-\frac{1}{\sigma} \right)^s \binom{M}{s}. \quad (5.16)$$

The norms alternate in sign; this is an exact signed orthogonality relation, not a positive orthogonality measure.

Proof. Multiply two generating functions and sum over n with the weight (5.2). The binomial theorem gives

$$\begin{aligned} \sum_{n=0}^M w(n)(1+z)^{M-n} \left(1 + \frac{z}{\sigma}\right)^n (1+t)^{M-n} \left(1 + \frac{t}{\sigma}\right)^n \\ = \left[(1+z)(1+t) - \sigma \left(1 + \frac{z}{\sigma}\right) \left(1 + \frac{t}{\sigma}\right) \right]^M \\ = (1-\sigma)^M \left(1 - \frac{zt}{\sigma}\right)^M. \end{aligned} \quad (5.17)$$

The coefficient of $z^r t^s$ vanishes unless $r = s$; the diagonal coefficient is (5.16). $\square$

Let T be the upper-triangular transition matrix defined by

$$K_s(n) = \sum_{b=0}^M p_b(n) T_{bs}, \quad T_{bs} = \begin{cases} \frac{x^{-b}}{b!} \binom{M-b}{s-b}, & b \leq s, \\ 0, & b > s. \end{cases} \quad (5.18)$$

For its leading $N \times N$ block,

$$T_{ss} = \frac{x^{-s}}{s!} \neq 0. \quad (5.19)$$

If $H_N = \text{diag}(h_0, \dots, h_{N-1})$, then (5.15) is equivalent to

$$T_N^T G T_N = H_N. \quad (5.20)$$

Consequently,

$$\det G = \prod_{s=0}^{N-1} h_s (s! x^s)^2. \quad (5.21)$$

This independently reproduces the all-pole determinant of [Exact identity 4.4](#); the detailed simplification is recorded in [Appendix B](#).

5.3 Occupation and complementary kernels

For $1 \leq N \leq M+1$, define the occupation kernel

$$K_N^{\text{occ}} = \Phi G^{-1} \Phi^T W, \quad (5.22)$$

that is,

$$K_{N,ij}^{\text{occ}} = w(j) \sum_{a,b=0}^{N-1} p_a(i) (G^{-1})_{ab} p_b(j). \quad (5.23)$$

Define its complement

$$Q_N = I - K_N^{\text{occ}}. \quad (5.24)$$

The nonsymmetric placement of the factor $w(j)$ is convenient for determinant insertions. The kernel is self-adjoint only with respect to the signed bilinear form induced by W .

Let P be the square evaluation matrix

$$P_{ns} = K_s(n), \quad 0 \leq n, s \leq M, \quad (5.25)$$

and let $H = \text{diag}(h_0, \dots, h_M)$. Exact orthogonality reads

$$P^T W P = H. \quad (5.26)$$

Since P is invertible, completeness gives

$$P H^{-1} P^T W = I. \quad (5.27)$$

Splitting the polynomial degrees at N yields the spectral-tail form of the complementary kernel.

Exact identity 5.3 (Complementary Krawtchouk tail). *For $0 \leq i, j \leq M$,*

$$Q_{N,ij} = w(j) \sum_{s=N}^M \frac{K_s(i) K_s(j)}{h_s}. \quad (5.28)$$

The empty sum for $N = M + 1$ is zero. Furthermore,

$$(K_N^{\text{occ}})^2 = K_N^{\text{occ}}, \quad Q_N^2 = Q_N, \quad (5.29)$$

with

$$\text{rank } K_N^{\text{occ}} = N, \quad \text{rank } Q_N = M + 1 - N. \quad (5.30)$$

The algebraic role of these kernels is expressed by multiplicative insertions. Let

$$\eta_a(S) = \mathbf{1}_{a \in S}, \quad \xi_a(S) = \mathbf{1}_{a \notin S} = 1 - \eta_a(S). \quad (5.31)$$

Exact identity 5.4 (Determinantal occupation and hole insertions). *For every finite $A \subset \{0, \dots, M\}$,*

$$\left\langle \prod_{a \in A} \eta_a \right\rangle_0 = \det K_{N,A}^{\text{occ}}, \quad (5.32)$$

$$\left\langle \prod_{a \in A} \xi_a \right\rangle_0 = \det Q_{N,A}. \quad (5.33)$$

Here $K_{N,A}^{\text{occ}}$ and $Q_{N,A}$ are principal submatrices with rows and columns indexed by A . These identities are exact even though the normalized average (5.3) is signed.

Proof. For a function f on the lattice, let $F = \text{diag}(f(0), \dots, f(M))$ and

$$G[f] = \Phi^T W F \Phi. \quad (5.34)$$

Finite Cauchy–Binet gives

$$\left\langle \prod_{n \in S} f(n) \right\rangle_0 = \frac{\det G[f]}{\det G}. \quad (5.35)$$

Using $\det(I + AB) = \det(I + BA)$,

$$\frac{\det G[f]}{\det G} = \det \left[I_N + G^{-1} \Phi^T W (F - I) \Phi \right]$$

$$= \det [I_{M+1} + (\mathbf{F} - I)K_N^{\text{occ}}]. \quad (5.36)$$

Take $f(a) = u_a$ on A and $f(n) = 1$ outside A . Expansion in the variables $u_a - 1$ gives (5.32). Setting every $u_a = 0$ gives

$$\left\langle \prod_{a \in A} (1 - \eta_a) \right\rangle_0 = \det(I_A - K_{N,A}^{\text{occ}}) = \det Q_{N,A},$$

which is (5.33). $\square$

Warning 5.5 (Algebraic determinantal process). The word “occupation” refers to the Boolean variables η_a , not to a positive determinantal point process. Neither $K_{N,aa}^{\text{occ}}$ nor $Q_{N,aa}$ is required to lie in $[0, 1]$. The positivity needed later emerges only after the explicit sign correction of Section 5.5.

5.4 Finite factorization of the lower restriction

Assume henceforth

$$1 \leq N \leq M, \quad 1 \leq k \leq N, \quad (5.37)$$

and let $Q_N^{(k)}$ denote the restriction of Q_N to rows and columns

$$0 \leq i, j < k. \quad (5.38)$$

Introduce the diagonal sign matrix

$$J_{jj} = (-1)^{N-j}, \quad 0 \leq j < k, \quad (5.39)$$

and the sign-adjusted matrix

$$\mathbf{K}^{(k)} = Q_N^{(k)} J. \quad (5.40)$$

The factorization below is the main exact output of the chapter.

Exact identity 5.6 (Finite complementary-kernel factorization; R2). *Let the intermediate indices be*

$$0 \leq b < k, \quad N \leq s \leq M. \quad (5.41)$$

Define

$$F_{ib} = (i)_{\underline{b}}, \quad 0 \leq i, b < k, \quad (5.42)$$

$$(D_1)_{bb} = x^{-b} \frac{(M-b)!}{b!(N-1-b)!}, \quad (5.43)$$

$$C_{bs} = \frac{1}{s-b}, \quad (5.44)$$

$$(D_2)_{ss} = x^s \frac{s!}{(M-s)!(s-N)!}, \quad (5.45)$$

$$B_{sj} = \frac{1}{(s-j)!}, \quad N \leq s \leq M, \quad 0 \leq j < k, \quad (5.46)$$

$$(D_3)_{jj} = \frac{1}{j!}. \quad (5.47)$$

Then

$$\boxed{\mathbf{K}^{(k)} = Q_N^{(k)} J = F D_1 C D_2 B D_3.} \quad (5.48)$$

Equivalently, for $0 \leq i, j < k$,

$$Q_{N,ij} = (-1)^{N-j} \sum_{b=0}^i \sum_{s=N}^M (i)_{\underline{b}} x^{-b} \frac{(M-b)!}{b!(N-1-b)!} \frac{1}{s-b} \\ \times x^s \frac{s!}{(M-s)!(s-N)!} \frac{1}{(s-j)!j!}. \quad (5.49)$$

Every summand in $(Q_N^{(k)} J)_{ij}$ is nonnegative, and at least one is positive.

Proof. We give the complete logical chain and defer only routine interpolation algebra to [Appendix C](#). Introduce the second polynomial basis

$$\psi_c(n) = (M-n)_{\underline{c}}, \quad 0 \leq c < N, \quad (5.50)$$

and the mixed moment matrix

$$\mathcal{M}_{bc} = \sum_{n=0}^M w(n) (n)_{\underline{b}} (M-n)_{\underline{c}}. \quad (5.51)$$

By [Lemma 4.2](#),

$$\mathcal{M}_{bc} = (-1)^b \sigma^b (1-\sigma)^{M-b-c} (M)_{\underline{b+c}}. \quad (5.52)$$

It factorizes as

$$\mathcal{M} = DTE, \quad (5.53)$$

where

$$D_{bb} = (-1)^b \sigma^b (1-\sigma)^{M-b} (M)_{\underline{b}}, \quad T_{bc} = (M-b)_{\underline{c}}, \quad E_{cc} = (1-\sigma)^{-c}. \quad (5.54)$$

The biorthogonal form of the occupation kernel is

$$K_{N,ij}^{\text{occ}} = w(j) \sum_{b,c=0}^{N-1} (i)_{\underline{b}} (\mathcal{M}^{-1})_{cb} (M-j)_{\underline{c}}. \quad (5.55)$$

Since $\mathcal{M}^{-1} = E^{-1}T^{-1}D^{-1}$, the inner sum is the polynomial

$$(\mathcal{R}_{1-\sigma} \ell_b)(M-j), \quad (5.56)$$

where ℓ_b is the Lagrange polynomial satisfying

$$\ell_b(M-r) = \delta_{br}, \quad 0 \leq r < N, \quad (5.57)$$

and $\mathcal{R}_q$ multiplies $(y)_{\underline{c}}$ by q^c . For an integer $L \geq 0$,

$$(\mathcal{R}_q f)(L) = \sum_{r=0}^L \binom{L}{r} q^r (1-q)^{L-r} f(r). \quad (5.58)$$

After the change of variable $s = M-r$, the values $0 \leq s < N$ reproduce the Newton expansion of δ_{ij} exactly and therefore cancel when $Q_N = I - K_N^{\text{occ}}$ is formed. The remaining values are

$$\ell_b(M-s) = (-1)^{N+b+1} \frac{s!}{b!(N-1-b)!(s-N)!(s-b)}, \quad s \geq N. \quad (5.59)$$

Substitution of this tail into (5.55), followed by cancellation of factorials and the identity

$$\sigma^{s-b} (1-\sigma)^{b-s} = x^{s-b}, \quad (5.60)$$

produces (5.49), hence (5.48). $\square$

The proof explains the origin of the nontrivial Cauchy factor. It is not an ansatz: the denominator $(s-b)^{-1}$ is the value of a Lagrange basis polynomial beyond its N interpolation nodes.

5.5 Corrected signs and positive principal minors

Let

$$A = \{a_1 < \cdots < a_m\} \subset \{0, \dots, k-1\}. \quad (5.61)$$

Define the sign-corrected principal minor

$$\mathfrak{P}(A) = (-1)^{mN + \sum_{a \in A} a + \binom{m}{2}} \det Q_{N,A}. \quad (5.62)$$

For a one-point set we write

$$P_a = \mathfrak{P}(\{a\}) = (-1)^{N+a} Q_{N,aa}. \quad (5.63)$$

Because $\mathbf{K}^{(k)} = Q_N^{(k)} J$,

$$\det \mathbf{K}_{A,A}^{(k)} = (-1)^{mN - \sum_{a \in A} a} \det Q_{N,A}, \quad (5.64)$$

and therefore

$$\mathfrak{P}(A) = (-1)^{\binom{m}{2}} \det \mathbf{K}_{A,A}^{(k)}. \quad (5.65)$$

The remaining sign is forced by the ordered Cauchy matrix. If

$$b_1 < \cdots < b_m < N \leq s_1 < \cdots < s_m, \quad (5.66)$$

then

$$\det \left[\frac{1}{s_j - b_i} \right]_{i,j=1}^m = (-1)^{\binom{m}{2}} \frac{\prod_{i < j} (b_j - b_i) \prod_{i < j} (s_j - s_i)}{\prod_{i,j} (s_j - b_i)}. \quad (5.67)$$

Thus the Cauchy factor in (5.48) is sign-regular rather than totally positive in the displayed ordering.

Exact identity 5.7 (Positive Cauchy–Binet expansion). *Let $m = |A|$. Then*

$$\mathfrak{P}(A) = \sum_{\substack{B_* = \{b_1 < \cdots < b_m\} \subset \{0, \dots, k-1\} \\ S_* = \{s_1 < \cdots < s_m\} \subset \{N, \dots, M\}}} \mathcal{W}(A; B_*, S_*; x), \quad (5.68)$$

where every nonzero summand is strictly positive. Explicitly,

$$\begin{aligned} \mathcal{W}(A; B_*, S_*; x) &= (-1)^{\binom{m}{2}} \det F_{A,B_*} \det C_{B_*,S_*} \det B_{S_*,A} \\ &\quad \times \prod_{b \in B_*} \left[x^{-b} \frac{(M-b)!}{b!(N-1-b)!} \right] \\ &\quad \times \prod_{s \in S_*} \left[x^s \frac{s!}{(M-s)!(s-N)!} \right] \prod_{a \in A} \frac{1}{a!}. \end{aligned} \quad (5.69)$$

Consequently,

$$\mathfrak{P}(A) \geq 0. \quad (5.70)$$

More precisely,

$$\mathfrak{P}(A) > 0 \iff |A| \leq M + 1 - N, \quad (5.71)$$

while larger principal minors vanish because $\text{rank } Q_N = M + 1 - N$. Also $\mathfrak{P}(\emptyset) = 1$, and $\mathfrak{P}(A)$ is a polynomial in x with nonnegative coefficients.

Proof. Apply Cauchy–Binet twice to (5.48). The minors of $F_{ib} = (i)_{\underline{b}} = b! \binom{i}{b}$ are nonnegative, as are the minors of

$$B_{sj} D_{3,jj} = \frac{1}{j!(s-j)!} = \frac{1}{s!} \binom{s}{j}. \quad (5.72)$$

Both statements follow from total nonnegativity of the Pascal matrix; see [Appendix D](#). The diagonal matrices D_1, D_2, D_3 are positive for $x > 0$. The Cauchy minor has the fixed sign (5.67), which is removed by (5.65). Hence every term in (5.69) is nonnegative. If $m \leq M + 1 - N$, strict positivity follows by choosing $B_* = A$ and an admissible consecutive S_* ; the corresponding Pascal minors are nonzero. If $m > M + 1 - N$, there is no m -element tail set S_* and the minor vanishes, consistently with the rank of Q_N . The only x dependence of a summand is

$$x^{\sum_{s \in S_*} s - \sum_{b \in B_*} b},$$

with a nonnegative integer exponent, proving the polynomial statement. $\square$

Corollary 5.8 (Degree-derivative bound). *For every integer $r \geq 0$,*

$$\boxed{|(x\partial_x)^r \det Q_{N,A}| \leq (mM)^r |\det Q_{N,A}|.} \quad (5.73)$$

Proof. After the fixed sign correction, the determinant is a polynomial with nonnegative coefficients. Every exponent in (5.69) is at most mM . The operator $(x\partial_x)^r$ multiplies a monomial of degree d by d^r , so the inequality holds term by term. $\square$

5.6 The exact one-point consequence

For $0 \leq a < N$, choose $k > a$ in [Exact identity 5.6](#) and set $i = j = a$. Since

$$\frac{(a)_{\underline{b}}}{a!} = \frac{1}{(a-b)!}, \quad (5.74)$$

we obtain the positive double sum

Exact identity 5.9 (One-point complementary minor). *For $0 \leq a < N$,*

$$\boxed{P_a = \sum_{b=0}^a \sum_{s=N}^M \frac{(M-b)! s! x^{s-b}}{(M-s)!(s-N)! b!(N-1-b)! (a-b)!(s-a)!(s-b)}}. \quad (5.75)$$

In particular, $P_a > 0$ for $x > 0$.

The logarithmic derivative has the exact positive-sum interpretation

$$x\partial_x \log P_a = \frac{\sum_{b,s} (s-b) T_{b,s}^{(a)}}{\sum_{b,s} T_{b,s}^{(a)}}, \quad T_{b,s}^{(a)} > 0. \quad (5.76)$$

This is a genuine positive average over the transformed double sum. It must not be confused with a positivity statement about the original signed ensemble. Its asymptotic analysis and its contour interpretation are deferred to [Chapters 6](#) and [7](#).

5.7 Logical output and verification record

The exact outputs used by later chapters are:

1. the moment determinant $Z_0 = \det G$;
2. the Krawtchouk-tail kernel (5.28);
3. the hole insertion identity (5.33);
4. the finite factorization (5.48);
5. the positive sign-corrected minors (5.68);
6. the exact one-point polynomial (5.75).

No saddle approximation and no branch identification enters this chapter.

Verification note 5.10 (Exact-rational kernel audit). The script

`code/verify_kernel.py`

independently constructs Q_N from the inverse moment matrix, from the full Krawtchouk tail, and from the factorization (5.48). It also checks signed orthogonality, completeness, idempotency, ranks, hole insertions by direct subset enumeration, the Cauchy–Binet expansion, and positivity of all tested sign-corrected principal minors. The archived output is

`code/kernel_verification_report.txt`.

Chapter 6

Pure contour sectors and excluded-pole periods

6.1 Pole-supported cycles and reduced residue periods

Let

$$\mathcal{L}_M = \{0, 1, \dots, M\} \quad (6.1)$$

be the lattice of pair-pole indices. For every $n \in \mathcal{L}_M$, let γ_n be a small positively oriented loop around the pole $2\varepsilon_n$ and no other pole. If $B \subset \mathcal{L}_M$, write

$$\gamma_B = \sum_{n \in B} \gamma_n \quad (6.2)$$

as an equality in one-variable homology. In particular,

$$\gamma_{\text{all}} = \gamma_{\mathcal{L}_M}, \quad \gamma_{\geq r} = \gamma_{\{r, r+1, \dots, M\}}. \quad (6.3)$$

The convention $\gamma_{\geq M+1} = 0$ will occasionally be useful.

Every product cycle whose one-variable factors are sums of the loops γ_n has the same geometric residue prefactor as the all-pole cycle. We therefore define its *reduced residue period* $\mathcal{Z}_\Gamma$ by

$$\boxed{Z_\Gamma = \mathcal{C}_{M,N,\rho} \exp\left(-\frac{2N\varepsilon_{F0}}{V}\right) \mathcal{Z}_\Gamma,} \quad (6.4)$$

where $\mathcal{C}_{M,N,\rho}$ is the coupling-independent factor in (4.9). Thus

$$\mathcal{Z}_{\text{all}} = z_{M,N} = N! Z_0. \quad (6.5)$$

No energy is assigned directly to $\mathcal{Z}_\Gamma$: stripping the exponential in (6.4) removes the additive contribution $2N\varepsilon_{F0}$. Ratios of periods in the same N -pair sector are unaffected by the common prefactor and may be computed entirely from the reduced periods.

For a residue subset $S \subset \mathcal{L}_M$, $|S| = N$, recall the hole indicators

$$\xi_a(S) = \mathbf{1}_{a \notin S}, \quad \xi_A(S) = \prod_{a \in A} \xi_a(S). \quad (6.6)$$

Since the variables are Boolean, $\xi_A \xi_B = \xi_{A \cup B}$.

6.2 Excluded-pole product cycles

Let $A \subset \mathcal{L}_M$. Remove every pole in A from every one-variable all-pole contour:

$$\gamma_{\text{all}}[A] = \gamma_{\mathcal{L}_M \setminus A}, \quad \Gamma_{\text{all}}[A] = \gamma_{\text{all}}[A]^{\times N}. \quad (6.7)$$

Denote the associated reduced period by $\mathcal{Z}_{\text{all}}[A]$. The residue subset selected by this product cycle must avoid A , and no other change is made to its signed Vandermonde weight.

Exact identity 6.1 (Excluded-pole period). *For every $A \subset \mathcal{L}_M$,*

$$\mathcal{Z}_{\text{all}}[A] = N! \sum_{|S|=N} W(S) \xi_A(S) = N! Z_0 \det Q_{N,A}. \quad (6.8)$$

Equivalently,

$$\frac{\mathcal{Z}_{\text{all}}[A]}{Z_{\text{all}}} = \frac{\mathcal{Z}_{\text{all}}[A]}{\mathcal{Z}_{\text{all}}} = \det Q_{N,A}. \quad (6.9)$$

The determinant is understood as 1 for $A = \emptyset$. If fewer than N poles remain, both sides vanish.

Proof. Take residues independently in the N labelled variables. A nonzero ordered residue configuration consists of N distinct indices in $\mathcal{L}_M \setminus A$. For a fixed unlabelled set S , its $N!$ assignments have the common weight $W(S)$. This gives the first equality in (6.8). The second is the hole-minor identity (5.33) multiplied by $N!Z_0$. Restoring the common prefactor in (6.4) proves (6.9). $\square$

The phrase *pure sector* will refer to an excluded-pole insertion ξ_A or to its period $\mathcal{Z}_{\text{all}}[A]$. It is a purity statement about the residue insertion, not a claim that the family of all such periods is linearly independent after restriction to $|S| = N$.

For a lower set $A \subset \{0, \dots, k-1\}$, the sign convention of (5.62) gives

$$\mathfrak{P}(A) = (-1)^{|A|N + \sum_{a \in A} a + \binom{|A|}{2}} \frac{\mathcal{Z}_{\text{all}}[A]}{Z_{\text{all}}}. \quad (6.10)$$

Thus the positive Cauchy–Binet polynomial of [Exact identity 5.7](#) is a sign-corrected ratio of genuine contour periods. The original excluded-pole period itself remains signed.

6.3 One restricted contour and the exact local projection

We first isolate the finite object underlying the earlier anomalous-gap calculation. Let one integration variable have the lower-cutoff contour $\gamma_{\geq a}$, while the remaining $N-1$ variables have all-pole contours:

$$\Gamma_a^{(1)} = \gamma_{\text{all}}^{\times(N-1)} \times \gamma_{\geq a}, \quad 0 \leq a \leq M+1. \quad (6.11)$$

Its reduced period will be denoted $\mathcal{Z}_a^{(1)}$.

For a fixed residue set S , define

$$X_a(S) = \#\{n \in S : n < a\}, \quad H_a(S) = \sum_{j=0}^{a-1} \xi_j(S) = a - X_a(S). \quad (6.12)$$

The distinguished variable has $N - X_a(S)$ allowed residues; the remaining residues can be assigned to the other labelled variables in $(N - 1)!$ ways. Hence

$$\mathcal{Z}_a^{(1)} = (N - 1)! \sum_{|S|=N} W(S)[N - X_a(S)]. \quad (6.13)$$

Using $X_a = a - H_a$ and (5.33),

$$\mathcal{Z}_a^{(1)} = (N - 1)! Z_0 \left[N - a + \sum_{j=0}^{a-1} Q_{N,jj} \right]. \quad (6.14)$$

The first term is an all-pole component. Define the projected cumulative period

$$\mathcal{A}_a := \mathcal{Z}_a^{(1)} - \frac{N - a}{N} \mathcal{Z}_{\text{all}}. \quad (6.15)$$

Then

$$\mathcal{A}_a = (N - 1)! Z_0 \sum_{j=0}^{a-1} Q_{N,jj}. \quad (6.16)$$

Its adjacent finite difference is local:

$$\mathcal{A}_{a+1} - \mathcal{A}_a = \mathcal{Z}_{a+1}^{(1)} - \mathcal{Z}_a^{(1)} + \frac{1}{N} \mathcal{Z}_{\text{all}} = (N - 1)! Z_0 Q_{N,aa}. \quad (6.17)$$

Exact identity 6.2 (Pure local marked period). *For $0 \leq a < N$, define*

$$\mathcal{Z}_{\text{mark},a} := (-1)^{N+a} (\mathcal{A}_{a+1} - \mathcal{A}_a). \quad (6.18)$$

Then

$$\mathcal{Z}_{\text{mark},a} = \frac{(-1)^{N+a}}{N} \mathcal{Z}_{\text{all}}[\{a\}] = (N - 1)! Z_0 P_a. \quad (6.19)$$

Consequently,

$$\frac{\mathcal{Z}_{\text{mark},a}}{\mathcal{Z}_{\text{all}}} = \frac{P_a}{N}. \quad (6.20)$$

Proof. Equation (6.17) gives the last expression in (6.19) after multiplication by $(-1)^{N+a}$ and use of $P_a = (-1)^{N+a} Q_{N,aa}$. The singleton specialization of (6.8) gives

$$\frac{1}{N} \mathcal{Z}_{\text{all}}[\{a\}] = (N - 1)! Z_0 Q_{N,aa},$$

which proves the middle expression. Division by (6.5) yields (6.20). $\square$

This result removes an ambiguity in the earlier construction. A single lower-cutoff contour is not a pure excited sector: it contains the explicitly displayed all-pole component. After the exact adjacent projection, however, the local factor P_a is not merely a formal kernel insertion. It is, up to a fixed sign and the trivial factor $1/N$, the period in which pole a is excluded from every integration contour.

Remark 6.3 (Relation to the cumulative anomalous insertion). If

$$\tilde{\mathcal{H}}_a := (-1)^{N+a+1} \sum_{j=0}^{a-1} Q_{N,jj}, \quad \tilde{\mathcal{H}}_0 = 0, \quad (6.21)$$

then

$$P_a = \tilde{\mathcal{H}}_{a+1} + \tilde{\mathcal{H}}_a. \quad (6.22)$$

The positive double-sum representation of $\tilde{\mathcal{H}}_a$ and the common WKB exponent of $\tilde{\mathcal{H}}_a$ and P_a will be derived in [Chapter 7](#). The finite identity (6.22) already shows exactly how the old cumulative object is converted into the local pure period.

6.4 Arbitrary nested lower cutoffs

Let

$$0 \leq r_1 < r_2 < \cdots < r_k \leq M, \quad 0 \leq k \leq N, \quad c = N - k. \quad (6.23)$$

The raw product cycle consists of c all-pole factors and k lower-cutoff factors:

$$\Gamma_{\mathbf{r}}^{\text{raw}} = \gamma_{\text{all}}^{\times c} \times \gamma_{\geq r_1} \times \cdots \times \gamma_{\geq r_k}. \quad (6.24)$$

The superscript “raw” emphasizes that no projection onto pure excluded-pole sectors has yet been performed.

For $r \in \mathcal{L}_M$, retain the cumulative occupation and hole counts

$$X_r(S) = \#\{n \in S : n < r\}, \quad H_r(S) = r - X_r(S) = \sum_{j=0}^{r-1} \xi_j(S). \quad (6.25)$$

Exact identity 6.4 (Arbitrary nested-cutoff multiplicity). *The reduced raw period of (6.24) is*

$$\mathcal{Z}_{\mathbf{r}}^{\text{raw}} = c! \sum_{|S|=N} W(S) \prod_{t=1}^k [c + t - X_{r_t}(S)]. \quad (6.26)$$

Equivalently, with

$$d_t = c + t - r_t, \quad (6.27)$$

we have

$$\mathcal{Z}_{\mathbf{r}}^{\text{raw}} = c! \sum_{|S|=N} W(S) \prod_{t=1}^k [d_t + H_{r_t}(S)]. \quad (6.28)$$

For the consecutive lower cascade $r_t = t$, this reduces to

$$\mathcal{Z}_k^{\text{cas}} = (N - k)! \sum_{|S|=N} W(S) \prod_{r=1}^k [N - k + H_r(S)]. \quad (6.29)$$

Proof. Fix S . Assign residues to the trapped contours from the largest cutoff downwards. Before assigning contour t , the $k - t$ higher-cutoff contours have already used residues that are also admissible for cutoff r_t . There are $N - X_{r_t}(S)$ residues of S at or above r_t , hence

$$N - X_{r_t}(S) - (k - t) = c + t - X_{r_t}(S)$$

choices at this step. After the k trapped variables have been assigned, the remaining c residues can be permuted among the all-pole variables in $c!$ ways. If a higher-cutoff assignment is impossible, the product contains a zero at the first obstruction; the formula still gives zero. Multiplying by $W(S)$ and summing over S proves (6.26). Equation (6.28) follows from $X_r = r - H_r$. $\square$

The same counting remains valid when a set $E \subset \mathcal{L}_M$ is removed from every contour that would otherwise enclose it. Define

$$\gamma_{\text{all}}[E] = \gamma_{\mathcal{L}_M \setminus E}, \quad \gamma_{\geq r}[E] = \sum_{\substack{n \geq r \\ n \notin E}} \gamma_n, \quad (6.30)$$

and let $\mathcal{Z}_r^{\text{raw}}[E]$ be the reduced period of the corresponding product cycle. Then

$$\boxed{\mathcal{Z}_r^{\text{raw}}[E] = c! \sum_{|S|=N} W(S) \xi_E(S) \prod_{t=1}^k [d_t + H_{r_t}(S)]}. \quad (6.31)$$

For $E = \{a\}$ this is the exact full marked-period object over an arbitrary nested background. Its asymptotic role in Case B will be studied only after the finite algebra has been specialized to the physical cascade.

6.5 Canonical Boolean–Möbius expansion

Let

$$R = \begin{cases} r_k, & k \geq 1, \\ 0, & k = 0, \end{cases} \quad I_t = \{0, 1, \dots, r_t - 1\}. \quad (6.32)$$

The multiplicity factor in (6.28) is the Boolean polynomial

$$F_r(\xi) = \prod_{t=1}^k \left[d_t + \sum_{j \in I_t} \xi_j \right]. \quad (6.33)$$

After imposing $\xi_j^2 = \xi_j$, it has a unique canonical multilinear expansion on the full Boolean cube:

$$\boxed{F_r(\xi) = \sum_{A \subset \{0, \dots, R-1\}} C_r(A) \xi_A}. \quad (6.34)$$

The coefficients are given by Boolean Möbius inversion.

Exact identity 6.5 (Möbius coefficients and pure-sector expansion). *For $A \subset \{0, \dots, R-1\}$,*

$$\boxed{C_r(A) = \sum_{B \subset A} (-1)^{|A| - |B|} \prod_{t=1}^k [d_t + |B \cap I_t|]}. \quad (6.35)$$

Consequently, for every excluded set $E \subset \mathcal{L}_M$,

$$\boxed{\mathcal{Z}_r^{\text{raw}}[E] = c! Z_0 \sum_{A \subset \{0, \dots, R-1\}} C_r(A) \det Q_{N, A \cup E}}. \quad (6.36)$$

Equivalently, in terms of pure excluded-pole periods,

$$\boxed{\mathcal{Z}_r^{\text{raw}}[E] = \frac{c!}{N!} \sum_{A \subset \{0, \dots, R-1\}} C_r(A) \mathcal{Z}_{\text{all}}[A \cup E]}. \quad (6.37)$$

Proof. For a function on R Boolean variables, the coefficient of the monomial ξ_A in its canonical multilinear representative is

$$\sum_{B \subset A} (-1)^{|A|-|B|} F(\mathbf{1}_B),$$

where $\mathbf{1}_B$ sets precisely the variables in B equal to 1. For (6.33),

$$F_r(\mathbf{1}_B) = \prod_{t=1}^k [d_t + |B \cap I_t|],$$

which proves (6.35). Insert (6.34) into (6.31), use $\xi_A \xi_E = \xi_{A \cup E}$, and apply (5.33); this gives (6.36). Equation (6.37) follows from (6.8). $\square$

The coefficients also admit an efficient triangular recursion. Let $C_t(A)$ be the canonical coefficient after the first t factors of (6.33), with

$$C_0(\emptyset) = 1, \quad C_0(A) = 0 \quad (A \neq \emptyset). \quad (6.38)$$

Then

$$\boxed{C_t(A) = [d_t + |A \cap I_t|] C_{t-1}(A) + \sum_{j \in A \cap I_t} C_{t-1}(A \setminus \{j\}).} \quad (6.39)$$

This recursion follows directly from Boolean multiplication: multiplying ξ_A by ξ_j leaves it unchanged when $j \in A$, while multiplying $\xi_{A \setminus \{j\}}$ by ξ_j creates ξ_A .

For the consecutive ground-state cascade $r_t = t$ and $c = N - k > 0$, every factor is

$$c + H_t = c + \sum_{j < t} \xi_j, \quad (6.40)$$

so all canonical coefficients are nonnegative. Writing

$$\prod_{t=1}^k (c + H_t) = c^k \left[1 + \sum_{\emptyset \neq A \subset \{0, \dots, k-1\}} \hat{C}_{k,c}(A) \xi_A \right], \quad (6.41)$$

we have

$$\hat{C}_{k,c}(A) \geq 0. \quad (6.42)$$

The singleton coefficient is exact:

$$\boxed{\hat{C}_{k,c}(\{a\}) = \left(1 + \frac{1}{c}\right)^{k-a} - 1, \quad 0 \leq a < k.} \quad (6.43)$$

Indeed, setting only $\xi_a = 1$ changes the final $k - a$ factors from c to $c + 1$.

Remark 6.6 (Positivity and cancellation are different issues). For the consecutive cascade, the Boolean coefficients in (6.41) are nonnegative. The uncorrected determinants $\det Q_{N,A}$ still carry the fixed signs described in Section 5.5. Thus nonnegativity of the Boolean expansion does not by itself exclude cancellation in the complete raw period. Later Case B estimates will use the sign-corrected Cauchy–Binet representation and the full algebraic prefactors, rather than a probabilistic argument.

6.6 A single determinant generating all pure sectors

The canonical expansion can be resummed before any asymptotic analysis. Let $E \subset \mathcal{L}_M$ be a finite excluded set and define the finite index set

$$\mathcal{I}_{r,E} = \{0, \dots, R-1\} \cup E. \quad (6.44)$$

For $j \in \mathcal{I}_{r,E}$, set

$$U_j(\mathbf{u}) = \sum_{t=1}^k \mathbf{1}_{j < r_t} u_t. \quad (6.45)$$

Introduce auxiliary variables z_e , $e \in E$, and the diagonal matrix

$$D_{jj}(\mathbf{u}, \mathbf{z}) = \exp[U_j(\mathbf{u})] \prod_{e \in E} (1 + z_e \mathbf{1}_{j=e}) - 1. \quad (6.46)$$

All matrices below are restricted to $\mathcal{I}_{r,E}$.

Exact identity 6.7 (Determinant Laplace transform of cumulative holes). *We have*

$$\left\langle \exp \left(\sum_{t=1}^k u_t H_{r_t} \right) \prod_{e \in E} (1 + z_e \xi_e) \right\rangle_0 = \det[I + D(\mathbf{u}, \mathbf{z}) Q_N]. \quad (6.47)$$

Consequently,

$$\frac{\mathcal{Z}_r^{\text{raw}}[E]}{c! Z_0} = \prod_{t=1}^k (d_t + \partial_{u_t}) \prod_{e \in E} \partial_{z_e} \left| \times \det[I + D(\mathbf{u}, \mathbf{z}) Q_N] \right|_{\mathbf{u}=0, \mathbf{z}=0} \quad (6.48)$$

For $E = \emptyset$ this generates the full raw ground period; for $E = \{a\}$ it generates the full excluded-pole marked period over the same nested background.

Proof. Since each ξ_j is Boolean,

$$\exp[U_j \xi_j] \prod_{e \in E} (1 + z_e \xi_e)^{\mathbf{1}_{j=e}} = 1 + D_{jj}(\mathbf{u}, \mathbf{z}) \xi_j.$$

Multiplication over $j \in \mathcal{I}_{r,E}$ and expansion in principal minors gives

$$\left\langle \prod_j [1 + D_{jj} \xi_j] \right\rangle_0 = \sum_A \left(\prod_{a \in A} D_{aa} \right) \det Q_{N,A} = \det(I + D Q_N),$$

where the first equality uses (5.33). This proves (6.47). The operator $d_t + \partial_{u_t}$ replaces the exponential by $d_t + H_{r_t}$ at $\mathbf{u} = 0$, while ∂_{z_e} selects ξ_e . Applying all derivatives proves (6.48). $\square$

For a singleton $E = \{a\}$, the derivative with respect to z_a may be evaluated before the u derivatives. Let

$$A(\mathbf{u}) = I + D(\mathbf{u}, 0) Q_N. \quad (6.49)$$

Then Jacobi's formula gives

$$\partial_{z_a} \det[I + D(\mathbf{u}, z_a) Q_N] \big|_{z_a=0} = \det A(\mathbf{u}) e^{U_a(\mathbf{u})} [Q_N A(\mathbf{u})^{-1}]_{aa}. \quad (6.50)$$

Thus the marked insertion above a fully resummed background is controlled by the diagonal of a deformed complementary kernel. This formula will be the finite starting point for the complete Case B marked ratio; no endpoint component has been selected in advance.

6.7 Logical output and verification record

The exact outputs of this chapter are:

1. every excluded-pole product cycle is the determinant minor in (6.8);
2. the pure local marked period is exactly the one-point factor P_a , as in (6.19);
3. arbitrary nested cutoffs have the exact multiplicity (6.26);
4. every raw period has the canonical Möbius expansion (6.37);
5. the complete family of ground and excluded-pole periods is generated by the single determinant (6.48).

No large- M limit, saddle localization, or physical branch identification enters these statements.

Verification note 6.8 (Exact-rational pure-contour audit). The script

`code/verify_pure_contours.py`

checks, in exact arithmetic, the excluded-pole ratios, the raw one-contour projection, the equality with the singleton excluded period, arbitrary-cutoff assignment counts by complete permutation enumeration, Boolean Möbius coefficients and their recursion, raw-period expansions into minors and pure periods, the consecutive-cascade singleton coefficient, and the determinant differential generator. The archived output is

`code/pure_contour_verification_report.txt.`

Part III

Case A and finite-seniority sectors

Chapter 7

Case A: one marked Richardson root

7.1 Exact positive objects and the period-energy difference

For a lower marked index

$$0 \leq a < N, \quad (7.1)$$

recall the sign-corrected one-point minor

$$P_a = (-1)^{N+a} Q_{N,aa} > 0. \quad (7.2)$$

By [Exact identity 5.9](#), it has the exact positive double-sum representation

$$P_a = \sum_{b=0}^a \sum_{s=N}^M T_{b,s}^{(a)}, \quad (7.3)$$

where

$$T_{b,s}^{(a)} = \frac{(M-b)! s! x^{s-b}}{(M-s)!(s-N)! b!(N-1-b)!(a-b)!(s-a)!(s-b)}. \quad (7.4)$$

Every summand is strictly positive for $x > 0$. Consequently,

$$x \partial_x \log P_a = \frac{\sum_{b,s} (s-b) T_{b,s}^{(a)}}{\sum_{b,s} T_{b,s}^{(a)}} \quad (7.5)$$

is a genuine positive average of the degree $s-b$ in the transformed double sum. No positivity is attributed to the original signed residue ensemble.

The cumulative object inherited from the earlier anomalous-gap calculation was defined in [\(6.21\)](#). It also admits a positive representation.

Exact identity 7.1 (Positive cumulative insertion). *For $a \geq 1$,*

$$\tilde{\mathcal{H}}_a = \sum_{b=0}^{a-1} \sum_{s=N}^M \frac{(M-b)! s! x^{s-b}}{(M-s)!(s-N)! b!(N-1-b)!(a-b-1)!(s-a)!(s-b)^2}. \quad (7.6)$$

Together with $\tilde{\mathcal{H}}_0 = 0$, it obeys the exact local relation

$$P_a = \tilde{\mathcal{H}}_{a+1} + \tilde{\mathcal{H}}_a. \quad (7.7)$$

Proof. Insert $Q_{N,jj} = (-1)^{N+j}P_j$ into (6.21), use (7.3), and interchange the sums over j, b, s . The only finite identity required is

$$\sum_{j=b}^{a-1} \frac{(-1)^{a+j+1}}{(j-b)!(s-j)!} = \frac{1}{(a-b-1)!(s-a)!(s-b)}. \quad (7.8)$$

This yields (7.6). Equation (7.7) is also immediate from the definition of $\tilde{\mathcal{H}}_a$, and agrees with (6.22). $\square$

The pure marked period was constructed exactly in [Exact identity 6.2](#). Since

$$\frac{\mathcal{Z}_{\text{mark},a}}{\mathcal{Z}_{\text{all}}} = \frac{P_a}{N}, \quad (7.9)$$

its period-energy difference is

$$\boxed{\Delta E_a^{\text{per}} := E_{\text{mark},a}^{\text{per}} - E_{\text{all}}^{\text{per}} = \frac{2}{\rho(1-\sigma)} x \partial_x \log P_a.} \quad (7.10)$$

The factor N^{-1} is coupling independent and drops out. Equation (7.10) is exact at finite M, N ; the remainder of this chapter evaluates it asymptotically.

7.2 Interior two-dimensional lattice saddle

Use the exact finite-size ratios

$$\nu = \frac{N}{M}, \quad \alpha = \frac{a}{M}, \quad \beta = \frac{b}{M}, \quad \tau = \frac{s}{M}. \quad (7.11)$$

The interior regime means that (α, ν, x) remains in a compact subset of

$$0 < \alpha < \nu < 1, \quad x > 0. \quad (7.12)$$

Write

$$L(y) = y \log y, \quad L(0) = 0. \quad (7.13)$$

Uniform Stirling expansion of (7.4) gives

$$T_{b,s}^{(a)} = \frac{\exp[M\Psi_{\alpha,\nu}(\beta, \tau)]}{(2\pi)^2 M^2} \mathcal{A}_{\alpha,\nu}(\beta, \tau) \left[1 + O(M^{-1})\right], \quad (7.14)$$

where

$$\begin{aligned} \Psi_{\alpha,\nu}(\beta, \tau) &= L(1-\beta) + L(\tau) - L(1-\tau) - L(\tau-\nu) \\ &\quad - L(\beta) - L(\nu-\beta) - L(\alpha-\beta) - L(\tau-\alpha) + (\tau-\beta) \log x, \end{aligned} \quad (7.15)$$

and

$$\boxed{\mathcal{A}_{\alpha,\nu}(\beta, \tau) = \frac{1}{\tau-\beta} \sqrt{\frac{(1-\beta)\tau(\nu-\beta)}{(1-\tau)(\tau-\nu)\beta(\alpha-\beta)(\tau-\alpha)}}.} \quad (7.16)$$

The derivation, including the shifts caused by $(N-1-b)!$, is given in [Appendix F](#).

The saddle equations are

$$(\alpha-\beta)(\nu-\beta) = x\beta(1-\beta), \quad (\tau-\nu)(\tau-\alpha) = x\tau(1-\tau). \quad (7.17)$$

Both coordinates are roots of

$$(1+x)z^2 - (\alpha + \nu + x)z + \alpha\nu = 0. \quad (7.18)$$

Define

$$D_\alpha = (\alpha + \nu + x)^2 - 4(1+x)\alpha\nu. \quad (7.19)$$

Then

$$\boxed{\beta_\alpha = \frac{\alpha + \nu + x - \sqrt{D_\alpha}}{2(1+x)}, \quad \tau_\alpha = \frac{\alpha + \nu + x + \sqrt{D_\alpha}}{2(1+x)}}. \quad (7.20)$$

The signs of the quadratic at $0, \alpha, \nu, 1$ show that

$$0 < \beta_\alpha < \alpha < \nu < \tau_\alpha < 1. \quad (7.21)$$

Thus the saddle is strictly interior throughout the domain (7.12).

Set

$$\Phi(\alpha; \nu, x) = \Psi_{\alpha, \nu}(\beta_\alpha, \tau_\alpha), \quad q(\alpha; \nu, x) = \tau_\alpha - \beta_\alpha = \frac{\sqrt{D_\alpha}}{1+x}. \quad (7.22)$$

The envelope derivatives are

$$\boxed{x\partial_x \Phi = q, \quad \partial_\alpha \Phi = \log \frac{\tau_\alpha - \alpha}{\alpha - \beta_\alpha}}. \quad (7.23)$$

The positive saddle curvatures are

$$h_\beta = -\partial_\beta^2 \Psi = \frac{(1+x)q}{x\beta_\alpha(1-\beta_\alpha)}, \quad h_\tau = -\partial_\tau^2 \Psi = \frac{(1+x)q}{x\tau_\alpha(1-\tau_\alpha)}. \quad (7.24)$$

There is no mixed second derivative because the action separates into a β -part and a τ -part.

Period asymptotics 7.2 (Interior one-point asymptotics). *Uniformly on compact subsets of (7.12),*

$$\boxed{P_a = \frac{C(\alpha, \nu, x)}{M} \exp[M\Phi(\alpha; \nu, x)] \left[1 + O(M^{-1})\right]}, \quad (7.25)$$

where

$$\boxed{C(\alpha, \nu, x) = \frac{\nu - \beta_\alpha}{2\pi(1+x)q^2} \sqrt{\frac{(1-\beta_\alpha)\tau_\alpha}{\beta_\alpha(1-\tau_\alpha)}}}. \quad (7.26)$$

Moreover,

$$\boxed{x\partial_x \log P_a = Mq(\alpha; \nu, x) + c(\alpha, \nu, x) + O(M^{-1})}, \quad (7.27)$$

with

$$\begin{aligned} c(\alpha, \nu, x) &:= x\partial_x \log C(\alpha, \nu, x) \\ &= \frac{\alpha - \beta_\alpha + x}{(1+x)q} - \frac{x}{1+x} \\ &\quad - \frac{2x\{\beta_\alpha(1-\beta_\alpha) + \tau_\alpha(1-\tau_\alpha)\}}{(1+x)q^2}. \end{aligned} \quad (7.28)$$

Proof. The summands are positive. On every compact subset of (7.12), Ψ has the unique nondegenerate maximum (7.20); outside a fixed neighborhood of that point the action is smaller by a positive amount. The two-dimensional lattice Gaussian gives

$$P_a = \frac{e^{M\Phi}}{2\pi M} \frac{\mathcal{A}(\beta_\alpha, \tau_\alpha)}{\sqrt{h_\beta h_\tau}} [1 + O(M^{-1})].$$

The saddle equations simplify this coefficient to (7.26). The expansion is differentiable uniformly for x in compact positive intervals. Equivalently, one may apply the same lattice Laplace expansion directly to the positive degree average (7.5). This gives (7.27); differentiating (7.26) and using the implicit derivatives of the two roots gives (7.28). Full algebra is recorded in Appendix F. $\square$

7.3 Square-root period dispersion

The discriminant has the completed-square form

$$\boxed{D_\alpha = (\alpha - \alpha_*)^2 + 4x(1+x)\nu(1-\nu)}, \quad (7.29)$$

where

$$\alpha_* = \nu(1+2x) - x = \frac{\nu(1+\sigma) - \sigma}{1-\sigma}. \quad (7.30)$$

Using the stable notation fixed in the front matter,

$$\tilde{\mu} = \Omega\alpha_*, \quad \Delta = 2\Omega\sqrt{x(1+x)\nu(1-\nu)}, \quad \xi = \frac{a}{\rho} = \Omega\alpha. \quad (7.31)$$

Hence

$$\Omega^2 D_\alpha = (\xi - \tilde{\mu})^2 + \Delta^2. \quad (7.32)$$

Period asymptotics 7.3 (Case A marked-period dispersion: lower chart). *Under the assumptions of Period asymptotics 7.2,*

$$\begin{aligned} \Delta E_a^{\text{per}} &= 2\sqrt{(\xi - \tilde{\mu})^2 + \Delta^2} \\ &\quad + \frac{2\Omega}{M(1-\sigma)} c(\alpha, \nu, x) + O(M^{-2}). \end{aligned} \quad (7.33)$$

In particular,

$$\boxed{\Delta E_a^{\text{per}} = 2\sqrt{(\xi - \tilde{\mu})^2 + \Delta^2} + O(M^{-1})}. \quad (7.34)$$

Proof. Insert (7.27) into the exact identity (7.10), use $M/\rho = \Omega$, and note that

$$\frac{q}{1-\sigma} = \sqrt{D_\alpha}$$

because $1+x = (1-\sigma)^{-1}$. Equation (7.32) gives the displayed square root. $\square$

This is a statement about the exactly defined period of Exact identity 6.2. Identification with a named finite- M Richardson branch is deferred to Chapter 14. No branch assumption enters Period asymptotics 7.3.

7.4 The exact lower edge and its endpoint expansion

At $a = 0$, the b -sum in (7.3) collapses:

$$P_0 = \frac{M!}{(N-1)!} \sum_{s=N}^M \frac{x^s}{s(M-s)!(s-N)!}. \quad (7.35)$$

Using $s = N + r$ and $(N+r)^{-1} = \int_0^1 t^{N+r-1} dt$, one obtains the exact one-dimensional integral

$$P_0 = \frac{M! x^N}{(N-1)!(M-N)!} \int_0^1 t^{N-1} (1+xt)^{M-N} dt. \quad (7.36)$$

The exponent is monotone and has its maximum at the endpoint $t = 1$.

Define

$$\Phi_0(\nu, x) = -\nu \log \nu - (1-\nu) \log(1-\nu) + \nu \log x + (1-\nu) \log(1+x), \quad (7.37)$$

$$\tau_0 = \frac{\nu+x}{1+x}, \quad C_0(\nu, x) = \frac{1}{\tau_0} \sqrt{\frac{\nu}{2\pi(1-\nu)}}. \quad (7.38)$$

The first logarithmic correction is

$$\delta_0^{\log} = \frac{1}{12} \left(1 - \frac{1}{\nu} - \frac{1}{1-\nu} \right) + \frac{x(1-\nu)}{(\nu+x)^2}. \quad (7.39)$$

Period asymptotics 7.4 (Lower-edge endpoint asymptotics). *For fixed $0 < \nu < 1$ and $x > 0$,*

$$P_0 = \frac{C_0(\nu, x)}{\sqrt{M}} e^{M\Phi_0(\nu, x)} \left[1 + \frac{\delta_0^{\log}}{M} + O(M^{-2}) \right]. \quad (7.40)$$

Furthermore,

$$x \partial_x \log P_0 = M\tau_0 + c_0 + \frac{d_0}{M} + O(M^{-2}), \quad (7.41)$$

where

$$c_0 = -\frac{x(1-\nu)}{(\nu+x)(1+x)}, \quad d_0 = \frac{x(1-\nu)(\nu-x)}{(\nu+x)^3}. \quad (7.42)$$

The proof by endpoint Watson expansion and Stirling's formula is given in [Appendix G](#). Since $D_0 = (\nu+x)^2$, the leading edge energy is the continuous $\alpha \downarrow 0$ specialization of [Period asymptotics 7.3](#):

$$\Delta E_0^{\text{per}} = 2\Omega(\nu+x) - \frac{2\Omega}{M} \frac{x(1-\nu)}{\nu+x} + O(M^{-2}). \quad (7.43)$$

The power of the prefactor changes from M^{-1} in the interior to $M^{-1/2}$ at the edge, but the logarithmic derivative and hence the energy are continuous.

7.5 The fixed-index edge layer

Let a remain fixed while $M \rightarrow \infty$. Define

$$\Lambda(\nu, x) = \frac{(\nu+x)^2}{x(1+x)}. \quad (7.44)$$

The next coefficient is

$$B_a(\nu, x) = -\frac{2ax(1-\nu)}{(\nu+x)^2} + \frac{a(a-1)}{2} \frac{\nu^2 - 2\nu x - 2\nu - x^2}{(\nu+x)^2}. \quad (7.45)$$

Period asymptotics 7.5 (Fixed- a lower-edge layer). *For every fixed nonnegative integer a ,*

$$\frac{P_a}{P_0} = \frac{(M\Lambda)^a}{a!} \left[1 + \frac{B_a(\nu, x)}{M} + O(M^{-2}) \right]. \quad (7.46)$$

Consequently,

$$\Delta E_a^{\text{per}} = 2\Omega(\nu+x) - \frac{2\Omega}{M} \frac{x(1-\nu) + a\alpha_*}{\nu+x} + O(M^{-2}). \quad (7.47)$$

Proof. Divide every summand of P_a by the corresponding $b=0$ summand of P_0 , expand the finite factorial ratios at fixed a, b , and average over the positive P_0 distribution. Only its mean and variance through orders M and M^0 are required; they follow from (7.41) and one additional $x\partial_x$. The binomial sums over b give (7.46). The derivative identity

$$x\partial_x \log \Lambda = -\frac{\alpha_*}{(1+x)(\nu+x)} \quad (7.48)$$

then gives (7.47). Details are in [Appendix G](#). $\square$

The fixed-index formula displays the crossing mechanism directly. If $\alpha_* < 0$, the energy rises with a , so the lower-edge minimum is at $a=0$. If $\alpha_* > 0$, the energy initially decreases and the minimum moves into the window. At $\alpha_* = 0$, all fixed a have the same energy through order M^{-1} ; their splitting begins at order M^{-2} .

7.6 Matching the edge and interior descriptions

The small- α expansion of the interior rate is

$$\Phi(\alpha) = \Phi_0 + \alpha \left[\log \frac{\Lambda}{\alpha} + 1 \right] + \frac{\Xi}{2} \alpha^2 + O(\alpha^3), \quad (7.49)$$

where

$$\Xi(\nu, x) = \frac{\nu^2 - 2\nu x - 2\nu - x^2}{(\nu+x)^2}. \quad (7.50)$$

The interior prefactor satisfies

$$C(\alpha, \nu, x) = \frac{C_0(\nu, x)}{\sqrt{2\pi\alpha}} [1 + O(\alpha)]. \quad (7.51)$$

Substitute $\alpha = a/M$ into (7.25), use Stirling's formula for $a!$, and compare with (7.46). If

$$a \longrightarrow \infty, \quad \frac{a^2}{M} \longrightarrow 0, \quad (7.52)$$

the two expressions have the same leading form:

$$P_a \sim \frac{C_0 e^{M\Phi_0}}{\sqrt{M}} \frac{(M\Lambda)^a}{a!}. \quad (7.53)$$

Thus the discrete layer $a = O(1)$, the overlap region $1 \ll a \ll \sqrt{M}$, and the macroscopic regime $a = \alpha M$ form a single matched asymptotic description. The detailed expansion is given in [Appendix G](#).

7.7 Finite-size staircase at the lower-edge crossing

The thermodynamic lower-edge crossing is

$$\boxed{\alpha_* = 0 \iff \nu = \nu_0 := \frac{x}{1+2x} = \frac{\sigma}{1+\sigma}.} \quad (7.54)$$

To resolve the finite-size motion of the minimum, take the double scaling

$$\nu = \nu_0 + \frac{\eta}{M}, \quad a = O(1), \quad \eta = O(1). \quad (7.55)$$

The leading even marked gap at the crossing is

$$\Delta_c^{\text{even}} = \frac{4\Omega x(1+x)}{1+2x} = \frac{4\Omega\sigma}{1-\sigma^2}. \quad (7.56)$$

Combining the two next orders of [Period asymptotics 7.4](#) and [7.5](#) yields the following finite-size expansion.

Period asymptotics 7.6 (Critical edge staircase). *Under (7.55),*

$$\boxed{\Delta E_a^{\text{per}} = \Delta_c^{\text{even}} + \frac{2\Omega}{M} \left(\eta - \frac{1}{2} \right) + \frac{\Omega}{M^2} \mathcal{G}_a(\eta, x) + O(M^{-3}),} \quad (7.57)$$

where

$$\begin{aligned} \mathcal{G}_a(\eta, x) &= \frac{1+2x}{2(1+x)} \left[\frac{1+2x}{x} \eta - 1 \right] \\ &\quad + \frac{a(1+2x)}{2x(1+x)} [a + 2x - 1 - 2(1+2x)\eta]. \end{aligned} \quad (7.58)$$

Therefore

$$\boxed{\Delta E_{a+1}^{\text{per}} - \Delta E_a^{\text{per}} = \frac{\Omega(1+2x)}{x(1+x)M^2} [a + x - (1+2x)\eta] + O(M^{-3}).} \quad (7.59)$$

The crossing between the minimizers a and $a+1$ occurs at

$$\boxed{\eta_a = \frac{a+x}{1+2x}.} \quad (7.60)$$

Thus the optimal integer moves through

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow \dots \quad (7.61)$$

as the filling is increased across an $O(M^{-1})$ window. Away from exact ties, it is the nearest nonnegative integer to

$$(1+2x)\eta - x + \frac{1}{2}. \quad (7.62)$$

The first displacement from the edge occurs at

$$\nu_{0 \rightarrow 1}^{(M)} = \nu_0 + \frac{\nu_0}{M} + O(M^{-2}). \quad (7.63)$$

The staircase heights are $O(M^{-2})$, so they disappear from the leading thermodynamic gap even though the marked periods themselves differ by large powers of M .

7.8 Lower-chart threshold and the absence of a new leading anomaly

For $0 < \nu \leq 1/2$, the unconstrained minimizer satisfies

$$\alpha_* - \nu = x(2\nu - 1) \leq 0. \quad (7.64)$$

Hence it lies either inside the lower chart $[0, \nu]$ or below its edge. Minimizing the leading result (7.34) gives

$$\Delta_{\text{even}, \downarrow}^{\text{per}} = \begin{cases} 2\Omega(\nu + x) + o(1), & 0 < \nu \leq \frac{x}{1+2x}, \\ 2\Delta + o(1), & \frac{x}{1+2x} \leq \nu \leq \frac{1}{2}. \end{cases} \quad (7.65)$$

The two branches and their first derivatives match at $\nu = \nu_0$; the second derivative changes because the constrained minimizer moves from the boundary into the interior. The matched period analysis shows that there is no additional leading contribution hidden in the edge layer. The upper chart and the full-window threshold follow by exact electron–hole duality in Chapter 8.

7.9 Logical output and verification record

The outputs used later are:

1. the exact positive period polynomial (7.3);
2. the complete interior asymptotics (7.25)–(7.28);
3. the square-root period dispersion (7.33);
4. the edge asymptotics (7.40)–(7.42);
5. the fixed-index layer (7.46);
6. the matched overlap (7.53);
7. the finite-size staircase (7.59).

No physical Richardson-branch identification and no Gaudin determinant is used in this chapter.

Verification note 7.7 (Case A positive-sum audit). The script

`code/verify_caseA.py`

checks the cumulative identity and $P_a = \tilde{\mathcal{H}}_{a+1} + \tilde{\mathcal{H}}_a$ in exact rational arithmetic. It then evaluates the exact positive sums by log-sum-exp and verifies the interior exponent, full prefactor, logarithmic derivative, endpoint expansion through order M^{-1} , fixed- a coefficient, edge/interior overlap, and the critical staircase through order M^{-2} . The archived output is

`code/caseA_verification_report.txt.`

Chapter 8

Exact electron–hole duality

8.1 Reflected complement of a residue set

Let

$$\mathcal{L}_M = \{0, 1, \dots, M\}, \quad L = M + 1, \quad (8.1)$$

and let the electron sector contain N pairs. Define the complementary pair number

$$\boxed{\overline{N} = L - N = M + 1 - N} \quad (8.2)$$

and the imbalance

$$\delta_N = N - \overline{N} = 2N - L. \quad (8.3)$$

For a subset $A \subset \mathcal{L}_M$, define its reflection

$$\mathcal{R}(A) = \{M - a : a \in A\}. \quad (8.4)$$

For an N -element residue set S , define the reflected complement

$$\boxed{\mathcal{D}(S) = \mathcal{R}(\mathcal{L}_M \setminus S)}. \quad (8.5)$$

Then $|\mathcal{D}(S)| = \overline{N}$ and

$$\mathcal{D}^2(S) = S. \quad (8.6)$$

Thus $\mathcal{D}$ is a bijection between the N - and $\overline{N}$ -element residue subsets.

The integer

$$\chi_{M,N} := \frac{M\delta_N}{2} \quad (8.7)$$

is always integral. If M is even this is immediate; if M is odd, then $L = M + 1$ and hence $\delta_N = 2N - L$ are even.

8.2 Exact duality of the signed Vandermonde weight

Recall

$$W(S) = \prod_{n \in S} \left[(-\sigma)^n \binom{M}{n} \right] \Delta(S)^2. \quad (8.8)$$

Although the same formula is used in every pair-number sector, its transformation under (8.5) depends on $|S| = N$.

Exact identity 8.1 (Reflected-complement weight). *For every $S \subset \mathcal{L}_M$ with $|S| = N$,*

$$\boxed{W(S) = \Theta_{M,N}(\sigma) W(\mathcal{D}(S))}, \quad (8.9)$$

where

$$\boxed{\Theta_{M,N}(\sigma) = (M!)^{\delta_N} (-\sigma)^{\chi_{M,N}}}. \quad (8.10)$$

The factor is independent of the particular subset S .

Proof. Let $T = \mathcal{L}_M \setminus S$ and

$$p_M(z) = \prod_{m=0}^M (z - m).$$

The complement identity for a finite Vandermonde set is

$$\Delta(S)^2 = \Delta(\mathcal{L}_M)^2 \frac{\Delta(T)^2}{\prod_{t \in T} p'_M(t)^2}. \quad (8.11)$$

Moreover,

$$p'_M(t) = (-1)^{M-t} t! (M-t)!, \quad (8.12)$$

and therefore

$$\frac{1}{(-\sigma)^t \binom{M}{t} p'_M(t)^2} = \frac{(-\sigma)^{-t}}{(M!)^2} \binom{M}{t}. \quad (8.13)$$

Using

$$\prod_{n=0}^M \binom{M}{n} \Delta(\mathcal{L}_M)^2 = (M!)^{M+1} \quad (8.14)$$

and reflecting $t = M - a$ converts the remaining T -weight into $W(\mathcal{D}(S))$. The powers of $M!$ and $-\sigma$ reduce respectively to δ_N and $\chi_{M,N}$. The detailed cancellation is given in [Appendix H](#). $\square$

Let $Z_0^{(N)}$ denote the unlabelled all-pole sum in the N -pair sector and let $z_{M,N} = N! Z_0^{(N)}$ be its ordered version. Summing (8.9) over all N -element sets gives

$$\boxed{Z_0^{(N)} = \Theta_{M,N}(\sigma) Z_0^{(\overline{N})}}, \quad (8.15)$$

$$\boxed{z_{M,N} = \frac{N!}{\overline{N}!} \Theta_{M,N}(\sigma) z_{M,\overline{N}}}. \quad (8.16)$$

These formulas also follow by direct comparison of the closed factorization in [Exact identity 4.4](#).

8.3 Full all-pole periods and their additive energy shift

The geometric residue factor in (4.10) satisfies

$$\frac{\mathcal{C}_{M,N,\rho}}{\mathcal{C}_{M,\overline{N},\rho}} = (-1)^{\delta_N} (M!)^{-\delta_N}, \quad (8.17)$$

because

$$N(N - M - 1) = \overline{N}(\overline{N} - M - 1) = -N\overline{N}. \quad (8.18)$$

Define the coupling-independent sign

$$\varsigma_{M,N} = (-1)^{\delta_N + \chi_{M,N}}. \quad (8.19)$$

Combining (8.16) with the full prefactors gives an exact identity for the meromorphic periods.

Exact identity 8.2 (All-pole period duality). *For $0 \leq N \leq M + 1$,*

$$\boxed{Z_{\text{all}}^{(N)} = \varsigma_{M,N} \frac{N!}{\bar{N}!} \sigma^{\chi_{M,N}} \exp\left(-\frac{2\delta_N \varepsilon_{F0}}{V}\right) Z_{\text{all}}^{(\bar{N})}.} \quad (8.20)$$

Consequently,

$$\boxed{E_{\text{all},N}^{\text{per}} - E_{\text{all},\bar{N}}^{\text{per}} = \delta_N(2\varepsilon_{F0} + \Omega).} \quad (8.21)$$

Proof. Only the factors $\sigma^{\chi_{M,N}}$ and $\exp(-2\delta_N \varepsilon_{F0}/V)$ in (8.20) depend on V . Since

$$\partial_{1/V} \log \sigma = -\frac{2}{\rho}, \quad \frac{M}{\rho} = \Omega,$$

application of $-\partial_{1/V}$ gives (8.21). Direct subtraction of the exact expression (4.33) gives the same result. $\square$

The finite transformation of the filling variable used in Chapters 4–7 is

$$\bar{\nu}_M := \frac{\bar{N}}{M} = 1 - \nu_M + \frac{1}{M}, \quad \nu_M = \frac{N}{M}. \quad (8.22)$$

The shifted variable

$$\boxed{\nu^\sharp = \frac{N - \frac{1}{2}}{M}} \quad (8.23)$$

is exactly symmetric:

$$\bar{\nu}^\sharp = 1 - \nu^\sharp. \quad (8.24)$$

We retain $\nu = N/M$ in asymptotic formulas because it is tied directly to the residue sums, but every upper-chart finite-size formula must use $\bar{\nu}_M$ rather than silently replacing it by $1 - \nu_M$.

The exact reflected-complement construction is related to earlier particle–hole formulations of integrable pairing models [4]; here it is implemented directly on residue weights, kernels, and pure contour sectors.

8.4 Krawtchouk reciprocity and kernel duality

The exact matrix form of electron–hole duality follows from a reciprocal identity for the Krawtchouk-type basis.

Lemma 8.3 (Reciprocal Krawtchouk identity). *For $0 \leq n, r \leq M$,*

$$\boxed{K_{M-r}(n) = \sigma^{r-n} K_r(M-n).} \quad (8.25)$$

Proof. If $G_n(z)$ denotes the generating polynomial in (5.11), then

$$\begin{aligned} z^M G_n(z^{-1}) &= (z+1)^{M-n} (z+\sigma^{-1})^n \\ &= \sigma^{-n} (1+z)^{M-n} (1+\sigma z)^n = \sigma^{-n} G_{M-n}(\sigma z). \end{aligned}$$

Comparison of the coefficient of z^r proves the claim. $\square$

Let $\mathbf{R}$ be the reflection permutation matrix and let $\mathbf{S}_\sigma$ be the diagonal gauge

$$(\mathbf{R})_{ij} = \delta_{i,M-j}, \quad \mathbf{S}_\sigma = \text{diag}(1, \sigma, \dots, \sigma^M). \quad (8.26)$$

Exact identity 8.4 (Complementary-kernel duality). *The canonical kernels of Section 5.3 satisfy*

$$Q_N = S_\sigma^{-1} R K_{\bar{N}}^{\text{occ}} R S_\sigma, \quad (8.27)$$

$$K_N^{\text{occ}} = S_\sigma^{-1} R Q_{\bar{N}} R S_\sigma. \quad (8.28)$$

Equivalently, entrywise,

$$Q_{N,ij} = \sigma^{j-i} K_{\bar{N},M-i,M-j}^{\text{occ}}. \quad (8.29)$$

For every finite $A \subset \mathcal{L}_M$,

$$\det Q_{N,A} = \det K_{\bar{N},\mathcal{R}(A)}^{\text{occ}}, \quad \det K_{N,A}^{\text{occ}} = \det Q_{\bar{N},\mathcal{R}(A)}. \quad (8.30)$$

Proof. Insert (8.25) into the Krawtchouk-tail representation (5.28) and set $r = M - s$. The range $s = N, \dots, M$ becomes $r = 0, \dots, \bar{N} - 1$. The ratio of weights and norms simplifies to

$$\frac{w(j) \sigma^{2r-i-j}}{h_{M-r}} = \sigma^{j-i} \frac{w(M-j)}{h_r},$$

which proves (8.29). The matrix formula follows. Since diagonal gauges and simultaneous row-column reflection do not change principal determinants, the first equality in (8.30) follows; the second follows by exchanging N and $\bar{N}$. $\square$

The same statement follows directly from the Boolean map. If $T = \mathcal{D}(S)$, then

$$\xi_a(S) = \eta_{M-a}(T), \quad \eta_a(S) = \xi_{M-a}(T). \quad (8.31)$$

Together with the constant weight relation (8.9), this gives

$$\left\langle \prod_{a \in A} \xi_a \right\rangle_N = \left\langle \prod_{b \in \mathcal{R}(A)} \eta_b \right\rangle_{\bar{N}}, \quad (8.32)$$

which is the insertion form of (8.30).

8.5 Excluded and occupation-supported pure sectors

The excluded-pole period of Chapter 6 is generated by ξ_A . Its dual insertion is an occupation constraint. Define the reduced occupation-supported sector

$$\mathcal{Z}_{\text{all}}^{\text{occ},(N)}[A] := N! \sum_{|S|=N} W(S) \eta_A(S) = N! Z_0^{(N)} \det K_{N,A}^{\text{occ}}. \quad (8.33)$$

It is a period of a finite linear combination of pole-supported product cycles: one sums over all assignments of the forced poles in A to labelled integration variables and leaves the remaining variables all-pole.

Exact identity 8.5 (Pure-sector electron-hole map). *For every $A \subset \mathcal{L}_M$,*

$$\mathcal{Z}_{\text{all}}^{(N)}[A] = \frac{N!}{\bar{N}!} \Theta_{M,N} \mathcal{Z}_{\text{all}}^{\text{occ},(\bar{N})}[\mathcal{R}(A)], \quad (8.34)$$

$$\mathcal{Z}_{\text{all}}^{\text{occ},(N)}[A] = \frac{N!}{\bar{N}!} \Theta_{M,N} \mathcal{Z}_{\text{all}}^{(\bar{N})}[\mathcal{R}(A)]. \quad (8.35)$$

After normalization by the corresponding all-pole periods, these identities reduce to (8.30).

This distinction is essential. An excluded electron pole becomes a forced occupied pole in the reflected hole ensemble. Therefore an upper electron marked chart is not obtained by applying the lower excluded-pole formula to a high electron index. It is obtained by reflecting a lower excluded-pole hole chart, hence by an electron occupation minor.

8.6 Lower and upper sign-corrected marked charts

For the lower chart retain the definition of Chapter 5:

$$P_a^{\downarrow, (N)} = (-1)^{N+a} Q_{N,aa}, \quad 0 \leq a < N. \quad (8.36)$$

For the upper chart define

$$P_a^{\uparrow, (N)} = (-1)^{N+a+1} K_{N,aa}^{\text{occ}}, \quad N \leq a \leq M. \quad (8.37)$$

Let

$$\bar{a} = M - a. \quad (8.38)$$

Then $0 \leq \bar{a} < \bar{N}$ precisely when $N \leq a \leq M$.

Exact identity 8.6 (Exact upper marked factor). *For $N \leq a \leq M$,*

$$P_a^{\uparrow, (N)} = P_{\bar{a}}^{\downarrow, (\bar{N})} > 0. \quad (8.39)$$

The associated reduced upper marked period can be written as

$$\mathcal{Z}_{\text{mark},a}^{\uparrow, (N)} := \frac{(-1)^{N+a+1}}{N} \mathcal{Z}_{\text{all}}^{\text{occ}, (N)}[\{a\}] = (N-1)! Z_0^{(N)} P_a^{\uparrow, (N)}. \quad (8.40)$$

Consequently,

$$\frac{\mathcal{Z}_{\text{mark},a}^{\uparrow, (N)}}{\mathcal{Z}_{\text{all}}^{(N)}} = \frac{P_a^{\uparrow, (N)}}{N}. \quad (8.41)$$

Proof. The diagonal specialization of (8.28) gives

$$K_{N,aa}^{\text{occ}} = Q_{\bar{N}, \bar{a} \bar{a}}.$$

The two sign exponents differ by an even integer:

$$(\bar{N} + \bar{a}) - (N + a + 1) = 2(M - N - a).$$

Thus (8.39) follows from the lower definition. The period identity is the singleton specialization of (8.33). $\square$

The multipoint version is also useful. Write $\mathfrak{P}_{\bar{N}}^{\downarrow}$ for the lower sign-corrected minor defined by (5.62) in the $\bar{N}$ -pair sector. For $A \subset \{N, N+1, \dots, M\}$, $m = |A|$, define

$$\mathfrak{P}_N^{\uparrow}(A) = (-1)^{m(N+1) + \sum_{a \in A} a + \binom{m}{2}} \det K_{N,A}^{\text{occ}}. \quad (8.42)$$

Then

$$\mathfrak{P}_N^{\uparrow}(A) = \mathfrak{P}_{\bar{N}}^{\downarrow}(\mathcal{R}(A)) \geq 0, \quad (8.43)$$

with strict positivity exactly when $m \leq N$; larger minors vanish because $\text{rank } K_N^{\text{occ}} = N$.

8.7 Upper WKB chart and full-window leading dispersion

The exact identity (8.39) transfers every lower-chart formula of Chapter 7 to the upper chart. At finite M use

$$\bar{\alpha} = \frac{\bar{a}}{M} = 1 - \alpha, \quad \bar{\nu}_M = \frac{\bar{N}}{M} = 1 - \nu_M + \frac{1}{M}. \quad (8.44)$$

For $\bar{\alpha}$ in a compact subset of $(0, \bar{\nu}_M)$,

$$P_a^{\uparrow, (N)} = \frac{C(\bar{\alpha}, \bar{\nu}_M, x)}{M} \exp[M\Phi(\bar{\alpha}; \bar{\nu}_M, x)][1 + O(M^{-1})]. \quad (8.45)$$

Also,

$$x\partial_x \log P_a^{\uparrow, (N)} = Mq(\bar{\alpha}; \bar{\nu}_M, x) + c(\bar{\alpha}, \bar{\nu}_M, x) + O(M^{-1}). \quad (8.46)$$

No new saddle calculation is required.

In the thermodynamic limit set

$$\bar{\nu} = 1 - \nu, \quad \bar{\alpha} = 1 - \alpha. \quad (8.47)$$

The saddle data transform as

$$\bar{\beta} = 1 - \tau_\alpha, \quad \bar{\tau} = 1 - \beta_\alpha, \quad (8.48)$$

which implies

$$D_{1-\alpha}(1 - \nu, x) = D_\alpha(\nu, x), \quad q(1 - \alpha; 1 - \nu, x) = q(\alpha; \nu, x), \quad (8.49)$$

$$\Phi(1 - \alpha; 1 - \nu, x) = \Phi(\alpha; \nu, x). \quad (8.50)$$

The prefactor is not itself invariant:

$$C(1 - \alpha, 1 - \nu, x) = \frac{\tau_\alpha - \nu}{\nu - \beta_\alpha} C(\alpha, \nu, x). \quad (8.51)$$

This is why the exact upper $O(M^{-1})$ coefficient must be obtained by reflection rather than copied from the lower chart.

The thermodynamic spectral parameters transform as

$$\bar{\tilde{\mu}} = \Omega - \tilde{\mu}, \quad \bar{\Delta} = \Delta, \quad \bar{\xi} = \Omega - \xi. \quad (8.52)$$

Hence

$$(\bar{\xi} - \bar{\tilde{\mu}})^2 + \bar{\Delta}^2 = (\xi - \tilde{\mu})^2 + \Delta^2. \quad (8.53)$$

Period asymptotics 8.7 (Case A marked-period dispersion: upper chart). *Let $a/M \rightarrow \alpha$ with $\nu < \alpha < 1$ and let the parameters remain in a compact interior domain. Then*

$$E_{\text{mark}, a}^{\text{per}} - E_{\text{all}}^{\text{per}} = 2\sqrt{(\xi - \tilde{\mu})^2 + \Delta^2} + O(M^{-1}). \quad (8.54)$$

Together with *Period asymptotics 7.3*, the same leading square root holds in both marked charts. The common interface $\alpha = \nu$ is obtained by one-sided continuation; the saddle remains nondegenerate there for fixed $x > 0$.

At the upper edge $a = M$ one has the exact identity

$$P_M^{\uparrow, (N)} = P_0^{\downarrow, (\bar{N})}. \quad (8.55)$$

Therefore the endpoint integral and its full expansion are obtained from [Equations \(7.36\)](#) and [\(7.41\)](#) by replacing N with $\bar{N}$. In particular,

$$E_{\text{mark}, M}^{\text{per}} - E_{\text{all}}^{\text{per}} = 2\Omega(1 - \nu + x) + O(M^{-1}) \quad (8.56)$$

in the thermodynamic limit. The upper edge is crossed by the chemical potential when

$$\boxed{\tilde{\mu} = \Omega \quad \Longleftrightarrow \quad \nu = \frac{1}{1 + \sigma}.} \quad (8.57)$$

This is the reflected image of the lower crossing $\nu = \sigma/(1 + \sigma)$.

8.8 Lower hole cascades and upper electron cascades

For a residue set S in the electron sector define the upper occupation count

$$U_r(S) := \sum_{j=M-r+1}^M \eta_j(S), \quad r = 1, \dots, M+1. \quad (8.58)$$

If $T = \mathcal{D}(S)$, then the lower hole count of [Chapter 6](#) obeys

$$\boxed{H_r(T) = U_r(S).} \quad (8.59)$$

Consequently every consecutive lower cascade in the $\bar{N}$ -pair hole ensemble is transformed into an upper electron cascade whose multiplicity polynomial is built from the U_r . This statement includes the full Boolean–Möbius expansion and its excluded/occupied pure sectors. [Chapter 11](#) will introduce the labelled upper contour normalization when the physical Case B cascade is specialized; no new finite algebra is required there.

8.9 Physical Richardson energies and the commutator term

The residue-period duality [\(8.21\)](#) is not yet the duality of eigenenergies of the reduced pairing Hamiltonian. The latter contains an additional commutator contribution.

In the seniority-zero pair space write

$$H = \sum_{n=0}^M 2\varepsilon_n b_n^\dagger b_n - V B^\dagger B, \quad B^\dagger = \sum_{n=0}^M b_n^\dagger. \quad (8.60)$$

Define reflected hole operators

$$\bar{b}_m^\dagger = b_{M-m}, \quad \bar{b}_m = b_{M-m}^\dagger, \quad \bar{n}_m = 1 - b_{M-m}^\dagger b_{M-m}. \quad (8.61)$$

Then an N -pair electron state is mapped to an $\bar{N}$ -pair hole state. Let $U_{\mathcal{D}}$ be the basis permutation defined by

$$U_{\mathcal{D}}|S\rangle = |\mathcal{D}(S)\rangle, \quad (8.62)$$

and let

$$E_{\text{ctr}} = 2\varepsilon_{F0} + \Omega \quad (8.63)$$

be the centre of the pair-energy window.

Exact identity 8.8 (Physical electron–hole duality). *On the seniority-zero sectors,*

$$H^{(N)} = \mathsf{U}_{\mathcal{D}}^\dagger H^{(\bar{N})} \mathsf{U}_{\mathcal{D}} + \delta_N (E_{\text{ctr}} - V) I. \quad (8.64)$$

Hence every paired Richardson eigenvalue satisfies

$$E_{\lambda, N}^{\text{RG}} - E_{\lambda, \bar{N}}^{\text{RG}} = \delta_N (2\varepsilon_{F0} + \Omega - V). \quad (8.65)$$

Proof. The kinetic term becomes

$$\sum_n 2\varepsilon_n b_n^\dagger b_n = \sum_m 2\varepsilon_m \bar{n}_m + \delta_N E_{\text{ctr}}.$$

Furthermore $B^\dagger = \bar{B}$ and $B = \bar{B}^\dagger$, so

$$B^\dagger B = \bar{B} \bar{B}^\dagger = \bar{B}^\dagger \bar{B} + [\bar{B}, \bar{B}^\dagger].$$

In the $\bar{N}$ -hole sector,

$$[\bar{B}, \bar{B}^\dagger] = L - 2\bar{N} = \delta_N.$$

The interaction therefore contributes the additional constant $-V\delta_N$, proving (8.64). $\square$

Comparing (8.21) and (8.65) shows exactly where the physical and period dualities differ:

$$(E_N^{\text{RG}} - E_N^{\text{per}}) - (E_{\bar{N}}^{\text{RG}} - E_{\bar{N}}^{\text{per}}) = -V\delta_N = V(\bar{N} - N). \quad (8.66)$$

Since $V = O(M^{-1})$ while $\delta_N = O(M)$, this is an $O(1)$ contribution. It will reappear in the Gaudin correction of Chapter 15; it must not be inserted into the residue-period energy by hand.

In thermodynamic variables,

$$V(\bar{N} - N) \longrightarrow v\Omega(1 - 2\nu). \quad (8.67)$$

The leading energy density and all excitation gaps remain electron–hole symmetric; the asymmetry is confined to the additive extensive-centre term and the corresponding $O(1)$ bridge correction.

8.10 Logical output and verification record

The exact outputs used by later chapters are:

1. the reflected-complement weight identity (8.9);
2. the exact all-pole period and period-energy transformations (8.20)–(8.21);
3. the matrix kernel similarities (8.27)–(8.28);
4. the excluded-to-occupied pure-sector map (8.34);
5. the upper marked factor (8.37) and its equality to a lower hole factor;
6. the upper-chart square-root period dispersion (8.54);
7. the physical Hamiltonian shift (8.65) and its distinction from period duality.

Verification note 8.9 (Exact-rational electron–hole audit). The script

`code/verify_eh_duality.py`

checks the reflected-complement weight for every subset in the stated finite range; compares unordered and ordered all-pole sums; verifies Krawtchouk reciprocity, the two matrix kernel similarities, principal-minor and pure-sector transformations, upper one-point factors and upper sign corrections; checks the all-pole period-energy shift symbolically; and verifies the physical Hamiltonian identity as an exact rational matrix equality in every pair-number sector tested. The archived output is

`code/eh_duality_verification_report.txt.`

Chapter 9

Blocked levels and odd sectors

9.1 Excitation terminology and one blocked level

Two operations must be kept distinct throughout the manuscript. A *real-root excitation* remains in the seniority-zero sector: the number of pairs is unchanged and one Richardson root is trapped between one-particle levels. A *blocked-level excitation* contains one or more singly occupied pair orbitals; every such orbital is removed from the pairing sums. The first operation was represented by the pure marked periods of Chapters 6–7. The second is treated here. This terminology avoids the accidental interchange of the two excitation labels in the legacy text.

Fix a lattice level $p \in \mathcal{L}_M = \{0, \dots, M\}$ and retain N paired electrons of each spin orientation, together with one unpaired electron in the orbital p . The paired subsystem is described by the meromorphic period

$$Z_N^{[p]} = \frac{1}{(2\pi i)^N} \int_{\Gamma_1} \cdots \int_{\Gamma_N} \frac{\prod_{i < j} (R_i - R_j)^2}{\prod_{j=1}^N \prod_{\substack{\ell=0 \\ \ell \neq p}}^M (2\varepsilon_\ell - R_j)} \exp\left(-\frac{1}{V} \sum_{j=1}^N R_j\right) \prod_{j=1}^N dR_j. \quad (9.1)$$

Equivalently, relative to the all-pole integrand,

$$S_N^{[p]}(\mathbf{R}) = S_N(\mathbf{R}) \prod_{j=1}^N (2\varepsilon_p - R_j). \quad (9.2)$$

The insertion is not the Boolean excluded-pole insertion of Chapter 6. It is a characteristic polynomial of the entire residue configuration.

For a residue subset $S \subset \mathcal{L}_M$, define

$$\chi_S(z) = \prod_{n \in S} (z - n). \quad (9.3)$$

At the residue $R_j = 2\varepsilon_{F0} + 2n_j/\rho$, the factor in (9.2) contributes

$$\left(\frac{2}{\rho}\right)^N \prod_{j=1}^N (p - n_j) = \left(\frac{2}{\rho}\right)^N \chi_S(p). \quad (9.4)$$

The factor vanishes whenever $p \in S$, which is precisely the blocking condition.

Let $Z_0^{(N)}$ denote the unlabelled N -residue sum of (4.13), with the pair number displayed when needed. Define the dimensionless blocked residue polynomial

$$\mathcal{B}_{N,p} = N! \sum_{\substack{S \subset \mathcal{L}_M \\ |S|=N}} W(S) \chi_S(p). \quad (9.5)$$

The full period has the exact reduction

$$Z_N^{[p]} = \mathcal{C}_{M,N,\rho} \left(\frac{2}{\rho} \right)^N \exp \left(-\frac{2N\varepsilon_{F0}}{V} \right) \mathcal{B}_{N,p}. \quad (9.6)$$

All extra powers of $2/\rho$ are independent of V and therefore disappear from every energy difference obtained by logarithmic differentiation.

9.2 Monic Krawtchouk polynomials and the Heine identity

Polynomial formulations of pairing equations through Heine–Stieltjes correspondences provide useful context for the construction below [34]. The present identity is instead obtained directly from the signed finite ensemble.

The Krawtchouk-type polynomials K_s were defined by (5.11). Their leading coefficient in the falling-factorial basis is $x^{-s}/s!$. We therefore introduce the monic family

$$\pi_s^{(M)}(z; x) = s! x^s K_s^{(M)}(z; \sigma). \quad (9.7)$$

Using (5.13),

$$\pi_s^{(M)}(z; x) = s! x^s \sum_{m=0}^s \binom{z}{m} \binom{M-m}{s-m} x^{-m}. \quad (9.8)$$

For an integer lattice point p , terms with $m > p$ vanish and every remaining term is positive for $x > 0$.

Exact identity 9.1 (One-characteristic-polynomial Heine identity). *For $0 \leq N \leq M$ and every complex z ,*

$$\frac{1}{Z_0^{(N)}} \sum_{|S|=N} W(S) \chi_S(z) = \pi_N^{(M)}(z; x). \quad (9.9)$$

Consequently,

$$\mathcal{B}_{N,p} = N! Z_0^{(N)} \pi_N^{(M)}(p; x), \quad (9.10)$$

$$\frac{Z_N^{[p]}}{Z_{\text{all}}^{(N)}} = \left(\frac{2}{\rho} \right)^N \pi_N^{(M)}(p; x). \quad (9.11)$$

A proof from finite Andreief algebra is given in Appendix I. The same identity can also be characterized intrinsically: the normalized average in (9.9) is monic of degree N and is orthogonal to every polynomial of degree below N in the signed bilinear form generated by $w(n)$.

The connection with the hypergeometric sum of Ch2 is now immediate. Since

$$\pi_N^{(M)}(0; x) = (M)_{\underline{N}} x^N, \quad (9.12)$$

we may write

$$\pi_N^{(M)}(p; x) = (M)_{\underline{N}} x^N \mathfrak{h}_{N,p}(x), \quad (9.13)$$

where

$$\mathfrak{h}_{N,p}(x) = \sum_{m=0}^{\min(N,p)} \binom{N}{m} \frac{(p)_{\underline{m}}}{(M)_{\underline{m}}} x^{-m}. \quad (9.14)$$

Because $x^{-1} = \sigma^{-1} - 1$, this is exactly the finite positive sum obtained in the legacy blocked-level calculation, now identified as a normalized monic orthogonal polynomial.

Let

$$T_m^{\text{bl}} = \binom{N}{m} \frac{(p)_{\underline{m}}}{(M)_{\underline{m}}} x^{-m}. \quad (9.15)$$

All T_m^{bl} are nonnegative, and

$$x \partial_x \log \pi_N^{(M)}(p; x) = \frac{\sum_m (N-m) T_m^{\text{bl}}}{\sum_m T_m^{\text{bl}}}. \quad (9.16)$$

Unlike the original signed residue ensemble, this transformed one-dimensional sum is a genuine positive object.

9.3 Positive one-dimensional WKB

Let

$$N = \nu M + O(1), \quad p = \alpha M + O(1), \quad 0 < \nu, \alpha < 1, \quad (9.17)$$

and write $m = \beta M + O(1)$. Introduce $L(y) = y \log y$, with $L(0) = 0$, and the normalized blocked action

$$\begin{aligned} \mathcal{U}_{\alpha, \nu}(\beta) &= L(\nu) + L(\alpha) + L(1 - \beta) - L(\beta) - L(\nu - \beta) \\ &\quad - L(\alpha - \beta) - \beta \log x. \end{aligned}$$

Stirling's formula gives

$$T_m^{\text{bl}} = \frac{1}{\sqrt{2\pi M}} \sqrt{\frac{\nu\alpha(1-\beta)}{\beta(\nu-\beta)(\alpha-\beta)}} \exp[M\mathcal{U}_{\alpha, \nu}(\beta)][1 + O(M^{-1})]. \quad (9.18)$$

The ratio of neighboring exact summands is

$$\frac{T_{m+1}^{\text{bl}}}{T_m^{\text{bl}}} = \frac{(N-m)(p-m)}{(m+1)(M-m)x}. \quad (9.19)$$

Hence the unique interior maximum satisfies

$$(\nu - \beta)(\alpha - \beta) = x\beta(1 - \beta). \quad (9.20)$$

The two roots of the corresponding quadratic are

$$\beta_\alpha = \frac{\alpha + \nu + x - \sqrt{D_\alpha}}{2(1+x)}, \quad \tau_\alpha = \frac{\alpha + \nu + x + \sqrt{D_\alpha}}{2(1+x)}, \quad (9.21)$$

with

$$D_\alpha = (\alpha + \nu + x)^2 - 4(1+x)\alpha\nu, \quad q(\alpha) = \tau_\alpha - \beta_\alpha. \quad (9.22)$$

For $0 < \alpha, \nu < 1$,

$$0 < \beta_\alpha < \min(\alpha, \nu), \quad (9.23)$$

so no boundary saddle is involved away from $\alpha = 0$.

Period asymptotics 9.2 (Blocked-polynomial WKB). *Uniformly when (α, ν, x) ranges over compact subsets of $(0, 1)^2 \times (0, \infty)$,*

$$\pi_N^{(M)}(p; x) = (M)_{\underline{N}} x^N \mathcal{B}(\alpha, \nu, x) \exp [M \mathcal{U}_{\alpha, \nu}(\beta_\alpha)] [1 + O(M^{-1})], \quad (9.24)$$

where

$$\mathcal{B}(\alpha, \nu, x) = \sqrt{\frac{\nu \alpha (1 - \beta_\alpha)}{\beta_\alpha (1 + x) q(\alpha)}}. \quad (9.25)$$

Moreover,

$$x \partial_x \log \pi_N^{(M)}(p; x) = M(\nu - \beta_\alpha) + b_{\text{bl}}(\alpha, \nu, x) + O(M^{-1}), \quad (9.26)$$

with

$$b_{\text{bl}} = \frac{x}{2(1+x)} \left[\frac{1}{q} - 1 - \frac{\beta_\alpha(1 - \beta_\alpha) + \tau_\alpha(1 - \tau_\alpha)}{q^2} \right]. \quad (9.27)$$

Proof. The action has a unique nondegenerate maximum at (9.20). Its curvature is

$$-\mathcal{U}''(\beta_\alpha) = \frac{(1+x)q(\alpha)}{x\beta_\alpha(1 - \beta_\alpha)}. \quad (9.28)$$

The lattice sum of (9.18) is therefore a one-dimensional Gaussian sum. Multiplying the local amplitude by the inverse square root of (9.28) gives (9.25). The envelope theorem yields

$$x \partial_x \mathcal{U}(\beta_\alpha) = -\beta_\alpha. \quad (9.29)$$

Together with the explicit factor x^N , this proves the leading term in (9.26). Differentiating (9.25), using

$$x \partial_x \beta_\alpha = -\frac{x\beta_\alpha(1 - \beta_\alpha)}{(1+x)q}, \quad x \partial_x \tau_\alpha = \frac{x\tau_\alpha(1 - \tau_\alpha)}{(1+x)q}, \quad (9.30)$$

produces (9.27). $\square$

At the lower lattice edge $p = 0$, the normalized sum contains only $m = 0$ and

$$\pi_N^{(M)}(0; x) = (M)_{\underline{N}} x^N, \quad x \partial_x \log \pi_N^{(M)}(0; x) = N \quad (9.31)$$

exactly. This agrees with the $\alpha \downarrow 0$ limit of the leading energy derived below, even though the Gaussian prefactor (9.25) is not uniform at that boundary.

9.4 Energy of the paired subsystem

Because the explicit factor $(2/\rho)^N$ in (9.11) is independent of V , the exact change of the paired period barycenter is

$$E_{N, \text{pair}}^{[p], \text{per}} - E_N^{\text{per}} = \frac{2}{\rho(1 - \sigma)} x \partial_x \log \pi_N^{(M)}(p; x). \quad (9.32)$$

Let

$$\xi_p = \frac{p}{\rho} = \Omega \alpha + O(M^{-1}), \quad \mu = \varepsilon_{F0} + \tilde{\mu}, \quad (9.33)$$

and recall

$$\mathcal{E}(\xi) = \sqrt{(\xi - \tilde{\mu})^2 + \Delta^2}. \quad (9.34)$$

The algebraic identity

$$2(1+x)(\nu - \beta_\alpha) = \sqrt{D_\alpha} - (\alpha - \alpha_*), \quad \alpha_* = \frac{\tilde{\mu}}{\Omega}, \quad (9.35)$$

converts (9.26) into the physical square root.

Period asymptotics 9.3 (Paired response to one blocked level). *For a macroscopically placed blocked level, and also at $p = 0$ by (9.31),*

$$E_{N,\text{pair}}^{[p],\text{per}} - E_N^{\text{per}} = \mathcal{E}(\xi_p) - (\xi_p - \tilde{\mu}) + O(M^{-1}). \quad (9.36)$$

The explicit $O(M^{-1})$ period correction in the interior is

$$\frac{2\Omega}{M(1-\sigma)} b_{\text{bl}}(\alpha, \nu, x). \quad (9.37)$$

The two terms in (9.36) have a useful interpretation. The square root is the quasiparticle contribution. The linear term compensates the kinetic energy that will be restored when the singly occupied orbital is added to the full odd state.

9.5 Odd particle and odd hole sectors

Let E_{2N}^{per} denote the all-pole period barycenter of the even N -pair reference sector. An odd state obtained by adding one electron to level p contains the same N pairs plus the kinetic energy $\varepsilon_p = \varepsilon_{F0} + \xi_p$. Therefore (9.36) gives

$$E_{2N+1,p}^{\text{per}} - E_{2N}^{\text{per}} = \mu + \mathcal{E}(\xi_p) + O(M^{-1}). \quad (9.38)$$

An odd state obtained by removing one electron contains $N - 1$ pairs and one singly occupied level. Applying the same blocked result in the $(N - 1)$ -pair sector and using the exact all-pole energy difference of Chapter 4 yields

$$E_{2N-1,p}^{\text{per}} - E_{2N}^{\text{per}} = -\mu + \mathcal{E}(\xi_p) + O(M^{-1}). \quad (9.39)$$

The $O(M^{-1})$ terms include the difference between the finite- M chemical work and its thermodynamic limit. Their complete physical values require the Gaussian branch bridge of Chapters 15 and 16.

Period asymptotics 9.4 (Odd/parity dispersion). *The chemical-work-subtracted odd particle and odd hole periods have the common leading dispersion*

$$E_{2N+1,p}^{\text{per}} - E_{2N}^{\text{per}} - \mu = E_{2N-1,p}^{\text{per}} - E_{2N}^{\text{per}} + \mu = \mathcal{E}(\xi_p) + O(M^{-1}). \quad (9.40)$$

Equivalently,

$$\frac{E_{2N+1,p}^{\text{per}} + E_{2N-1,p}^{\text{per}} - 2E_{2N}^{\text{per}}}{2} = \mathcal{E}(\xi_p) + O(M^{-1}). \quad (9.41)$$

Minimization over the Debye window gives the period quasiparticle threshold

$$\Delta_{\text{qp}}^{\text{per}} = \min_{0 \leq \xi \leq \Omega} \mathcal{E}(\xi) + o(1). \quad (9.42)$$

Using $x = \sigma/(1 - \sigma)$,

$$\Delta_{\text{qp}}^{\text{per}} = \begin{cases} \Omega(\nu + x) + o(1), & 0 < \nu \leq \frac{\sigma}{1 + \sigma}, \\ 2\Omega\sqrt{x(1+x)\nu(1-\nu)} + o(1), & \frac{\sigma}{1 + \sigma} \leq \nu \leq \frac{1}{1 + \sigma}, \\ \Omega(1 - \nu + x) + o(1), & \frac{1}{1 + \sigma} \leq \nu < 1. \end{cases} \quad (9.43)$$

The three branches are respectively the dilute-pair edge, the interior BCS-like minimum, and the dilute-hole edge. The physical interpretation follows after branch identification; the period calculation itself uses only the positive polynomial (9.8).

9.6 Christoffel deformation and a blocked complementary kernel

The blocked residue ensemble can be normalized directly:

$$\langle F \rangle_{[p]} = \frac{\langle \chi_S(p) F(S) \rangle_0}{\pi_N^{(M)}(p; x)}. \quad (9.44)$$

Its one-particle weight is the Christoffel deformation

$$w_{[p]}(n) = (p - n)w(n). \quad (9.45)$$

Let $\mathfrak{h}_j$ be the signed squared norm of the monic polynomial $\pi_j^{(M)}$. The monic polynomials for (9.45) are

$$\pi_j^{[p]}(z) = \frac{\pi_{j+1}^{(M)}(z) - \frac{\pi_{j+1}^{(M)}(p)}{\pi_j^{(M)}(p)} \pi_j^{(M)}(z)}{z - p}, \quad (9.46)$$

and their norms are

$$\mathfrak{h}_j^{[p]} = \mathfrak{h}_j \frac{\pi_{j+1}^{(M)}(p)}{\pi_j^{(M)}(p)}. \quad (9.47)$$

Because (9.8) is positive at every integer p for $x > 0$, no exceptional zero occurs in these ratios.

The corresponding occupation kernel and its complement are

$$K_{N,ij}^{[p],\text{occ}} = (p - j)w(j) \sum_{s=0}^{N-1} \frac{\pi_s^{[p]}(i)\pi_s^{[p]}(j)}{\mathfrak{h}_s^{[p]}}, \quad Q_N^{[p]} = I - K_N^{[p],\text{occ}}. \quad (9.48)$$

They generate hole insertions in the blocked ensemble:

$$\langle \xi_A \rangle_{[p]} = \det Q_{N,A}^{[p]}. \quad (9.49)$$

This exact finite construction is the input needed when blocked sectors are placed over the Case B nested background.

9.7 Verification output

Verification note 9.5 (Blocked-sector audit). The script

`code/verify_seniority.py`

checks the Heine identity by direct subset enumeration, the relation to the Ch2 hypergeometric sum, the Christoffel polynomial and norm formulas, and the blocked complementary-kernel minors in exact rational arithmetic. It also evaluates the positive one-dimensional sum by log-sum-exp and verifies the full prefactor, the constant logarithmic-derivative term, the pair-subsystem square root, and the odd particle/hole bookkeeping along finite-size sequences. The archived output is

`code/seniority_verification_report.txt`.

Chapter 10

Relations among the excitation sectors

10.1 Stable classification of the sectors

We use the following names throughout the remainder of the manuscript.

Sector R . A seniority-zero state with the same number of pairs as the reference state and one Richardson root trapped between one-particle levels. Its pure residue representative is the marked period of Chapter 7.

Sector B_r . A state with r singly occupied pair orbitals. Those r levels are blocked and the number of active pairs is reduced accordingly. The case $r = 1$ is odd; $r = 2$ is the physically broken-pair sector.

The sectors are different finite- M eigenspaces. Equality of their thermodynamic thresholds does not identify their states, root topologies, matrix elements, or subleading finite-size corrections.

10.2 Exact finite-seniority periods

Let n be the number of active pairs and let

$$\mathbf{p} = (p_1, \dots, p_r), \quad 0 \leq p_1 < \dots < p_r \leq M, \quad n + r \leq M + 1, \quad (10.1)$$

be a set of r blocked levels. For a residue subset S , define

$$\chi_S(\mathbf{p}) = \prod_{i=1}^r \chi_S(p_i), \quad \Delta(\mathbf{p}) = \prod_{1 \leq i < j \leq r} (p_j - p_i). \quad (10.2)$$

The reduced finite-seniority residue polynomial is

$$\mathcal{B}_{n;\mathbf{p}} = n! \sum_{\substack{S \subset \mathcal{L}_M \\ |S|=n}} W(S) \chi_S(\mathbf{p}). \quad (10.3)$$

The corresponding full period contains the coupling-independent scale factor $(2/\rho)^{nr}$:

$$Z_n^{[\mathbf{p}]} = \mathcal{C}_{M,n,\rho} \left(\frac{2}{\rho} \right)^{nr} \exp \left(-\frac{2n\varepsilon_{F0}}{V} \right) \mathcal{B}_{n;\mathbf{p}}. \quad (10.4)$$

Exact identity 10.1 (Finite Heine–Christoffel determinant). *For fixed finite r and $n + r \leq M + 1$,*

$$\boxed{\frac{\mathcal{B}_{n;\mathbf{p}}}{n!Z_0^{(n)}} = \frac{\det \left[\pi_{n+j-1}^{(M)}(p_i; x) \right]_{i,j=1}^r}{\Delta(\mathbf{p})}}. \quad (10.5)$$

For $r = 1$, this reduces to (9.9). For $r = 2$,

$$\boxed{\frac{\mathcal{B}_{n;p,q}}{n!Z_0^{(n)}} = \frac{\pi_n^{(M)}(p)\pi_{n+1}^{(M)}(q) - \pi_{n+1}^{(M)}(p)\pi_n^{(M)}(q)}{q - p}}. \quad (10.6)$$

The proof in Appendix I uses only finite Andreief algebra and the monicity of the π_j . Positivity of the one-point polynomial is not required for the determinant identity. When blocked lattice points approach the same macroscopic energy, the determinant has a confluent discrete-Wronskian limit; its leading exponential rate remains the sum of the one-point rates.

10.2.1 Christoffel-deformed pure sectors

The normalized finite-seniority ensemble is

$$\langle F \rangle_{[\mathbf{p}]} = \frac{\langle \chi_S(\mathbf{p}) F(S) \rangle_0}{\langle \chi_S(\mathbf{p}) \rangle_0}. \quad (10.7)$$

It is again a finite polynomial ensemble, with lattice weight

$$w_{[\mathbf{p}]}(m) = w(m) \prod_{i=1}^r (p_i - m). \quad (10.8)$$

Consequently there is a complementary kernel $Q_n^{[\mathbf{p}]}$ such that

$$\boxed{\langle \xi_A \rangle_{[\mathbf{p}]} = \det Q_{n,A}^{[\mathbf{p}]}}, \quad (10.9)$$

This formula gives the exact finite-seniority insertion over any later nested-contour background: the Boolean cascade polynomial is averaged with the Christoffel-deformed kernel rather than with Q_n itself.

10.3 Thermodynamic additivity for finite seniority

Let r remain fixed as $M \rightarrow \infty$, and let

$$p_i = \alpha_i M + O(1), \quad 0 \leq \alpha_i \leq 1. \quad (10.10)$$

For separated α_i , each row of the determinant in (10.5) has the one-point exponential rate of Chapter 9. The finite determinant and the Vandermonde modify only the algebraic prefactor. If several α_i coincide, the confluent determinant replaces values by a finite number of derivatives and again changes only the prefactor. Therefore the logarithmic derivative is additive at order M .

Period asymptotics 10.2 (Finite-seniority additivity). *For a fixed finite blocked set $\mathbf{p}$ in an n -pair background,*

$$\boxed{-\partial_{1/V} \log \frac{Z_n^{[\mathbf{p}]}}{Z_{\text{all}}^{(n)}} = \sum_{i=1}^r [\mathcal{E}_n(\xi_{p_i}) - (\xi_{p_i} - \tilde{\mu}_n)] + O(M^{-1})}. \quad (10.11)$$

Here $\tilde{\mu}_n$ and $\mathcal{E}_n$ are evaluated at filling n/M . After the kinetic energies of the singly occupied levels are restored,

$$E_{2n+r,\mathbf{p}}^{\text{per}} - E_{2n}^{\text{per}} = r\mu_n + \sum_{i=1}^r \mathcal{E}_n(\xi_{p_i}) + O(M^{-1}). \quad (10.12)$$

For a broken pair at fixed total particle number $2N$, the active pair number is $n = N - 1$ and $r = 2$. The chemical work needed to compare the $(N - 1)$ -pair and N -pair reference sectors cancels the two terms 2μ . Thus

$$E_{B_2;p,q}^{\text{per}} - E_{2N}^{\text{per}} = \mathcal{E}(\xi_p) + \mathcal{E}(\xi_q) + O(M^{-1}). \quad (10.13)$$

The complete $O(M^{-1})$ comparison with the real-root sector is postponed to Chapters 16 and 17.

10.4 Exact particle–hole convolution for the marked period

We now connect the seniority-zero marked factor $P_a^\downarrow$ to the blocked polynomials of Chapter 9. For clarity, write

$$\pi_s^{(M)}(z; y) = s!y^s \sum_{m=0}^s \binom{z}{m} \binom{M-m}{s-m} y^{-m} \quad (10.14)$$

when the coupling variable is y rather than x .

Exact identity 10.3 (Particle–hole convolution of the marked factor). *For $0 \leq a < N \leq M$,*

$$P_a^\downarrow = \frac{N(M-N+1)x}{a!(M-a)!} \int_0^1 \pi_{N-1}^{(M)}(a; xt) \times \pi_{M-N}^{(M)}(M-a; xt) dt. \quad (10.15)$$

The first polynomial is the blocked polynomial of an $(N - 1)$ -pair electron sector at level a . The second is the blocked polynomial of an $(M - N)$ -pair hole sector at the reflected level $M - a$. Since

$$M - N = \bar{N} - 1, \quad (10.16)$$

the two factors are precisely the odd removal and odd addition sectors surrounding the even N -pair reference state. The integral couples them through a single positive endpoint variable.

Proof. Expand both polynomials in (10.15). In the first polynomial use the summation index b . In the second set its summation index to $M - s$. The product then contains t^{s-b-1} , and

$$\int_0^1 t^{s-b-1} dt = \frac{1}{s-b}. \quad (10.17)$$

The factorial coefficient outside the integral cancels the remaining binomial normalizations and reproduces term by term the positive double sum (7.3). The complete coefficient calculation is recorded in Appendix I. $\square$

The convolution provides an independent finite-dimensional explanation of the factor two in the marked energy. It is stronger than a comparison of the final square roots: it relates the exact finite- M period to the two neighboring odd sectors before any saddle approximation.

10.5 Endpoint Laplace asymptotics and the factor two

Every coefficient of the two polynomials in (10.15) is positive for $x, t > 0$. Hence the exponent of the t -integrand is increasing and its maximum is at $t = 1$. The logarithmic slope there is

$$\begin{aligned} t \partial_t \log \left[\pi_{N-1}^{(M)}(a; xt) \pi_{M-N}^{(M)}(M-a; xt) \right] \Big|_{t=1} \\ = Mq(\alpha) + O(1). \end{aligned} \quad (10.18)$$

Indeed, the electron polynomial contributes

$$M(\nu - \beta_\alpha) + O(1), \quad (10.19)$$

and the reflected hole polynomial contributes

$$M(\tau_\alpha - \nu) + O(1). \quad (10.20)$$

Their sum is $Mq(\alpha) + O(1)$.

One-sided Laplace asymptotics therefore gives

$$\boxed{P_a^\downarrow = \frac{N(M-N+1)x}{a!(M-a)!} \frac{\pi_{N-1}^{(M)}(a; x) \pi_{M-N}^{(M)}(M-a; x)}{Mq(\alpha)} \times [1 + O(M^{-1})].} \quad (10.21)$$

The prefactor outside the two polynomials and the endpoint factor $1/[Mq(\alpha)]$ contribute only $O(1)$ to $x \partial_x \log P_a$ and hence only $O(M^{-1})$ to the energy.

The electron-side logarithmic derivative is

$$\frac{2}{\rho(1-\sigma)} x \partial_x \log \pi_{N-1}^{(M)}(a; x) = \mathcal{E}(\xi) - (\xi - \tilde{\mu}) + O(M^{-1}), \quad (10.22)$$

while exact electron–hole duality gives

$$\frac{2}{\rho(1-\sigma)} x \partial_x \log \pi_{M-N}^{(M)}(M-a; x) = \mathcal{E}(\xi) + (\xi - \tilde{\mu}) + O(M^{-1}). \quad (10.23)$$

The linear terms cancel.

Period asymptotics 10.4 (Origin of the marked factor two). *The exact convolution (10.15) implies*

$$\boxed{-\partial_{1/V} \log P_a^\downarrow = 2\mathcal{E}(\xi) + O(M^{-1}).} \quad (10.24)$$

At leading order, the real-root marked period is the sum of one odd-particle and one odd-hole quasiparticle contribution at the same macroscopic energy.

This result agrees with, but does not assume, the direct two-dimensional saddle of Chapter 7. The convolution gives a second derivation based entirely on finite-seniority polynomials.

10.6 Equality of the leading even thresholds

The two even sectors have the leading dispersions

$$\Delta_R^{(0)}(\xi) = 2\mathcal{E}(\xi), \quad (10.25)$$

$$\Delta_{B_2}^{(0)}(\xi_1, \xi_2) = \mathcal{E}(\xi_1) + \mathcal{E}(\xi_2). \quad (10.26)$$

At finite M , the two blocked levels must be distinct. In the thermodynamic limit their spacing tends to zero, so they can both approach the minimizer of $\mathcal{E}$.

Period asymptotics 10.5 (Common leading neutral threshold). *The real-root and broken-pair residue sectors have the same leading neutral threshold:*

$$\boxed{\min_{\xi} \Delta_R^{(0)}(\xi) = \min_{\xi_1 \neq \xi_2} \Delta_{B_2}^{(0)}(\xi_1, \xi_2) = 2\Delta_{\text{qp}}.} \quad (10.27)$$

Thus the leading even gap is

$$\boxed{\Delta_{\text{even}} = 2\Delta_{\text{qp}}.} \quad (10.28)$$

The equality is only a threshold statement. The sectors have different seniority, different finite root patterns, and generally different operator matrix elements. When the minimum of $\mathcal{E}$ lies in the interior of the Debye window, their first physical finite-size corrections coincide. When the minimum is pinned to an edge, they split at order $d = \Omega/M$, with an additional $d/\log M$ soft-root scale. Those effects are treated in Chapters 16 and 17.

A useful exact edge identity, proved in Appendix I, is

$$\boxed{\pi_n^{(M)}(0)\pi_{n+1}^{(M)}(1) - \pi_{n+1}^{(M)}(0)\pi_n^{(M)}(1) = \frac{\pi_n^{(M)}(0)\pi_{n+1}^{(M)}(0)}{Mx}.} \quad (10.29)$$

It is the finite algebraic source of the especially simple two-blocked-level period at the lower edge.

10.7 Verification output

Verification note 10.6 (Finite-seniority and convolution audit). The script

`code/verify_seniority.py`

checks the one- and multi-characteristic-polynomial identities by direct subset enumeration, the two-level and general determinant formulas, one-step Christoffel transformations, the exact marked convolution, and the special edge identity in exact rational arithmetic. Independent large- M tests verify the one-dimensional blocked prefactor, the additive degree densities of the electron and hole polynomials, and the convergence of both even sectors to the common threshold. The archived output is

`code/seniority_verification_report.txt.`

Part IV

Case B and the contour-topology transition

Chapter 11

Exact consecutive-cascade periods in Case B

11.1 The lower consecutive cascade

Fix an active pair number N and an integer

$$0 \leq k < N, \quad c = N - k > 0. \quad (11.1)$$

The integer k is the finite topological label of the lower cascade. The corresponding product cycle contains c all-pole contours and the k consecutive lower-cutoff contours

$$\gamma_{\geq 1}, \gamma_{\geq 2}, \dots, \gamma_{\geq k} : \quad \Gamma_{N,k}^{\downarrow} = \gamma_{\text{all}}^{\times c} \times \gamma_{\geq 1} \times \dots \times \gamma_{\geq k}. \quad (11.2)$$

Its full period is written

$$Z_{N,k}^{\downarrow} = \mathcal{C}_{M,N,\rho} \exp\left(-\frac{2N\varepsilon_{F0}}{V}\right) \mathfrak{Z}_{N,k}^{\downarrow}, \quad (11.3)$$

where $\mathcal{C}_{M,N,\rho}$ is the common geometric factor in (4.9). Thus $\mathfrak{Z}_{N,k}^{\downarrow}$ is the reduced residue period. At $k = 0$,

$$\mathfrak{Z}_{N,0}^{\downarrow} = N!Z_0 = \mathcal{Z}_{\text{all}}. \quad (11.4)$$

For a residue set $S \subset \mathcal{L}_M$, $|S| = N$, retain

$$H_r(S) = \sum_{j=0}^{r-1} \xi_j(S) = r - \#\{n \in S : n < r\}. \quad (11.5)$$

The variable H_r counts holes among the first r lattice nodes.

Exact identity 11.1 (Physical consecutive-cascade period). *The reduced lower-cascade period is*

$$\boxed{\mathfrak{Z}_{N,k}^{\downarrow} = c! \sum_{\substack{S \subset \mathcal{L}_M \\ |S|=N}} W(S) \prod_{r=1}^k [c + H_r(S)]}. \quad (11.6)$$

Proof. For fixed S , assign residues to the trapped variables from the largest cutoff downwards. Before the contour with cutoff r is assigned, the $k - r$ higher-cutoff variables have already occupied admissible residues. There are

$$N - \#\{n \in S : n < r\} - (k - r) = c + H_r(S)$$

remaining choices. The c residues left after all trapped variables are assigned may be permuted among the all-pole variables in $c!$ ways. Multiplication over r gives the multiplicity in (11.6). This is the specialization $r_t = t$ of [Exact identity 6.4](#). $\square$

The restriction $k < N$ is natural for the thermodynamic Case B branch studied below: a nonzero macroscopic set of collective contours remains. The unnormalized counting formula itself also has a $c = 0$ limit, but the normalized expansions used in the endpoint analysis divide by c^k and will always assume $c > 0$.

Warning 11.2 (The cascade label is held fixed in an energy derivative). For every finite system, k labels a homology class. Therefore k and $c = N - k$ are held fixed when $-\partial_{1/V} \log Z_{N,k}^\downarrow$ is evaluated. In the thermodynamic analysis one may select a sequence $k = k(M; \nu, \sigma)$ after the finite derivative has been defined, but one must not differentiate the integer k with respect to the coupling.

11.2 Positive Boolean coefficients

Let

$$\mathcal{I}_k = \{0, 1, \dots, k-1\}. \quad (11.7)$$

Normalize the multiplicity polynomial by its value on the no-hole configuration:

$$\boxed{\prod_{r=1}^k (c + H_r) = c^k \sum_{A \subset \mathcal{I}_k} \hat{C}_{k,c}(A) \xi_A,} \quad \hat{C}_{k,c}(\emptyset) = 1. \quad (11.8)$$

The coefficients are the specialization of the canonical Boolean–Möbius coefficients of Chapter 6. For later asymptotics it is useful to record a manifestly positive representation.

For $A \subset \mathcal{I}_k$, let $\mathcal{S}_k(A)$ be the set of pairs (T, f) such that

$$T \subset \{1, \dots, k\}, \quad f : T \longrightarrow A \quad (11.9)$$

is surjective and satisfies

$$f(r) < r \quad (r \in T). \quad (11.10)$$

For $A = \emptyset$, $\mathcal{S}_k(\emptyset)$ contains the single empty selection.

Exact identity 11.3 (Surjective-selection formula). *For every $A \subset \mathcal{I}_k$,*

$$\boxed{\hat{C}_{k,c}(A) = \sum_{(T,f) \in \mathcal{S}_k(A)} c^{-|T|}.} \quad (11.11)$$

Consequently,

$$\boxed{\hat{C}_{k,c}(A) \geq 0.} \quad (11.12)$$

The singleton coefficient is

$$\boxed{\hat{C}_{k,c}(\{a\}) = \left(1 + \frac{1}{c}\right)^{k-a} - 1, \quad 0 \leq a < k.} \quad (11.13)$$

Moreover,

$$\boxed{0 \leq \hat{C}_{k,c}(A) \leq \prod_{a \in A} \left[\left(1 + \frac{1}{c}\right)^{k-a} - 1 \right].} \quad (11.14)$$

Proof. Expand

$$\prod_{r=1}^k \left(1 + \frac{1}{c} \sum_{j < r} \xi_j \right).$$

Each factor either contributes 1 or selects one allowed variable ξ_j with $j < r$. After imposing $\xi_j^2 = \xi_j$, a selection has support exactly A if and only if the selected factors form a set T and their assignments define a surjection $f : T \rightarrow A$ satisfying (11.10). Every used factor contributes c^{-1} , proving (11.11). For a singleton $A = \{a\}$, choose a nonempty subset of the $k - a$ factors with $r > a$; summing their weights gives (11.13). Finally, allow every $a \in A$ to choose an arbitrary nonempty subset of its admissible factors independently. This drops the constraint that one factor cannot select two different variables and therefore gives the upper bound (11.14). $\square$

The exact Möbius form remains

$$\widehat{C}_{k,c}(A) = c^{-k} \sum_{B \subset A} (-1)^{|A|-|B|} \prod_{r=1}^k [c + |B \cap \{0, \dots, r-1\}|]. \quad (11.15)$$

The positive selection form and the alternating Möbius form are complementary: the former is adapted to bounds, while the latter is convenient for exact symbolic generation.

11.3 The full ground period as a pure-sector sum

Define the normalized ground cascade factor

$$\mathcal{R}_{N,k}^\downarrow := \frac{\mathfrak{Z}_{N,k}^\downarrow}{c!c^k Z_0}. \quad (11.16)$$

Combining (11.8) with the hole-minor identity gives the complete finite support sum.

Exact identity 11.4 (Ground cascade as a sum of pure sectors). *We have*

$$\mathcal{R}_{N,k}^\downarrow = \sum_{A \subset \mathcal{I}_k} \widehat{C}_{k,c}(A) \det Q_{N,A}. \quad (11.17)$$

Equivalently,

$$\mathcal{R}_{N,k}^\downarrow = 1 + \sum_{\emptyset \neq A \subset \mathcal{I}_k} s_N(A) \widehat{C}_{k,c}(A) \mathfrak{P}_N(A), \quad (11.18)$$

where

$$s_N(A) = (-1)^{|A|N + \sum_{a \in A} a + \binom{|A|}{2}} \quad (11.19)$$

and $\mathfrak{P}_N(A) \geq 0$ is the sign-corrected minor of Chapter 5.

Proof. Insert (11.8) into (11.6), divide by $c!c^k Z_0$, and use

$$\langle \xi_A \rangle_0 = \det Q_{N,A}.$$

Equation (11.18) is the definition of the sign-corrected minors rewritten as an identity for the uncorrected determinants. $\square$

The coefficients in (11.17) are nonnegative, but the determinants carry the fixed signs $s_N(A)$. Therefore the complete ground period is not a positive sum. The positive Cauchy–Binet expansion controls each $\mathfrak{P}_N(A)$ separately; possible cancellation among different supports must be estimated only after the algebraic prefactors have been retained. This is the task of Chapter 12, not an assumption of the present chapter.

11.4 Complete marked periods over the cascade

Let $E \subset \mathcal{L}_M$ be a finite set of poles excluded from every factor of the product cycle (11.2). The corresponding reduced period is

$$\mathfrak{Z}_{N,k}^\downarrow[E] = c! \sum_{|S|=N} W(S) \xi_E(S) \prod_{r=1}^k [c + H_r(S)]. \quad (11.20)$$

Define

$$\mathcal{R}_{N,k}^\downarrow[E] := \sum_{A \subset \mathcal{I}_k} \hat{C}_{k,c}(A) \det Q_{N,A \cup E}. \quad (11.21)$$

Thus $\mathcal{R}_{N,k}^\downarrow[\emptyset] = \mathcal{R}_{N,k}^\downarrow$.

Exact identity 11.5 (Complete marked period). *For every finite $E \subset \mathcal{L}_M$,*

$$\mathfrak{Z}_{N,k}^\downarrow[E] = c! c^k Z_0 \mathcal{R}_{N,k}^\downarrow[E]. \quad (11.22)$$

Consequently,

$$\frac{Z_{N,k}^\downarrow[E]}{Z_{N,k}^\downarrow} = \frac{\mathcal{R}_{N,k}^\downarrow[E]}{\mathcal{R}_{N,k}^\downarrow}. \quad (11.23)$$

For a single lower-chart marked level $a < N$, $Q_{N,aa} \neq 0$ and it is useful to isolate the Case A one-point factor:

$$\mathcal{M}_{N,k}^\downarrow(a) := \frac{\mathcal{R}_{N,k}^\downarrow[\{a\}]}{Q_{N,aa}}. \quad (11.24)$$

Then the exact marked ratio is

$$\frac{Z_{N,k}^\downarrow[\{a\}]}{Z_{N,k}^\downarrow} = Q_{N,aa} \frac{\mathcal{M}_{N,k}^\downarrow(a)}{\mathcal{R}_{N,k}^\downarrow}. \quad (11.25)$$

After the fixed sign is removed,

$$(-1)^{N+a} \frac{Z_{N,k}^\downarrow[\{a\}]}{Z_{N,k}^\downarrow} = P_a^\downarrow \frac{\mathcal{M}_{N,k}^\downarrow(a)}{\mathcal{R}_{N,k}^\downarrow}. \quad (11.26)$$

At $k = 0$, both correction factors equal 1 and the Case A period of Chapter 7 is recovered. In the lower Case B threshold problem the separated marked index will satisfy

$$k \leq a < N. \quad (11.27)$$

The exact formulas remain meaningful outside this range, but a marked point inside the endpoint cluster requires a different asymptotic scaling.

11.5 A determinant generator for the complete cascade

The sums over all supports in (11.17) and (11.21) can be resummed before any large- M approximation. For $j \in \mathcal{I}_k \cup E$, define

$$U_j(\mathbf{u}) = \sum_{r=1}^k \mathbf{1}_{j < r} u_r. \quad (11.28)$$

Introduce variables z_e , $e \in E$, and the diagonal matrix

$$D_{jj}(\mathbf{u}, \mathbf{z}) = \exp[U_j(\mathbf{u})] \prod_{e \in E} (1 + z_e \mathbf{1}_{j=e}) - 1. \quad (11.29)$$

All matrices in this section are restricted to $\mathcal{I}_k \cup E$.

Exact identity 11.6 (Physical-cascade determinant generator). *The complete excluded-pole period obeys*

$$\boxed{\frac{\mathfrak{Z}_{N,k}^\downarrow[E]}{c!Z_0} = \prod_{r=1}^k (c + \partial_{u_r}) \prod_{e \in E} \partial_{z_e} \times \det[I + D(\mathbf{u}, \mathbf{z})Q_N]|_{\mathbf{u}=0, \mathbf{z}=0}.} \quad (11.30)$$

For a singleton $E = \{a\}$, set

$$A(\mathbf{u}) = I + D(\mathbf{u}, 0)Q_N. \quad (11.31)$$

Jacobi's formula gives

$$\boxed{\partial_{z_a} \det[I + D(\mathbf{u}, z_a)Q_N]|_{z_a=0} = \det A(\mathbf{u}) e^{U_a(\mathbf{u})} [Q_N A(\mathbf{u})^{-1}]_{aa}.} \quad (11.32)$$

For a separated point $a \geq k$, $U_a = 0$. Equation (11.32) is the exact finite precursor of the Schur-complement separation used in the Case B asymptotics.

11.6 Finite-seniority sectors over the cascade

The same physical cascade can be placed over a sector with a finite blocked set. Let n be the number of active pairs,

$$\mathbf{p} = (p_1, \dots, p_s), \quad 0 \leq p_1 < \dots < p_s \leq M, \quad n + s \leq M + 1, \quad (11.33)$$

let $0 \leq k < n$, and put

$$c_n = n - k. \quad (11.34)$$

The blocked characteristic insertion is

$$\chi_S(\mathbf{p}) = \prod_{i=1}^s \prod_{m \in S} (p_i - m). \quad (11.35)$$

For an additional excluded set E , define the reduced cascade period

$$\boxed{\mathfrak{Z}_{n,k}^{\downarrow, [\mathbf{p}]}[E] = c_n! \sum_{|S|=n} W_n(S) \chi_S(\mathbf{p}) \xi_E(S) \prod_{r=1}^k [c_n + H_r(S)].} \quad (11.36)$$

Here W_n denotes the same signed Vandermonde weight with the subset size displayed. The corresponding full period is

$$\boxed{Z_{n,k}^{\downarrow, [\mathbf{p}]}[E] = \mathcal{C}_{M,n,\rho} \left(\frac{2}{\rho} \right)^{ns} e^{-2n\varepsilon_{F0}/V} \mathfrak{Z}_{n,k}^{\downarrow, [\mathbf{p}]}[E].} \quad (11.37)$$

Normalize the finite-seniority insertion by

$$\mathcal{H}_{n;\mathbf{p}} := \frac{1}{Z_0^{(n)}} \sum_{|S|=n} W_n(S) \chi_S(\mathbf{p}) = \frac{\det[\pi_{n+j-1}^{(M)}(p_i; x)]_{i,j=1}^s}{\Delta(\mathbf{p})}. \quad (11.38)$$

For $s = 0$, set $\mathcal{H}_{n;\emptyset} = 1$. Let $Q_n^{[\mathbf{p}]}$ be the complementary kernel of the Christoffel-deformed ensemble introduced in (10.9). Define

$$\mathcal{R}_{n,k}^{\downarrow, [\mathbf{p}]}[E] := \sum_{A \subset \mathcal{I}_k} \hat{C}_{k,c_n}(A) \det Q_{n,A \cup E}^{[\mathbf{p}]} \quad (11.39)$$

Exact identity 11.7 (Finite seniority over the consecutive cascade). *For every fixed blocked set $\mathbf{p}$ and excluded set E ,*

$$\mathfrak{Z}_{n,k}^{\downarrow, [\mathbf{p}]}[E] = c_n! c_n^k Z_0^{(n)} \mathcal{H}_{n;\mathbf{p}} \mathcal{R}_{n,k}^{\downarrow, [\mathbf{p}]}[E]. \quad (11.40)$$

In particular,

$$\frac{Z_{n,k}^{\downarrow, [\mathbf{p}]}}{Z_{n,k}^{\downarrow}} = \left(\frac{2}{\rho}\right)^{ns} \mathcal{H}_{n;\mathbf{p}} \frac{\mathcal{R}_{n,k}^{\downarrow, [\mathbf{p}]}[\emptyset]}{\mathcal{R}_{n,k}^{\downarrow}[\emptyset]}. \quad (11.41)$$

Within the same blocked background,

$$\frac{Z_{n,k}^{\downarrow, [\mathbf{p}]}[E]}{Z_{n,k}^{\downarrow, [\mathbf{p}]}} = \frac{\mathcal{R}_{n,k}^{\downarrow, [\mathbf{p}]}[E]}{\mathcal{R}_{n,k}^{\downarrow, [\mathbf{p}]}[\emptyset]}. \quad (11.42)$$

Proof. Divide the signed measure $W_n(S)\chi_S(\mathbf{p})$ by its total $Z_0^{(n)}\mathcal{H}_{n;\mathbf{p}}$. The resulting finite polynomial ensemble has the hole-minor identity

$$\langle \xi_A \rangle_{[\mathbf{p}]} = \det Q_{n,A}^{[\mathbf{p}]}.$$

Expanding the same Boolean cascade polynomial (11.8), now with $c = c_n$, gives (11.40). Restoring the coupling-independent factor $(2/\rho)^{ns}$ gives (11.41); division within a fixed blocked sector gives (11.42). $\square$

The blocked level may lie inside the lower endpoint region. The factor $\chi_S(\mathbf{p})$ then annihilates residues that occupy it, while H_r counts it as a hole in every cumulative interval that contains it. No modification of the counting formula is required.

The determinant generator also survives unchanged after replacing Q_n by $Q_n^{[\mathbf{p}]}$:

$$\begin{aligned} & \frac{\mathfrak{Z}_{n,k}^{\downarrow, [\mathbf{p}]}[E]}{c_n! Z_0^{(n)} \mathcal{H}_{n;\mathbf{p}}} \\ &= \prod_{r=1}^k (c_n + \partial_{u_r}) \prod_{e \in E} \partial_{z_e} \det[I + D(\mathbf{u}, \mathbf{z}) Q_n^{[\mathbf{p}]}] \Big|_0. \end{aligned} \quad (11.43)$$

Thus ground, odd, broken-pair, and additionally marked finite-seniority sectors are generated by one finite determinant after the appropriate Christoffel deformation.

11.7 Exact period-energy bookkeeping

The coupling-independent factorials and powers of $2/\rho$ in the preceding formulas are retained for exact normalization but drop out of logarithmic energy derivatives. With k fixed as in Warning 11.2,

$$E_{N,k}^{\downarrow, \text{per}} = 2N\varepsilon_{F0} - \partial_{1/V} \log \mathfrak{Z}_{N,k}^{\downarrow}. \quad (11.44)$$

For an excluded marked set E in the same pair sector,

$$\Delta E_{N,k}^{\downarrow, \text{per}}[E] = -\partial_{1/V} \log \frac{\mathfrak{Z}_{N,k}^{\downarrow}[E]}{\mathfrak{Z}_{N,k}^{\downarrow}}. \quad (11.45)$$

For a finite-seniority background, the pair-subsystem barycenter is

$$E_{n,k,\text{pair}}^{\downarrow, [\mathbf{p}], \text{per}} = 2n\varepsilon_{F0} - \partial_{1/V} \log \mathfrak{Z}_{n,k}^{\downarrow, [\mathbf{p}]}. \quad (11.46)$$

The total many-body energy additionally contains the kinetic energies of the singly occupied orbitals:

$$E_{2n+s,k}^{\downarrow, [\mathbf{p}], \text{per}} = E_{n,k,\text{pair}}^{\downarrow, [\mathbf{p}], \text{per}} + \sum_{i=1}^s \varepsilon_{p_i}. \quad (11.47)$$

These identities contain no approximation. Chapters 12 and 13 will determine which sequences of k represent the thermodynamic Case B branch and will estimate the exact ratios in (11.25) and (11.41).

11.8 Upper cascades from exact electron–hole duality

No independent upper-endpoint counting is necessary. Let

$$\overline{N} = M + 1 - N, \quad \overline{c} = \overline{N} - k, \quad (11.48)$$

with $0 \leq k < \overline{N}$, and define the upper occupation counts

$$U_r(S) = \sum_{j=M-r+1}^M \mathbf{1}_{j \in S}, \quad r = 1, \dots, k. \quad (11.49)$$

The reflected-complement identity of Chapter 8 gives

$$H_r(\mathcal{D}(S)) = U_r(S). \quad (11.50)$$

Define the reduced upper electron period by the exact dual normalization

$$\begin{aligned} \mathfrak{Z}_{N,k}^{\uparrow} &:= \frac{N!}{\overline{N}!} \Theta_{M,N}(\sigma) \mathfrak{Z}_{\overline{N},k}^{\downarrow} \\ &= \frac{N!}{\overline{N}!} \overline{c}! \sum_{|S|=N} W_N(S) \prod_{r=1}^k [\overline{c} + U_r(S)]. \end{aligned} \quad (11.51)$$

At $k = 0$, this reduces to $N!Z_0^{(N)}$, as required.

Reindex the upper nodes by $a \mapsto M - a$. The same Boolean coefficients appear, but hole minors are replaced by occupation minors:

$$\begin{aligned} \frac{\mathfrak{Z}_{N,k}^{\uparrow}}{N!Z_0^{(N)}} &= \frac{\overline{c}! \overline{c}^k}{\overline{N}!} \\ &\quad \times \sum_{A \subset \mathcal{I}_k} \widehat{C}_{k,\overline{c}}(A) \det K_{N,\mathcal{R}(A)}^{\text{occ}}. \end{aligned} \quad (11.52)$$

Equation (8.30) converts each occupation minor to the corresponding lower hole minor in the $\overline{N}$ sector. Upper marked and finite-seniority periods are obtained in exactly the same way: reflect the marked or blocked data and use the lower formulas with $N \mapsto \overline{N}$.

Restoring full periods yields

$$Z_{N,k}^{\uparrow} = \varsigma_{M,N} \frac{N!}{\overline{N}!} \sigma^{\chi_{M,N}} \exp\left(-\frac{2\delta_N \varepsilon_{F0}}{V}\right) Z_{\overline{N},k}^{\downarrow}. \quad (11.53)$$

Thus the upper Case B sector is not a second independent construction. It is the exact occupation-supported image of the lower hole cascade.

11.9 What is and is not established here

The present chapter establishes the complete finite- M algebra needed for Case B:

1. the physical consecutive ground cascade (11.6);
2. its complete pure-sector sum (11.17);
3. the full marked ratio (11.23), without fixing an endpoint support;
4. all finite-seniority backgrounds through (11.40);
5. one determinant generator for every one of these sectors;
6. the exact upper-cascade image.

No statement has yet been made about the size of the support sums. In particular, the positivity of the coefficients $\widehat{C}_{k,c}(A)$ does not imply positivity of the full period, and no endpoint component has been assumed to dominate. Chapter 12 will combine the positive Cauchy–Binet representation of each sign-corrected minor with its full algebraic prefactor and then perform the complete support summation.

11.10 Verification output

Verification note 11.8 (Finite Case B exact-period audit). The script

`code/verify_caseB_exact.py`

checks, in exact rational arithmetic, the consecutive-cascade multiplicity by complete assignment enumeration; the positive surjective-selection formula, singleton coefficient, and collision-free coefficient bound; the ground and excluded-pole cascade periods against their Q -minor expansions; the determinant differential generator; one- and two-blocked Christoffel-deformed cascade periods; marked insertions over blocked backgrounds; and the reflected-complement upper-cascade formula. The archived output is

`code/caseB_exact_verification_report.txt.`

Chapter 12

Thermodynamic Case B and complete endpoint resummation

12.1 Compact lower Case B

We first work on the strict lower electron branch

$$0 < \nu < \frac{1}{2}. \quad (12.1)$$

The upper branch is obtained exactly by electron–hole duality in [Section 12.8](#). Throughout this chapter $\nu = N/M$ denotes the exact finite-system ratio. Every endpoint, action, and prefactor is evaluated at this ratio before a large- M expansion is taken. This convention matters for the coefficient of the leading finite correction, even when N/M differs from a limiting filling by only $O(M^{-1})$.

Recall the one-point rate $\Phi(\alpha; \nu, x)$ and the saddle roots $\beta_\alpha, \tau_\alpha$ from [Equations \(7.20\)](#) and [\(7.22\)](#). On the lower branch,

$$\partial_\alpha \Phi = \log \frac{\tau_\alpha - \alpha}{\alpha - \beta_\alpha} > 0, \quad 0 < \alpha < \nu. \quad (12.2)$$

Indeed, a zero of the derivative would require

$$\tau_\alpha + \beta_\alpha = 2\alpha, \quad (12.3)$$

which gives

$$\alpha = \frac{\nu + x}{1 + 2x} \geq \nu \quad (12.4)$$

when $\nu \leq 1/2$. Since $\partial_\alpha \Phi \rightarrow +\infty$ as $\alpha \downarrow 0$, the derivative is positive throughout the lower chart.

Definition 12.1 (Compact lower Case B interior). *A compact lower Case B parameter set is a compact set of pairs (ν, x) for which there is a unique endpoint*

$$\kappa = \kappa(\nu, x) \in (0, \nu) \quad (12.5)$$

satisfying

$$\boxed{\Phi(\kappa; \nu, x) = 0}, \quad (12.6)$$

and constants $\eta, d_0 > 0$, independent of the point in the compact set, such that

$$\eta \leq \kappa \leq \nu - \eta, \quad \partial_\alpha \Phi(\kappa; \nu, x) \geq d_0. \quad (12.7)$$

For each finite system we choose an integer cascade length

$$k = \kappa M + \delta_M, \quad |\delta_M| \leq C_\delta, \quad (12.8)$$

and put $c = N - k$. The integer k is held fixed in every derivative with respect to $1/V$, as required by [Warning 11.2](#).

Monotonicity implies

$$\Phi(\alpha) < 0 \quad (0 \leq \alpha < \kappa), \quad \Phi(\alpha) > 0 \quad (\kappa < \alpha < \nu). \quad (12.9)$$

The compactness assumptions are deliberately stronger than necessary. They isolate the macroscopic Case B interior; the nonuniform limit $\kappa \downarrow 0$ is treated separately in [Chapter 13](#).

12.2 Full off-diagonal kernel asymptotics

Let

$$\mathbf{K}_{ij} = (-1)^{N-j} Q_{N,ij} \quad (12.10)$$

be the positive sign-gauged lower kernel of [Equation \(5.40\)](#). For

$$i = \alpha M, \quad j = \gamma M, \quad 0 < \alpha, \gamma < \nu, \quad (12.11)$$

define the separated actions

$$U_{\alpha,\nu}(\beta) = L(\alpha) - L(\alpha - \beta) + L(1 - \beta) - L(\beta) - L(\nu - \beta) - \beta \log x, \quad (12.12)$$

$$V_{\gamma,\nu}(\tau) = L(\tau) - L(1 - \tau) - L(\tau - \nu) - L(\tau - \gamma) - L(\gamma) + \tau \log x, \quad (12.13)$$

where $L(y) = y \log y$, $L(0) = 0$. Put

$$u(\alpha) = U_{\alpha,\nu}(\beta_\alpha), \quad v(\gamma) = V_{\gamma,\nu}(\tau_\gamma). \quad (12.14)$$

On the diagonal,

$$u(\alpha) + v(\alpha) = \Phi(\alpha). \quad (12.15)$$

Period asymptotics 12.2 (Off-diagonal complementary-kernel WKB). *Uniformly for α and γ in compact subintervals of $(0, \nu)$,*

$$\boxed{\mathbf{K}_{ij} = \frac{\mathcal{G}(\alpha, \gamma; \nu, x)}{M} \exp\{M[u(\alpha) + v(\gamma)]\} \left[1 + O(M^{-1})\right]}, \quad (12.16)$$

where

$$\begin{aligned} \mathcal{G}(\alpha, \gamma; \nu, x) &= \frac{x}{2\pi(1+x)} \frac{(1 - \beta_\alpha)\tau_\gamma}{(\tau_\gamma - \beta_\alpha)\sqrt{q(\alpha)q(\gamma)}} \\ &\quad \times \sqrt{\frac{\alpha(\nu - \beta_\alpha)}{\gamma(\alpha - \beta_\alpha)(\tau_\gamma - \nu)(\tau_\gamma - \gamma)}}. \end{aligned} \quad (12.17)$$

The diagonal specialization is exactly the one-point prefactor of [Equation \(7.26\)](#):

$$\boxed{\mathcal{G}(\alpha, \alpha; \nu, x) = C(\alpha, \nu, x)}. \quad (12.18)$$

Proof. Insert the exact positive double sum (5.49) into (12.10). With $b = \beta M$, $s = \tau M$, Stirling's formula gives

$$T_{b,s}^{(i,j)} = \frac{e^{M[U_{\alpha,\nu}(\beta) + V_{\gamma,\nu}(\tau)]}}{(2\pi)^2 M^2} \mathcal{A}_{\alpha,\gamma}(\beta, \tau) [1 + O(M^{-1})], \quad (12.19)$$

where

$$\mathcal{A}_{\alpha,\gamma}(\beta, \tau) = \frac{1}{\tau - \beta} \sqrt{\frac{\alpha(1 - \beta)(\nu - \beta)\tau}{\gamma(\alpha - \beta)\beta(1 - \tau)(\tau - \nu)(\tau - \gamma)}}. \quad (12.20)$$

The two saddle equations are the lower-root equation at α and the upper-root equation at γ . Their curvatures are the corresponding diagonal curvatures (7.24). The two-dimensional lattice Gaussian supplies $2\pi M / \sqrt{h_\beta(\alpha)h_\tau(\gamma)}$. Simplification by the saddle equations yields (12.17). The complete calculation is recorded in Appendix J.1. $\square$

The factor M^{-1} in (12.16) is the first ingredient missed by an exponent-only endpoint estimate. The second ingredient is the confluent vanishing of a determinant when several source indices approach the same macroscopic endpoint.

12.3 Confluent endpoint minors

Write

$$C_\kappa = C(\kappa, \nu, x), \quad q_\kappa = q(\kappa; \nu, x), \quad d_\kappa = \partial_\alpha \Phi(\kappa; \nu, x) > 0. \quad (12.21)$$

The derivatives of the two saddle roots are

$$\beta'_\kappa = \frac{\nu - \beta_\kappa}{(1 + x)q_\kappa}, \quad \tau'_\kappa = \frac{\tau_\kappa - \nu}{(1 + x)q_\kappa}. \quad (12.22)$$

Let m be fixed and choose distinct indices

$$a_r = \kappa M + \delta_r(M), \quad r = 1, \dots, m, \quad (12.23)$$

where the offsets are uniformly bounded and their pairwise differences are fixed nonzero integers. Define

$$\mathcal{V}(\boldsymbol{\delta}) = \prod_{r < s} (\delta_s - \delta_r). \quad (12.24)$$

Period asymptotics 12.3 (Confluent endpoint minor). For $A_M = \{a_1, \dots, a_m\}$,

$$\begin{aligned} \mathfrak{P}(A_M) &= \frac{C_\kappa^m}{M^{m^2}} \left(\frac{\beta'_\kappa \tau'_\kappa}{q_\kappa^2} \right)^{\binom{m}{2}} \mathcal{V}(\boldsymbol{\delta})^2 \\ &\quad \times \exp \left[d_\kappa \sum_{r=1}^m \delta_r(M) \right] [1 + O(M^{-1})]. \end{aligned} \quad (12.25)$$

At fixed lattice indices in the logarithmic derivative,

$$x \partial_x \log \mathfrak{P}(A_M) = m M q_\kappa + O(1). \quad (12.26)$$

Proof. The prefactor in (12.17) factorizes as

$$\mathcal{G}(\alpha, \gamma) = \frac{\ell(\alpha)r(\gamma)}{\tau_\gamma - \beta_\alpha}. \quad (12.27)$$

After extracting row factors $M^{-1}e^{Mu(\alpha_r)}\ell(\alpha_r)$ and column factors $e^{Mv(\alpha_s)}r(\alpha_s)$, the leading normalized determinant is Cauchy. Because this determinant is confluent, the entrywise remainder in (12.16) is not sufficient by itself: subleading terms could otherwise be larger than the Vandermonde-suppressed leading determinant. The localized expansion in Appendix J.2 retains only a smooth saddle neighborhood strictly below the termination boundary $b = i$. At integer source indices the discarded positive sum, after row and column normalization, is exponentially small. Divided differences are applied to this localized smooth kernel, where every subleading term retains both Vandermonde factors. Determinant multilinearity then transfers the result to the exact lattice: the exponentially small error stays negligible relative to the polynomially small confluent determinant. This proves a relative correction of $O(M^{-1})$ without an unrestricted off-lattice continuation. The two leading Vandermonde factors behave as

$$\Delta(\beta_{\alpha_1}, \dots, \beta_{\alpha_m}) \sim M^{-\binom{m}{2}} (\beta'_\kappa)^{\binom{m}{2}} \mathcal{V}(\delta), \quad (12.28)$$

and similarly for τ . The Cauchy denominator tends to $q_\kappa^{m^2}$, while $\ell(\kappa)r(\kappa) = C_\kappa q_\kappa$. This gives (12.25). The localized expansion and transfer estimate also hold after one coupling derivative at fixed lattice indices. Differentiating the factored formula, with $x\partial_x\Phi = q$, therefore gives the asserted logarithmic derivative. Details are given in Appendix J.3. $\square$

Thus a cluster of m endpoint indices carries the algebraic price

$$\boxed{\mathfrak{P}(A_M) = O(M^{-m^2})}. \quad (12.29)$$

The factor M^{-m} comes from the individual Gaussian prefactors; the additional $M^{-m(m-1)}$ comes from the two confluent Vandermonde factors.

12.4 Uniform minor envelopes and Boolean suppression

For the complete support sum, fixed-size asymptotics are not enough: supports of all sizes must be bounded simultaneously. Choose a small parameter-independent strip width $\epsilon > 0$ such that

$$[\kappa - \epsilon, \kappa] \subset (0, \nu) \quad (12.30)$$

throughout the compact Case B set. The off-diagonal estimate and a crude positive-sum Laplace bound give weights $\omega_M(a)$ satisfying

$$\mathfrak{P}(A) \leq |A|! \prod_{a \in A} \omega_M(a), \quad (12.31)$$

where one may take

$$\omega_M(a) = \begin{cases} \frac{C}{M} e^{M\Phi(a/M)}, & a/M \in [\kappa - \epsilon, \kappa + O(M^{-1})], \\ CM^B e^{M\Phi(a/M)}, & 0 \leq a/M \leq \kappa - \epsilon. \end{cases} \quad (12.32)$$

Here B and C are uniform constants. The polynomial loss in the deep region is harmless because Φ is bounded there by a strictly negative constant.

The Boolean product bound (11.14) introduces the singleton weights

$$g_{M,k}(a) = \left(1 + \frac{1}{c}\right)^{k-a} - 1. \quad (12.33)$$

Define

$$S_M = \sum_{a=0}^{k-1} g_{M,k}(a) \omega_M(a). \quad (12.34)$$

Lemma 12.4 (Endpoint summability). *Uniformly on compact lower Case B sets,*

$$\boxed{S_M = O(M^{-2})}. \quad (12.35)$$

Proof. In the endpoint strip write $a = k - r$, $r \geq 1$. Since $c \geq c_0 M$,

$$g_{M,k}(k - r) \leq C \frac{r}{M}. \quad (12.36)$$

The derivative of Φ is bounded below in the strip, while $k = \kappa M + O(1)$; hence

$$M\Phi((k - r)/M) \leq C_0 - d_1 r. \quad (12.37)$$

Together with the first line of (12.32), this gives

$$\sum_{a/M \geq \kappa - \epsilon} g_{M,k}(a) \omega_M(a) \leq \frac{C}{M^2} \sum_{r \geq 1} r e^{-d_1 r} = O(M^{-2}). \quad (12.38)$$

In the deep region, $g_{M,k}(a) \leq C$ and $\Phi(a/M) \leq -c_1 < 0$, so its total contribution is exponentially small. The full proof, including uniform constants, is in [Appendix J.4](#). $\square$

The two factors in the endpoint estimate have distinct origins:

$$\underbrace{M^{-1}}_{\text{kernel saddle}} \times \underbrace{M^{-1}}_{\text{Boolean coefficient}} = M^{-2}. \quad (12.39)$$

The confluent law (12.29) makes multi-index endpoint clusters smaller still.

12.5 Complete ground-period resummation

Recall the exact normalized ground factor

$$\mathcal{R}_{N,k}^\downarrow = 1 + \sum_{A \neq \emptyset} \hat{C}_{k,c}(A) \det Q_{N,A} \quad (12.40)$$

from [Exact identity 11.4](#).

Period asymptotics 12.5 (Complete Case B ground resummation; R9). *For the sequence of compact lower Case B cycles in [Definition 12.1](#),*

$$\boxed{\mathcal{R}_{N,k}^\downarrow = 1 + O(M^{-2})}, \quad (12.41)$$

$$\boxed{x \partial_x \mathcal{R}_{N,k}^\downarrow = O(M^{-1}), \quad x \partial_x \log \mathcal{R}_{N,k}^\downarrow = O(M^{-1})}. \quad (12.42)$$

Consequently,

$$\boxed{E_{N,k}^{\downarrow, \text{per}} - E_{\text{all}}^{\text{per}} = O(M^{-2})}. \quad (12.43)$$

Proof. Let $m = |A|$. By (11.14) and (12.31),

$$\hat{C}_{k,c}(A) |\det Q_{N,A}| \leq m! \prod_{a \in A} [g_{M,k}(a) \omega_M(a)]. \quad (12.44)$$

If e_m denotes the elementary symmetric polynomial, then

$$m! e_m(w_0, \dots, w_{k-1}) \leq \left(\sum_a w_a \right)^m. \quad (12.45)$$

Therefore

$$|\mathcal{R}_{N,k}^\downarrow - 1| \leq \sum_{m \geq 1} S_M^m = O(M^{-2}). \quad (12.46)$$

For the derivative use the exact degree bound (5.73); the Boolean coefficients are independent of x when the finite homology label k is held fixed. Thus

$$|x \partial_x \mathcal{R}_{N,k}^\downarrow| \leq M \sum_{m \geq 1} m S_M^m = O(M^{-1}). \quad (12.47)$$

Since $\mathcal{R} = 1 + O(M^{-2})$, the same estimate holds for the logarithmic derivative. Finally, $(N - k)!(N - k)^k$ is independent of V , and the factor Z_0 is common to the cascade and all-pole sectors. Applying (3.15) gives (12.43). $\square$

The leading finite correction can also be extracted. With the exact-ratio convention $\nu = N/M$, put $\kappa = \kappa(\nu, x)$ and write $\delta_M = k - \kappa M$. Then the singleton supports give

$$\mathcal{R}_{N,k}^\downarrow - 1 = (-1)^{N+k+1} \frac{C_\kappa e^{d_\kappa \delta_M}}{(\nu - \kappa) M^2} \frac{e^{-d_\kappa}}{(1 + e^{-d_\kappa})^2} + o(M^{-2}). \quad (12.48)$$

The parity and the bounded endpoint offset cause a small finite-size oscillation. If instead one centers k at $M\kappa_\infty$, where $\kappa_\infty = \kappa(\nu_\infty, x)$ and $N - \nu_\infty M = O(1)$, and evaluates the action and prefactors at the limiting filling ν_∞ , the leading coefficient acquires the additional factor $\exp[(N - \nu_\infty M) \partial_\nu \Phi(\kappa_\infty; \nu_\infty, x)]$. Evaluating all quantities at N/M already includes this effect. Equation (12.48) is a pointwise asymptotic at fixed parameter values; it must not be differentiated after replacing the fixed integer k by $\kappa(x)M + \delta_M$. The energy statement is instead obtained from the uniform derivative estimate (12.42).

12.6 Complete separated marked-period resummation

Let

$$a = \alpha M, \quad \kappa + \eta_1 \leq \alpha \leq \nu - \eta_1 \quad (12.49)$$

for some fixed $\eta_1 > 0$. Recall from (11.24) that

$$\mathcal{M}_{N,k}^\downarrow(a) = \frac{\mathcal{R}_{N,k}^\downarrow[\{a\}]}{Q_{N,aa}}. \quad (12.50)$$

Period asymptotics 12.6 (Complete separated marked resummation). *Uniformly in the range (12.49),*

$$\boxed{\mathcal{M}_{N,k}^\downarrow(a) = 1 + O(M^{-2})}, \quad (12.51)$$

$$\boxed{x \partial_x \log \mathcal{M}_{N,k}^\downarrow(a) = O(M^{-1})}. \quad (12.52)$$

Hence

$$\boxed{\begin{aligned} -\partial_{1/V} \log \frac{Z_{N,k}^\downarrow[\{a\}]}{Z_{N,k}^\downarrow} \\ = -\partial_{1/V} \log P_a^\downarrow + O(M^{-2}). \end{aligned}} \quad (12.53)$$

In particular, the full Case B marked period has the same leading and $O(M^{-1})$ dispersion as the Case A one-point period:

$$\boxed{\begin{aligned} \Delta E_{N,k}^{\downarrow, \text{per}}(\xi) &= 2\sqrt{(\xi - \tilde{\mu})^2 + \Delta^2} \\ &\quad + \frac{2\Omega}{M(1 - \sigma)} c(\alpha, \nu, x) + O(M^{-2}). \end{aligned}} \quad (12.54)$$

Proof. For $A \subset \{0, \dots, k-1\}$, the permutation expansion of the enlarged principal minor and [Period asymptotics 12.2](#) give

$$\left| \frac{\det Q_{N, A \cup \{a\}}}{Q_{N, aa}} \right| \leq (m+1)! \prod_{b \in A} \omega_M(b), \quad m = |A|, \quad (12.55)$$

uniformly for separated a . The lower bound on P_a needed in the division follows from [Period asymptotics 7.2](#). Combining (12.55) with the Boolean product bound gives

$$|\mathcal{M}_{N,k}^\downarrow(a) - 1| \leq \sum_{m \geq 1} (m+1) S_M^m = O(M^{-2}). \quad (12.56)$$

The quotient derivative is bounded by $(m+2)M$ times the absolute quotient, using [Corollary 5.8](#) for numerator and denominator. Hence (12.52). Insert the exact factorization (11.26) and use [Period asymptotics 7.3](#) and 12.5. The details are given in [Appendix J.6](#). $\square$

Thus the physical consecutive cascade first enters the separated marked-period energy only at order M^{-2} . The universal $O(M^{-1})$ term is already present in the one-point factor P_a .

12.7 The first period threshold in compact Case B

The square-root discriminant is minimized at

$$\alpha_* = \frac{\tilde{\mu}}{\Omega} = \nu(1+2x) - x. \quad (12.57)$$

For the strict lower branch, $\alpha_* < \nu$ because

$$\alpha_* - \nu = x(2\nu - 1) < 0. \quad (12.58)$$

The endpoint lies strictly below this minimum.

Lemma 12.7 (Separation of the Case B endpoint and the spectral minimum). *In every compact lower Case B interior,*

$$\boxed{0 < \kappa < \alpha_* < \nu.} \quad (12.59)$$

Proof. The lower Debye-edge crossing $\alpha_* = 0$ occurs at

$$\nu_0 = \frac{x}{1+2x} = \frac{\sigma}{1+\sigma}. \quad (12.60)$$

At this point the edge rate is

$$\Phi(0; \nu_0, x) = \log(1+2x) > 0. \quad (12.61)$$

A lower Case B point has $\Phi(0) < 0$, and therefore lies at $\nu > \nu_0$, so $\alpha_* > 0$. At $\alpha = \alpha_*$, the two roots satisfy

$$\beta_{\alpha_*} + \tau_{\alpha_*} = 2\nu, \quad (12.62)$$

which implies

$$\partial_\nu \Phi(\alpha_*, \nu) = \log \frac{\tau_{\alpha_*} - \nu}{\nu - \beta_{\alpha_*}} = 0. \quad (12.63)$$

Along $\alpha_*(\nu) = \nu(1+2x) - x$,

$$\frac{d}{d\nu} \Phi(\alpha_*(\nu), \nu) = (1+2x) \partial_\alpha \Phi(\alpha_*) > 0. \quad (12.64)$$

Starting from the positive value (12.61), one has $\Phi(\alpha_*) > 0$ throughout the lower Case B branch. Since $\Phi(\kappa) = 0$ and Φ is increasing, $\kappa < \alpha_*$. The last inequality is (12.58). $\square$

The minimum is therefore in the separated range of [Period asymptotics 12.6](#). Since

$$D_{\alpha_*} = 4x(1+x)\nu(1-\nu), \quad (12.65)$$

we obtain the lower Case B marked threshold

$$\Delta_{R,B}^{\downarrow, \text{per}} = 2\Delta + \frac{2\Omega}{M(1-\sigma)}c(\alpha_*, \nu, x) + O(M^{-2}). \quad (12.66)$$

In particular,

$$\Delta_{R,B}^{\downarrow, \text{per}} = 2\Delta + O(M^{-1}). \quad (12.67)$$

Pure endpoint sectors are higher at leading order. If a fixed set of m excluded indices remains within $O(1)$ lattice spacings of κM , [Equation \(12.26\)](#) gives

$$\Delta E_{\text{endpoint}}^{(m), \text{per}} = 2m\Omega\sqrt{D_\kappa} + O(M^{-1}). \quad (12.68)$$

Because D_α is a convex quadratic minimized at α_* ,

$$D_\kappa > D_{\alpha_*}, \quad (12.69)$$

and no finite endpoint cluster lowers the first marked-period threshold.

The exact identification of these period sectors with named Richardson eigenbranches is postponed to [Chapter 14](#). The present conclusion is entirely internal to the residue-period problem.

12.8 Upper Case B and the half-filled junction

For a compact upper branch with $\nu > 1/2$, apply the exact transformation of [Chapter 8](#):

$$\bar{\nu}_M = \frac{M+1-N}{M} = 1-\nu + \frac{1}{M}, \quad \bar{\alpha} = 1-\alpha, \quad \bar{\xi} = \Omega - \xi. \quad (12.70)$$

The upper occupation-supported cascade is the reflected image of a lower hole cascade. The lower-branch estimates therefore apply with the exact hole filling $\bar{\nu}_M$. At leading thermodynamic order the familiar reflection identities are

$$D_{1-\alpha}(1-\nu, x) = D_\alpha(\nu, x), \quad \Phi(1-\alpha; 1-\nu, x) = \Phi(\alpha; \nu, x). \quad (12.71)$$

Hence the upper compact Case B ground period and separated marked period satisfy the transformed analogues of [\(12.43\)](#) and [\(12.54\)](#), evaluated at $\bar{\nu}_M$. Replacing this ratio by $1-\nu$ before extracting the $O(M^{-1})$ coefficient would discard the finite-level-count shift; the leading reflection identities alone do not justify that replacement.

The thermodynamic half-filled point is the junction of the lower and upper marked charts. The leading result follows by either one-sided limit. A uniform treatment of the $O(M^{-1})$ correction at the chart junction is included later with the finite-size spectrum in [Chapter 16](#); it is not needed for the compact strict-branch theorem proved here.

12.9 Status and verification

The output of this chapter is stronger than an exponent-only buffered estimate:

$$\boxed{E_{0,B}^{\text{per}} = E_{\text{all}}^{\text{per}} + O(M^{-2})}, \quad (12.72)$$

$$\boxed{\Delta E_B^{\text{per}}(\xi) = 2\mathcal{E}(\xi) + \frac{2\Omega}{M(1-\sigma)}c(\alpha, \nu, x) + O(M^{-2})} \quad (12.73)$$

for a separated marked point. The complete endpoint sum has been performed; no endpoint component has been selected by assumption.

Verification note 12.8 (Independent lattice audit and rejected-continuation regression). The revision checker `code/verify_revision01.py` uses exact integer and rational arithmetic for finite kernel sums, determinants, and fixed-label coupling derivatives. It does not import the original verification modules. The recorded checks comprise:

1. 96 small-system comparisons of direct residue-subset sums with kernel minors and their coupling derivatives;
2. 24 confluent-minor and derivative comparisons at $M = 100, 200, 400$, with one, two, and three marked indices, including nonconsecutive offsets;
3. a negative regression showing that the discarded unrestricted noninteger-source continuation fails, while its positive truncated core approaches the expected saddle contribution.

The large-system tests use both a strict lower Case B endpoint and an interior source point. The ratios to the leading asymptote approach unity across the tested size range, and the fixed-label derivative residuals are consistent with the stated $O(1)$ order. The report `revision/verification_revision01.txt` ends with

Overall revision-01 verification: PASS.

The original complete-support enumeration and off-diagonal comparisons are retained with the archived Module 18 reports. None of these finite numerical checks substitutes for the uniform support estimate or the localized confluent proof above.

Chapter 13

Birth of the cascade and the Case A–B transition

13.1 Scope: a topology crossover distinct from the Debye-edge crossing

Chapter 12 treated compact Case B interiors, where the endpoint density

$$\kappa \in (0, \nu) \tag{13.1}$$

is bounded away from zero. The present chapter removes that restriction and studies the nonuniform limit in which the consecutive cascade is born from the lower Debye edge. The relevant one-point rate is the function $\Phi(\alpha; \nu, x)$ of Equation (7.22). Its lower-edge value is

$$\Phi_0(\nu, x) := \Phi(0; \nu, x). \tag{13.2}$$

Two parameter loci must be kept separate throughout:

1. the chemical-potential crossing

$$\tilde{\mu} = 0 \iff \nu = \nu_0 := \frac{x}{1+2x} = \frac{\sigma}{1+\sigma}, \tag{13.3}$$

already resolved by the discrete marked-root layer in Section 7.7;

2. the contour-topology transition

$$\Phi_0(\nu, x) = 0, \tag{13.4}$$

at which a positive solution of $\Phi(\kappa) = 0$ is born.

The first crossing changes the constrained minimizer of the quasiparticle dispersion. The second is the birth line for a positive endpoint of the period rate function and hence for the chosen cascade-contour prescription. Its identification with a thermodynamic change in the density of real Richardson roots is conjectural; branch naming alone does not prove it. The two equations are distinct.

Throughout this chapter, κ means the nonnegative parameter defined by the rate equation $\Phi(\kappa; \nu, x) = 0$ on the Case B side and by $\kappa = 0$ on the Case A side. Statements proved from this equation concern that parameter. Physical root counts are denoted separately by N_{real} and are discussed only in Section 13.9.

13.2 The exact edge rate and the transition line

The exact endpoint integral (7.36) gives

$$\Phi_0(\nu, x) = -\nu \log \nu - (1 - \nu) \log(1 - \nu) + \nu \log x + (1 - \nu) \log(1 + x). \quad (13.5)$$

Using $x = \sigma/(1 - \sigma)$, the same function is

$$\Phi_0(\nu, \sigma) = \nu \log \sigma - \log(1 - \sigma) - \nu \log \nu - (1 - \nu) \log(1 - \nu). \quad (13.6)$$

The continuous endpoint values at $\nu = 0, 1$ are understood in the usual entropy convention.

Lemma 13.1 (Shape of the edge rate). *For fixed $0 < \sigma < 1$,*

$$\partial_\nu \Phi_0 = \log \frac{\sigma(1 - \nu)}{\nu}, \quad \partial_\nu^2 \Phi_0 = -\frac{1}{\nu(1 - \nu)} < 0. \quad (13.7)$$

Thus Φ_0 is strictly concave and has its unique maximum at

$$\nu_0 = \frac{\sigma}{1 + \sigma}. \quad (13.8)$$

At this point

$$\Phi_0(\nu_0, \sigma) = \log \frac{1 + \sigma}{1 - \sigma} = \log(1 + 2\sigma) > 0. \quad (13.9)$$

The maximum of the endpoint rate therefore coincides with the crossing $\tilde{\mu} = 0$. In particular, that crossing occurs at the point where the all-pole side is most strongly separated from the birth of a lower cascade.

On the strict lower chart $0 < \nu \leq 1/2$, the full one-point rate is increasing in α by (12.2). Hence the sign of Φ_0 completely determines whether a positive endpoint exists.

Period asymptotics 13.2 (Lower Case A–B transition). *For fixed $0 < \nu \leq 1/2$ and $x > 0$:*

1. *if $\Phi_0(\nu, x) \geq 0$, then $\Phi(\alpha) > 0$ for every $\alpha \in (0, \nu)$ and no positive lower-cascade endpoint exists;*
2. *if $\Phi_0(\nu, x) < 0$, then there is a unique*

$$\kappa(\nu, x) \in (0, \alpha_*) \quad (13.10)$$

satisfying

$$\Phi(\kappa; \nu, x) = 0. \quad (13.11)$$

The transition line is therefore

$$\sigma^\nu = (1 - \sigma)\nu^\nu(1 - \nu)^{1-\nu}. \quad (13.12)$$

Proof. Strict monotonicity in α gives the first statement. If $\Phi_0 < 0$, then necessarily $\nu > \nu_0$ and hence $\alpha_* > 0$. The proof of Lemma 12.7 shows that $\Phi(\alpha_*) > 0$. The intermediate-value theorem and monotonicity give a unique root in $(0, \alpha_*)$. Equation (13.12) is simply $\Phi_0 = 0$ written using (13.6). $\square$

13.2.1 Global rate diagram and its critical coupling

At half filling,

$$\Phi_0\left(\frac{1}{2}, x\right) = \frac{1}{2} \log[4x(1+x)]. \quad (13.13)$$

The critical values are therefore

$$x_c = \frac{\sqrt{2}-1}{2}, \quad \sigma_c = 3 - 2\sqrt{2} = (\sqrt{2}-1)^2, \quad (13.14)$$

$$v_c = -\frac{2}{\log \sigma_c} = \frac{1}{\log(1+\sqrt{2})} = 1.134592657\dots \quad (13.15)$$

If $\sigma \geq \sigma_c$ (equivalently $v \geq v_c$), the endpoint rate is nonnegative throughout the lower half of the band and no macroscopic Case B occurs. If $0 < \sigma < \sigma_c$, there is one lower transition filling

$$\nu_B(\sigma) \in (\nu_0, 1/2) \quad (13.16)$$

solving (13.12). Exact electron-hole duality gives the upper boundary $1 - \nu_B$. The Case B interval is therefore

$$\nu_B(\sigma) < \nu < 1 - \nu_B(\sigma), \quad (13.17)$$

with a lower cascade for $\nu < 1/2$ and its upper occupation-supported dual for $\nu > 1/2$.

13.2.2 Weak-coupling location of the transition

For $\sigma \downarrow 0$, set $\nu_B = c\sigma + O(\sigma^2)$. Expanding (13.6) gives

$$\Phi_0 = \sigma[1 + c - c \log c] + O(\sigma^2). \quad (13.18)$$

The root condition is $c(\log c - 1) = 1$. Hence

$$\nu_B(\sigma) = \frac{\sigma}{W(1/e)} + O(\sigma^2) = 3.591121477\dots \sigma + O(\sigma^2). \quad (13.19)$$

By contrast,

$$\nu_0 = \sigma + O(\sigma^2). \quad (13.20)$$

Thus the two crossovers remain separated even in weak coupling:

$$\nu_B - \nu_0 = 2.591121477\dots \sigma + O(\sigma^2). \quad (13.21)$$

13.3 Small- α structure of the one-point rate

The edge/interior matching in Section 7.6 already established

$$\Phi(\alpha) = \Phi_0 + \alpha \left[\log \frac{\Lambda}{\alpha} + 1 \right] + \frac{\Xi}{2} \alpha^2 + O(\alpha^3), \quad (13.22)$$

where

$$\Lambda(\nu, x) = \frac{(\nu+x)^2}{x(1+x)} = \frac{[\sigma + \nu(1-\sigma)]^2}{\sigma}, \quad (13.23)$$

$$\Xi(\nu, x) = \frac{\nu^2 - 2\nu x - 2\nu - x^2}{(\nu+x)^2}. \quad (13.24)$$

For later use, differentiation gives

$$\partial_\alpha \Phi(\alpha) = \log \frac{\Lambda}{\alpha} + \Xi \alpha + O(\alpha^2). \quad (13.25)$$

The logarithm is the source of the nonanalytic birth law and of the logarithmically enlarged finite-size transition window.

For completeness, the saddle coordinates have the expansions

$$\beta_\alpha = \frac{\nu}{\nu+x} \alpha - \frac{\nu x(1-\nu)}{(\nu+x)^3} \alpha^2 + O(\alpha^3), \quad (13.26)$$

$$\tau_\alpha = \frac{\nu+x}{1+x} + \frac{x(1-\nu)}{(1+x)(\nu+x)} \alpha + \frac{\nu x(1-\nu)}{(\nu+x)^3} \alpha^2 + O(\alpha^3). \quad (13.27)$$

13.4 Lambert- W birth law

On the Case B side define the positive edge deficit

$$\delta_\Phi := -\Phi_0(\nu, x) > 0. \quad (13.28)$$

The endpoint condition (13.11) and (13.22) give

$$\delta_\Phi = \kappa \log \frac{e\Lambda}{\kappa} + \frac{\Xi}{2} \kappa^2 + O(\kappa^3). \quad (13.29)$$

Period asymptotics 13.3 (Lambert- W birth of the rate-defined cascade endpoint). *As $\delta_\Phi \downarrow 0$ on the Case B side, let*

$$\kappa_0 := -\frac{\delta_\Phi}{W_{-1}[-\delta_\Phi/(e\Lambda)]}, \quad (13.30)$$

where W_{-1} is the negative real branch of Lambert's function. Then

$$\kappa = \kappa_0 - \frac{\Xi \kappa_0^2}{2 \log(\Lambda/\kappa_0)} + O\left(\frac{\kappa_0^3}{\log(1/\kappa_0)}\right). \quad (13.31)$$

In particular,

$$\kappa = -\frac{\delta_\Phi}{W_{-1}[-\delta_\Phi/(e\Lambda)]} \left[1 + O\left(\frac{\delta_\Phi}{\log^2(1/\delta_\Phi)}\right) \right], \quad (13.32)$$

so that

$$\kappa \sim \frac{\delta_\Phi}{\log(1/\delta_\Phi)}. \quad (13.33)$$

Proof. Dropping the quadratic term in (13.29) gives

$$\delta_\Phi = \kappa \log(e\Lambda/\kappa).$$

Writing $t = \delta_\Phi/\kappa$ yields $-t = W_{-1}[-\delta_\Phi/(e\Lambda)]$, which proves (13.30). The derivative of the leading implicit function at κ_0 is $\log(\Lambda/\kappa_0)$. One Newton correction gives the displayed Ξ term; the cubic remainder gives the stated error. $\square$

Define

$$L_\Phi := \log \frac{e\Lambda}{\delta_\Phi}. \quad (13.34)$$

The standard expansion of W_{-1} gives

$$\kappa = \frac{\delta_\Phi}{L_\Phi + \log L_\Phi + O(\log L_\Phi / L_\Phi)}. \quad (13.35)$$

If κ is extended by zero to the Case A side, then

$$\partial_{\delta_\Phi} \kappa \longrightarrow 0 \quad (\delta_\Phi \downarrow 0), \quad (13.36)$$

whereas

$$\partial_{\delta_\Phi}^2 \kappa \asymp \frac{1}{\delta_\Phi \log^2(1/\delta_\Phi)}. \quad (13.37)$$

The rate-defined endpoint κ , extended by zero to the other side, is therefore C^1 but not C^2 at its birth. No statement about the differentiability of a physical root-density limit follows without proving the identification discussed below.

13.4.1 Approach in filling or coupling

At fixed σ , write $\nu = \nu_B + \delta\nu$ with $\delta\nu > 0$. Since

$$-\partial_\nu \Phi_0(\nu_B, \sigma) = \log \frac{\nu_B}{\sigma(1 - \nu_B)} =: A_\nu > 0, \quad (13.38)$$

we have

$$\delta_\Phi = A_\nu \delta\nu + O(\delta\nu^2), \quad (13.39)$$

whence

$$\kappa \sim \frac{A_\nu(\nu - \nu_B)}{\log[1/(\nu - \nu_B)]}. \quad (13.40)$$

At fixed ν , let $\sigma_B(\nu)$ solve (13.12). Case B lies at $\sigma < \sigma_B$, and

$$\delta_\Phi = A_\sigma(\sigma_B - \sigma) + O[(\sigma_B - \sigma)^2], \quad (13.41)$$

where

$$A_\sigma = \frac{\nu}{\sigma_B} + \frac{1}{1 - \sigma_B}. \quad (13.42)$$

Since $\sigma = e^{-2/v}$,

$$\delta_\Phi = A_v(v_B - v) + O[(v_B - v)^2], \quad (13.43)$$

with

$$A_v = \frac{2\sigma_B}{v_B^2} \left(\frac{\nu}{\sigma_B} + \frac{1}{1 - \sigma_B} \right). \quad (13.44)$$

13.5 Finite-size crossover and mesoscopic cascades

For the period-contour prescription the integer cascade label is chosen so that

$$k \simeq M\kappa. \quad (13.45)$$

The inversion below is evaluated on sequences with $\kappa = k/M$; if an integer is obtained by rounding a prescribed $M\kappa$, its offset must be retained at subleading order. These choices define a period label and do not measure the physical real-root count. For $k = o(M)$, the endpoint equation can be inverted directly. When a parameter path crosses the rate transition, write

$$\Lambda_B := \Lambda(\nu_B, x_B), \quad \Xi_B := \Xi(\nu_B, x_B) \quad (13.46)$$

where (ν_B, x_B) is the point at which the chosen parameter path meets the transition line.

$$\delta_\Phi = \frac{k}{M} \log \frac{e\Lambda M}{k} + \frac{\Xi}{2} \frac{k^2}{M^2} + O\left(\frac{k^3}{M^3}\right). \quad (13.47)$$

Here Λ and Ξ are evaluated at the actual parameter point. For leading scaling laws they may be replaced by their transition values.

Corollary 13.4 (Finite-size width of the rate-defined cascade crossover). *If $k = K = O(1)$ is fixed, then*

$$\delta_\Phi = \frac{K}{M} \left[\log M + \log \frac{e\Lambda_B}{K} \right] + O\left(\frac{\log M}{M^2}\right). \quad (13.48)$$

Consequently the window in which the prescribed period label first becomes of order one has width

$$\delta_\Phi^{(M)} \asymp \frac{\log M}{M}. \quad (13.49)$$

At fixed coupling the corresponding filling width is

$$\delta\nu_{A-B}^{(M)} \asymp \frac{\log M}{A_\nu M}. \quad (13.50)$$

This scale is parametrically broader than the $O(M^{-1})$ staircase at $\tilde{\mu} = 0$ found in Section 7.7. The centers of the two windows are nevertheless separated by the finite thermodynamic distance $\nu_B - \nu_0$.

The inversion (13.47) distinguishes four regimes:

Regime	Cascade size
Finite-size Case A	$M\delta_\Phi / \log(1/\delta_\Phi) \rightarrow 0$, so $k \rightarrow 0$
First nonzero period labels	$\delta_\Phi \asymp (\log M)/M$, so $k = O(1)$
Mesoscopic cascade	$(\log M)/M \ll \delta_\Phi \ll 1$, so $k \rightarrow \infty$ but $k/M \rightarrow 0$
Macroscopic Case B	fixed $\delta_\Phi > 0$, so $k = \kappa M + o(M)$

If

$$k \sim cM^\gamma, \quad 0 < \gamma < 1, \quad (13.51)$$

then

$$\delta_\Phi \sim cM^{\gamma-1} \left[(1-\gamma) \log M + \log \frac{e\Lambda_B}{c} \right]. \quad (13.52)$$

No particular power such as $\sqrt{M}$ is selected by this period-label onset. The scale $k \sim \sqrt{M}$ only marks where the quadratic correction in the local rate and the simplest fixed-index matching cease to be uniformly negligible.

13.6 Matching the one-point prefactor at the birth point

The transition is nonuniform not only in the exponent but also in the algebraic prefactor. In compact Case B,

$$P_{\kappa M} = \frac{C(\kappa, \nu, x)}{M} [1 + O(M^{-1})], \quad \Phi(\kappa) = 0. \quad (13.53)$$

As $\kappa \downarrow 0$, (7.51) gives

$$C(\kappa, \nu, x) = \frac{C_0(\nu, x)}{\sqrt{2\pi\kappa}} [1 + O(\kappa)]. \quad (13.54)$$

Therefore

$$P_{\kappa M} \sim \frac{C_0}{\sqrt{2\pi\kappa} M}. \quad (13.55)$$

On the edge side, suppose

$$k \rightarrow \infty, \quad \frac{k^2}{M} \rightarrow 0, \quad \Phi(k/M) = 0. \quad (13.56)$$

The matched formula (7.53) and the endpoint equation imply

$$P_k = \frac{C_0}{\sqrt{2\pi k M}} \left[1 + O\left(\frac{1}{k} + \frac{k^2}{M}\right) \right]. \quad (13.57)$$

Since $k = \kappa M$, equations (13.55) and (13.57) coincide. Thus the edge prefactor $M^{-1/2}$ and the compact-Case B prefactor M^{-1} are two limits of one matched expression.

The endpoint decay rate also has a useful singular expansion:

$$d_\kappa := \Phi'(\kappa) = \log \frac{\Lambda}{\kappa} + \Xi\kappa + O(\kappa^2), \quad (13.58)$$

so

$$e^{-d_\kappa} = \frac{\kappa}{\Lambda} [1 + O(\kappa)]. \quad (13.59)$$

13.7 Uniform smallness of the complete cascade correction

The compact Case B singleton formula (12.48) can now be matched to the birth regime. Choose the canonical finite sequence

$$k_M = \lfloor \kappa M \rfloor, \quad (13.60)$$

so that the bounded endpoint offset is nonpositive. Combining (13.54) and (13.59) with the exact Boolean singleton coefficient gives the following uniform interpolation.

Period asymptotics 13.5 (Cascade correction through the birth window). *Let $1 \leq k_M = o(M)$ be the canonical sequence (13.60), with parameters chosen so that k_M/M follows the positive endpoint. Then*

$$\mathcal{R}_{N, k_M}^\downarrow - 1 = O\left(\frac{\sqrt{k_M}}{M^{5/2}}\right), \quad (13.61)$$

$$x \partial_x \log \mathcal{R}_{N, k_M}^\downarrow = O\left(\frac{\sqrt{k_M}}{M^{3/2}}\right), \quad (13.62)$$

and hence

$$E_{N,k_M}^{\downarrow,\text{per}} - E_{\text{all}}^{\text{per}} = O\left(\frac{\sqrt{k_M}}{M^{5/2}}\right). \quad (13.63)$$

For a marked point separated by a fixed macroscopic distance from the endpoint, the same orders hold for the quotient background factor $\mathcal{M}_{N,k_M}^{\downarrow}(a)$.

Proof. The largest admissible singleton support is $a = k_M - 1$. Its estimate is obtained by splitting the birth regime into the same matched ranges as Chapter 7. For fixed k_M , the exact fixed-index formula (7.46) and the endpoint relation (13.47) give $P_{k_M-1} = O(M^{-3/2})$. In the overlap $1 \ll k_M \ll \sqrt{M}$, equations (13.57) and (13.59) give

$$P_{k_M-1} = O\left(\frac{\sqrt{k_M}}{M^{3/2}}\right).$$

For $\sqrt{M} \lesssim k_M = o(M)$, the interior formula (13.53), the singular prefactor (13.54), and the one-lattice-step rate $e^{-d_\kappa} = O(k_M/M)$ give the same bound. Its Boolean coefficient is $O(M^{-1})$. The remaining singleton supports form a geometric tail with ratio $e^{-d_\kappa} = O(k_M/M)$, while every additional support index carries both another Boolean factor and the positive-minor suppression of Chapter 12. This gives (13.61). The exact degree bound supplies at most one factor M to the logarithmic derivative, proving (13.62); conversion by (3.15) gives the energy estimate. The mixed-minor estimate used for the marked quotient is unchanged for a separated point. $\square$

For fixed k , the correction is $O(M^{-5/2})$. For $k \sim \kappa M$ it becomes $O(M^{-2})$, reproducing the compact result of Period asymptotics 12.5. The topology crossover therefore creates no intermediate enhancement of the ground-period energy or of a separated marked gap.

13.8 The spectral minimum remains separated at the transition

At the lower transition filling ν_B , the endpoint is born at $\kappa = 0$, while

$$\alpha_{*,B} = \nu_B(1 + 2x) - x = (1 + 2x)(\nu_B - \nu_0) > 0. \quad (13.64)$$

Thus the minimum of the square-root dispersion remains a finite macroscopic distance from the newborn cascade.

At the transition, a local endpoint quantum has leading energy

$$2\Omega\sqrt{D_0} = 2\sqrt{\tilde{\mu}_B^2 + \Delta_B^2}, \quad (13.65)$$

whereas the first separated even threshold is

$$2\Delta_B. \quad (13.66)$$

Their difference is strictly positive:

$$2\left[\sqrt{\tilde{\mu}_B^2 + \Delta_B^2} - \Delta_B\right] > 0. \quad (13.67)$$

Consequently the newborn endpoint in the period construction is separated from the minimum of its candidate square-root dispersion. Within the period asymptotics it does not replace that minimum. Promotion of this statement to a physical gap requires a contour-to-eigenbranch bridge; the calculation does not by itself identify the physical root-topology birth.

13.9 Finite Richardson-root topology: numerical evidence

The analytic results above concern the rate-defined endpoint κ and the chosen period label k . Neither the residue identities nor the weak-label continuation in [Chapter 14](#) proves that they count real Richardson roots. The following conjecture states the proposed additional identification explicitly.

Conjecture 13.6 (Physical real-root count and the period endpoint). *On a fixed lower-chart thermodynamic sequence, away from additional excluded boundaries, the number of real roots of the physical ground branch obeys*

$$N_{\text{real}}(M; \nu, v) = M\kappa(\nu, v) + O(1), \quad (13.68)$$

where the remainder is bounded uniformly in M . Identification of the first finite root-collision window with $\delta_\Phi \asymp (\log M)/M$ would additionally require control uniform as $\kappa \downarrow 0$; it is not a consequence of a compact-interior bound alone.

The fixed-size examples below are consistent with this conjecture. They do not constitute a size-scaling test, a uniform error bound, or a proof of the physical transition line or Lambert exponent.

For the half-filled sequence

$$M = 20, \quad N = 10, \quad \Omega = 1, \quad (13.69)$$

the ground branch was continued in regular eigenvalue-based variables and the Baxter polynomial was reconstructed. The table compares the continuum prediction $M\kappa$ with the number of numerically real roots.

σ	$M\kappa$	real roots	below window	interlevel
0.1400	0.506683	0	0	0
0.1200	0.969912	0	0	0
0.1175	1.035176	2	1	1
0.1000	1.542046	2	1	1
0.0800	2.244455	2	1	1
0.0500	3.641638	4	1	3
0.0200	5.832218	6	1	5

In the displayed samples the first real pair is absent at $M\kappa = 0.969912$ and present at $M\kappa = 1.035176$: one root lies below the lower edge and one lies in the first interlevel interval. Further examples have more real roots, with the parity constraint imposed by complex conjugation. These are observations at $M = 20$, not an asymptotic statement as $M \rightarrow \infty$. The script checks that the discrepancy from $\text{round}(M\kappa)$ is at most two for these seven samples. The continuation equations and reconstruction procedure of [Chapter 14](#) improve the identification of the finite branches; they do not promote this finite table to a proof of [Conjecture 13.6](#).

13.10 Physical interpretation and logical output

The established period transition has two equivalent descriptions:

1. the edge rate Φ_0 changes sign;
2. a positive solution of $\Phi(\kappa) = 0$ is born with the Lambert law [\(13.32\)](#).

The conjectural third description is the birth of a nonzero physical real-root density. Proving it requires more than the branch labels. Likewise, calling the rate line a Stokes line of the physical thimble decomposition requires a separate analysis of that decomposition. The period formulas suggest no new singularity in the candidate leading expressions for

$$\frac{E_0}{M}, \quad \Delta_{\text{qp}}, \quad \Delta_{\text{even}} \quad (13.70)$$

on either side of the rate line; this is not an unconditional contour-based theorem about the corresponding physical eigenenergies. The period correction is in fact uniformly smaller in the birth window than in compact Case B, within [Period asymptotics 13.5](#) and its matching assumptions.

The outputs used later are:

1. the exact transition line ([13.12](#));
2. the critical coupling ([13.15](#));
3. the weak-coupling boundary ([13.19](#));
4. the Lambert birth law ([13.32](#));
5. the finite-size inversion ([13.47](#)) and the width ([13.49](#));
6. the matched endpoint prefactor ([13.57](#));
7. the uniform cascade estimate ([13.63](#));
8. the finite root-topology evidence summarized in ([13.68](#)).

Verification note 13.7 (Transition-law and root-topology audit). The script `code/verify_AB_transition.py` checks:

1. the transition line, strict-concavity formulas, and the exact half-filled critical coupling;
2. the weak-coupling asymptotic $\nu_B \sim \sigma/W(1/e)$;
3. the leading and Ξ -improved Lambert formulas over several decades in δ_Φ ;
4. the finite-size inversion and the $\log M/M$ onset scale;
5. the matching $P_k \sqrt{2\pi k M}/C_0 \rightarrow 1$ with the coupling tuned so that $\Phi(k/M) = 0$ exactly;
6. the complete support sum in the birth window and its $\sqrt{k}/M^{5/2}$ scaling;
7. the ground-branch real-root counts in the table above, using eigenvalue-based continuation and Chebyshev reconstruction of the Baxter polynomial.

The archived report ends with

A-B transition verification: PASS.

Part V

From residue periods to physical Richardson branches

Chapter 14

Identification and continuation of Richardson branches

14.1 Why a separate branch module is needed

The preceding chapters solved two exact finite problems and their thermodynamic asymptotics:

1. a relative homology class was converted into an explicit residue period;
2. the logarithmic derivative of that period was evaluated at large M .

Those results do not by themselves name an eigenstate of the reduced pairing Hamiltonian. A period may contain more than one steepest-descent component, and the exact quantity

$$E_{\Gamma}^{\text{per}} = -\partial_{1/V} \log Z_{\Gamma} \quad (14.1)$$

is a contour barycenter before it is a physical branch energy. The purpose of this chapter is to assign permanent Richardson labels to the sectors already constructed and to give a regular continuation method through collisions of the individual Richardson roots.

The hierarchy of statements will be kept explicit:

1. the leading weak-coupling residue subset is an exact combinatorial fact;
2. the corresponding eigenvalue-based branch exists analytically near $V = 0$;
3. the ground branch is globally unique for every $V > 0$;
4. the excited branches used below are defined by continuous continuation of their complete eigenvalue-based data, not by ordering their total energies;
5. numerical continuation and Baxter reconstruction provide consistency diagnostics for the parameter paths and finite sequences used later; these checks are not a proof of contour localization or a rigorous numerical certification of the entire path.

A theorem that every chosen eigenvalue-based coordinate chart remains nonsingular for every positive coupling is neither needed nor claimed; the underlying finite Hermitian eigenbranches nevertheless exist globally by [Branch identification 14.5](#).

14.2 Paired Hamiltonian and zero-coupling labels

Write the pair poles as

$$\mathbf{E}_n := 2\varepsilon_n = 2\varepsilon_{F0} + \frac{2n}{\rho}, \quad n = 0, \dots, M, \quad (14.2)$$

and let

$$b_n^\dagger = c_{n\uparrow}^\dagger c_{n\downarrow}^\dagger \quad (14.3)$$

be the hard-core pair creation operator. In the seniority-zero sector with N pairs,

$$H_N = \sum_{n=0}^M \mathbf{E}_n b_n^\dagger b_n - V \sum_{n,m=0}^M b_n^\dagger b_m. \quad (14.4)$$

Use the subset basis

$$|S\rangle = \prod_{n \in S} b_n^\dagger |0\rangle, \quad S \subset \mathcal{L}_M, \quad |S| = N. \quad (14.5)$$

Its matrix elements are

$$\langle S | H_N | S \rangle = \sum_{n \in S} \mathbf{E}_n - VN, \quad (14.6)$$

while

$$\langle T | H_N | S \rangle = -V \quad (14.7)$$

if T is obtained from S by moving one pair from an occupied to an unoccupied level, and the matrix element is zero otherwise.

At $V = 0$ every subset is an exact eigenstate. We use the subset itself as the permanent weak-coupling label. The ground label is

$$S_0 = \{0, 1, \dots, N-1\}. \quad (14.8)$$

For a lower marked level $0 \leq a < N$, define

$$S_a = \{0, 1, \dots, N\} \setminus \{a\}. \quad (14.9)$$

It differs from S_0 by promoting the pair on level a to the first empty level N , so

$$\sum_{n \in S_a} n - \sum_{n \in S_0} n = N - a. \quad (14.10)$$

Equal spacing can make different subsets degenerate in their *total* $V = 0$ energy. The subset label is nevertheless unambiguous because the complete binary occupation vector is different.

14.3 Weak-coupling labels of the exact residue sectors

Let

$$\deg(S) := \sum_{n \in S} n. \quad (14.11)$$

As $\sigma \downarrow 0$, the signed residue weight has the form

$$W(S) = \mathfrak{w}(S) \sigma^{\deg(S)} [1 + O(\sigma)], \quad \mathfrak{w}(S) \neq 0. \quad (14.12)$$

Therefore the leading weak-coupling term of a pure sector is determined by the unique admissible subset with minimal degree whenever such a subset is unique.

Exact identity 14.1 (Weak-coupling labels of the pure sectors). *The exact sectors constructed in Chapters 6–11 have the following leading labels.*

1. *The all-pole period has the unique leading subset S_0 .*
2. *The pure lower marked period with $0 \leq a < N$ has the unique leading subset S_a and an additional power σ^{N-a} relative to the all-pole period.*
3. *For a consecutive lower cascade with $k \leq a < N$, the ground cascade has leading subset S_0 , while the complete cascade period with pole a excluded has leading subset S_a .*
4. *A one-blocked-level period with active pair number n and blocked set P has leading active subset consisting of the lowest n nodes of $\mathcal{L}_M \setminus P$. The same greedy rule applies to every fixed finite-seniority Heine–Christoffel sector.*

Proof. For the all-pole sector, S_0 is the unique N -element subset of minimal integer sum. If $a < N$ is forbidden, the unique minimum is obtained by replacing a with N , which gives S_a and (14.10). For the consecutive cascade, the multiplicity factor on S_0 is

$$(N - k)!(N - k)^k \neq 0, \quad (14.13)$$

and, when $k \leq a$, the same factor is nonzero on S_a because S_a has no hole strictly below any cutoff $r \leq k$. Thus the cascade does not change either minimal subset. Finally, a blocked characteristic polynomial vanishes whenever an active residue subset contains a blocked node. The first nonzero subset is therefore obtained by taking the lowest allowed nodes. Uniqueness follows from the strict ordering of the lattice indices. $\square$

In particular, the exact identity

$$\mathcal{Z}_{\text{mark},a} = (N - 1)!Z_0P_a \quad (14.14)$$

from [Exact identity 6.2](#) and the positive sum of [Chapter 7](#) have the weak label S_a . Likewise, the complete Case B ground and marked periods of [Exact identities 11.1](#) and [11.5](#) have labels S_0 and S_a , respectively. These labels name the candidate physical branches. They do not show that the corresponding period is dominated by the named saddle at finite coupling or in the thermodynamic limit.

Proposition 14.2 (A weak label does not determine period asymptotics). *For each $j = 1, \dots, N$, let γ_j be a small positively oriented circle containing only the pole E_{j-1} , and let $\Gamma_{S_0} = \prod_j \gamma_j$. This is an admissible contour in [Equation \(3.3\)](#). Its period has the unique weak label S_0 , but*

$$Z_{\Gamma_{S_0}}(u) = C_{M,N} \exp \left[-u \sum_{n=0}^{N-1} E_n \right], \quad C_{M,N} \neq 0, \quad (14.15)$$

and therefore

$$E_{\Gamma_{S_0}}^{\text{per}} = 2N\varepsilon_{F0} + \frac{N(N-1)}{\rho} \quad (14.16)$$

for every $V > 0$. At fixed $\nu \in (0, 1)$ and $v > 1$, this barycenter exceeds the physical ground energy by at least an amount of order M .

Proof. There is exactly one residue tuple on this product contour. Its poles are distinct, so the Vandermonde factor does not vanish, and all its coupling dependence is the displayed exponential. This proves (14.15)–(14.16). To show that the discrepancy can be extensive

without assuming any thermodynamic formula under review, use the normalized uniform superposition of all N -pair subsets. In this symmetric state,

$$\left\langle \sum_n \mathbf{E}_n b_n^\dagger b_n \right\rangle = 2N\varepsilon_{F0} + N\Omega, \quad \left\langle \sum_{n,m} b_n^\dagger b_m \right\rangle = N(M - N + 2). \quad (14.17)$$

The variational principle gives $E_0^{\text{RG}} \leq E_D$, where

$$E_D = 2N\varepsilon_{F0} + N\Omega - VN(M - N + 2). \quad (14.18)$$

With $V = v\Omega/M$ and $N/M = \nu$,

$$E_D - E_{\Gamma_{S_0}}^{\text{per}} = -\Omega M \nu (1 - \nu)(v - 1) + O(1). \quad (14.19)$$

This is negative and extensive for fixed $v > 1$. $\square$

The counterexample does not disprove the particular all-pole or cascade asymptotics. It shows that their promotion to physical energies needs an independent contour-to-saddle argument even at leading order. Perron–Frobenius uniqueness of the physical ground state cannot supply that argument.

14.4 Regular eigenvalue-based variables

Regular variables that replace singular Bethe roots by quadratic equations have been developed for Richardson–Gaudin models and their extensions [5, 7, 35]. We adapt that representation to the normalization and branch labels of the present model.

Let $R_1, \dots, R_N$ be a Richardson solution and define

$$\Lambda_n := \sum_{j=1}^N \frac{1}{\mathbf{E}_n - R_j}, \quad \lambda_n := V\Lambda_n. \quad (14.20)$$

The individual roots may collide and leave the real axis, but the values λ_n remain finite and real for a conjugation-invariant physical solution.

Exact identity 14.3 (Quadratic eigenvalue-based equations). *The variables (14.20) obey*

$$\boxed{\lambda_n^2 - \lambda_n - V \sum_{m \neq n} \frac{\lambda_n - \lambda_m}{\mathbf{E}_n - \mathbf{E}_m} = 0, \quad n = 0, \dots, M.} \quad (14.21)$$

For a solution represented by N Richardson roots,

$$\boxed{\sum_{n=0}^M \lambda_n = N.} \quad (14.22)$$

Its energy is

$$\boxed{E^{\text{RG}} = \sum_{n=0}^M \mathbf{E}_n \lambda_n - VN(M - N + 2).} \quad (14.23)$$

More generally, if the active level set has cardinality L_{act} , the last term is $-VN(L_{\text{act}} - N + 1)$.

Proof. Multiply the Richardson equation for root R_j by $(E_n - R_j)^{-1}$ and sum over j . The double-root term combines pairwise into

$$2 \sum_{j < \ell} \frac{1}{(E_n - R_j)(E_n - R_\ell)},$$

while the contribution of a different pole m is

$$\sum_j \frac{1}{(E_n - R_j)(E_m - R_j)} = \frac{\Lambda_m - \Lambda_n}{E_n - E_m}.$$

This yields (14.21). Summing the original Richardson equations over the pole index gives (14.22), since the root–root term is antisymmetric. Finally,

$$\begin{aligned} \sum_n E_n \lambda_n &= V \sum_{j,n} \frac{E_n}{E_n - R_j} \\ &= VN(M+1) + \sum_j R_j - 2V \sum_{j \neq \ell} \frac{R_j}{R_j - R_\ell}. \end{aligned}$$

Each unordered pair contributes one to the last double sum, giving (14.23). $\square$

The energy formula is especially valuable numerically: it gives the physical eigenenergy without reconstructing ill-conditioned high-degree roots.

14.5 Local analytic branches at weak coupling

For a subset $S \subset \mathcal{L}_M$, $|S| = N$, define its indicator vector

$$\chi_n^S = \mathbf{1}_{n \in S}. \quad (14.24)$$

At $V = 0$, every binary vector solves (14.21). The Jacobian with respect to the λ variables is

$$\left. \frac{\partial F_n}{\partial \lambda_m} \right|_{V=0, \lambda=\chi^S} = (2\chi_n^S - 1)\delta_{nm}, \quad (14.25)$$

which is invertible.

Branch identification 14.4 (Unique local branch for every residue subset). *For every N -element subset S there exists $V_S > 0$ and a unique real-analytic solution*

$$\lambda^S(V), \quad 0 \leq V < V_S, \quad (14.26)$$

of (14.21) satisfying

$$\lambda^S(0) = \chi^S. \quad (14.27)$$

Near zero coupling the associated Richardson roots obey, after a suitable ordering,

$$R_j = E_{s_j} - V + O(V^2), \quad S = \{s_1, \dots, s_N\}. \quad (14.28)$$

Thus the binary subset gives a branch label even when several branches have the same total energy at $V = 0$.

Proof. Equation (14.25) and the implicit-function theorem give the unique analytic solution of the quadratic equations. To establish that it is a physical Richardson branch rather than use only necessity of those equations, write $S = \{s_1, \dots, s_N\}$ and set

$$R_j = E_{s_j} - V t_j. \quad (14.29)$$

The Richardson equations multiplied by V become

$$1 = \frac{1}{t_j} + V \sum_{n \neq s_j} \frac{1}{E_n - E_{s_j} + V t_j} + 2V \sum_{\ell \neq j} \frac{1}{E_{s_j} - E_{s_\ell} - V(t_j - t_\ell)}. \quad (14.30)$$

All denominators other than t_j stay nonzero near $V = 0$, $\mathbf{t} = \mathbf{1}$. Bringing the right side to the left gives the identity Jacobian with respect to $\mathbf{t}$ at that point. A second application of the implicit-function theorem therefore gives analytic roots with $t_j = 1 + O(V)$ and the expansion (14.28). Their λ vector tends to χ^S and solves the quadratic equations, so it equals the locally unique vector already constructed. After removing the scalar singular normalization, the Richardson Bethe vector tends to $|S\rangle$; it is nonzero near $V = 0$ and supplies the corresponding physical eigenstate. $\square$

We denote the maximal continuation of this local solution by

$$\mathcal{B}_S. \quad (14.31)$$

This definition uses the full vector λ , not the rank of the total energy among neighbouring eigenvalues. Energy crossings therefore do not exchange the labels.

14.6 Global finite-dimensional continuation and the two bridge conditions

The local eigenvalue-based construction is enough to identify the weak-coupling label, but a physical statement at finite positive coupling also requires that the corresponding Hamiltonian eigenbranch exists along the real coupling path. This existence is a finite-dimensional fact and should be separated from the stronger contour-to-thimble statement used later.

Branch identification 14.5 (Global existence of finite Hamiltonian eigenbranches). *For every finite M, N and every compact real interval $I \subset [0, \infty)$, the analytic Hermitian family $H_N(V)$ admits a real-analytic parameterization of its eigenvalues and an analytic orthonormal eigenbasis on I . Consequently every local branch selected by [Branch identification 14.4](#) extends along the positive real coupling axis as a physical Hamiltonian eigenbranch. At an exact degeneracy the analytic continuation, rather than the ordinal position of the energy, fixes the label.*

Proof. The matrix entries of $H_N(V)$ are affine functions of the real parameter V and $H_N(V)$ is Hermitian. The finite-dimensional Rellich theorem therefore provides real-analytic eigenvalues and eigenvectors on every compact real interval. The locally analytic Bethe/eigenvalue-based branch fixes one of those analytic germs near $V = 0$; analytic continuation patches the germs through isolated degeneracies. The continuation is not determined by energy ordering at a crossing, which is why the complete weak label and the simultaneous eigenvalue-based data remain the operative identifiers. $\square$

The preceding result establishes existence of the physical eigenbranch. It does not prove that an exact multidimensional contour period is represented by the thimble attached to that branch, even at leading order; see [Proposition 14.2](#). The following are hypotheses needed for a contour-based derivation of the stated physical asymptotics. They are not consequences of branch naming, of Rellich's theorem, or of the finite numerical checks below.

Assumption 14.6 (Ground contour and local Gaussian bridge B_0). *On every compact fixed- v , fixed-filling parameter domain to which a ground-state one-loop formula is applied, assume the following two statements separately.*

1. $B_{0,\text{sel}}$: let $T_0(u)$ denote the contribution of the thimble attached to the ground saddle, including its nonzero intersection coefficient. The full all-pole or chosen cascade period and T_0 are nonzero on the domain, and

$$-\partial_u \log \frac{Z_\Gamma(u)}{T_0(u)} = o(1) \quad (14.32)$$

uniformly as $M \rightarrow \infty$. This condition controls the full signed sum, including possible cancellations between thimbles.

2. $B_{0,\text{loc}}$: on that thimble, a Gaussian expansion about the named ground saddle is valid uniformly, with the Hessian normalization appropriate to the integration variables. The logarithmic derivative of its non-Gaussian remainder contributes $O(M^{-1})$ to the energy. In particular, growth of the number of integration variables has been controlled rather than inferred from a fixed-dimensional saddle formula.

Neither statement is proved here. In this manuscript B_0 means their conjunction. The further identification of a finite Hessian contribution with an auxiliary spectral-shift expression is a distinct hypothesis, G , stated in [Assumption 15.2](#); it is not included implicitly in saddle selection or branch naming.

Warning 14.7 (Control of the remainder alone does not exclude cancellation). Writing $Z_\Gamma = T_0 + T_{\text{rest}}$, the weaker condition $\partial_u \log T_{\text{rest}} - \partial_u \log T_0 = o(1)$ would not imply (14.32). For example, take $u > 0$, $c > 0$, $\eta_M \rightarrow 0$, and

$$T_0 = e^{-A_M u}, \quad T_{\text{rest}} = -e^{-A_M u}(1 + \eta_M e^{-cu}). \quad (14.33)$$

The two separate logarithmic derivatives differ by $o(1)$. Their sum is

$$Z_\Gamma = -\eta_M e^{-(A_M+c)u},$$

and its barycenter differs from that of T_0 by c . This remains true for u of order M . Thus a nonzero ground coefficient and similar derivatives of separate contributions do not suffice for a signed contour integral.

Assumption 14.8 (Finite-defect thimble bridge B_{def}). *For each marked or finite-seniority sector a used in Chapters 16–17, let Z_a be the exact period and T_a the thimble contribution attached to its candidate eigenbranch. Assume that the relevant full ratios are nonzero and*

$$-\partial_u \log \left[\frac{Z_a/Z_0}{T_a/T_0} \right] = O(d^2) \quad (14.34)$$

in compact interior domains; at a compact edge require control through the remainder stated for the particular edge formula. This is a full-ratio selection hypothesis, including cancellation control, not a consequence of the common weak labels.

In addition, assume that the local finite-defect Gaussian expansions are uniform. In the comparison of an interior marked-root Schur complement with the two-blocked-sector determinant, assume that the collective-sea contributions to the energies coincide through order d , so that the logarithmic derivative of their residual determinant ratio is $O(d^2)$. This equality is an unresolved input: it includes the cancellation needed for the proposed equality of the two interior thresholds and cannot be counted as an independent proof of that cancellation.

At a compact Debye edge, assume that removal of the singular fixed-level harmonic part leaves a uniformly bounded sea response. Each of these hypotheses must be checked separately for the physical branch under consideration.

For the physical edge-gap formulas, assume also a separate energy-response matching statement. After subtracting the explicit dispersion displacement $2v_-r_1^{\text{RG}}$ from the lower marked-sector physical-minus-period energy difference, or $2v_+s_1^{\text{RG}}$ at the upper edge, the remaining sea correction agrees with the two-blocked-sector correction through order d . In conjunction with G , that common term is $d\mathcal{H}_-$ or $d\mathcal{H}_+$, with remainder $O(d^2)$ in the blocked sector and $O(d/\log^2 M + d^2)$ in the marked sector, as in Equations (17.34), (17.35), (17.38) and (17.39). This energy matching and its remainders are additional unproved inputs; they do not follow from the bounded dressed-root equation alone.

These bridge conditions are not used in the exact finite- M residue algebra or in statements confined to period asymptotics. They enter when a period or a Gaussian period ratio is promoted to a physical eigenbranch asymptotic. In particular, the present branch-label results alone do not supply an unconditional leading-order contour-to-energy theorem. The archived continuation, exact-spectrum, Baxter, and determinant calculations test selected consequences and normalizations; agreement on finite examples is evidence, not a proof of either bridge condition.

14.7 Global identification of the paired ground branch

The graph whose vertices are N -element subsets and whose edges move one pair is connected. For $V > 0$, all off-diagonal matrix elements (14.7) are strictly negative on this graph.

Branch identification 14.9 (Global ground branch). *For every $V > 0$, the seniority-zero N -pair Hamiltonian (14.4) has a unique ground state. It is the global continuation of the weak-coupling label S_0 . Thus the candidate physical ground branch named by the Case A and Case B period labels is the same branch*

$$\mathcal{B}_{S_0}. \quad (14.35)$$

Any collision or change in the root topology of this branch does not change its eigenstate label. This theorem does not locate those collisions or equate their counting function with the period parameter κ .

Proof. Choose a constant C large enough that $CI - H_N$ is entrywise nonnegative. Its off-diagonal graph is connected, hence the matrix is irreducible. The Perron–Frobenius theorem gives a simple largest eigenvalue of $CI - H_N$, equivalently a simple lowest eigenvalue of H_N . At $V = 0$ the unique lowest subset is S_0 , so analytic continuation identifies the unique ground state with (14.35). $\square$

For excited branches no analogous ordering theorem is invoked. The branch $\mathcal{B}_{S_a}$ is followed by its complete λ vector through root collisions. The numerical continuation used in this manuscript is described in Appendix N and archived in `code/verify_branch_identification.py`.

14.8 Baxter-polynomial reconstruction

Define the monic Baxter polynomial

$$\mathcal{Q}(z) = \prod_{j=1}^N (z - R_j). \quad (14.36)$$

Then

$$\frac{\mathcal{Q}'(E_n)}{\mathcal{Q}(E_n)} = \Lambda_n = \frac{\lambda_n}{V}. \quad (14.37)$$

Equivalently,

$$\boxed{\mathcal{Q}'(E_n) - \frac{\lambda_n}{V} \mathcal{Q}(E_n) = 0, \quad n = 0, \dots, M.} \quad (14.38)$$

This is an overdetermined homogeneous linear system for the $N+1$ polynomial coefficients. On an exact branch it has a one-dimensional null space. For numerical stability the implementation maps the pair window to $[-1, 1]$ and uses a Chebyshev basis rather than monomials.

The reconstruction is audited by four independent residuals:

1. the smallest singular value of (14.38);
2. the interpolation residual at every pair pole;
3. the energy identity

$$\sum_j R_j = \sum_n E_n \lambda_n - VN(M - N + 2); \quad (14.39)$$

4. direct substitution of the reconstructed roots in the Richardson equations.

Root reconstruction is used for topology and visualization. Energies at larger N are computed from (14.23), which is considerably better conditioned.

14.9 Branch assignment of the sectors used in this work

Combining the exact weak labels with the local analytic branches gives the permanent assignments

$$\text{all-pole ground and lower Case B ground} \longleftrightarrow \mathcal{B}_{S_0}, \quad (14.40)$$

$$\text{pure marked level } a \text{ and its complete cascade version} \longleftrightarrow \mathcal{B}_{S_a}, \quad 0 \leq a < N, \quad (14.41)$$

with the upper chart supplied by exact electron–hole duality. A blocked set P is assigned to the branch whose active binary label is the set of lowest allowed levels in $\mathcal{L}_M \setminus P$. Multi-blocked sectors follow the same rule.

Branch identification 14.10 (Branch identification used by the manuscript). *The period sectors entering Results R4, R6, R7, and R9 have the same unique local weak-coupling labels as the Richardson branches stated in (14.40)–(14.41) and their blocked analogues. The ground assignment is global by Branch identification 14.9. The marked and blocked assignments are their maximal continuous eigenvalue-based continuations; the paths and parameter sequences used in the later physical comparisons pass the continuation, sector-sum, Baxter, Hamiltonian-spectrum, and Richardson-residual diagnostics of Verification note 14.11.*

This establishes a common labeling convention and records numerical diagnostics for the selected eigenbranches. It does not establish a contour-to-saddle bridge at finite M or at leading thermodynamic order. The next numerical comparison is consistent with an $O(1)$ difference between the ground energy and the all-pole barycenter; assigning that difference to a Gaussian determinant is a separate claim requiring the bridge conditions and the checks in Chapter 15.

14.10 Representative root topology

Consider

$$M = 12, \quad N = 4, \quad a = 2, \quad \Omega = 1. \quad (14.42)$$

The ground label is $S_0 = \{0, 1, 2, 3\}$ and the marked label is $S_2 = \{0, 1, 3, 4\}$. The reconstructed root counts are shown in Table 14.1. An “inside” root is real and lies in one of the pair-pole intervals.

Table 14.1: Root topology of continuously labelled ground and marked branches for (14.42).

v	branch	real	below	inside	above	occupied intervals
0.2	ground	4	1	3	0	0, 1, 2
0.2	marked	4	1	3	0	0, 2, 3
0.5	ground	2	1	1	0	0
0.5	marked	4	1	3	0	0, 2, 3
1.0	ground	0	0	0	0	–
1.0	marked	2	1	1	0	2
1.5	ground	0	0	0	0	–
1.5	marked	2	1	1	0	2

At $v = 1.5$ the ground roots are

$$-0.54377239 \pm 0.26657917i, \quad -0.30308509 \pm 0.84447671i, \quad (14.43)$$

so the finite system is a representative of Case A. The marked branch has

$$-0.36635657, \quad -0.18452971 \pm 0.53715810i, \quad 0.40092909. \quad (14.44)$$

Since

$$E_2 = \frac{1}{3}, \quad E_3 = \frac{1}{2}, \quad (14.45)$$

the last root is trapped in the intended interval (E_2, E_3) . The additional real root below the window is compatible with conjugation symmetry of the remaining collective roots; the branch is identified by its complete weak label and continuation, not by requiring exactly one real number in the entire complex plane.

The fixed-size ground-branch root counts displayed in Chapter 13 are close to $M\kappa$ in the listed examples. The present construction supplies a permanent branch label for those computations. It does not prove a bound uniform in M or the conjectural identification of the physical root-density birth with the rate-defined Lambert law.

14.11 Direct spectral and thermodynamic checks

For small systems the λ -branch energies were compared with direct diagonalization of (14.4). Representative results are given in Table 14.2. The marked branch need not have a fixed ordinal position in the spectrum because weak-coupling degeneracies split differently, but its energy agrees with one exact eigenvalue.

Next take the finite sequence

$$\nu = \frac{1}{3}, \quad \alpha = \frac{1}{6}, \quad v = 1, \quad \Omega = 1. \quad (14.46)$$

Table 14.2: Comparison of eigenvalue-based branch energies with exact paired-Hamiltonian eigenvalues.

M	N	a	v	branch	E_λ	exact E	index
6	2	1	1.0	ground	-0.582932871242	-0.582932871242	0
6	2	1	1.0	marked	0.223837604752	0.223837604752	1
8	3	1	1.0	ground	-0.393774174568	-0.393774174568	0
8	3	1	1.0	marked	0.541052574546	0.541052574546	2
8	3	2	1.5	ground	-1.472682811755	-1.472682811756	0
8	3	2	1.5	marked	-0.065652747437	-0.065652747437	1

Table 14.3: Finite-size comparisons of the physical marked branch, the exact period, and the candidate thermodynamic limit for (14.46).

M	$E_0^{\text{RG}} - E_{\text{all}}^{\text{per}}$	physical gap	period gap	$2\sqrt{D_\alpha}$	$M(\Delta_{\text{phys}} - \Delta_{\text{per}})$
12	-0.3939175753	0.8223243931	0.7680787147	0.8342937004	0.65094814
18	-0.3936172663	0.8283297461	0.7909911808	0.8342937004	0.67209418
24	-0.3934496675	0.8307434662	0.8022215850	0.8342937004	0.68452515
36	-0.3932585184	0.8325563427	0.8132011400	0.8342937004	0.69678730

The physical branch energies are obtained from (14.23). The exact period gap uses the positive finite sum P_a , and the thermodynamic value is $2\sqrt{D_\alpha}$. The results are shown in Table 14.3.

The displayed finite sequences are consistent with two different scalings.

1. The absolute ground energy appears to differ from the all-pole period barycenter by a finite $O(1)$ limit. The table does not identify its origin or prove the Gaussian formula considered in Chapter 15.
2. The physical marked gap and its exact period counterpart have a difference consistent with $O(M^{-1})$, and both approach the square-root expression of Period asymptotics 7.3 and Period asymptotics 12.6 over the displayed range. A finite table does not prove this common limit or its stated error uniformly.

The same check was carried out for one blocked level. With the same ν , α , and v , the physical odd energy difference, exact blocked-period difference, and thermodynamic value are

M	physical odd difference	period difference	thermodynamic value	$M(\text{phys} - \text{per})$
12	0.6809895020	0.6866735526	0.6983076359	-0.06820861
18	0.6864501363	0.6906252302	0.6983076359	-0.07515169
24	0.6893497270	0.6925873230	0.6983076359	-0.07770230
36	0.6922937299	0.6945265698	0.6983076359	-0.08038224

This supports the proposed asymptotic interpretation of the Heine polynomial sector of Period asymptotics 9.4. It does not prove the full-ratio bridge or the physical $O(M^{-1})$ correction required in the finite-defect Gaudin analysis.

14.12 Logical output and verification record

The outputs used in the remainder of the manuscript are:

1. the exact subset labels (14.8)–(14.9);
2. the regular equations (14.21);
3. the local analytic branches of [Branch identification 14.4](#);
4. the globally unique ground assignment of [Branch identification 14.9](#);
5. the energy formula (14.23);
6. the Baxter reconstruction (14.38);
7. the sector assignments of [Branch identification 14.10](#);
8. the numerical evidence that the named physical branches approach the already derived period asymptotics.

Verification note 14.11 (Branch-continuation and reconstruction audit). The script

`code/verify_branch_identification.py`

performs:

1. 1524 exact combinatorial checks of all-pole, excluded-pole, cascade, and blocked weak labels;
2. local small- V continuation from nontrivial binary subsets;
3. direct paired-Hamiltonian diagonalization for ground and marked branches and one blocked sector;
4. Baxter reconstruction, root classification, interpolation residuals, energy residuals, and direct Richardson residuals;
5. physical/period/thermodynamic convergence sequences for the ground, marked, and odd sectors.

The archived report ends with

Branch-identification verification: PASS.

Implementation details and all numerical conventions are recorded in [Appendix N](#).

Chapter 15

Gaussian bridge and the subextensive ground-state energy

15.1 The Gaudin Hessian and the one-loop bridge

Gaudin-type determinants are standard finite-dimensional norm/Hessian objects associated with Bethe states [14, 36]. Under the ground bridge stated in Chapter 14, their logarithmic derivative supplies the difference between the period barycenter and the physical saddle through the stated order.

It is convenient to remove the inert lower-edge energy and the factor of two in the pair poles. Put

$$R_j = 2\varepsilon_{F0} + 2r_j, \quad \xi_n = \frac{n}{\rho}, \quad 0 \leq n \leq M. \quad (15.1)$$

Up to a coupling-independent Jacobian, the logarithm of the meromorphic integrand is

$$\mathcal{S}(\mathbf{r}; u) = 2 \sum_{i < j} \log(r_i - r_j) - \sum_{i=1}^N \sum_{n=0}^M \log(\xi_n - r_i) - 2u \sum_{i=1}^N r_i, \quad u = \frac{1}{V}. \quad (15.2)$$

Its stationary equations are the scaled Richardson equations. On the branch certified in Chapter 14, denote the saddle by $\mathbf{r}_*$. The Hessian is

$$\mathcal{G}_{ii} = \sum_{n=0}^M \frac{1}{(\xi_n - r_{i,*})^2} - 2 \sum_{j \neq i} \frac{1}{(r_{i,*} - r_{j,*})^2}, \quad (15.3)$$

$$\mathcal{G}_{ij} = \frac{2}{(r_{i,*} - r_{j,*})^2}, \quad i \neq j. \quad (15.4)$$

The overall constant relating this matrix to the Hessian written in the unscaled variables R_j is independent of V and therefore disappears from every logarithmic derivative below.

The ground-state saddle is invariant under complex conjugation. We choose the continuous branch of $\log \det(-\mathcal{G})$ along the physical continuation path. A constant orientation phase of the thimble and its integer intersection coefficient are independent of u and likewise drop out of the energy.

Branch identification 15.1 (Gaussian bridge for the certified ground branch, under B_0). *Assume the ground thimble-selection bridge Assumption 14.6. On every compact parameter set with fixed $v > 0$, $0 < \nu < 1$, and away from the joint mesoscopic limit $M\sigma = O(1)$, the ground-state period and the physical Richardson branch satisfy*

$$E_0^{\text{RG}} - E_0^{\text{per}} = -\frac{1}{2} \frac{\partial}{\partial(1/V)} \log \det(-\mathcal{G}_0) + o(1). \quad (15.5)$$

The same formula holds in Cases A and B. In Case B, E_0^{per} may be taken to be the physical consecutive-cascade period; by [Period asymptotics 12.5](#) it differs from the all-pole period by only $O(M^{-2})$.

Proof. Under [Assumption 14.6](#), expand the selected ground thimble around $\mathbf{r}_*$. With the standard steepest-descent orientation,

$$\log Z_0^{\text{th}} = \mathcal{S}(\mathbf{r}_*; u) - \frac{1}{2} \log \det(-\mathcal{G}_0) + \frac{N}{2} \log(2\pi) + \log \mathbf{n}_0 + \mathcal{R}_M(u), \quad (15.6)$$

where $\mathbf{n}_0$ is the locally constant intersection coefficient and $\partial_u \mathcal{R}_M = O(M^{-1})$ by $\mathbf{B}_{0,\text{loc}}$. Stationarity removes the implicit derivative of $\mathbf{r}_*$, and

$$-\partial_u \mathcal{S}(\mathbf{r}_*; u) = 2 \sum_j r_{j,*} = E_0^{\text{RG}} - 2N\varepsilon_{F0}. \quad (15.7)$$

The constant Jacobian, $N \log(2\pi)/2$, the thimble phase, and $\mathbf{n}_0$ have zero u -derivative. The $O(M^{-1})$ energy bound on higher loops, including differentiation of the remainder, is part of [Assumption 14.6](#); it is not inferred from the dimension of the Hessian. The branch assignment is supplied by [Branch identification 14.9](#). Appendix K records the normalization and the unresolved requirements for uniform loop control. $\square$

Equation (15.5), when its hypothesis holds, relates the observed $O(1)$ difference to a Hessian logarithmic derivative. Cancellation of the common part in finite-defect ratios requires additional uniformity estimates; these are stated explicitly below and used in Chapters 16–17.

15.2 Finite spectral curve and interlacing auxiliary zeros

The leading large-system saddle is described by two finite-system parameters $\tilde{\mu}_M$ and $\Delta_M > 0$, defined by

$$\boxed{\frac{2}{V} = \sum_{n=0}^M \frac{1}{\mathcal{E}_{n,M}}}, \quad \boxed{M+1-2N = \sum_{n=0}^M \frac{\xi_n - \tilde{\mu}_M}{\mathcal{E}_{n,M}}}, \quad (15.8)$$

where

$$\mathcal{E}_{n,M} = \sqrt{(\xi_n - \tilde{\mu}_M)^2 + \Delta_M^2}. \quad (15.9)$$

For fixed v and ν ,

$$\tilde{\mu}_M = \tilde{\mu} + O(M^{-1}), \quad \Delta_M = \Delta + O(M^{-1}). \quad (15.10)$$

No mean-field variational argument is being introduced here: (15.8) is the two-parameter form of the leading spectral curve obtained from the large- M Richardson saddle.

Define the rational function

$$\boxed{\mathcal{F}_M(z) = \sum_{n=0}^M \frac{1}{(z - \xi_n) \mathcal{E}_{n,M}}}. \quad (15.11)$$

Because all residues $\mathcal{E}_{n,M}^{-1}$ are positive,

$$\mathcal{F}'_M(z) = - \sum_{n=0}^M \frac{1}{(z - \xi_n)^2 \mathcal{E}_{n,M}} < 0 \quad (15.12)$$

inside every open interval free of poles. Hence there is exactly one zero

$$y_\ell \in (\xi_{\ell-1}, \xi_\ell), \quad \ell = 1, \dots, M. \quad (15.13)$$

The product representation is

$$\mathcal{F}_M(z) = \frac{2}{V} \frac{\prod_{\ell=1}^M (z - y_\ell)}{\prod_{n=0}^M (z - \xi_n)}. \quad (15.14)$$

The coefficient follows from $\mathcal{F}_M(z) \sim z^{-1} \sum_n \mathcal{E}_{n,M}^{-1} = 2/(Vz)$ at infinity.

The numbers y_ℓ are auxiliary zeros of the spectral curve. They are neither Richardson roots nor the eigenvalues of the $N \times N$ matrix $\mathcal{G}$: there are M such zeros, generally different from N . An augmented response representation would be needed to justify any interpretation as a diagonalization of that Hessian.

Assumption 15.2 (Hessian–spectral-shift bridge $\mathbf{G}$). *On the compact parameter sets where it is used, the logarithmic derivative of the exact ground-saddle Hessian has the auxiliary spectral-shift representation (15.15), with a uniform $O(M^{-1})$ remainder. With $d = \Omega/M$, for the finite-defect applications in Chapters 16–17, the change of that remainder is $O(d^2)$, and the first variation of the finite spectral shift is represented by $d\mathcal{H}(\xi) + O(d^2)$ in the indicated interior or compact-edge regime. These finite-defect requirements are additional to the ground-state assertion. This hypothesis is independent of the thimble-selection and Gaussian-remainder hypothesis $\mathbf{B}_0$.*

Period asymptotics 15.3 (Conditional finite Gaudin spectral shift, under $\mathbf{G}$). *Under Assumption 15.2, for the ground branch,*

$$\begin{aligned} -\frac{1}{2} \frac{\partial}{\partial(1/V)} \log \det(-\mathcal{G}_0) &= V \left(\frac{M+1}{2} - N \right) \\ &\quad + \sum_{\ell=1}^M \mathcal{E}_M(y_\ell) - \sum_{n=0}^M \mathcal{E}_{n,M} + O(M^{-1}), \end{aligned} \quad (15.15)$$

where

$$\mathcal{E}_M(z) = \sqrt{(z - \tilde{\mu}_M)^2 + \Delta_M^2} \quad (15.16)$$

uses the branch continuous from positive real $z > \Omega$.

The displayed identity is the ground-state content of $\mathbf{G}$, not a newly proved determinant factorization. The right-hand side is the standard auxiliary expression for the first Richardson resolvent correction [23]. Section K.3 derives its coefficient from that known resolvent formula and identifies the missing step needed to connect it to $\det \mathcal{G}$.

The formula is already computationally useful: it requires one monotone scalar root in each one-particle interval rather than the coupled Richardson roots. It will also make the continuum limit transparent.

15.3 Continuum spectral shift

Write

$$\mathcal{E}(\epsilon) = \sqrt{(\epsilon - \tilde{\mu})^2 + \Delta^2}, \quad 0 \leq \epsilon \leq \Omega, \quad (15.17)$$

and define the continuum Cauchy transform

$$\mathcal{F}(z) = \int_0^\Omega \frac{d\epsilon}{(z - \epsilon)\mathcal{E}(\epsilon)}. \quad (15.18)$$

For $0 < \epsilon < \Omega$,

$$\mathcal{F}(\epsilon + i0) = \text{PV} \int_0^\Omega \frac{d\epsilon'}{(\epsilon - \epsilon')\mathcal{E}(\epsilon')} - \frac{i\pi}{\mathcal{E}(\epsilon)}. \quad (15.19)$$

Choose the continuous phase with values in $[-\pi, 0]$ and define

$$a(\epsilon) = -\frac{1}{\pi} \arg \mathcal{F}(\epsilon + i0). \quad (15.20)$$

Then

$$a(0+) = 1, \quad a(\Omega-) = 0. \quad (15.21)$$

If

$$y_\ell = \xi_\ell - d a(\xi_\ell) + o(d), \quad d = \rho^{-1}, \quad (15.22)$$

then the auxiliary zero is measured from the upper pole of its interval. The phase formula (15.20) is the continuum form of this local displacement.

Lemma 15.4 (Continuum zero-minus-pole sum). *For every smooth function g on the closed window,*

$$\sum_{\ell=1}^M g(y_\ell) - \sum_{n=0}^M g(\xi_n) = -g(0) - \int_0^\Omega g'(\epsilon) a(\epsilon) d\epsilon + o(1). \quad (15.23)$$

In particular,

$$\lim_{M \rightarrow \infty} \left[\sum_{\ell=1}^M \mathcal{E}_M(y_\ell) - \sum_{n=0}^M \mathcal{E}_{n,M} \right] = -\mathcal{E}(0) - \int_0^\Omega \frac{\epsilon - \tilde{\mu}}{\mathcal{E}(\epsilon)} a(\epsilon) d\epsilon. \quad (15.24)$$

Proof. Pair y_ℓ with the upper pole ξ_ℓ and use (15.22). The missing lower pole gives $-g(0)$, while the remaining first-order differences form a Riemann sum. The logarithmic endpoint approach of the first and last zero changes the result only by $o(1)$. A contour proof from (15.14) is given in [Appendix K.4](#). $\square$

15.4 Explicit phase and the closed one-dimensional formula

Introduce the hyperbolic coordinate

$$\epsilon = \tilde{\mu} + \Delta \sinh u, \quad \mathcal{E}(\epsilon) = \Delta \cosh u, \quad (15.25)$$

and the endpoints

$$u_- = \text{arsinh} \frac{-\tilde{\mu}}{\Delta}, \quad u_+ = \text{arsinh} \frac{\Omega - \tilde{\mu}}{\Delta}. \quad (15.26)$$

The gap equation is

$$u_+ - u_- = \frac{2}{v}. \quad (15.27)$$

A direct principal-value integration gives

$$\mathcal{F}(\epsilon(u) + i0) = \frac{\mathcal{L}(u) - i\pi}{\Delta \cosh u}, \quad (15.28)$$

where

$$\mathcal{L}(u) = \log \left| \frac{\sinh \frac{u-u_-}{2} \cosh \frac{u+u_+}{2}}{\cosh \frac{u+u_-}{2} \sinh \frac{u_+-u}{2}} \right|. \quad (15.29)$$

Therefore

$$a(u) = \frac{1}{2} - \frac{1}{\pi} \arctan \frac{\mathcal{L}(u)}{\pi}. \quad (15.30)$$

The endpoint limits in (15.21) follow from $\mathcal{L}(u_-) = -\infty$ and $\mathcal{L}(u_+) = +\infty$.

Physical consequence 15.5 (Subextensive ground-energy correction, under $\mathbf{B}_0, \mathbf{G}$). *Assume Assumptions 14.6 and 15.2. Let*

$$\delta E_{\text{fl}}(\nu, v) = \lim_{M \rightarrow \infty} (E_0^{\text{RG}} - E_0^{\text{per}}). \quad (15.31)$$

Then

$$\begin{aligned} \delta E_{\text{fl}}(\nu, v) = v\Omega \left(\frac{1}{2} - \nu \right) - \mathcal{E}(0) \\ - \int_0^\Omega \frac{\epsilon - \tilde{\mu}}{\mathcal{E}(\epsilon)} a(\epsilon) \, d\epsilon, \end{aligned} \quad (15.32)$$

where a is defined by (15.20). Equivalently,

$$\delta E_{\text{fl}} = v\Omega \left(\frac{1}{2} - \nu \right) - \mathcal{E}(0) - \Delta \int_{u_-}^{u_+} \sinh u \, a(u) \, du. \quad (15.33)$$

Consequently,

$$E_0^{\text{RG}} = E_{\text{all}}^{\text{per}} + \delta E_{\text{fl}}(\nu, v) + o(1), \quad (15.34)$$

throughout Cases A and B.

Proof. The explicit term in (15.15) satisfies

$$V \left(\frac{M+1}{2} - N \right) \rightarrow v\Omega \left(\frac{1}{2} - \nu \right). \quad (15.35)$$

Combine this limit with Lemma 15.4, the Hessian identification assumed in $\mathbf{G}$, and Branch identification 15.1. The equality of the all-pole and physical Case B period energies through the required order follows from Period asymptotics 12.5. $\square$

For the exact finite sequence $N = \nu M$, the complete expansion may be written

$$\begin{aligned} E_0^{\text{RG}} = 2N\varepsilon_{F0} + M\Omega \left[\nu^2 - 2x\nu(1-\nu) \right] \\ - \Omega\nu(1+2x) + \delta E_{\text{fl}}(\nu, v) + o(1). \end{aligned} \quad (15.36)$$

The third term is already contained in the exact all-pole period. The last term is the genuinely Gaussian difference between the physical branch and the period barycenter.

15.5 Smoothness and electron–hole covariance

For fixed $v > 0$ and $0 < \nu < 1$,

$$\Delta > 0, \quad \mathcal{E}(0) = \Omega(\nu + x), \quad \mathcal{E}(\Omega) = \Omega(1 - \nu + x). \quad (15.37)$$

The logarithmic singularities of $\mathcal{L}(u)$ at $u_\pm$ are integrable and occur for every parameter value. They do not become stronger when $\tilde{\mu}$ reaches either edge.

Physical consequence 15.6 (Smooth spectral-shift functional; physical consequence under B_0, G). *The function $\delta E_{\text{fl}}(\nu, v)$ is smooth on every compact subset of*

$$0 < \nu < 1, \quad v > 0. \quad (15.38)$$

In particular, it is smooth across

$$\tilde{\mu} = 0, \quad \tilde{\mu} = \Omega, \quad (15.39)$$

and across the Case A–B line

$$\Phi_0(\nu, \sigma) = 0. \quad (15.40)$$

The corresponding statement about the complete physical $O(M^0)$ ground energy follows under B_0 and G ; smoothness of the displayed integral alone does not establish either bridge.

Proof. At $\tilde{\mu} = 0$, the lower hyperbolic endpoint simply passes through $u_- = 0$; all denominators remain nonzero because $\Delta > 0$. The same argument applies at the upper edge. The Case A–B condition involves the residue rate Φ_0 , whereas (15.33) depends only on the smooth spectral-curve parameters $\tilde{\mu}$ and Δ . Parameter derivatives may be passed under the integral after isolating the universal logarithmic endpoint behavior. Details are given in [Appendix K.7](#). $\square$

The exact physical electron–hole transformation of [Exact identity 8.8](#) requires a non-symmetric fluctuation term. The spectral-shift formula gives

$$\boxed{\delta E_{\text{fl}}(\nu, v) - \delta E_{\text{fl}}(1 - \nu, v) = v\Omega(1 - 2\nu).} \quad (15.41)$$

This is precisely the thermodynamic limit of

$$-V\delta_N = V(\bar{N} - N) \quad (15.42)$$

found in [Equation \(8.66\)](#). The corresponding phase functions satisfy

$$a_{1-\nu}(\epsilon) = 1 - a_\nu(\Omega - \epsilon). \quad (15.43)$$

15.6 Half filling

At $\nu = 1/2$, put

$$D = \frac{\Omega}{2}, \quad t = \frac{1}{v}. \quad (15.44)$$

Then

$$\tilde{\mu} = D, \quad \Delta = \frac{D}{\sinh t}, \quad \mathcal{E}(0) = D \coth t, \quad u_{\pm} = \pm t. \quad (15.45)$$

The general one-dimensional formula reduces to

$$\boxed{\delta E_{\text{fl}}\left(\frac{1}{2}, v\right) = -D \coth \frac{1}{v} \varphi(v),} \quad (15.46)$$

where

$$\boxed{\varphi(v) = 2 \int_0^\infty \frac{ds}{\pi(1+s^2)} \frac{\cosh(\pi s/2)}{\sqrt{\cosh^2(\pi s/2) + \sinh^2(1/v)}}.} \quad (15.47)$$

The reduction is derived in [Appendix K.6](#). This representation is particularly useful for independent comparison with the traditional finite-size expansion, although the derivation here originates from the Gaussian determinant of the Selberg–residue period.

15.7 Direct verification

The finite spectral-shift formula, the continuum integral, and direct Richardson continuation were compared in parameter regions representing all relevant topologies. Energies are quoted in units $\Omega = 1$, with $\varepsilon_{F0} = 0$.

Table 15.1: Physical ground energy minus the exact all-pole period, the finite auxiliary zero formula (15.15), and the continuum limit (15.32).

regime	(ν, v)	M	$E_0^{\text{RG}} - E_{\text{all}}^{\text{per}}$	finite shift	continuum
Case A	$(1/3, 1)$	12	-0.39391758	-0.39528019	-0.39287449
		18	-0.39361727	-0.39456049	-0.39287449
		24	-0.39344967	-0.39417020	-0.39287449
		36	-0.39327018	-0.39375929	-0.39287449
Case B	$(1/2, 0.8)$	12	-0.47596501	-0.47888336	-0.47324617
		20	-0.47500401	-0.47673560	-0.47324617
		28	-0.47454183	-0.47577275	-0.47324617
Case A	$(1/2, 1.5)$	12	-0.80308309	-0.80356013	-0.79989020
		20	-0.80188264	-0.80220395	-0.79989020
		28	-0.80133802	-0.80157855	-0.79989020
$\tilde{\mu} < 0$	$(0.1, 1)$	20	-0.12611070	-0.12621092	-0.12618481
		40	-0.12615321	-0.12624580	-0.12618481

The physical sequence and the finite spectral-shift sequence approach the same continuum value with ordinary $O(M^{-1})$ residuals. The table includes a macroscopic Case B, two Case A regimes, and a state with the chemical potential below the interaction window.

A second test differentiates the exact finite Gaudin determinant reconstructed from the Richardson roots. Representative values are

M	(ν, v)	$-\frac{1}{2}\partial_{1/V} \log \det \mathcal{G} $	physical-period	finite shift
12	$(1/3, 1)$	-0.39700528	-0.39391758	-0.39528019
18	$(1/3, 1)$	-0.39577045	-0.39361727	-0.39456049
12	$(1/2, 0.8)$	-0.47840976	-0.47596501	-0.47888336
12	$(1/2, 1.5)$	-0.80632137	-0.80308309	-0.80356013

The differences are of the expected next-loop order.

At half filling the real-axis integral and (15.46) agree to the displayed numerical precision:

v	general formula	half-filled formula
0.8	-0.473246172742	-0.473246172742
1.0	-0.565997920677	-0.565997920677
1.5	-0.799890203779	-0.799890203779

Verification note 15.7 (Ground-state Gaudin and spectral-shift audit). The script

`code/verify_gaudin_ground.py`

performs:

1. finite gap/number solves and one-zero-per-interval interlacing checks;
2. complete finite zero-minus-pole sums in Cases A and B and below the Debye edge;

3. constrained eigenvalue-based continuation of the physical ground branch;
4. direct finite differences of $\log |\det \mathcal{G}|$ reconstructed from Baxter roots;
5. evaluation of the real-axis and hyperbolic continuum formulas;
6. half-filled identity checks, electron–hole covariance, and smooth parameter scans through $\tilde{\mu} = 0$ and the A–B line.

The archived report ends with

Ground-state Gaudin correction verification: PASS.

Part VI

Finite-size excitation spectrum

Chapter 16

Interior finite-size corrections to the excitation spectrum

16.1 Interior regime and finite-size bookkeeping

Put

$$d = \frac{1}{\rho} = \frac{\Omega}{M}. \quad (16.1)$$

Throughout this chapter the parameters remain in a compact subset of the interior-minimum regime

$$0 < \tilde{\mu} < \Omega, \quad \text{dist}(\tilde{\mu}, \{0, \Omega\}) \geq \eta\Omega, \quad \Delta \geq \eta\Omega \quad (16.2)$$

for some fixed $\eta > 0$. The quasiparticle dispersion

$$\mathcal{E}(\xi) = \sqrt{(\xi - \tilde{\mu})^2 + \Delta^2} \quad (16.3)$$

has its unconstrained minimum at

$$\xi_* = \tilde{\mu}, \quad \mathcal{E}(\xi_*) = \Delta. \quad (16.4)$$

The nearest lattice levels differ from $\tilde{\mu}$ by $O(d)$. Since $\mathcal{E}'(\tilde{\mu}) = 0$, their placement changes the sum of two quasiparticle energies only by $O(d^2)$.

We compare two distinct even sectors:

1. the seniority-zero real-root sector R , represented by the pure marked period and by the branch $\mathcal{B}_{S_a}$ of [Chapter 14](#);
2. the seniority-two broken-pair sector B_2 , with two neighboring blocked levels and $N - 1$ active pairs.

Their thermodynamic thresholds agree by [Chapter 10](#). The purpose here is to record the candidate first correction, derive the explicit response formulas entering it, and separate the still-required uniform determinant estimates.

16.2 Residue-prefactor correction at the minimum

Let

$$\alpha_* = \frac{\tilde{\mu}}{\Omega} = \nu(1 + 2x) - x. \quad (16.5)$$

For the lower chart, $\nu \leq 1/2$, the two saddle coordinates of Equation (7.20) simplify at $\alpha = \alpha_*$ to

$$\beta_* = \nu - r, \quad \tau_* = \nu + r, \quad r = \sqrt{\sigma\nu(1-\nu)}. \quad (16.6)$$

Substitution into the exact prefactor derivative (7.28) gives

$$c_*^{\text{eff}} = \sqrt{\frac{\sigma\nu}{1-\nu}} - \frac{1+\sigma}{2}. \quad (16.7)$$

For the upper chart one must first use the exact dual filling

$$\bar{\nu}_M = 1 - \nu + \frac{1}{M}. \quad (16.8)$$

The raw lower-hole prefactor is evaluated at $\bar{\nu}_M$, and the M^{-1} shift of the leading term $2\Delta(\bar{\nu}_M)$ contributes at the same order d . After the two contributions are combined, the effective coefficient is again (16.7). Thus the compact formula below is valid in both marked charts under the fixed convention $\nu = N/M$. This finite-duality correction is also displayed explicitly in Section 17.3.

Period asymptotics 16.1 (Interior marked-period threshold through order d). *For the marked period whose interval contains $\tilde{\mu}$,*

$$\Delta_R^{\text{per}} = 2\Delta + d \left[-(1+2x) + \frac{\Delta}{\Omega(1-\nu)} \right] + O(d^2). \quad (16.9)$$

The estimate is uniform on compact sets satisfying (16.2). The same formula holds in compact Case B; the complete cascade changes the marked period only by $O(d^2)$ according to Period asymptotics 12.6.

Proof. Evaluate (7.33) at α_* . Since $d = \Omega/M$,

$$\frac{2\Omega}{M(1-\sigma)} c_*^{\text{eff}} = d \left[-(1+2x) + \frac{\Delta}{\Omega(1-\nu)} \right]. \quad (16.10)$$

For the upper chart, the raw reflected prefactor and the M^{-1} displacement of the exact dual filling must be expanded together. The latter changes the leading 2Δ term by $O(d)$ and is precisely what restores the coefficient displayed above. $\square$

The first contribution, $-d(1+2x)$, will reappear as the relaxation of the spectral curve after two levels are removed. The second term is the discrete residue-prefactor contribution, including the finite-duality displacement in the upper chart. The next sections show that the broken-pair period has the same two terms at the interior minimum.

16.3 Linear response of the spectral curve to two blocked levels

Let two neighboring blocked levels coalesce macroscopically at a point $\xi \in (0, \Omega)$. The active pair number changes from N to $N-1$, while the number of active one-particle levels changes from $M+1$ to $M-1$. Consequently

$$(M-1) - 2(N-1) = M+1 - 2N, \quad (16.11)$$

so the right-hand side of the spectral-curve number equation is unchanged.

Write the perturbed parameters as

$$\tilde{\mu}_{[\xi]} = \tilde{\mu} + d\dot{\mu}_\xi + O(d^2), \quad \Delta_{[\xi]} = \Delta + d\dot{\Delta}_\xi + O(d^2). \quad (16.12)$$

Define

$$\eta_\xi = \xi - \tilde{\mu}, \quad \mathcal{E}_\xi = \sqrt{\eta_\xi^2 + \Delta^2}, \quad (16.13)$$

and

$$A = \frac{1}{\mathcal{E}(0)} - \frac{1}{\mathcal{E}(\Omega)}, \quad (16.14)$$

$$B = -\frac{1}{\Delta} \left[\frac{\Omega - \tilde{\mu}}{\mathcal{E}(\Omega)} + \frac{\tilde{\mu}}{\mathcal{E}(0)} \right]. \quad (16.15)$$

Lemma 16.2 (Finite-defect response equations). *The first response of the spectral curve is determined by*

$$\begin{pmatrix} A & B \\ \Delta B & -\Delta A \end{pmatrix} \begin{pmatrix} \dot{\mu}_\xi \\ \dot{\Delta}_\xi \end{pmatrix} = \begin{pmatrix} 2/\mathcal{E}_\xi \\ 2\eta_\xi/\mathcal{E}_\xi \end{pmatrix}. \quad (16.16)$$

Equivalently,

$$\dot{\mu}_\xi = \frac{2(A + B\eta_\xi/\Delta)}{\mathcal{E}_\xi(A^2 + B^2)}, \quad (16.17)$$

$$\dot{\Delta}_\xi = \frac{2(B - A\eta_\xi/\Delta)}{\mathcal{E}_\xi(A^2 + B^2)}. \quad (16.18)$$

Proof. Linearize the two continuum spectral-curve equations after subtracting the two blocked levels. The required integrals are

$$\int_0^\Omega \frac{\epsilon - \tilde{\mu}}{\mathcal{E}(\epsilon)^3} d\epsilon = A, \quad -\Delta \int_0^\Omega \frac{d\epsilon}{\mathcal{E}(\epsilon)^3} = B. \quad (16.19)$$

The gap equation gives the first row of (16.16); the number equation gives the second. The determinant is $-\Delta(A^2 + B^2) \neq 0$ under (16.2). Details are given in [Appendix L.1](#). $\square$

At the minimum $\xi = \tilde{\mu}$, these expressions simplify completely.

Corollary 16.3 (Response at the quasiparticle minimum). *At $\xi = \tilde{\mu}$,*

$$\dot{\Delta}_* = -(1 + 2x), \quad (16.20)$$

$$\dot{\mu}_* = -(2\nu - 1) \sqrt{\frac{x(1+x)}{\nu(1-\nu)}}. \quad (16.21)$$

The filling-independent value of $\dot{\Delta}_*$ reproduces the first term in the candidate blocked-period coefficient. Obtaining its other term from the exact two-level Heine–Christoffel determinant requires a confluent asymptotic calculation whose full coefficients and uniform remainder are not supplied here. We therefore isolate the estimate used below.

Assumption 16.4 (Confluent finite-defect period estimate C_{per}). *For two neighboring blocked levels at the interior quasiparticle minimum, the exact seniority-two period and the marked period have the same energy coefficient through order d , uniformly on the compact sets (16.2). This estimate includes the finite upper-chart filling shift and, when used, the compact Case B resummation.*

Under this estimate,

$$\Delta_{B_2}^{\text{per}} = \Delta_R^{\text{per}} + O(d^2) \quad (16.22)$$

in the interior-minimum regime. Appendix L records the proposed confluent route and the missing remainder estimate; the finite- M comparisons provide evidence for $\mathbb{C}_{\text{per}}$.

16.4 Finite-defect Gaudin response

The auxiliary spectral-shift functional of Chapter 15 can be written, up to its explicit uniform term, as

$$\mathcal{I}[\mathcal{E}, \mathcal{F}] = \frac{1}{2\pi i} \oint_{\mathcal{C}} \mathcal{E}(z) \partial_z \log \mathcal{F}(z) dz, \quad (16.23)$$

where

$$\mathcal{F}(z) = \int_0^\Omega \frac{d\epsilon}{(z - \epsilon)\mathcal{E}(\epsilon)}. \quad (16.24)$$

The contour surrounds $[0, \Omega]$ but excludes the two branch points $\tilde{\mu} \pm i\Delta$.

The explicit uniform contribution is

$$V\left(\frac{L}{2} - N\right), \quad L = M + 1. \quad (16.25)$$

For a broken pair, $L \mapsto L - 2$ and $N \mapsto N - 1$, so this term is unchanged exactly. Only the spectral functional (16.23) responds.

Define

$$\dot{\mathcal{E}}_\xi(z) = \frac{-(z - \tilde{\mu})\dot{\mu}_\xi + \Delta\dot{\Delta}_\xi}{\mathcal{E}(z)}, \quad (16.26)$$

and

$$\dot{\mathcal{F}}_\xi(z) = \int_0^\Omega \frac{(\epsilon - \tilde{\mu})\dot{\mu}_\xi - \Delta\dot{\Delta}_\xi}{(z - \epsilon)\mathcal{E}(\epsilon)^3} d\epsilon - \frac{2}{(z - \xi)\mathcal{E}_\xi}. \quad (16.27)$$

Equivalently,

$$\dot{\mathcal{F}}_\xi = (\dot{\mu}_\xi \partial_{\tilde{\mu}} + \dot{\Delta}_\xi \partial_\Delta) \mathcal{F} - \frac{2}{(z - \xi)\mathcal{E}_\xi}. \quad (16.28)$$

Physical consequence 16.5 (Finite-defect Gaudin response, under $\mathbb{B}_0, \mathbb{G}, \mathbb{B}_{\text{def}}$). *The first variation of the auxiliary functional is the contour expression below. Its identification with the physical-minus-period energy difference additionally assumes Assumptions 14.6, 14.8 and 15.2.*

$$\mathcal{H}(\xi) = \frac{1}{2\pi i} \oint_{\mathcal{C}} \left[\dot{\mathcal{E}}_\xi(z) \partial_z \log \mathcal{F}(z) - \mathcal{E}'(z) \frac{\dot{\mathcal{F}}_\xi(z)}{\mathcal{F}(z)} \right] dz. \quad (16.29)$$

For the broken-pair branch,

$$\Delta_{B_2}^{\text{RG}} - \Delta_{B_2}^{\text{per}} = d \mathcal{H}(\xi) + O(d^2). \quad (16.30)$$

Proof. Vary (16.23). The first variation is

$$\frac{1}{2\pi i} \oint_{\mathcal{C}} \left[\dot{\mathcal{E}}_\xi \partial_z \log \mathcal{F} + \mathcal{E} \partial_z \left(\frac{\dot{\mathcal{F}}_\xi}{\mathcal{F}} \right) \right] dz. \quad (16.31)$$

Integration by parts on the closed contour gives (16.29). The response equations preserve the coupling and number constraints to first order. The remainder of this smooth functional variation is quadratic in the defect. Passing that estimate to physical energy differences, including the change of the finite- M and higher-loop remainders, uses $\mathbf{G}$ and $\mathbf{B}_{\text{def}}$; it is not established by differentiating the contour functional alone. Appendix L gives the contour normalization and real-axis form. $\square$

16.5 Real-axis form

For $0 < \epsilon < \Omega$, write

$$\mathcal{F}(\epsilon + i0) = \frac{\mathcal{L}(\epsilon) - i\pi}{\mathcal{E}(\epsilon)}, \quad (16.32)$$

where $\mathcal{L}$ is the explicit logarithm of Equation (15.29). Define

$$a(\epsilon) = \frac{1}{2} - \frac{1}{\pi} \arctan \frac{\mathcal{L}(\epsilon)}{\pi}, \quad (16.33)$$

$$\dot{a}_\xi(\epsilon) = -\frac{1}{\pi} \text{Im} \frac{\dot{\mathcal{F}}_\xi(\epsilon + i0)}{\mathcal{F}(\epsilon + i0)}. \quad (16.34)$$

The second equation is understood distributionally because the direct defect source has a pole at $\epsilon = \xi$.

Proposition 16.6 (Real-axis defect functional). *The contour response is equivalently*

$$\boxed{\mathcal{H}(\xi) = -\dot{\mathcal{E}}_\xi(0) - \int_0^\Omega \left[\partial_\epsilon \dot{\mathcal{E}}_\xi(\epsilon) a(\epsilon) + \mathcal{E}'(\epsilon) \dot{a}_\xi(\epsilon) \right] d\epsilon.} \quad (16.35)$$

The distribution in (16.34) has the explicit decomposition

$$\begin{aligned} \dot{a}_\xi(\epsilon) = & -\frac{\dot{\mathcal{L}}_\xi(\epsilon)}{\mathcal{L}(\epsilon)^2 + \pi^2} \\ & + \frac{2\mathcal{E}(\epsilon)}{\mathcal{E}_\xi[\mathcal{L}(\epsilon)^2 + \pi^2]} \text{PV} \frac{1}{\epsilon - \xi} \\ & - \frac{2\mathcal{L}(\xi)}{\mathcal{L}(\xi)^2 + \pi^2} \delta(\epsilon - \xi), \end{aligned} \quad (16.36)$$

where

$$\dot{\mathcal{L}}_\xi = (\dot{\mu}_\xi \partial_\mu + \dot{\Delta}_\xi \partial_\Delta) \mathcal{L} \quad (16.37)$$

is taken at fixed ϵ .

Equation (16.35) is numerically preferable to the formally integrated expression containing $a'(\epsilon)$: the endpoint approach of a is logarithmically slow, whereas the displayed integrand is absolutely integrable without subtracting large endpoint terms. A principal-value formula without distributions is written in Appendix L.4.

16.6 Equality of the two physical even gaps

For the broken-pair sector, Physical consequence 16.5 gives the response of the collective sea. The real-root sector has one additional explicit root and an $(N - 1)$ -dimensional sea. Its Gaudin matrix has the block form

$$\mathcal{G}_R = \begin{pmatrix} g_{rr} & \mathbf{b}^\top \\ \mathbf{b} & \mathcal{G}_{\text{sea}} \end{pmatrix}. \quad (16.38)$$

The Schur complement

$$g_{rr} - \mathbf{b}^\top \mathcal{G}_{\text{sea}}^{-1} \mathbf{b} \quad (16.39)$$

is the derivative of the dressed equation for the separated root whenever that elimination is regular. The exact particle-hole convolution motivates comparison with the blocked determinant, but does not by itself identify the Gaudin sea blocks. Equality of their Gaussian responses through order d is an explicit clause of $\mathbf{B}_{\text{def}}$. The missing matrix identity and uniform remainder estimate are recorded in Appendix L.

Physical consequence 16.7 (Common interior finite-size gap, under $\mathbf{B}_0, \mathbf{G}, \mathbf{B}_{\text{def}}, \mathbf{C}_{\text{per}}$). *Under (16.2), the bridge conditions Assumptions 14.6, 14.8 and 15.2, and the confluent estimate Assumption 16.4, the two physical even thresholds share the following conditional expansion. Define*

$$\Delta_{\text{int}}^{(1)} := 2\Delta + d \left[-(1 + 2x) + \frac{\Delta}{\Omega(1 - \nu)} + \mathcal{H}(\tilde{\mu}) \right]. \quad (16.40)$$

Then

$$\boxed{\Delta_R^{\text{RG}} = \Delta_{\text{int}}^{(1)} + O(d^2), \quad \Delta_{B_2}^{\text{RG}} = \Delta_{\text{int}}^{(1)} + O(d^2),} \quad (16.41)$$

and hence

$$\boxed{\Delta_{B_2}^{\text{RG}} - \Delta_R^{\text{RG}} = O(d^2).} \quad (16.42)$$

In compact Case B, the same result holds: the complete cascade modifies both finite-defect ratios only at $O(d^2)$.

Proof. The residue-period equality through order d is the assumed estimate $\mathbf{C}_{\text{per}}$. The broken-pair one-loop change is $d\mathcal{H}(\tilde{\mu})$ under $\mathbf{G}$ and the finite-defect bridges. The assumed matching clause of $\mathbf{B}_{\text{def}}$ gives the same term for the real-root branch. Displacements of the nearest level and separated root enter the leading square root quadratically because $\mathcal{E}'(\tilde{\mu}) = 0$. The compact Case B background estimate is supplied by Chapter 12. This combines the stated inputs; it does not independently prove either determinant comparison. $\square$

The equality is specific to an interior stationary minimum. At a Debye edge, $\mathcal{E}'$ does not vanish; the two blocked levels and the soft real root then sample the boundary differently already at order d . That regime is treated separately in Chapter 17.

16.7 Half-filled reduction

At half filling,

$$\nu = \frac{1}{2}, \quad D = \frac{\Omega}{2}, \quad \tilde{\mu} = D, \quad \Delta = \frac{D}{\sinh(1/v)}. \quad (16.43)$$

Equation (16.7) gives

$$\frac{2c_*^{\text{eff}}}{1 - \sigma} = -\tanh \frac{1}{2v}. \quad (16.44)$$

The response functional reduces to

$$\boxed{\mathcal{H}(D) = \frac{\Delta}{D} \varphi(v) = \frac{\varphi(v)}{\sinh(1/v)},} \quad (16.45)$$

where the function $\varphi(v)$ is the positive integral defined in (15.47). Hence

$$\boxed{\Delta_R^{\text{RG}} = \Delta_{B_2}^{\text{RG}} = 2\Delta + d \left[-\tanh \frac{1}{2v} + \frac{\varphi(v)}{\sinh(1/v)} \right] + O(d^2).} \quad (16.46)$$

Equivalently,

$$\Delta_R^{\text{RG}} = \Delta_{B_2}^{\text{RG}} = 2\Delta - d\sqrt{1 + \frac{\Delta^2}{D^2}} + d\frac{\Delta}{D}[1 + \varphi(v)] + O(d^2). \quad (16.47)$$

This reduction is used as an independent normalization check of both the residue prefactor and the contour orientation of (16.29).

16.8 Logical output and verification record

The outputs used later are:

1. the unified period coefficient (16.9);
2. the finite-defect response equations (16.16);
3. the contour and real-axis definitions (16.29) and (16.35);
4. the common physical interior gap (16.41);
5. the half-filled reduction (16.46).

Verification note 16.8 (Finite-defect response audit). The script

`code/verify_finite_defect.py`

performs the following independent checks:

1. symbolic and numerical checks of c_*^{eff} , the finite upper-duality shift, $\dot{\mu}_*$, and $\dot{\Delta}_*$;
2. exact-period sequences showing that marked and two-blocked thresholds have the same coefficient of d and differ only by $O(d^2)$;
3. direct contour evaluation of (16.29);
4. the real-axis principal-value representation (16.35);
5. direct finite variation of the complete one-loop contour functional;
6. finite interlacing-mode spectra before and after blocking two levels;
7. the half-filled identity (16.45);
8. physical Richardson-branch comparisons with the full prediction (16.41).

The archived output is

`code/finite_defect_verification_report.txt.`

Chapter 17

Edge excitations and the soft $d/\log M$ scale

17.1 Scope and edge notation

Put

$$d = \frac{1}{\rho} = \frac{\Omega}{M}, \quad (17.1)$$

and retain

$$\mathcal{E}(\xi) = \sqrt{(\xi - \tilde{\mu})^2 + \Delta^2}. \quad (17.2)$$

The lower-edge regime is

$$\tilde{\mu} < 0, \quad \mathcal{E}_- := \mathcal{E}(0) = \Omega(\nu + x), \quad v_- := \mathcal{E}'(0) = \frac{-\tilde{\mu}}{\mathcal{E}_-} > 0. \quad (17.3)$$

The upper-edge regime is

$$\tilde{\mu} > \Omega, \quad \mathcal{E}_+ := \mathcal{E}(\Omega) = \Omega(1 - \nu + x), \quad v_+ := -\mathcal{E}'(\Omega) = \frac{\tilde{\mu} - \Omega}{\mathcal{E}_+} > 0. \quad (17.4)$$

At a constrained edge minimum the dispersion has nonzero slope. Therefore the two even sectors sample the discrete boundary differently:

1. the broken-pair sector must block two distinct lattice levels;
2. the real-root sector contains one separated Richardson root inside the first or last interlevel interval.

The difference is already $O(d)$. In addition, the separated root approaches the edge only as $d/\log M$, producing an intermediate scale

$$d \gg \frac{d}{\log M} \gg d^2. \quad (17.5)$$

17.2 Exact lower-edge periods

17.2.1 Two blocked levels

For a broken pair at the lower edge, the two singly occupied levels are $p = 0$ and $q = 1$, and the active paired subsystem contains $n = N - 1$ pairs. The exact determinant identity (10.29) gives

$$\frac{\mathcal{B}_{N-1;0,1}}{(N-1)!Z_0^{(N-1)}} = \frac{\pi_{N-1}(0)\pi_N(0)}{Mx}. \quad (17.6)$$

Its logarithmic degree is exactly

$$x\partial_x \log \frac{\pi_{N-1}(0)\pi_N(0)}{Mx} = 2(N-1). \quad (17.7)$$

Combining this with the exact all-pole energies and adding the one-electron kinetic energy $\xi_0 + \xi_1 = d$ yields the following finite identity.

Exact identity 17.1 (Exact lower-edge broken-pair period). *For every $2 \leq N \leq M$,*

$$\Delta_{B_2,-}^{\text{per}} = \frac{\Omega}{M}(2Mx + 2N - 1) = 2\mathcal{E}_- - d. \quad (17.8)$$

Here $\nu = N/M$ is kept as the exact finite-system filling convention used throughout the manuscript.

17.2.2 Real-root period

The lower real-root threshold is generated by P_0 . From the endpoint expansion (7.41),

$$\Delta_{R,-}^{\text{per}} = 2\mathcal{E}_- - 2d \frac{x(1-\nu)}{\nu+x} + O(d^2). \quad (17.9)$$

Consequently,

$$\Delta_{B_2,-}^{\text{per}} - \Delta_{R,-}^{\text{per}} = d \frac{-\tilde{\mu}}{\mathcal{E}_-} + O(d^2) = dv_- + O(d^2). \quad (17.10)$$

The equality follows from

$$-1 + \frac{2x(1-\nu)}{\nu+x} = -\frac{\alpha_*}{\nu+x} = \frac{-\tilde{\mu}}{\mathcal{E}_-}. \quad (17.11)$$

Thus the period splitting is the edge slope multiplied by one level spacing.

17.3 Exact upper-edge convention

The finite electron-hole transform uses

$$\bar{N} = M + 1 - N, \quad \bar{\nu}_M = 1 - \nu + \frac{1}{M}. \quad (17.12)$$

This M^{-1} shift contributes at order d and must be retained. Reflecting (17.8) gives

$$\Delta_{B_2,+}^{\text{per}} = 2\mathcal{E}_+ + d. \quad (17.13)$$

The upper marked factor is $P_M^{\uparrow,(N)} = P_0^{\downarrow,(\bar{N})}$. Expanding the lower-hole endpoint formula at (17.12) gives

$$\Delta_{R,+}^{\text{per}} = 2\mathcal{E}_+ + d \left[2 - \frac{2x\nu}{1-\nu+x} \right] + O(d^2). \quad (17.14)$$

Therefore

$$\Delta_{B_2,+}^{\text{per}} - \Delta_{R,+}^{\text{per}} = d \frac{\tilde{\mu} - \Omega}{\mathcal{E}_+} + O(d^2) = dv_+ + O(d^2). \quad (17.15)$$

Remark 17.2 (Why the individual upper coefficients differ from the lower ones). The leading thermodynamic square root is exactly reflection invariant. The individual $O(d)$ coefficients are not obtained by the substitution $\nu \mapsto 1-\nu$ at fixed $\nu = N/M$; one must use $\bar{\nu}_M = 1-\nu + M^{-1}$. The extra derivative of the leading term $2\Delta(\bar{\nu}_M)$ accounts for the finite shift. The physically relevant splitting (17.15) remains the reflected image of (17.10).

17.4 The first auxiliary Gaudin zero

Let $y_1^{(F)}$ be the first zero of the auxiliary function (15.11):

$$0 < y_1^{(F)} < d. \quad (17.16)$$

Put

$$y_1^{(F)} = d\lambda_M^{(F)}, \quad 0 < \lambda_M^{(F)} < 1. \quad (17.17)$$

After separating the pole at $\xi_0 = 0$, the zero equation is

$$\frac{1}{\lambda_M^{(F)}} = \sum_{n=1}^M \frac{\mathcal{E}_{0,M}}{\mathcal{E}_{n,M}} \frac{1}{n - \lambda_M^{(F)}}. \quad (17.18)$$

Define the convergent edge constant

$$\mathcal{C}_- = \int_0^\Omega \frac{\mathcal{E}_-/\mathcal{E}(\epsilon) - 1}{\epsilon} d\epsilon, \quad (17.19)$$

and

$$L_-(M) = \log M + \gamma + \mathcal{C}_-. \quad (17.20)$$

Here γ is Euler's constant.

Period asymptotics 17.3 (Auxiliary soft-zero expansion). *On compact lower-edge parameter sets with fixed $v > 0$,*

$$\frac{y_1^{(F)}}{d} = \frac{1}{L_-(M)} - \frac{\pi^2}{6L_-(M)^3} + O(L_-(M)^{-4}). \quad (17.21)$$

In particular,

$$y_1^{(F)} \sim \frac{d}{\log M}. \quad (17.22)$$

The proof is given in [Appendix M.3](#). The $\pi^2/6$ term comes from the second harmonic moment

$$\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}. \quad (17.23)$$

The upper auxiliary zero $y_M^{(F)} \in (\Omega - d, \Omega)$ is described by the edge distance

$$s_M^{(F)} = \Omega - y_M^{(F)}. \quad (17.24)$$

With

$$\mathcal{C}_+ = \int_0^\Omega \frac{\mathcal{E}_+/\mathcal{E}(\epsilon) - 1}{\Omega - \epsilon} d\epsilon, \quad (17.25)$$

$$L_+(M) = \log M + \gamma + \mathcal{C}_+, \quad (17.26)$$

one has

$$\frac{s_M^{(F)}}{d} = \frac{1}{L_+(M)} - \frac{\pi^2}{6L_+(M)^3} + O(L_+^{-4}). \quad (17.27)$$

Exact reflection gives

$$\mathcal{C}_+(\nu, v) = \mathcal{C}_-(1 - \nu, v). \quad (17.28)$$

17.5 The physical soft Richardson root

Let r_1^{RG} denote the real Richardson root of the continuation-certified lower marked branch in the first pair interval. In the scaled half-pair coordinate of (15.1),

$$0 < r_1^{\text{RG}} < d. \quad (17.29)$$

To transfer the auxiliary estimate to this physical root one needs a dressed Richardson equation with the same leading harmonic part and a uniformly bounded remainder. This is an explicit compact-edge clause of $\mathbf{B}_{\text{def}}$. Appendix M derives the soft-root law from that clause; it does not prove the sea-resolvent bound.

Branch identification 17.4 (Physical lower soft-root law, under $\mathbf{B}_{\text{def}}$). *Assume the compact-edge bounded-sea clause of Assumption 14.8. For the branch labeled by $S_{a=0} = \{1, \dots, N\}$ and continued as in Chapter 14,*

$$r_1^{\text{RG}} = \frac{d}{L_-(M)} + O\left(\frac{d}{L_-(M)^2}\right). \quad (17.30)$$

The conservative remainder is uniform on compact lower-edge parameter sets. Numerically, $r_1^{\text{RG}} - y_1^{(F)}$ is substantially smaller than this bound along the tested sequences, but no stronger general remainder is claimed.

The upper physical root is described by its distance from the upper edge,

$$s_1^{\text{RG}} = \Omega - r_M^{\text{RG}}, \quad (17.31)$$

and obeys

$$s_1^{\text{RG}} = \frac{d}{L_+(M)} + O\left(\frac{d}{L_+(M)^2}\right). \quad (17.32)$$

17.6 Physical edge gaps and refined splitting

The proposed physical edge expansion uses the same auxiliary response $\mathcal{H}(\xi)$ as Chapter 16, together with an additional soft-coordinate term for the marked sector. Equality of the remaining physical sea response and control of the displayed remainders are edge applications of $\mathbf{G}$ and $\mathbf{B}_{\text{def}}$; the bounded-root estimate alone does not establish them. Define

$$\mathcal{H}_- := \mathcal{H}(0), \quad \mathcal{H}_+ := \mathcal{H}(\Omega). \quad (17.33)$$

Physical consequence 17.5 (Lower-edge gaps, under $\mathbf{B}_0, \mathbf{G}, \mathbf{B}_{\text{def}}$). *Assume the ground, Hessian and edge finite-defect clauses of Assumptions 14.6, 14.8 and 15.2. For $\tilde{\mu} < 0$,*

$$\Delta_R^{\text{RG}} = 2\mathcal{E}_- + d \left[-\frac{2x(1-\nu)}{\nu+x} + \mathcal{H}_- \right] + 2v_- r_1^{\text{RG}} + O\left(\frac{d}{\log^2 M} + d^2\right), \quad (17.34)$$

$$\Delta_{B_2}^{\text{RG}} = 2\mathcal{E}_- + d[-1 + \mathcal{H}_-] + O(d^2). \quad (17.35)$$

Consequently,

$$\Delta_{B_2}^{\text{RG}} - \Delta_R^{\text{RG}} = v_- [d - 2r_1^{\text{RG}}] + O\left(\frac{d}{\log^2 M} + d^2\right). \quad (17.36)$$

Using (17.30),

$$\Delta_{B_2}^{\text{RG}} - \Delta_R^{\text{RG}} = dv_- \left[1 - \frac{2}{L_-(M)} \right] + O\left(\frac{d}{L_-(M)^2} + d^2\right). \quad (17.37)$$

The limiting coefficient v_- is therefore approached only logarithmically. Moderate systems can lie far from the asymptotic coefficient even when all ordinary $1/M$ corrections are already small.

Physical consequence 17.6 (Upper-edge gaps, under B_0, G, B_{def}). *Under the same hypotheses and exact reflection, for $\tilde{\mu} > \Omega$,*

$$\Delta_R^{\text{RG}} = 2\mathcal{E}_+ + d \left[2 - \frac{2x\nu}{1-\nu+x} + \mathcal{H}_+ \right] + 2v_+ s_1^{\text{RG}} + O\left(\frac{d}{\log^2 M} + d^2\right), \quad (17.38)$$

$$\Delta_{B_2}^{\text{RG}} = 2\mathcal{E}_+ + d[1 + \mathcal{H}_+] + O(d^2). \quad (17.39)$$

Hence

$$\Delta_{B_2}^{\text{RG}} - \Delta_R^{\text{RG}} = dv_+ \left[1 - \frac{2}{L_+(M)} \right] + O\left(\frac{d}{L_+(M)^2} + d^2\right). \quad (17.40)$$

17.7 Matching at the Debye-edge crossings

At the lower crossing,

$$\tilde{\mu} = 0 \iff \nu = \frac{\sigma}{1+\sigma}, \quad (17.41)$$

one has $v_- = 0$. Therefore the order- d splitting and the $d/\log M$ correction vanish simultaneously. The energy shift due to the soft coordinate begins quadratically,

$$\mathcal{E}(r_1^{\text{RG}}) - \mathcal{E}(0) = O\left(\frac{d^2}{\Delta \log^2 M}\right). \quad (17.42)$$

The displayed coefficients match continuously with the interior expression of [Physical consequence 16.7](#). This coefficient matching does not supply a uniform remainder in an M -dependent neighborhood of the crossing. The same qualification holds at $\tilde{\mu} = \Omega$ by reflection.

The edge crossover therefore leaves a finite-size signature, but not a new thermodynamic anomaly:

the leading even gap remains $2\Delta_{\text{qp}}$,
the sector splitting is $O(d)$,
and its approach contains the scale $d/\log M$.

(17.43)

17.8 Verification output

Verification note 17.7 (Soft-root and edge-splitting audit). The script

`code/verify_edge_soft_scale.py`

checks the exact lower-edge determinant and closed broken-pair period, the reflected upper formula, the auxiliary zero through the $\pi^2/(6L^3)$ term, upper/lower auxiliary reflection, continuation-certified physical soft roots, physical real-root and broken-pair gaps, separation of the common Gaudin response, and the vanishing of the order- d splitting at both Debye-edge crossings. The archived output is

`code/edge_soft_scale_verification_report.txt.`

For the representative lower-edge sequence $\nu = 0.1$, $v = 1$, the refined formula using $1 - 2\lambda_M^{(F)}$ reduces the splitting residual monotonically, while $\lambda_M^{\text{RG}} - \lambda_M^{(F)}$ decreases over the continuation-certified sizes.

Part VII

Physical synthesis

Chapter 18

Physical synthesis: energy, gaps, and pairing regimes

Remark 18.1 (Scope of this synthesis in Revision 01). This chapter retains the physical formulas as conditional consequences. It does not close the remaining proof obligations: full contour selection, uniform Gaussian control, the Hessian-to-spectral-shift relation, and the finite-defect/prefactor comparisons. In particular, the arguments below may use Assumptions 14.6, 15.2, and 14.8, together with Assumption 16.4. Statements about physical root counts also require the separate correspondence conjecture of Chapter 13.

18.1 Assumptions and hierarchy of statements

The physical conclusions below concern the equally spaced reduced pairing model with

$$\varepsilon_n = \varepsilon_{F0} + \frac{n}{\rho}, \quad n = 0, \dots, M, \quad \frac{M}{\rho} = \Omega, \quad (18.1)$$

and the thermodynamic sequence

$$M, N, \rho \longrightarrow \infty, \quad \nu = \frac{N}{M} \in (0, 1), \quad v = \rho V > 0 \quad (18.2)$$

with fixed ν, v, Ω . We use

$$\sigma = e^{-2/v}, \quad x = \frac{\sigma}{1 - \sigma}, \quad (18.3)$$

$$\boxed{\tilde{\mu} = \Omega[\nu(1 + 2x) - x], \quad \Delta = 2\Omega\sqrt{x(1+x)\nu(1-\nu)},} \quad (18.4)$$

and

$$\boxed{\mathcal{E}(\xi) = \sqrt{(\xi - \tilde{\mu})^2 + \Delta^2}, \quad 0 \leq \xi \leq \Omega.} \quad (18.5)$$

Three levels of accuracy must be kept distinct.

1. The leading ground energy is extensive, $E_0 = O(M)$.
2. The complete physical ground energy contains a nonzero Gaussian term of order M^0 .
3. The excitation thresholds are $O(M^0)$, while their first finite-size corrections are generally of order

$$d = \frac{1}{\rho} = \frac{\Omega}{M}. \quad (18.6)$$

At a Debye edge an additional non-power scale $d/\log M$ appears.

The exact finite residue identities and period asymptotics do not require a thimble hypothesis. Their promotion to physical energies, including leading excitation energies, is a separate issue. The physical statements below use the bridge conditions [Assumptions 14.6, 14.8, 15.2 and 16.4](#). The ground eigenbranch itself exists globally; the excited statements apply to the analytic continuations of the stated weak labels on the connected parameter domains under consideration.

The claims are uniform on compact subsets with fixed $v > 0$. The joint mesoscopic weak-coupling limit $M\sigma = O(1)$ is stated separately in [Section 18.7](#) and is not absorbed into the fixed- v theorem.

18.2 Main physical theorem

Physical consequence 18.2 (Ground energy and spectral thresholds, under the bridge conditions; R15). *Under (18.1)–(18.2) and bridge conditions [Assumptions 14.6, 14.8, 15.2 and 16.4](#), the physical Richardson ground state and the threshold-relevant analytic continuations satisfy the following statements in both Cases A and B.*

(i) *The full ground energy through order M^0 is*

$$\boxed{E_0^{\text{RG}} = 2N\varepsilon_{F0} + \frac{N(N-1)}{\rho} - \frac{2N(M-N+1)}{\rho}x + \delta E_{\text{H}}(\nu, v) + o(1),} \quad (18.7)$$

where δE_{H} is the one-dimensional Gaudin spectral-shift functional of [Equation \(15.32\)](#). Equivalently,

$$\boxed{\frac{E_0^{\text{RG}}}{M} = 2\nu\varepsilon_{F0} + \Omega\nu^2 - 2\Omega x\nu(1-\nu) + o(1).} \quad (18.8)$$

(ii) *After subtraction of the chemical work, odd particle and odd hole branches have the common leading dispersion*

$$\boxed{E_{2N+1,p}^{\text{RG}} - E_{2N}^{\text{RG}} - \mu = E_{2N-1,p}^{\text{RG}} - E_{2N}^{\text{RG}} + \mu = \mathcal{E}(\xi_p) + o(1),} \quad (18.9)$$

where $\mu = \varepsilon_{F0} + \tilde{\mu}$.

(iii) *The odd or parity threshold is*

$$\boxed{\Delta_{\text{qp}} = \min_{0 \leq \xi \leq \Omega} \mathcal{E}(\xi),} \quad (18.10)$$

and the leading neutral even threshold is

$$\boxed{\Delta_{\text{even}} = 2\Delta_{\text{qp}}.} \quad (18.11)$$

It can be reached either by a seniority-zero real-root branch or, in the thermodynamic limit, by a seniority-two broken-pair branch.

(iv) *The leading density (18.8), the dispersion (18.5), and the thresholds (18.10)–(18.11) are the same in Cases A and B. The rate-defined A–B change alters the contour description without changing these period-derived formulas. Its identification with physical root topology requires the stated correspondence conjecture.*

Proof. Equation (18.7) is the exact all-pole barycenter of [Exact identity 4.6](#) plus the conditional Gaussian correction of [Physical consequence 15.5](#). Complete Case B support resummation changes the relevant ground period only by $O(M^{-2})$, by [Period asymptotics 12.5](#). The odd dispersion is the Heine blocked-period result of [Period asymptotics 9.4](#), assigned a branch label by [Branch identification 14.10](#); a subleading period-to-branch difference is an additional bridge input, not a consequence of that label. The common even threshold is the exact particle–hole convolution and finite-seniority result of [Period asymptotics 10.5](#), with the complete Case B marked background controlled by [Period asymptotics 12.6](#). The same branch is followed through the A–B root-topology change by [Branch identification 14.9](#) and [Chapter 13](#). $\square$

18.3 Ground-state energy: extensive and subextensive pieces

The exact period contribution in (18.7) may be separated as

$$E_{\text{all}}^{\text{per}} = M e_{\infty}(\nu, v) + e_{\text{per}}^{(0)}(\nu, v), \quad (18.12)$$

where

$$e_{\infty}(\nu, v) = 2\nu\varepsilon_{F0} + \Omega\nu^2 - 2\Omega x\nu(1 - \nu), \quad (18.13)$$

$$e_{\text{per}}^{(0)}(\nu, v) = -\Omega\nu(1 + 2x). \quad (18.14)$$

The physical constant term is therefore

$$E_0^{(0)}(\nu, v) = -\Omega\nu(1 + 2x) + \delta E_{\text{fl}}(\nu, v). \quad (18.15)$$

For reference, the Gaussian part is

$$\begin{aligned} \delta E_{\text{fl}}(\nu, v) = v\Omega \left(\frac{1}{2} - \nu \right) - \mathcal{E}(0) \\ - \int_0^{\Omega} \frac{\epsilon - \tilde{\mu}}{\mathcal{E}(\epsilon)} a(\epsilon) \, d\epsilon, \end{aligned} \quad (18.16)$$

where $a(\epsilon)$ is the auxiliary Gaudin phase shift of [Equation \(15.20\)](#). The formula is smooth for every fixed $v > 0$ and $0 < \nu < 1$, including both Debye-edge crossings and the A–B line.

The leading interaction term is electron–hole symmetric. More precisely,

$$e_{\infty}(\nu, v) - e_{\infty}(1 - \nu, v) = (2\nu - 1)(2\varepsilon_{F0} + \Omega), \quad (18.17)$$

while the fluctuation term obeys

$$\delta E_{\text{fl}}(\nu, v) - \delta E_{\text{fl}}(1 - \nu, v) = v\Omega(1 - 2\nu). \quad (18.18)$$

Their combination reproduces the exact physical commutator shift of [Chapter 8](#).

18.4 The three quasiparticle regimes

The evolution from a dilute bound-pair regime to an interior BCS-like minimum is the finite-window analogue of the familiar weak-to-strong coupling crossover in attractive Fermi systems [37, 38]. In the present exactly solvable model the three branches arise directly by constrained minimization of (18.5).

The unconstrained minimum of $\mathcal{E}(\xi)$ is at

$$\xi_* = \tilde{\mu}. \quad (18.19)$$

Confinement to the Debye window creates three regimes. Define

$$\nu_0 = \frac{\sigma}{1+\sigma} = \frac{x}{1+2x}, \quad 1 - \nu_0 = \frac{1}{1+\sigma}. \quad (18.20)$$

Then

$$\Delta_{\text{qp}} = \begin{cases} \Omega(\nu + x), & 0 < \nu \leq \nu_0, \\ \Delta, & \nu_0 \leq \nu \leq 1 - \nu_0, \\ \Omega(1 - \nu + x), & 1 - \nu_0 \leq \nu < 1. \end{cases} \quad (18.21)$$

The even threshold is twice every entry in (18.21).

Regime	Position of $\tilde{\mu}$	Minimizer	Leading physical interpretation
Dilute electron pairs	$\tilde{\mu} < 0$	$\xi = 0$	Breaking a dilute bound pair; the edge binding scale controls the threshold.
Interior paired regime	$0 < \tilde{\mu} < \Omega$	$\xi = \tilde{\mu}$	Collective BCS-like minimum with $\Delta_{\text{qp}} = \Delta$.
Dilute hole pairs	$\tilde{\mu} > \Omega$	$\xi = \Omega$	Electron-hole image of the dilute-pair regime.

The thermodynamic isolated-pair binding energy in this window is

$$\varepsilon_c = 2\Omega x = \frac{2\Omega\sigma}{1-\sigma}. \quad (18.22)$$

Thus

$$\nu \downarrow 0 : \quad \Delta_{\text{qp}} \longrightarrow \frac{\varepsilon_c}{2}, \quad \Delta_{\text{even}} \longrightarrow \varepsilon_c, \quad (18.23)$$

and the same statement holds for dilute holes as $\nu \uparrow 1$.

The joining of the first and second branches at $\nu = \nu_0$ is continuous through the first derivative:

$$\left. \frac{\partial \Delta_{\text{qp}}}{\partial \nu} \right|_{\nu_0-} = \left. \frac{\partial \Delta_{\text{qp}}}{\partial \nu} \right|_{\nu_0+} = \Omega. \quad (18.24)$$

The second derivative changes from zero on the edge branch to

$$\left. \frac{\partial^2 \Delta_{\text{qp}}}{\partial \nu^2} \right|_{\nu_0+} = -\frac{\Omega(1+\sigma)^3}{2\sigma(1-\sigma)}. \quad (18.25)$$

The upper crossing is the reflected counterpart. Hence the Debye-edge crossing is a C^1 , but not C^2 , constrained-minimum crossover. It is not an additional root-topology transition.

18.5 Case A and Case B: topology versus observables

The lower edge rate controlling the cascade is

$$\Phi_0(\nu, \sigma) = \nu \log \sigma - \log(1 - \sigma) - \nu \log \nu - (1 - \nu) \log(1 - \nu). \quad (18.26)$$

For $0 < \nu \leq 1/2$:

$$\Phi_0 \geq 0 \iff \text{Case A}, \quad \Phi_0 < 0 \iff \text{lower Case B}. \quad (18.27)$$

In Case B the density κ of the consecutive cascade is the unique solution of

$$\Phi(\kappa) = 0, \quad 0 < \kappa < \alpha_* := \frac{\tilde{\mu}}{\Omega}. \quad (18.28)$$

The upper half is obtained by exact electron–hole reflection.

At the chemical-potential crossing,

$$\Phi_0(\nu_0, \sigma) = \log \frac{1 + \sigma}{1 - \sigma} > 0. \quad (18.29)$$

Thus $\tilde{\mu} = 0$ lies strictly inside Case A. The root-topology line is instead

$$\sigma^\nu = (1 - \sigma)\nu^\nu(1 - \nu)^{1-\nu}. \quad (18.30)$$

At half filling this line ends at

$$\sigma_c = 3 - 2\sqrt{2}, \quad v_c = \frac{1}{\log(1 + \sqrt{2})}. \quad (18.31)$$

If $v \geq v_c$, every filling is in Case A. If $v < v_c$, let $\nu_B \in (\nu_0, 1/2)$ solve (18.30); then

$$\nu_B < \nu < 1 - \nu_B \quad (18.32)$$

is the Case B interval.

For $v < v_c$, the full lower-to-upper sequence is therefore:

Filling interval	Quasiparticle minimum	Root topology	Description
$0 < \nu < \nu_0$	lower edge	Case A	dilute electron pairs
$\nu_0 < \nu < \nu_B$	interior	Case A	interior paired state without a macroscopic real-root cascade
$\nu_B < \nu < 1/2$	interior	lower Case B	lower cascade of trapped roots
$1/2 < \nu < 1 - \nu_B$	interior	upper Case B	reflected upper cascade
$1 - \nu_B < \nu < 1 - \nu_0$	interior	Case A	interior paired state after the cascade disappears
$1 - \nu_0 < \nu < 1$	upper edge	Case A	dilute hole pairs

This table makes the logical separation explicit. The three pairing regimes are determined by the position of $\tilde{\mu}$. Cases A and B are determined by Φ_0 and describe the topology of the exact roots. The two classifications overlap but are not the same.

Physical consequence 18.3 (Absence of a leading A–B anomaly). *Across the line (18.30),*

$$\boxed{e_{\infty}^A = e_{\infty}^B, \quad \delta E_{\text{fl}}^A = \delta E_{\text{fl}}^B, \quad \Delta_{\text{qp}}^A = \Delta_{\text{qp}}^B, \quad \Delta_{\text{even}}^A = \Delta_{\text{even}}^B.} \quad (18.33)$$

At the level established here this is a crossover of period rates and contour organization. Its identification with a physical root-topology change is conjectural; the period-derived energy formulas have no singularity on that line.

The cascade itself is nonanalytic. On the Case B side, with

$$\delta_{\Phi} = -\Phi_0 > 0, \quad (18.34)$$

one has

$$\boxed{\kappa = -\frac{\delta_{\Phi}}{W_{-1}[-\delta_{\Phi}/(e\Lambda)]} \sim \frac{\delta_{\Phi}}{\log(1/\delta_{\Phi})}.} \quad (18.35)$$

Extended by zero into Case A, κ is C^1 but not C^2 . This nonanalyticity is real, but it belongs to a root-topology diagnostic rather than to the leading energy or gap.

18.6 Finite-size spectrum: what survives beyond the thermodynamic limit

The leading equality $\Delta_{\text{even}} = 2\Delta_{\text{qp}}$ does not mean that the two even sectors remain identical at every finite size.

18.6.1 Interior minimum

For $0 < \tilde{\mu} < \Omega$, under $B_0, G, B_{\text{def}}, C_{\text{per}}$, the real-root and broken-pair physical gaps coincide through order d . The last condition is the pending period estimate in Assumption 16.4:

$$\boxed{\begin{aligned} \Delta_R^{\text{RG}} &= 2\Delta + d \left[-(1+2x) + \frac{\Delta}{\Omega(1-\nu)} + \mathcal{H}(\tilde{\mu}) \right] + O(d^2), \\ \Delta_{B_2}^{\text{RG}} &= 2\Delta + d \left[-(1+2x) + \frac{\Delta}{\Omega(1-\nu)} + \mathcal{H}(\tilde{\mu}) \right] + O(d^2). \end{aligned}} \quad (18.36)$$

The exact finite convention $\nu = N/M$ is used in this formula. The apparently asymmetric coefficient is electron–hole covariant only after the exact shift $\bar{N}/M = 1 - \nu + M^{-1}$ is retained.

At an interior stationary minimum,

$$\boxed{\Delta_{B_2}^{\text{RG}} - \Delta_R^{\text{RG}} = O(d^2).} \quad (18.37)$$

The two sectors remain distinct eigenstates, but their first physical finite-size corrections agree.

18.6.2 Debye-edge minimum

At the lower edge, define

$$\mathcal{E}_{-} = \mathcal{E}(0) = \Omega(\nu + x), \quad v_{-} = -\frac{\tilde{\mu}}{\mathcal{E}_{-}} > 0, \quad (18.38)$$

and

$$L_{-}(M) = \log M + \gamma + \mathcal{C}_{-}. \quad (18.39)$$

Then

$$\Delta_{B_2}^{\text{RG}} - \Delta_R^{\text{RG}} = dv_- \left[1 - \frac{2}{L_-(M)} \right] + O \left(\frac{d}{L_-(M)^2} + d^2 \right). \quad (18.40)$$

The upper formula is

$$\Delta_{B_2}^{\text{RG}} - \Delta_R^{\text{RG}} = dv_+ \left[1 - \frac{2}{L_+(M)} \right] + O \left(\frac{d}{L_+(M)^2} + d^2 \right), \quad (18.41)$$

where

$$v_+ = \frac{\tilde{\mu} - \Omega}{\mathcal{E}(\Omega)} > 0. \quad (18.42)$$

Thus the finite-size hierarchy at an edge is

$$d \gg \frac{d}{\log M} \gg d^2. \quad (18.43)$$

The limiting $O(d)$ splitting is reached only logarithmically. At $\tilde{\mu} = 0$ or $\tilde{\mu} = \Omega$, $v_{\mp} = 0$, and both the $O(d)$ splitting and the $d/\log M$ correction vanish.

18.6.3 Two finite-size crossover widths

The two thermodynamic crossovers have different finite-size widths.

Crossover	Control condition	Parameter width	Principal finite-size signature
Debye-edge minimum	$\tilde{\mu} = 0$ or $\tilde{\mu} = \Omega$	$O(M^{-1})$ in filling	Discrete staircase of the minimizing level; energy steps $O(M^{-2})$.
Case A–B topology	$\Phi_0 = 0$	$O((\log M)/M)$	First trapped roots, then a mesoscopic cascade $1 \ll k \ll M$.

The centers of these windows remain separated by the finite interval $\nu_B - \nu_0$ at fixed coupling.

18.7 Four independent criteria

Several statements often described informally as a “breakdown of BCS” refer to different comparisons. In the present solution they can be separated cleanly.

Criterion	Dimensionless comparison	What changes	What it does not imply
Collective discreteness	d/Δ	Whether many levels resolve the collective gap and whether ordinary power-series finite-size corrections are small.	It does not determine the position of $\tilde{\mu}$ or the topology of the Richardson roots.

Criterion	Dimensionless comparison	What changes	What it does not imply
Isolated-pair resolution	$\varepsilon_c/d = 2Mx$	Whether the binding of an isolated pair is resolved against the level spacing. The nonuniform regime is $M\sigma = O(1)$.	It is not the same as the chemical-potential crossing, although both involve the dilute-pair scale.
Constrained minimum	$\tilde{\mu}/\Omega$ relative to $[0, 1]$	Whether Δ_{qp} is controlled by the lower edge, the interior spectral parameter Δ , or the upper edge.	It does not by itself create or remove the Case B cascade.
Root/contour topology	sign of Φ_0 , or κ	Whether the physical ground branch is represented by an all-detached root pattern or by a macroscopic trapped-root cascade.	It creates no additional leading anomaly in E_0 , Δ_{qp} , or Δ_{even} .

The first two criteria are finite-size comparisons. The third and fourth remain meaningful in the strict thermodynamic limit, but govern different structures. In particular,

$$\boxed{\tilde{\mu} = 0 \quad \text{and} \quad \Phi_0 = 0} \quad \text{are distinct equations.} \quad (18.44)$$

The fixed- ν results of this manuscript assume

$$M\sigma \longrightarrow \infty. \quad (18.45)$$

If instead $M\sigma = O(1)$, then

$$\frac{\varepsilon_c}{d} = 2Mx = O(1), \quad (18.46)$$

and the isolated-pair binding competes directly with the level spacing. The collective gap remains parametrically larger at fixed filling. This joint limit is the natural place to seek deviations that are absent at fixed ν ; it is outside the uniform claims of [Physical consequence 18.2](#).

18.7.1 A joint limit with distinct collective and pair scales

If $\sigma = c/M$, with $c > 0$ and $0 < \nu < 1$ fixed, the same parameter formulas imply

$$\frac{\varepsilon_c}{d} \longrightarrow 2c, \quad \frac{\Delta}{d} \sim 2\sqrt{c\nu(1-\nu)M} \longrightarrow \infty. \quad (18.47)$$

Thus $\varepsilon_c \sim d \ll \Delta$. The comparison of all three scales requires an additional dilute limit, such as $\nu = O(M^{-1})$, outside the present fixed-filling expansion.

18.7.2 Meaning of the active-window edge

The model contains only the specified active orbitals. A physical band bottom, where no further orbitals exist, differs from a Debye interaction cutoff, outside which fermionic orbitals

may remain. If such excluded orbitals are present and the chemical potential enters their region, the minimum over the active window need not be the gap of the full system. A frozen lower Fermi sea is an explicit constraint, not an automatic equilibrium conclusion of the active Hamiltonian.

18.8 What is and is not anomalous

The word “anomaly” must be qualified by the observable and the order in M .

Physical consequence 18.4 (Complete crossover summary). *For fixed $v > 0$ and $0 < \nu < 1$:*

1. *The leading ground-energy density is analytic across both Debye-edge crossings and the A–B line.*
2. *The complete $O(M^0)$ ground-energy term is smooth across the same loci.*
3. *The quasiparticle threshold is C^1 , but not C^2 , when its constrained minimum enters or leaves the Debye window.*
4. *The A–B line produces a nonanalytic root-topology density $\kappa \sim \delta_\Phi / \log(1/\delta_\Phi)$, but no additional leading anomaly in the energy or gaps.*
5. *At finite size, the two even sectors split at order d only when the minimum is pinned to an edge, and the approach to that coefficient contains the softer scale $d/\log M$.*

The answer to the motivating question is therefore precise. Passage of the chemical potential through the lower Debye edge does not reveal a missing thermodynamic correction to the BCS square-root spectrum within the fixed- v thermodynamic limit. What it does reveal is a change of constrained minimizer and, at finite size, a change in how the real-root and broken-pair sectors approach their common thermodynamic threshold. The separate A–B transition changes the internal topology of the exact Richardson solution but is likewise invisible in the leading energy and gap.

18.9 Verification output

Verification note 18.5 (Physical-synthesis consistency audit). The script

`code/verify_physical_synthesis.py`

checks symbolically and numerically: the equivalence of the x - and σ -parametrizations; the edge values $\mathcal{E}(0)$ and $\mathcal{E}(\Omega)$; continuity and first-derivative matching of (18.21); the second-derivative jump; electron–hole covariance of the ground density and gaps; the isolated-pair limits; positivity of (18.29); the critical coupling and the ordering $\nu_0 < \nu_B < 1/2$; the disappearance of the edge sector splitting at both crossings; and the hierarchy of the finite-size crossover scales. The archived output is

`code/physical_synthesis_verification_report.txt.`

18.10 Comparison with the direct self-dual route

The conditional synthesis in this chapter concerns the period construction at general filling. The fourth edition supplies a separate physical route at half filling: Chapters 23–25 start from finite self-dual Richardson polynomials and control leading energy and gap limits without a

continuum root-support ansatz. Their finite-size comparison distinguishes the condensation energy from the spectral gap. This is independent support for familiar physical scales in the common setting, but not a proof of the general-filling period prefactor, Hessian, or contour-selection estimates used here.

Chapter 19

Conclusions and the next proof obligations

19.1 What the finite construction establishes

The exact finite- M part of the method consists of the all-pole Selberg–Nørlund–Rice factorization, the complementary signed kernel, its positive sign-corrected minors, and the determinant generators for marked, blocked, and nested contours. These objects are well-defined periods before they are assigned any physical interpretation.

Revision 01 repairs one concrete asymptotic defect. Continuing the full source-index sum by gamma functions adds a spurious off-lattice tail. The replacement in Appendix J first localizes the original positive lattice sum near its saddles. A smooth interpolation is used only for that localized sum; its error relative to the exact lattice kernel is exponentially bounded. The divided-difference argument then gives the confluent fixed- m minor law, including the factor M^{-m^2} . This repair does not introduce a new physical assumption.

The complete Case B support bound concerns the exact chosen cascade period. It gives $\mathcal{R} = 1 + O(M^{-2})$ and the analogous separated marked estimate in compact Case B domains. A physical inference from these estimates still requires control of the contour-to-saddle relation.

The Case A extension added in Chapter 20 raises the Vandermonde power to four while preserving the stationary Richardson equations. An exact moment Pfaffian and a positive compact integral give additional control of the full period, including its nonvanishing. Fixed-spectrum coupling expansions quantify the remaining error of a two-temperature energy extrapolation. These results enlarge the exact period layer; they do not establish the physical saddle bridge or a Pfaffian treatment of the Case B cascade.

19.2 Physical formulas retained under explicit conditions

The candidate ground-state expansion is

$$E_0^{\text{RG}} = 2N\varepsilon_{F0} + \frac{N(N-1)}{\rho} - \frac{2N(M-N+1)}{\rho} \frac{\sigma}{1-\sigma} + \delta E_{\text{fl}}(\nu, v) + o(1). \quad (19.1)$$

The exact period supplies the first three terms. The spectral-shift functional for δE_{fl} agrees with the standard Richardson expansion [23]; its independent Gaudin-Hessian derivation within this period method remains Assumption 15.2. Selection of the full ground contour and the local energy remainder are the separate Assumption 14.6.

The spectral scale and candidate thresholds are

$$\mathcal{E}(\xi) = \sqrt{(\xi - \tilde{\mu})^2 + \Delta^2}, \quad (19.2)$$

$$\Delta_{\text{qp}} = \min_{[0, \Omega]} \mathcal{E}, \quad \Delta_{\text{even}} = 2\Delta_{\text{qp}}. \quad (19.3)$$

The period calculations produce these expressions. Weak-coupling branch labels alone do not prove that their logarithmic derivatives are the corresponding physical excitation energies. The conditional statements in Chapters 14–18 specify the required additional inputs.

The chemical-potential edge crossing changes the constrained minimizer. Its gap is C^1 , but not C^2 . The A–B endpoint is defined by a different equation, $\Phi(\kappa) = 0$, and obeys

$$\kappa \sim \frac{-\Phi_0}{\log[1/(-\Phi_0)]}. \quad (19.4)$$

This is an analytic law for the period endpoint. Its identification with a physical real-root density is still a correspondence conjecture, not a consequence of the Lambert inversion or of a single finite-size root plot.

19.3 Finite-size results and their conditions

The proposed common interior correction to the real-root and broken-pair gaps requires both a confluent blocked-period prefactor estimate and a comparison of the physical finite-defect determinants. These are explicitly distinguished in Chapter 16 and Appendix L. Their equality has not been derived merely by writing a particle–hole convolution for periods.

For an edge, the auxiliary interlacing Gaudin zero admits a harmonic-sum expansion. The corresponding physical root law,

$$r_1^{\text{RG}} = \frac{d}{\log M + \gamma + \mathcal{C}_-} + O\left(\frac{d}{\log^2 M}\right), \quad (19.5)$$

additionally requires the bounded sea response stated in the finite-defect bridge. These two derivations have different proof status. At fixed nonzero macroscopic energy scales, the proposed hierarchy is

$$d \gg d/\log M \gg d^2/\Omega. \quad (19.6)$$

The middle scale describes a coordinate displacement and a correction to a finite excitation threshold; it is not evidence of a new gapless mode.

19.4 The methodological task that remains

The intended chain is

$$\text{exact period} \longrightarrow \text{period asymptotics} \xrightarrow{\text{selection and response estimates}} \text{eigenenergy}. \quad (19.7)$$

The first arrow is established for the sectors treated here, with the repaired lattice argument for the confluent Case B minors. The second arrow remains the central task. Perron–Frobenius and analytic continuation establish facts about eigenbranches, not the absence of cancellation between contour contributions.

A productive next step is to construct a ground contour representation whose full ratio to the relevant saddle cycle can be controlled, or to derive an exact identity that supplies the same energy estimate. Once that is done, the growing-dimensional Gaussian remainder and the proposed Hessian spectral-shift identity can be tested in a definite setting. Replacing these tasks by stronger assumptions would improve logical wording but would not finish the method.

19.5 Scope and later extensions

The principal sequence fixes $v > 0$, $\nu \in (0, 1)$, and Ω . For the separate weak-coupling sequence $\sigma = c/M$,

$$\frac{\varepsilon_c}{d} \rightarrow 2c, \quad \frac{\Delta}{d} \sim 2\sqrt{c\nu(1-\nu)M} \rightarrow \infty. \quad (19.8)$$

Thus isolated-pair binding and level spacing can compete while the collective gap remains well resolved. A simultaneous dilute limit is a different problem. A physical band bottom must also be distinguished from a cutoff in the interaction alone.

Nonuniform level densities, multiband or nonconstant couplings, finite-temperature sums over many sectors, and higher finite-size coefficients remain interesting extensions. None resolves the contour-selection problem of the present model automatically. Hyperbolic and multicomponent Richardson–Gaudin models may have genuine transitions, so any relation between contour topology and thermodynamic singularities must be checked anew.

19.6 Status of the unified text

The full manuscript is retained as a working reference so that exact definitions, valid algebra, repaired estimates, comparison formulas and pending bridges can be inspected together. Revision 01 closes the source-index interpolation defect in the lattice-minor proof and corrects the presentation of the remaining logical dependencies. It does not yet claim a completed independent analytic derivation of the full physical spectrum.

19.7 Results added in version 1.2

The full-period route now has an additional physical endpoint: Chapter 21 proves its approach to the Hamiltonian ground energy at fixed finite L, N when $p \rightarrow \infty$. This is obtained from a positive vector evolution and an exact scalar projection, rather than a scalar thimble estimate. It narrows the earlier open question in that order of limits; it does not solve the fixed- p , growing-dimensional problem formulated in this chapter. Chapter 22 also closes the all-order *formal* extensive cancellation problem for the one-pair Gaussian expansion. Analytic summation and uniformity, and the stated blocked Case B transfer, remain distinct questions. No new ground-energy formula or new physical gap is claimed merely because these familiar expressions are recovered.

19.8 Duality as a second route

The fourth-edition extension, developed in Chapters 23–25, changes the range of methods available to the project. Finite duality equations preserve direct access to Richardson roots, while a polynomial formulation passes through their collisions on the physical ground branch. The real boundary problem and its energy functional provide a controlled leading BCS limit in the half-filled self-dual setting. A finite defect and a regularized excited response then address selected gaps and finite-size crossover behaviour. These conclusions use finite spectral arguments and comparison estimates, without postulating the macroscopic root arc.

The future tasks are consequently twofold. The period route still requires the estimates listed above in its broader sectors. The duality route can be developed through rigorous excited crossover limits, comparison with the full pair-preserving minimum, efficient certified root reconstruction, and extensions to general symmetric spectra or mutually dual sectors. The present evidence for a short response iteration makes these concrete computational questions, but does not by itself settle their uniformity.

Part VIII

Higher inverse temperature: Case A

Chapter 20

Higher auxiliary inverse temperature: the Case A Pfaffian

20.1 Scope and normalization

This chapter adds a fourth-power Vandermonde calculation to the *common all-pole contour* used for Case A. Its exact statements concern that mathematically specified period. They do not require a root-topology assumption, and hence remain identities when the physical roots change their arrangement. This fact does not identify the same period with the physical ground-state saddle outside Case A, or establish that identification within Case A. The separate contour-to-spectrum obligations of Chapter 14 remain in force. No result about a mixed-contour or nested-contour extension at the increased auxiliary inverse temperature is included here.

Write $L = M + 1$ for the *number* of active levels, and set

$$a_n = 2\varepsilon_n, \quad q(z) = \prod_{n=0}^{L-1} (z - a_n), \quad D(\mathbf{z}) = \prod_{i < j} (z_j - z_i), \quad u = V^{-1}. \quad (20.1)$$

The algebra below holds for arbitrary distinct real a_n ; the spectrum of the preceding chapters is recovered by $a_n = 2\varepsilon_{F0} + 2n/\rho$. We use a new letter p for the multiplier of the auxiliary inverse temperature, to avoid the cascade-index notation elsewhere in the manuscript. For a positive integer p , define

$$\mathcal{Z}_p(u) = \frac{1}{N!} \oint_{\gamma_{\text{all}}^{\times N}} D(\mathbf{z})^{2p} \prod_{j=1}^N \frac{e^{-puz_j} dz_j}{q(z_j)^p 2\pi i}, \quad E_p(u) = -\frac{1}{p} \frac{\partial_u \mathcal{Z}_p}{\mathcal{Z}_p}. \quad (20.2)$$

Every contour is counterclockwise and encloses every pole once. The factor $1/N!$ is included here. Thus, with the orientation and denominator of Chapter 4,

$$\mathcal{Z}_1 = \frac{(-1)^{LN}}{N!} Z_{\text{all}}, \quad E_1 = E_{\text{all}}^{\text{per}}. \quad (20.3)$$

The constant sign and factorial do not affect logarithmic derivatives. We calculate $p = 2$, for which the log-gas convention is $\beta = 2p = 4$, and use $p = 1$ as the explicitly defined comparison. General higher-power Vandermonde algebra exists [39]; a comparable evaluation at other values of p is not asserted by this chapter.

The logarithm of the *entire* original weight is multiplied by p . Consequently its stationary equations remain

$$u = \sum_{n=0}^{L-1} \frac{1}{a_n - z_i} + 2 \sum_{j \neq i} \frac{1}{z_i - z_j}. \quad (20.4)$$

Changing only the Vandermonde power would change these equations. Although the stationary points are unchanged, their relative contributions and the fluctuations integrated around them need not be unchanged. In particular, E_p is a *period energy*, not an eigenvalue by definition. The exact positive representation proved below is a special property of the full $p = 2$ period; it is not a consequence of merely squaring a complex weight.

20.2 The finite Pfaffian and its residue moments

Put $t = 2u$ and introduce the moments

$$\mu_s(t) = \oint_{\gamma_{\text{all}}} \frac{z^s e^{-tz}}{q(z)^2} \frac{dz}{2\pi i}, \quad s \geq 0. \quad (20.5)$$

Here and below $\mathcal{Z}_2(u)$ and $\mathcal{Z}_2(t/2)$ denote the same period with the indicated independent variable.

Exact identity 20.1 (Common-contour Pfaffian). *For $1 \leq N \leq L$,*

$$\boxed{\mathcal{Z}_2(u) = \text{Pf } \mathbf{A}(t), \quad \mathbf{A}_{ij}(t) = (j - i)\mu_{i+j-1}(t), \quad 0 \leq i, j \leq 2N - 1.} \quad (20.6)$$

The diagonal is zero. The moment μ_{-1} is not used; the largest index required for the Pfaffian is $4N - 4$.

Proof. In a $2N \times 2N$ determinant, place the columns z_α^i and iz_α^{i-1} consecutively for each $\alpha = 1, \dots, N$, with row index $i = 0, \dots, 2N - 1$. The confluent Vandermonde identity gives exactly $D(\mathbf{z})^4$ with this column ordering. The determinant integration identity of de Bruijn [40] gives

$$\oint D(\mathbf{z})^4 \prod_{\alpha=1}^N w(z_\alpha) \frac{dz_\alpha}{2\pi i} = N! \text{Pf} \left[\oint (z^i j z^{j-1} - iz^{i-1} z^j) w(z) \frac{dz}{2\pi i} \right]. \quad (20.7)$$

It can also be proved directly by expanding the determinant and grouping permutations into pairings of its rows. It is a finite algebraic identity of linear functionals; positivity of the functional is unnecessary. Take $w(z) = e^{-tz}/q(z)^2$ and cancel $N!$. $\square$

For a nonsingular skew matrix, $\partial_t \log \text{Pf } \mathbf{A} = \frac{1}{2} \text{tr}(\mathbf{A}^{-1} \partial_t \mathbf{A})$. Since $\partial_t \mu_s = -\mu_{s+1}$, this yields

$$\boxed{E_2 = -\partial_t \log \mathcal{Z}_2(t/2) = \frac{1}{2} \text{tr}(\mathbf{A}^{-1} \mathbf{C}), \quad \mathbf{C}_{ij} = (j - i)\mu_{i+j}.} \quad (20.8)$$

The energy requires moments through $4N - 3$. Absence of singularity for real levels and $t > 0$ will follow from Theorem 20.3. For comparison, the familiar $p = 1$ determinant is

$$\mathcal{Z}_1 = \det \mathbf{B}, \quad \mathbf{B}_{ij} = \nu_{i+j}, \quad \nu_s = \oint_{\gamma_{\text{all}}} \frac{z^s e^{-uz}}{q(z)} \frac{dz}{2\pi i}, \quad (20.9)$$

$$E_1 = \text{tr}(\mathbf{B}^{-1} \mathbf{B}_+), \quad (\mathbf{B}_+)_{ij} = \nu_{i+j+1}, \quad 0 \leq i, j < N. \quad (20.10)$$

20.2.1 Explicit finite sums

For distinct poles let $q_n(z) = q(z)/(z - a_n)$. Differentiating the regular factor at a double pole gives

$$\mu_s(t) = \sum_{n=0}^{L-1} \frac{d}{dz} \left(\frac{z^s e^{-tz}}{q_n(z)^2} \right) \Big|_{z=a_n}$$

$$= \sum_{n=0}^{L-1} \frac{e^{-ta_n}}{q'(a_n)^2} \left[sa_n^{s-1} - \left(t + 2 \sum_{m \neq n} \frac{1}{a_n - a_m} \right) a_n^s \right]. \quad (20.11)$$

For $s = 0$ the first term is identically zero, including when $a_n = 0$. For $a_n = a_0 + hn$ with $h > 0$,

$$q'(a_n)^2 = h^{2(L-1)} [n!(L-1-n)!]^2, \quad \sum_{m \neq n} \frac{1}{a_n - a_m} = \frac{H_n - H_{L-1-n}}{h}, \quad H_n = \sum_{j=1}^n \frac{1}{j}, \quad H_0 = 0. \quad (20.12)$$

These formulas evaluate every entry of the Pfaffian. They are not a claim that its value always admits a short product factorization.

Moment formation and matrix elimination require a polynomial number of arithmetic operations. Cancellation between moments can nevertheless require many digits. In numerical use it is preferable to center and scale the spectrum, compute with adjustable precision, and verify stability under an increase in precision. If $a_n = a_0 + h\tilde{a}_n$, the transformation of the period energy is $E_p = Na_0 + h\tilde{E}_p$, with the coupling also rescaled to V/h . Algebraic nonvanishing does not imply numerical good conditioning.

20.3 An independent residue expansion by matchings

The following construction supplies an independent algebraic way of testing the Pfaffian, without expanding its rows. For a subset $I \subset \{0, \dots, L-1\}$ of size N , define

$$A_I = \sum_{i \in I} a_i, \quad \omega_I = \frac{\prod_{i < j, i, j \in I} (a_i - a_j)^4}{\prod_{i \in I} q'(a_i)^2}, \quad (20.13)$$

$$b_i^{(I)} = 4 \sum_{j \in I \setminus \{i\}} \frac{1}{a_i - a_j} - 2 \sum_{m \neq i} \frac{1}{a_i - a_m}, \quad c_{ij} = \frac{4}{(a_i - a_j)^2}. \quad (20.14)$$

Let $\mathfrak{M}(I)$ be the set of matchings of the complete graph on I , including the empty matching. A matching consists of disjoint unordered pairs. Write $U(\mathfrak{m})$ for its unmatched vertices.

Exact identity 20.2 (Matching expansion). *The all-pole period is*

$$\mathcal{Z}_2(t/2) = \sum_{|I|=N} e^{-tA_I} \omega_I \sum_{\mathfrak{m} \in \mathfrak{M}(I)} \prod_{i \in U(\mathfrak{m})} (-t + b_i^{(I)}) \prod_{\{i,j\} \in \mathfrak{m}} c_{ij}. \quad (20.15)$$

Proof. When different variables are assigned different poles, the multiple residue requires one derivative in each variable of the regular part of the integrand. The logarithm of that regular part is a sum of one-variable and two-variable functions. Its first derivatives at the assigned poles are $-t + b_i^{(I)}$, its mixed second derivatives are c_{ij} , and derivatives with respect to three or more distinct variables vanish. The derivative formula for an exponential is a sum over set partitions of the derivative indices. Only singleton and two-element blocks survive, precisely giving the matching sum in (20.15).

If $r \geq 2$ variables are assigned to the same pole, their mutual Vandermonde fourth power has total degree $2r(r-1)$ in those local variables. The residue supplies total derivative order only r . Since $2r(r-1) > r$, that assignment vanishes. Thus only distinct assignments remain, and their $N!$ permutations cancel the normalization in (20.2). $\square$

Individual subset contributions in (20.15) can have either sign. The representation is exact but generally more expensive than a Pfaffian evaluation. Its main role here is a separate finite formula and a transparent way of extracting endpoint expansions.

20.4 A positive quaternion Grassmann representation

Let $\mathcal{A} = \text{diag}(a_0, \dots, a_{L-1})$ and let $\text{Gr}_{\mathbb{H}}(N, L)$ denote the space of orthogonal quaternion projectors of rank N in $\mathbb{H}^L$. Its invariant probability measure is denoted by $d\mu(P)$. For a quaternion Hermitian matrix the trace $\text{tr}_{\mathbb{H}}$ is the real sum of its eigenvalues; in the usual doubled complex representation it is *half* the complex trace. Put

$$K = N[2(L - N) + 1],$$

$$C_{L,N} = \prod_{j=0}^{N-1} \frac{(2j+1)!}{(2L-2j-1)!} = \prod_{j=1}^N \frac{\Gamma(2j)}{\Gamma(2(L-N+j))}, \quad (20.16)$$

$$I_{L,N}(t) = \int_{\text{Gr}_{\mathbb{H}}(N,L)} e^{-t \text{tr}_{\mathbb{H}}(\mathcal{A}P)} d\mu(P). \quad (20.17)$$

Theorem 20.3 (Positive full-period representation). *For $1 \leq N \leq L$, real levels and $t > 0$,*

$$\boxed{\mathcal{Z}_2(t/2) = (-1)^N C_{L,N} t^K I_{L,N}(t)}. \quad (20.18)$$

In particular, $(-1)^N \mathcal{Z}_2(t/2) > 0$. The representation extends continuously to coincident real levels when the left side is defined by the original contour integral.

Proof. We give the transform argument, including its convergence condition and the regularization needed when that condition fails. The underlying quaternion matrix integration and polar-coordinate formulas belong to the standard matrix-Jacobian framework [41]; all constants that can depend on measure conventions will be fixed by a direct contour calculation at the end.

Choose $c < \min_n a_n$. Let X be quaternion Hermitian of size N and let $Q \in \mathbb{H}^{N \times L}$. A quaternion has four real coordinates, so its scalar Gaussian integral is

$$\int_{\mathbb{R}^4} e^{-b|q|^2} dq = \pi^2 b^{-2}, \quad \text{Re } b > 0. \quad (20.19)$$

Diagonalizing X with real eigenvalues $x_1, \dots, x_N$, one obtains

$$\begin{aligned} \pi^{-2NL} \int \exp \left[-\text{tr}_{\mathbb{H}} Q(\mathcal{A} - c\mathbf{1})Q^\dagger + i \text{tr}_{\mathbb{H}} XQQ^\dagger \right] dQ \\ = \prod_{n=0}^{L-1} \prod_{j=1}^N (a_n - c - ix_j)^{-2}. \end{aligned} \quad (20.20)$$

The symbol i in this Fourier transform is an external commuting complex scalar, not a quaternion unit. There is no extra factor two in the exponent with the stated trace convention.

Let $g_c(S)$ be the density on the positive Hermitian cone obtained by pushing forward the positive measure on the left of (20.20) under $Q \mapsto S = QQ^\dagger$. In the polar decomposition $Q = S^{1/2}H$, where $HH^\dagger = \mathbf{1}_N$, the quaternion Jacobian gives

$$g_c(S) = k_{L,N} \det_{\mathbb{H}}(S)^{2(L-N)+1} \int_{\text{St}_{\mathbb{H}}(N,L)} \exp \left[-\text{tr}_{\mathbb{H}} S^{1/2}H(\mathcal{A} - c\mathbf{1})H^\dagger S^{1/2} \right] d\mu(H). \quad (20.21)$$

Here $k_{L,N} > 0$ depends only on dimensions and Lebesgue-measure conventions. The power $2(L - N) + 1$, including its final $+1$, is essential. At $S = t\mathbf{1}_N$ this is

$$g_c(t\mathbf{1}_N) = k_{L,N} t^K e^{tcN} I_{L,N}(t), \quad (20.22)$$

since $P = H^\dagger H$ has the invariant Grassmann distribution.

The real dimension of the Hermitian matrix space is $2N^2 - N$. In its Fourier inversion at $t\mathbf{1}_N$, diagonalizing X supplies the spectral Jacobian $D(\mathbf{x})^4$; its eigenvector integral is constant. Writing $z_j = c + ix_j$ therefore relates this inverse transform, up to a positive dimension-dependent constant, to

$$\mathcal{B}(t) = \frac{1}{N!} \int_{(c+i\mathbb{R})^N} D(\mathbf{z})^4 \prod_j \frac{e^{-tz_j}}{q(z_j)^2} \frac{dz_j}{2\pi i}. \quad (20.23)$$

There is no Vandermonde phase: $D(i\mathbf{x})^4 = D(\mathbf{x})^4$. The scalar factor e^{-tcN} from the exponential cancels the factor e^{tcN} in (20.22). The upward line to the left of all poles closes to the right for $t > 0$; this closure is clockwise. Thus each variable supplies a minus sign and $\mathcal{B}(t) = (-1)^N \mathcal{Z}_2(t/2)$ whenever the inversion is absolutely convergent.

Absolute convergence. Since c is strictly separated from every level,

$$|D(\mathbf{x})|^4 \leq C \prod_j (1 + x_j^2)^{2(N-1)}, \quad \prod_n |a_n - c - ix_j|^2 \geq c_0 (1 + x_j^2)^L \quad (20.24)$$

with $c_0 > 0$. The spectral integral is consequently absolutely integrable if $L - 2(N - 1) > 1/2$, which for integer dimensions is $L \geq 2N - 1$. In this range the preceding steps are ordinary Fourier inversion and one-variable residue closure.

Regularization for $N \leq L < 2N - 1$. Insert $\exp[-\epsilon \operatorname{tr}_{\mathbb{H}}(X^2)]$ in the Fourier integral and let $\epsilon \downarrow 0$. In matrix coordinates this is Gaussian smoothing of g_c . Formula (20.21) shows that g_c is smooth in a neighborhood of $t\mathbf{1}_N$ for every $t > 0$; hence the regularized inverse converges there to $g_c(t\mathbf{1}_N)$. On the spectral side, expand $D(\mathbf{z})^4$ into its finite monomial sum. The Gaussian cutoff then factors into the N scalar variables. For each factor, polynomial division of $z^m/q(z)^2$ leaves a polynomial and a proper rational function. The inverse Fourier transform of the polynomial is a finite sum of derivatives of the delta distribution at $t = 0$; its Gaussian regularization tends to zero at any fixed $t > 0$. The proper rational part is a finite sum of simple and double poles. Its inverse transform for $t > 0$ is the negative sum of the residues of $z^m e^{-tz}/q(z)^2$. Gaussian regularization converges to this smooth function away from $t = 0$. Taking products and the finite monomial sum thus again gives $(-1)^N \mathcal{Z}_2(t/2)$. This argument does not assert absolute convergence in the range where it is unavailable.

We have established (20.18) with some positive constant depending only on L, N . To determine it, let all levels coalesce at zero. Then $I_{L,N}(t) = 1$ and the contour coefficient calculation gives

$$\mathcal{Z}_2(t/2) = (-t)^K \operatorname{Pf} \left[\frac{j-i}{\Gamma(2L-i-j+1)} \right]_{i,j=0}^{2N-1}. \quad (20.25)$$

We interpret $1/\Gamma(0), 1/\Gamma(-1), \dots$ as zero. The elementary Pfaffian product identity

$$\operatorname{Pf} [(j-i)(a)_{i+j}]_{i,j=0}^{2N-1} = \prod_{j=0}^{N-1} (2j+1)!(a)_{2j+1} \quad (20.26)$$

uses the rising factorial $(a)_m = a(a+1) \cdots (a+m-1)$. For $a > 0$ it follows by applying (20.7) to $x^a e^{-x} dx / \Gamma(a)$ and evaluating the Laguerre Selberg integral. Both sides are polynomials in

a , so the identity holds for all a ; related explicit Pfaffian evaluations are discussed in Ref. [42]. For completeness, the specialization in the present problem follows from

$$\frac{(-1)^m(-b)_m}{\Gamma(b+1)} = \frac{1}{\Gamma(b-m+1)}, \quad m = 0, 1, \dots, \quad (20.27)$$

and then set $b = 2L$. Applying (20.26) shows that the Pfaffian in (20.25) is $C_{L,N}$ in (20.16). Since $K \equiv N \pmod{2}$, the sign is $(-1)^N$. This fixes the full normalization without leaving a matrix-volume constant unspecified. Continuity of the contour integral on a common contour, and of the compact Grassmann integral, permits coalescence and its reverse throughout the argument. $\square$

20.4.1 Nonvanishing, fluctuation identities, and full filling

Let $s(P) = \text{tr}_{\mathbb{H}}(\mathcal{A}P)$, and write $\langle \cdot \rangle_t$ for expectation in the explicitly normalized positive measure proportional to $e^{-ts(P)} d\mu(P)$. Theorem 20.3 gives

$$E_2(u) = -\frac{K}{2u} + \langle s \rangle_{2u}, \quad (20.28)$$

$$\partial_u \left(E_2(u) + \frac{K}{2u} \right) = -2 \text{Var}_{2u}(s) \leq 0. \quad (20.29)$$

The normalized quantity $(-1)^N \mathcal{Z}_2(t/2)/(C_{L,N}t^K) = I_{L,N}(t)$ is an entire function of t . If all $a_n \geq 0$, it is completely monotone for $t > 0$. For an arbitrary real spectrum this statement applies after subtracting the lowest level. In particular, the full $p = 2$ period has no zero on the positive real coupling axis, and its Pfaffian matrix is nonsingular there. The min–max bounds for the trace of a rank- N projector imply

$$\sum_{n=0}^{N-1} a_n - \frac{K}{2u} \leq E_2(u) \leq \sum_{n=L-N}^{L-1} a_n - \frac{K}{2u} \quad (20.30)$$

when the levels are ordered. These are bounds on a *period energy*, not variational bounds on a Hamiltonian eigenvalue.

At full filling, the Grassmann space has a single projector, $P = \mathbf{1}_L$, and $C_{L,L} = 1$, $K = L$. Hence

$$\mathcal{Z}_2(t/2) = (-t)^L e^{-t \sum_n a_n}, \quad E_2 = \sum_n a_n - \frac{L}{2} V. \quad (20.31)$$

The physical seniority-zero sector then has one basis state with energy $\sum_n a_n - LV$, so even this exact positive period is not the physical energy itself.

20.5 A controlled finite-spectrum comparison of period energies

Define, wherever the two period energies are finite,

$$\hat{E} = 2E_2 - E_1. \quad (20.32)$$

The motivation is cancellation of a leading inverse-temperature correction. At $p = 1, 2$, however, $1/p$ is not an asymptotic small parameter. The statements below are derived directly at the two coupling endpoints. They concern the same *full* period throughout; the weak-coupling comparison is an algebraic diagnostic and not an increased-temperature calculation in the Case B contour geometry.

20.5.1 Weak coupling at fixed size

Fix L, N and distinct ordered levels, and take $V \rightarrow 0^+$. Let $I_0 = \{0, \dots, N-1\}$ and $A_0 = \sum_{i \in I_0} a_i$. For $N < L$ define

$$\begin{aligned} h_i &= \sum_{b \notin I_0} \frac{1}{a_b - a_i}, & d_i &= \sum_{j \in I_0 \setminus \{i\}} \frac{1}{a_i - a_j}, \\ S &= \sum_{i \in I_0} h_i, & Q &= \sum_{i \in I_0} \sum_{b \notin I_0} \frac{1}{(a_b - a_i)^2}, \\ T &= \sum_{i \in I_0} (h_i^2 + 2d_i h_i). \end{aligned} \quad (20.33)$$

Empty sums are zero at full filling.

Proposition 20.4 (First physical defect at weak coupling). *For the fixed finite spectrum,*

$$\begin{aligned} E_1 &= A_0 + \text{exponentially small terms}, \\ E_2 &= A_0 - \frac{N}{2}V - \frac{S}{2}V^2 - \frac{T}{2}V^3 + O(V^4), \end{aligned} \quad (20.34)$$

$$E_0^{\text{RG}} = A_0 - NV - SV^2 - (T - Q)V^3 + O(V^4). \quad (20.35)$$

Consequently, for $0 < N < L$,

$$\boxed{\hat{E} - E_0^{\text{RG}} = -QV^3 + O(V^4), \quad Q > 0.} \quad (20.36)$$

Proof. The smallest subset energy A_I occurs uniquely at I_0 . For $p = 1$ its residue has a coupling-independent prefactor, so there are no algebraic corrections to A_0 . For $p = 2$, $b_i^{(I_0)} = 2(d_i + h_i)$, and the matching polynomial is

$$(-2u)^N \left[1 - \frac{\sum_i b_i}{2u} + \frac{\sum_{i < j} (b_i b_j + c_{ij})}{4u^2} + O(u^{-3}) \right]. \quad (20.37)$$

Pair antisymmetry gives $\sum_i d_i = 0$. Cancellation of the contributions of each unordered triple of occupied indices gives

$$\sum_i d_i^2 = 2 \sum_{i < j} \frac{1}{(a_i - a_j)^2}. \quad (20.38)$$

It follows that the logarithm of the bracket in (20.37) is $-S/u - T/(2u^2) + O(u^{-3})$. Differentiating this finite rational asymptotic expansion with $-\frac{1}{2}\partial_u$ proves (20.34). For $N < L$, other subsets have an energy excess at least $\delta = a_N - a_{N-1} > 0$ and give exponentially small terms, possibly with polynomial prefactors; at full filling there are no other subsets. Such terms are absorbed in the stated $O(V^4)$.

For the physical comparison, work directly with $H_N = \sum_n a_n b_n^\dagger b_n - V \sum_{n,m} b_n^\dagger b_m$. Its diagonal interaction is $-NV$ throughout the N -pair sector. After removing that constant, the second-order perturbation energy is $-V^2 \sum_{i,b} (a_b - a_i)^{-1} = -SV^2$. At third order, the two intermediate one-hop configurations must share an occupied hole or an empty particle index. Their ordered sum is

$$R = \sum_i \sum_{\substack{b,c \notin I_0 \\ b \neq c}} \frac{1}{(a_b - a_i)(a_c - a_i)} + \sum_{b \notin I_0} \sum_{\substack{i,j \in I_0 \\ i \neq j}} \frac{1}{(a_b - a_i)(a_b - a_j)}. \quad (20.39)$$

The diagonal normalization term cancels after the constant $-NV$ is removed. The identity $\sum_i d_i h_i = \sum_{i < j, b} [(a_b - a_i)(a_b - a_j)]^{-1}$ gives $R = T - Q$. Thus physical perturbation theory yields (20.35), without invoking a contour-saddle identification. Subtraction proves (20.36). $\square$

The cancellation through V^2 is therefore exact in this finite-spectrum limit, while the first defect is known and nonzero. For sufficiently weak coupling $\hat{E}$ lies *below* the physical ground energy; it is not a variational upper bound. None of these fixed-size estimates licenses an exchange with the thermodynamic sequence $V \sim L^{-1}$.

20.5.2 Strong coupling at fixed size

Now fix L, N and the spectrum and take $V \rightarrow \infty$. For $L > 1$ put

$$\bar{a} = \frac{1}{L} \sum_n a_n, \quad A_2 = \sum_n (a_n - \bar{a})^2, \quad B = \frac{N(L-N)}{L^2(L-1)} A_2. \quad (20.40)$$

Proposition 20.5 (First physical defect at strong coupling). *For $p = 1, 2$ and the fixed finite spectrum,*

$$E_p = - \left[N(L - N + 1) - \frac{N}{p} \right] V + N\bar{a} - \frac{B}{V} \frac{pL}{pL + 1} + O(V^{-2}), \quad (20.41)$$

$$E_0^{\text{RG}} = -N(L - N + 1)V + N\bar{a} - \frac{B}{V} + O(V^{-2}). \quad (20.42)$$

In particular,

$$\boxed{\hat{E} - E_0^{\text{RG}} = \frac{B}{(2L+1)(L+1)V} + O(V^{-2})}. \quad (20.43)$$

If the spectrum is symmetric about $\bar{a}$, the remainders displayed in this proposition improve to $O(V^{-3})$.

Proof. Shift the spectrum so that $\bar{a} = 0$, and rescale $w_i = uz_i$. With $d_p = pN(L - N + 1) - N$,

$$\mathcal{Z}_p(u) = C_p u^{d_p} \left[1 + \frac{pA_2}{2} u^2 \langle p_2 \rangle_c + O(u^3) \right], \quad p_1 = \sum_i w_i^{-1}, \quad p_2 = \sum_i w_i^{-2}. \quad (20.44)$$

The brackets are normalized contour averages for the coalescent weight

$$W_p(\mathbf{w}) = D(\mathbf{w})^{2p} \prod_i w_i^{-pL} e^{-p \sum_i w_i}. \quad (20.45)$$

One may take fixed circles enclosing zero and all the small scaled poles ua_n . Hence (20.44) is an ordinary Taylor expansion of a finite contour integral; no saddle approximation is involved.

The normalizing coalescent coefficient is nonzero. For $p = 2$ this follows from (20.16) with $t = 2u$. For $p = 1$ the coalescent moment determinant gives

$$|C_1| = \prod_{j=0}^{N-1} \frac{j!}{(L-j-1)!} > 0. \quad (20.46)$$

An elementary determinant proof multiplies row i by $(L-i-1)!$ in the matrix $[1/\Gamma(L-i-j)]$: its entries become the falling-factorial polynomials $(L-i-1)_j$. Their determinant is their Vandermonde determinant, since each polynomial is monic of degree j . Taking its magnitude gives (20.46), including the zero entries specified by reciprocal gamma factors.

Integration of the three total derivatives $\sum_i \partial_i W_p$, $\sum_i \partial_i (p_1 W_p)$ and $\sum_i \partial_i (w_i^{-1} W_p)$ gives, respectively,

$$\begin{aligned} \langle p_1 \rangle_c &= -\frac{N}{L}, \\ 0 &= -\langle p_2 \rangle_c - pN \langle p_1 \rangle_c - pL \langle p_1^2 \rangle_c, \end{aligned}$$

$$0 = -p\langle p_1 \rangle_c - [p(L-1) + 1]\langle p_2 \rangle_c - p\langle p_1^2 \rangle_c. \quad (20.47)$$

These are identities on closed meromorphic contours. Apparent fractions on the diagonals are canceled by the integer Vandermonde powers; they do not introduce boundary terms. Solving the last two identities yields

$$\langle p_2 \rangle_c = \frac{pN(L-N)}{L(L-1)(pL+1)}. \quad (20.48)$$

Apply $-p^{-1}\partial_u \log$ to (20.44), then restore $\bar{a}$. This proves (20.41).

For the physical energy, the dominant interaction is $-VS^+S^-$. Its ground state at fixed N is the fully symmetric spin- $L/2$ state and has energy $-VN(L-N+1)$. The kinetic term has expectation $N\bar{a}$ and variance $N(L-N)A_2/[L(L-1)]$ in that state. After its mean is removed, its action lies in the spin- $L/2 - 1$ subspace, separated in interaction energy by VL . Second-order perturbation theory therefore gives the correction $-B/V$ in (20.42). Combining the two period expansions proves (20.43). Finally, for a centered reflection-symmetric spectrum all odd centered spectral corrections vanish. Equivalently, reflection of all levels is implemented by a permutation of the levels, so the centered expansion is even before its logarithmic derivative; the next energy term is V^{-3} . $\square$

At $0 < N < L$ and a nonconstant spectrum, $B > 0$. Thus the sign of the first strong-coupling error is opposite to the weak-coupling error. The small factor $[(2L+1)(L+1)]^{-1}$ describes a *coefficient* in a fixed-size expansion. It is not by itself a uniform error bound at fixed dimensionless coupling in the thermodynamic limit.

20.5.3 Two analytic checks

For one pair in two levels, $a_0 = 0$, $a_1 = h > 0$, let $\alpha = h/2$ and $x = h/V$. Direct residues and diagonalization of the physical 2×2 Hamiltonian give

$$\begin{aligned} E_1 &= \alpha - \alpha \coth(x/2), \\ E_2 &= \alpha - \alpha \frac{x \sinh x}{x \cosh x - \sinh x}, \\ E_0^{\text{RG}} &= \alpha - V - \sqrt{\alpha^2 + V^2}. \end{aligned} \quad (20.49)$$

Thus the first weak-coupling defect is $-V^3/h^2$, even though the physical cubic coefficient in this example is zero. At strong coupling the defect is $h^2/(120V) + O(V^{-3})$.

At $N = L$, the $p = 1$ period has only the full subset of simple-pole residues, so $E_1 = \sum_n a_n$. Combining with (20.31) gives

$$\hat{E} = \sum_n a_n - LV = E_0^{\text{RG}} \quad (N = L) \quad (20.50)$$

for every $V > 0$. This also covers $L = N = 1$ directly, without using formulas that divide by $L - 1$.

20.6 Reproducibility and what the extension establishes

The accompanying `code/pfaffian_caseA/` material implements the full period only. Its checks compare the moment Pfaffian with the separate matching expansion, test coalescent normalizations and analytic small systems, and check the period coefficients in fixed-size coupling limits. The physical-energy comparisons in this chapter are derived analytically from the Hamiltonian and are not certified by those period-only checks. Verification counts and

precision settings belong to the accompanying execution report. Agreement of computational routes checks their implementation; it does not remove the authorship and verification qualification stated at the beginning of this manuscript.

The Pfaffian reduction uses known determinant-integration algebra, and the positive representation uses known quaternion Gaussian and polar-coordinate methods. The contribution recorded here is their explicit application, normalization, and endpoint error analysis for the stated Richardson meromorphic weight. Priority for these particular application formulas has not been established by a comprehensive literature investigation.

The exact finite period, its nonvanishing on $V > 0$, and its logarithmic derivative are now available for the Case A all-pole construction at $\beta = 4$. The combination $2E_2 - E_1$ has justified cancellations and explicit first defects in two narrow coupling limits. These results do not prove a convergent inverse-temperature extrapolation, do not control a simultaneous large-size limit, and do not select the physical saddle. The positive auxiliary projector measure also does not turn the individual signed residues into physical probabilities. Establishing the remaining contour-to-spectrum relation would require a further argument beyond the finite Pfaffian identities proved here.

Third-edition continuation. The limitations just stated concern the two finite temperatures and the arguments of this chapter. Chapter 21 supplies a separate exact vector argument for the limit $p \rightarrow \infty$ at fixed system size. It does not make the two-temperature extrapolation exact or provide a uniform thermodynamic error bound.

Chapter 21

Case A at every even auxiliary inverse temperature

21.1 The finite period and its hypergeometric energy

This chapter extends the common all-pole construction of Chapter 20 from $p = 2$ to every positive integer p , with auxiliary inverse temperature $\beta = 2p$. The result is a multivariable Jack hypergeometric function, not an ordinary scalar ${}_1F_1$. In addition, an exact vector-valued period supplies a finite-system proof that its energy approaches the physical ground energy as $p \rightarrow \infty$. This latter argument is independent of the continuum Richardson–Gaudin construction. It concerns the common contour and does not construct the mixed contours assigned to Case B elsewhere.

We retain $L = M + 1$, $a_j = 2\varepsilon_j$, $q(z) = \prod_{j=0}^{L-1} (z - a_j)$, $D(\mathbf{z}) = \prod_{i < j} (z_j - z_i)$ and $u = V^{-1} > 0$. All L levels are real and initially distinct; $1 \leq N \leq L$. Put $t = pu$ and

$$\mathcal{Z}_p(u) = \frac{1}{N!} \oint_{\gamma_{\text{all}}^{\times N}} D(\mathbf{z})^{2p} \prod_{i=1}^N \frac{e^{-pu z_i} dz_i}{q(z_i)^p 2\pi i}, \quad E_p = -\frac{1}{p} \partial_u \log \mathcal{Z}_p, \quad (21.1)$$

$$c = N(L - N + 1),$$

$$K_p = pc - N,$$

$$C_{L,N,p} = \prod_{j=1}^N \frac{\Gamma(pj)}{\Gamma(p)\Gamma(p(L - N + j))}, \quad s_{L,N,p} = (-1)^{K_p + pN(N-1)/2}. \quad (21.2)$$

Every contour encloses all poles counterclockwise once. For integer p there are no branch choices or poles on the diagonals $z_i = z_j$.

The Jack parameter used here is $\alpha = 1/p$. For a partition λ and $k = |\lambda|$, define

$$\begin{aligned} [b]_{\lambda}^{(\alpha)} &= \prod_{i \geq 1} (b - (i - 1)/\alpha)_{\lambda_i}, \\ c_{\lambda}(\alpha) &= \prod_{s \in \lambda} [\alpha a(s) + \ell(s) + 1], & c'_{\lambda}(\alpha) &= \prod_{s \in \lambda} [\alpha(a(s) + 1) + \ell(s)], \\ C_{\lambda}^{(\alpha)}(x) &= \frac{\alpha^k k!}{c'_{\lambda}(\alpha)} P_{\lambda}^{(\alpha)}(x). \end{aligned} \quad (21.3)$$

Here P_{λ} is monic in the monomial symmetric basis, $a(s)$ and $\ell(s)$ are the arm and leg lengths, and $(b)_m$ is the ordinary rising factorial. Thus $\sum_{|\lambda|=k} C_{\lambda}^{(\alpha)}(x) = (\sum_j x_j)^k$. Our precise convention is

$${}_1F_1^{(\alpha)}(b; d; x) = \sum_{\ell(\lambda) \leq L} \frac{[b]_{\lambda}^{(\alpha)} C_{\lambda}^{(\alpha)}(x)}{[d]_{\lambda}^{(\alpha)} |\lambda|!}. \quad (21.4)$$

The Jack series and its Selberg identities are established special-function tools [43, 44]. When $b = pN, d = pL, \alpha = 1/p$, partitions longer than N vanish; all remaining denominator factors are positive. This fixes a convention that is sometimes written with the reciprocal parameter in the literature.

Theorem 21.1 (Exact even-temperature all-pole evaluation). *For every integer $p \geq 1$,*

$$\boxed{\mathcal{Z}_p(u) = s_{L,N,p} C_{L,N,p}(pu)^{K_p} F_p(u)}, \quad F_p(u) = {}_1F_1^{(1/p)}(pN; pL; -pu\mathbf{a}), \quad (21.5)$$

$$\boxed{E_p(u) = -\frac{K_p}{pu} - \frac{1}{p} \partial_u \log F_p(u)}. \quad (21.6)$$

The derivative is along the vector argument $-pu\mathbf{a}$; it cannot in general be replaced by the elementary scalar contiguous-function ratio. The function F_p is entire in u , with $F_p(0) = 1$.

21.1.1 Derivation and the complete normalization

We spell out the confluent constant-term argument to make the integer- p contour and its sign unambiguous. The symmetric Morris–Jack identity following from the constant-term integration theory of Ref. [44] reads

$$W_{A,B}(x) = \prod_{i=1}^N (1 - x_i)^A (1 - x_i^{-1})^B \prod_{i \neq j} (1 - x_i/x_j)^p, \\ \frac{\text{CT}[P_\lambda^{(1/p)}(x^{-1})W_{A,B}(x)]}{\text{CT}[W_{A,B}(x)]} = \frac{[-A]_\lambda^{(1/p)}}{[1 + B + (N - 1)p]_\lambda^{(1/p)}} P_\lambda^{(1/p)}(1^N), \quad (21.7)$$

$$\text{CT}[W_{A,B}] = \prod_{j=0}^{N-1} \frac{\Gamma(1 + A + B + jp)\Gamma(1 + (j + 1)p)}{\Gamma(1 + A + jp)\Gamma(1 + B + jp)\Gamma(1 + p)}. \quad (21.8)$$

It suffices here to use nonnegative integers A, B, p , so every constant term is an ordinary finite Laurent-polynomial coefficient. Set $B = p(L - N + 1) - 1$, so $NB = K_p$, and rescale $x_i = tz_i/A$. On fixed circles in the z_i planes, as integer $A \rightarrow \infty$,

$$(1 - tz_i/A)^A \longrightarrow e^{-tz_i}, \quad (1 - A/(tz_i))^B \sim (-A/t)^B z_i^{-B}, \\ \prod_{i \neq j} (1 - z_i/z_j)^p = (-1)^{pN(N-1)/2} D(\mathbf{z})^{2p} \prod_i z_i^{-p(N-1)}. \quad (21.9)$$

The convergence after the indicated scalar normalization is uniform on these compact contours. Introduce

$$\mathcal{L}_t[f] = \frac{1}{N!} \oint D(\mathbf{z})^{2p} \prod_i z_i^{-pL} e^{-tz_i} f(\mathbf{z}) \frac{dz_i}{2\pi i}. \quad (21.10)$$

Equations (21.7)–(21.9) give

$$\mathcal{L}_t[1] = s_{L,N,p} C_{L,N,p} t^{K_p}, \quad (21.11)$$

$$\frac{\mathcal{L}_t[P_\lambda^{(1/p)}(z^{-1})]}{\mathcal{L}_t[1]} = (-t)^{|\lambda|} \frac{P_\lambda^{(1/p)}(1^N)}{[pL]_\lambda^{(1/p)}}. \quad (21.12)$$

Indeed $(t/A)^{|\lambda|} [-A]_\lambda^{(1/p)} \rightarrow (-t)^{|\lambda|}$; the gamma-product limit in (21.8) supplies A^{K_p} . Its remaining $N!$ cancels the contour normalization, because $\Gamma(1 + pj)/\Gamma(1 + p) = j\Gamma(pj)/\Gamma(p)$.

Choose all circles to have radius greater than $\max_j |a_j|$. The Jack Cauchy identity gives the convergent expansion

$$\prod_{i,j} (1 - a_j/z_i)^{-p} = \sum_{\lambda} \frac{c_{\lambda}(1/p)}{c'_{\lambda}(1/p)} P_{\lambda}^{(1/p)}(\mathbf{a}) P_{\lambda}^{(1/p)}(z^{-1}). \quad (21.13)$$

Use (21.12) and $P_{\lambda}^{(\alpha)}(1^N) = \alpha^{|\lambda|} [N/\alpha]_{\lambda}^{(\alpha)} / c_{\lambda}(\alpha)$. Each term becomes exactly the corresponding term of (21.4). This proves (21.5), including its normalization. Differentiation proves (21.6). The contour integrand is entire in u ; the proved power-series factor removes its zero of order K_p at zero. The same compact-contour argument permits coalescent levels. No unproved analytic continuation from $p = 1, 2$ has been used.

21.1.2 Simple limits and the meaning of the energy

At $p = 1$, $K_1 = N(L - N)$ and the formula reduces to the all-pole determinant. At $p = 2$, $s_{L,N,2} = (-1)^N$ and $C_{L,N,2} = \prod_{j=1}^N \Gamma(2j) / \Gamma(2(L - N + j))$, recovering the Pfaffian and quaternion-projector result of Chapter 20. At $N = 1$ there is the elementary Dirichlet representation

$$F_p(u) = \mathbb{E} \exp \left[-pu \sum_{j=0}^{L-1} a_j X_j \right], \quad (X_0, \dots, X_{L-1}) \sim \text{Dirichlet}(p, \dots, p). \quad (21.14)$$

For $L = 2$, $a_0 = 0$, $a_1 = h$, this is the ordinary scalar function ${}_1F_1(p; 2p; -puh)$. The reduction at full filling is

$$\begin{aligned} \mathcal{Z}_p(u) &= \frac{(-1)^{L(p-1)+pL(L-1)/2}}{\Gamma(p)^L} (pu)^{L(p-1)} e^{-pu \sum_j a_j}, \\ E_p(u) &= \sum_j a_j - L(1 - 1/p)V. \end{aligned} \quad (21.15)$$

The physical energy of this one-dimensional sector is $\sum_j a_j - LV$. Thus the exact special-function result remains a *period energy at finite p*. Its physical ground-state limit is proved next; it is not an interpretation imposed on the definition.

21.2 An exact imaginary-time bridge to the pairing Hamiltonian

In the fixed- N hard-core-pair sector let

$$H(u) = \sum_j a_j b_j^{\dagger} b_j - u^{-1} S^+ S^-, \quad S^+ = \sum_j b_j^{\dagger}, \quad S^- = \sum_j b_j, \quad (21.16)$$

including the diagonal interaction $-NV$. Off-shell Bethe integrals and inverse-coupling evolution for the BCS Hamiltonian are established constructions [45]; here we derive the particular scalar projection and positive branch needed for (21.1) explicitly.

Write $W_p = e^{-pu} \sum_{\alpha} z_{\alpha} D^{2p} \prod_{\alpha} q(z_{\alpha})^{-p}$ and define

$$\begin{aligned} B(z) &= \sum_j \frac{b_j^{\dagger}}{a_j - z}, & |B_J\rangle &= \prod_{\alpha \in J} B(z_{\alpha}) |0\rangle, \\ |\Psi_p(u)\rangle &= \frac{1}{N!} \oint W_p |B_{\{1, \dots, N\}}\rangle \prod_{\alpha} \frac{dz_{\alpha}}{2\pi i}, \end{aligned}$$

$$R_\alpha = u + \sum_j \frac{1}{z_\alpha - a_j} - 2 \sum_{\beta \neq \alpha} \frac{1}{z_\alpha - z_\beta}, \quad \partial_{z_\alpha} W_p = -p R_\alpha W_p. \quad (21.17)$$

The hard-core commutators give

$$\begin{aligned} S^- |B_{\{1, \dots, N\}}\rangle &= \sum_\alpha (u - R_\alpha) |B_{\{1, \dots, N\} \setminus \{\alpha\}}\rangle, \\ H(u) |B_{\{1, \dots, N\}}\rangle &= \left(\sum_\alpha z_\alpha \right) |B_{\{1, \dots, N\}}\rangle + u^{-1} \sum_\alpha R_\alpha S^+ |B_{\{1, \dots, N\} \setminus \{\alpha\}}\rangle. \end{aligned} \quad (21.18)$$

Each vector in the final sum is independent of z_α . Its $W_p R_\alpha$ integral is consequently a closed-contour total derivative and vanishes. Therefore

$$\boxed{\partial_u |\Psi_p(u)\rangle = -p H(u) |\Psi_p(u)\rangle.} \quad (21.19)$$

This is auxiliary imaginary-time evolution in u , not the physical time evolution or a physical finite-temperature partition function.

21.2.1 The scalar projection and its essential correction

Let $\langle \mathbf{1} |$ be the row vector with all occupation-basis entries equal to one. The exact relation is

$$\boxed{\langle \mathbf{1} | \Psi_p(u) \rangle = u^N \mathcal{Z}_p(u).} \quad (21.20)$$

For completeness, integrate $W_p \sum_{|J|=m} |B_J\rangle$ and call the result $|T_m\rangle$, with the same $1/N!$ normalization. Lowering a term with $|J| = m$ replaces its coefficient by

$$S^- |B_J\rangle = \sum_{\alpha \in J} \left[u - R_\alpha - 2 \sum_{\beta \notin J} \frac{1}{z_\alpha - z_\beta} \right] |B_{J \setminus \{\alpha\}}\rangle. \quad (21.21)$$

The total derivatives again vanish. For each remaining subset, the double sum over its omitted indices cancels pairwise under $\alpha \leftrightarrow \beta$. Each subset occurs $N - m + 1$ times, giving $S^- |T_m\rangle = (N - m + 1)u |T_{m-1}\rangle$. Iteration yields $(S^-)^N |\Psi_p\rangle = N! u^N \mathcal{Z}_p |0\rangle$. Since $(S^-)^N |I\rangle = N! |0\rangle$ for every occupied subset I , (21.20) follows. Differentiation gives

$$\boxed{E_p(u) = \frac{\langle \mathbf{1} | H(u) | \Psi_p(u) \rangle}{\langle \mathbf{1} | \Psi_p(u) \rangle} + \frac{N}{pu}.} \quad (21.22)$$

The fraction is a mixed matrix element, not a Rayleigh quotient. No variational upper-bound property is asserted for E_p .

21.2.2 Selection of a positive regular branch

At simultaneous infinity the weight has homogeneous degree $-pc$ and the N Bethe factors supply degree $-N$. Contour coefficient extraction therefore requires at least pc powers from the exponential in the vector period. At that first order all levels can be put to zero and $B(z) = -S^+/z$; hence all occupation components are equal. Their common coefficient is nonzero by (21.20) and (21.11). After a fixed scalar sign is chosen,

$$|\psi_p(u)\rangle = \text{const } u^{pc} [|D_N\rangle + O(u)], \quad \text{const} > 0, \quad (21.23)$$

where $|D_N\rangle$ is the positive uniform Dicke vector. The off-diagonal entries of $-pH(u)$ are nonnegative and the pair-hop graph is connected for $0 < N < L$. Forward evolution preserves positivity; thus all components of ψ_p are positive for $u > 0$. In particular,

$$s_{L,N,p} \mathcal{Z}_p(u) > 0, \quad F_p(u) > 0 \quad (u > 0). \quad (21.24)$$

No scalar zeros obstruct the energy on the attractive real axis.

To see that the branch has no unspecified admixture, put $T = S^+ S^-$ and $Q = c\mathbf{1} - T$. The real symmetric matrix Q is the Laplacian of the pair-hop graph and is positive semidefinite. With ψ_p normalized as $u^{pc} f(u)$, $f(0) = \mathbf{1}$, the Taylor coefficients obey

$$[k\mathbf{1} + pQ]f_k = -pAf_{k-1}, \quad f_0 = \mathbf{1}, \quad A_{II} = \sum_{j \in I} a_j. \quad (21.25)$$

Every matrix on the left is invertible for $k \geq 1, p > 0$. Moreover $\|f_k\|_2 \leq (p\|A\|_2)^k \|\mathbf{1}\|_2/k!$. The regular branch is unique and the series is entire in u .

21.3 A controlled physical ground-state limit at finite size

Theorem 21.2 (Ground-state limit of the Case A full period). *Fix finite L, N , the real spectrum, and $0 < u_{\min} \leq U < \infty$. Then, as positive integer $p \rightarrow \infty$,*

$$\boxed{E_p(u) = E_0^{\text{RG}}(u) + O(p^{-1})} \quad (21.26)$$

uniformly for $u \in [u_{\min}, U]$. The constants may depend on the finite spectrum, $L, N, u_{\min}, U$. No estimate uniform in system size or in either coupling endpoint is implied.

Proof. The one-dimensional sectors are immediate. For $0 < N < L$, choose $0 < u_0 < u_{\min}$. Irreducibility and nonpositive off-diagonal entries of $H(u)$ imply a simple ground eigenvalue and a normalized strictly positive eigenvector $v(u)$. On $[u_0, U]$ the spectral gap has positive minimum δ , and v is smooth. Set

$$m = \min_{u, I} v_I(u) > 0, \quad B_v = \max_u \|v'(u)\|_2, \quad a_p = v^\top \psi_p > 0, \quad w = \psi_p/a_p - v, \quad R = \|w\|_2. \quad (21.27)$$

Positivity gives $a_p \geq m\|\psi_p\|_1 \geq m\|\psi_p\|_2$ and therefore $R \leq R_{\max} = \sqrt{m^{-2} - 1}$, uniformly in p . Using $v^\top w = 0$ and $v^\top v' = 0$, direct differentiation of (21.19) gives

$$w' = -p(H - E_0^{\text{RG}})w - v' - (v + w)(v'^\top w). \quad (21.28)$$

Taking its scalar product with w , and using the gap on the space orthogonal to v , yields the upper-right-derivative bound

$$\begin{aligned} R' &\leq -p\delta R + B_v(1 + R_{\max}^2), \\ R(u) &\leq R_{\max} e^{-p\delta(u-u_0)} + \frac{B_v(1 + R_{\max}^2)}{p\delta}. \end{aligned} \quad (21.29)$$

This proves $R = O(p^{-1})$ uniformly after $u_{\min}$, without a p -dependent assumption on the initial overlap. Since $v + w$ is entrywise positive and has Euclidean norm at least one, $\langle \mathbf{1} | v + w \rangle \geq 1$. With $d = \binom{L}{N}$,

$$\left| \frac{\langle \mathbf{1} | H | \psi_p \rangle}{\langle \mathbf{1} | \psi_p \rangle} - E_0^{\text{RG}} \right| \leq \sqrt{d} \|H - E_0^{\text{RG}}\|_2 R. \quad (21.30)$$

The operator norm is bounded on the compact interval. Adding the exact $N/(pu)$ term in (21.22) proves the claim. $\square$

This proof does not inspect the geometry of Richardson roots. Root collisions or complex conjugation of individual roots do not affect the positive finite-dimensional evolution. Accordingly the physical limit holds for the common all-pole family at every attractive finite coupling, even where calling the physical root geometry “Case A” is inappropriate. This statement does not identify a finite- p period with an eigenvalue, nor does it extend the separate Case B contour construction to higher p .

21.4 A positive finite- p formulation and its practical scope

Normalize the vector by its component sum, $P_I = \psi_{p,I} / \sum_J \psi_{p,J}$. Each configuration has $D_0 = N(L - N)$ neighbors, written $J \sim I$, and

$$\begin{aligned} \frac{dP_I}{du} &= -p(A_I - \bar{A})P_I + \frac{p}{u} \left(\sum_{J \sim I} P_J - D_0 P_I \right), & \bar{A} &= \sum_I A_I P_I, \\ P_I(0) &= \binom{L}{N}^{-1}, & E_p(u) &= \bar{A}(u) - \frac{c - N/p}{u}. \end{aligned} \quad (21.31)$$

The initial value is understood through the regular limiting branch. The normalized weights remain positive and sum to one. They are auxiliary amplitudes normalized in the 1-norm, not occupation probabilities obtained by squaring a quantum wavefunction. The matrix formulation has dimension $\binom{L}{N}$, so its role here is a proof, a finite-size benchmark, and a possible starting point for controlled approximations. It is not automatically an efficient large-system algorithm.

The same formulas define a continuation to all real $p > 0$. More precisely, the entire solution of (21.25) satisfies

$$\frac{1}{\binom{L}{N}} \langle \mathbf{1} | f(u) \rangle = {}_1F_1^{(1/p)}(pN; pL; -pua). \quad (21.32)$$

For integer p this follows from the contour and scalar projection. For each fixed Taylor degree both sides are rational functions of p : on the left by (21.25), on the right by the finite partition sum and the rational Jack coefficients. Equality at every positive integer thus proves equality of these rational functions. There are no poles for $p > 0$ in the relevant coefficients, and both series are entire; therefore the equality holds for all real $p > 0$. Positivity and the proof of Theorem 21.2 apply to this continuation. For noninteger p it must not be represented as an ordinary product of unbranched meromorphic contours.

At finite p a compact special-function formula has replaced the Vandermonde coefficient algebra. At fixed finite size its physical ground-state limit is now justified. These two conclusions still do not bound the error of an extrapolation from only $p = 1, 2, 3$, or allow the $p \rightarrow \infty$ and thermodynamic limits to be exchanged. The formal thermodynamic anomaly calculations from a different work branch are not included in this chapter.

21.5 Verification and provenance

The accompanying `code/even_beta_caseA/` checks three independent computational constructions. Direct order- p residues combined with a monomial Vandermonde expansion are compared with the entire vector evolution. Independently generated monic Jack polynomials are obtained from their triangular differential eigenproblem; their hypergeometric Taylor coefficients are compared exactly with the vector recurrence. Full filling supplies a closed-form boundary check. The execution report records the spectra, values of p , precision, and agreement. These checks test implementation and normalization; the general statements rest on the proofs above.

The special-function ingredients and off-shell inverse-coupling evolution have explicit prior literature. The contribution recorded here is their application to this normalized scalar Richardson period, the exact scalar projection, and the resulting finite-size ground-state control. No literature-wide priority claim is made. As elsewhere in this manuscript, these are LLM-assisted derivations; the author has not independently verified every calculation.

Part IX

The one-pair saddle route at auxiliary $\beta = 2$

Chapter 22

The one-pair saddle route at Vandermonde power two

This chapter develops the alternative integral route initiated in Section IV and Appendix B of [1]. Its new component is an all-order *formal* explanation of the cancellation of higher extensive ground-energy coefficients. A polynomial insertion and a Toda identity then produce the blocked-level square root. A finite certificate transfers the ground-period answer to a prescribed Case B cascade. No continuum Richardson-root distribution is used.

Here *inverse temperature two* means the Dyson exponent $\beta_D = 2$, namely a Vandermonde square. In the multiplier convention of Chapter 20 this is $p = 1$, $T_{\text{eff}} = V$, not the $p = 2$, power-four Pfaffian problem. All these temperatures are auxiliary. Case A denotes the common all-pole contour; Case B denotes a specified cascade of distinct contours. Neither name denotes a filling regime.

22.1 Objects, scaling, and status

Keep the manuscript's $M + 1$ levels and thermodynamic conventions:

$$\begin{aligned} \varepsilon_j &= \varepsilon_{F0} + j/\rho, \quad j = 0, \dots, M, & \Omega &= M/\rho, \quad \nu = N/M, \\ u &= 1/V, \quad \sigma = e^{-2u/\rho}, & x &= \frac{\sigma}{1 - \sigma}. \end{aligned}$$

Hold $0 < \sigma < 1$, $0 < \nu < 1$, and Ω fixed in the limit. A logarithmic derivative holds the original levels, N, M, ρ , and every integer contour label fixed:

$$-\partial_u = \frac{2}{\rho(1 - \sigma)} x \partial_x. \tag{22.1}$$

The exact quantity $E^{\text{per}} = -\partial_u \log Z$ is a period barycenter, not the exact finite-size Richardson eigenenergy.

Statement	Status in this chapter
Higher extensive ground coefficients	All-order coefficientwise theorem
Case A ground period	Exact finite-lattice trace identity
Case A one blocked level	Exact Heine–Toda identity and positive-coefficient limit
Two separated blocked levels	Finite determinant and nonvanishing leading prefactor
Case B ground period	Exact finite certificate and compact-endpoint asymptotics
Separated Case B marked period	Existing mixed-minor result of Chapter 12
Case B blocked sectors	Conditional on Christoffel-deformed cascade control
Physical excitation energies	Conditional on a branch bridge with $o(1)$ control of differences

An $o(M)$ error in a ground energy does not control a gap of order one.

22.2 All-order formal cancellation

Set

$$\epsilon_c = \frac{2\Omega\sigma}{1-\sigma}, \quad \delta = (\rho\Omega\sigma)^{-1/2}, \quad c_m = \frac{(-i)^m}{m(m-1)} \frac{1-\sigma^{m-1}}{1-\sigma}. \quad (22.2)$$

Use X_j for local coordinates, reserving x for the coupling variable. The one-pair expansion is the normalized formal average

$$m_N(\delta) = \frac{\int X_1 \Delta(X)^2 \exp\{\sum_j [-X_j^2/2 + \sum_{m \geq 3} c_m \delta^{m-2} X_j^m]\} d^N X}{\int \Delta(X)^2 \exp\{\sum_j [-X_j^2/2 + \sum_{m \geq 3} c_m \delta^{m-2} X_j^m]\} d^N X}. \quad (22.3)$$

Both integrals mean coefficientwise Gaussian integration followed by formal division. No convergent infinite Taylor series is inserted under an integral over the full real axis. The corresponding continuum-weight auxiliary energy is

$$E_{\text{loc}} = 2N\varepsilon_{F0} - N\epsilon_c - iN\epsilon_c\delta m_N(\delta). \quad (22.4)$$

Theorem 22.1 (Cancellation of every higher extensive coefficient). *For each $k \geq 1$,*

$$[\delta^{2k-1}]m_N = \sum_{g=0}^{\lfloor k/2 \rfloor} C_{k,g}(\sigma) N^{k-2g}, \quad C_{1,0} = \frac{i(1+\sigma)}{2}, \quad C_{k,0} = 0 \quad (k \geq 2). \quad (22.5)$$

This is a formal theorem, symbolic in σ ; it asserts neither uniform convergence in filling nor dominance of a physical saddle.

22.2.1 The extensive coefficient and polynomial recurrences

The Gaussian Vandermonde-square integral is the eigenvalue form of a formal Hermitian matrix integral with covariance $\langle X_{ab} X_{cd} \rangle_0 = \delta_{ad} \delta_{bc}$. Normalization retains connected ribbon graphs attached to the marked $\text{Tr } X$. With v_m interaction vertices of valence m , at order δ^{2k-1} ,

$$\sum_{m \geq 3} (m-2)v_m = 2k-1, \quad E = \frac{1 + \sum_m m v_m}{2}, \quad V = 1 + \sum_m v_m.$$

Euler's formula gives $F = k + 1 - 2g$ index loops. Division of the marked trace by N gives N^{k-2g} . In (22.4) this term scales as N^{1-2g} at fixed N/ρ , so $g = 0$ selects precisely the extensive part.

Put $h = \delta^2$, $y = \delta X$, $t = Nh = \nu/\sigma$. The formal weight is $\Delta(y)^2 \prod_j e^{-U(y_j)/h}$, where

$$U'(y) = -\frac{i}{1-\sigma} \log \frac{1+iy}{1+i\sigma y} = y + O(y^2). \quad (22.6)$$

Let $p_n(y)$ be its monic orthogonal polynomials, with squared norms H_n , and write

$$yp_n = p_{n+1} + b_n p_n + a_n p_{n-1}, \quad a_n = H_n/H_{n-1}.$$

Gaussian nondegeneracy defines these objects order by order. Integration by parts and orthogonality give the exact formal identities

$$H_n^{-1} \int U' p_n^2 e^{-U/h} dy = 0, \quad (22.7)$$

$$H_{n-1}^{-1} \int U' p_n p_{n-1} e^{-U/h} dy = nh. \quad (22.8)$$

In the second identity p'_n projects to np_{n-1} , while p'_{n-1} is orthogonal to p_n . The polynomial kernel gives the exact trace identity

$$\left\langle \sum_{j=1}^N y_j \right\rangle = \sum_{n=0}^{N-1} b_n. \quad (22.9)$$

At each formal order keep the highest powers of n , set $s = nh$, and denote the leading recurrence symbols by $B(s) = O(s)$, $A(s) = s + O(s^2)$. A Jacobi-matrix power is a sum over finitely long nearest-neighbor paths. Fixed index shifts change only lower powers of n ; all recurrence coefficients on a path can therefore be replaced by the same leading symbols. With $Y(z) = B + \sqrt{A}(z + z^{-1})$, constant-term extraction yields

$$\text{CT } U'(Y) = 0, \quad \text{CT } Y U'(Y) = 2s. \quad (22.10)$$

The factor two comes from the two adjacent matrix elements. The Laurent variable labels recurrence steps, not a Richardson-root arc.

22.2.2 Closed solution and uniqueness

For $\eta = 1, \sigma$, factor on the Gaussian branch

$$1 + i\eta Y = C_\eta(1 + r_\eta z)(1 + r_\eta z^{-1}), \quad r_\eta = \frac{i\eta\sqrt{A}}{C_\eta}, \quad C_\eta - \frac{\eta^2 A}{C_\eta} = 1 + i\eta B. \quad (22.11)$$

Here $C_\eta = 1 + O(s)$, so coefficientwise logarithms give $\text{CT } \log(1 + i\eta Y) = \log C_\eta$. The first string equation implies $C_1 = C_\sigma = C$. Its $z^{\pm 1}$ coefficients are r_η , and the second equation gives $A/C = s$. Subtracting the constant-factor equations gives

$$\boxed{B(s) = i(1 + \sigma)s, \quad A(s) = s(1 - \sigma s), \quad C(s) = 1 - \sigma s.} \quad (22.12)$$

The Jacobian of the two string equations in (B, A) at the Gaussian point is $\begin{pmatrix} 1 & -i(1+\sigma) \\ 0 & 2 \end{pmatrix}$. Its nonzero determinant selects a unique formal solution.

The leading-degree part of (22.9) is

$$\langle y_1 \rangle_{\text{planar}} = \frac{1}{t} \int_0^t B(s) ds = \frac{i(1 + \sigma)}{2} t. \quad (22.13)$$

The integral here records the leading coefficient of the finite power sum $\sum_{n < N} n^k$. It is not a continuum-root approximation. This proves (22.5).

Not every finite- N coefficient vanishes. The first residual is

$$m_N(\delta) = \frac{iN(1+\sigma)}{2}\delta - \frac{i(1+\sigma)(1-\sigma)^2}{24}\delta^3 + O(\delta^5). \quad (22.14)$$

It is genus one and contributes $O(\rho^{-1})$ to E_{loc} . Restoring the original finite level spacing also changes subextensive terms; these are not determined by the continuum potential alone. In particular, interpreting the factorization as convergent Fourier series would impose additional restrictions, which are unnecessary for the coefficientwise theorem and are not physical phase boundaries.

22.3 Ground energy and independent finite-lattice control

The extensive formal answer is

$$E_{\text{ext}} = 2N\varepsilon_{F0} - \frac{2N\Omega\sigma}{1-\sigma} + \frac{N^2}{\rho} \frac{1+\sigma}{1-\sigma}. \quad (22.15)$$

This energy formula is known. The new result closes the specific unfinished integral expansion of [1]; it should be compared explicitly with the direct-Richardson all-order analysis of [46].

For an independent global check, let $P_n(R)$ be the monic polynomials for the exact all-pole weight

$$w(R; u) = \frac{e^{-uR}}{\prod_{j=0}^M (2\varepsilon_j - R)}.$$

Residues give the signed binomial weight $(-\sigma)^j \binom{M}{j}$, whose exact recurrence coefficients are

$$\mathbf{b}_n = 2\varepsilon_{F0} + \frac{2}{\rho} \frac{(1+\sigma)n - \sigma M}{1-\sigma}, \quad (22.16)$$

$$\mathbf{a}_n = -\frac{4}{\rho^2} \frac{\sigma n(M-n+1)}{(1-\sigma)^2}. \quad (22.17)$$

The trace sum $\sum_{n < N} \mathbf{b}_n$ gives exactly

$$E_A^{\text{per}} = 2N\varepsilon_{F0} + \frac{N(N-1)}{\rho} - 2N \left(\Omega - \frac{N-1}{\rho} \right) \frac{\sigma}{1-\sigma}. \quad (22.18)$$

Thus $E_{\text{ext}} - E_A^{\text{per}} = N(1+\sigma)/[\rho(1-\sigma)]$ is of order one. The global agreement of extensive terms follows from a separate exact calculation; it does not turn the formal cancellation theorem into a convergence theorem.

The original-coordinate transformation $R = 2\varepsilon_{F0} - \epsilon_c - i\epsilon_c y$ sends (22.12), at $s = \nu/\sigma$, to $\mathbf{b}_N \rightarrow 2\mu$, $\mathbf{a}_N \rightarrow -\Delta^2$, where

$$\mu = \varepsilon_{F0} + \Omega \frac{\nu(1+\sigma) - \sigma}{1-\sigma}, \quad \Delta = \frac{2\Omega\sqrt{\sigma\nu(1-\nu)}}{1-\sigma}. \quad (22.19)$$

Also $\mu = \frac{1}{2}\partial_N E_{\text{ext}}$. The value of Δ comes from the off-diagonal polynomial recurrence, not from the ground energy alone.

22.4 Blocked level: exact Toda response and its branch

Removing the pole of level p inserts a characteristic polynomial. The exact Heine identity is

$$\frac{Z_N^{[p]}}{Z_N} = \left\langle \prod_{j=1}^N (2\varepsilon_p - R_j) \right\rangle_A = P_N(2\varepsilon_p). \quad (22.20)$$

Differentiating orthogonality with $\partial_u w = -Rw$, at fixed original argument R , gives the exact Toda identity

$$\partial_u P_n(R) = a_n P_{n-1}(R). \quad (22.21)$$

All projections below degree $n-1$ vanish; the remaining projection is the ratio of consecutive norms. Therefore

$$D_{N,p}^{\text{per}} := E_{N,\text{pairs}}^{[p],\text{per}} - E_N^{\text{per}} = -a_N \frac{P_{N-1}(2\varepsilon_p)}{P_N(2\varepsilon_p)}. \quad (22.22)$$

Holding the rescaled y , which depends on coupling, fixed instead of R in this derivative would be incorrect.

For $r_N(z) = P_N(z)/P_{N-1}(z)$, the leading smooth recurrence is

$$r^2 - (z - 2\mu)r - \Delta^2 = 0. \quad (22.23)$$

Its sign is fixed by an exact positive representation. In lattice coordinates the monic polynomial at the actual blocked point is

$$Q_N(p) = N! x^N [\zeta^N] (1 + \zeta/\sigma)^p (1 + \zeta)^{M-p} > 0. \quad (22.24)$$

For $\alpha = p/M$, the positive coefficient saddle solves

$$\nu = \alpha \frac{\zeta_*}{\sigma + \zeta_*} + (1 - \alpha) \frac{\zeta_*}{1 + \zeta_*}. \quad (22.25)$$

The right-hand side increases strictly from zero to one. Exponential tilting turns the coefficient into a central probability of a sum of two groups of Bernoulli variables. Their variance is $O(M)$, including at $\alpha = 0, 1$, for fixed positive coupling and interior filling. The local central-limit estimate gives the ratio of adjacent coefficients as $\zeta_*^{-1}[1 + O(M^{-1})]$. Hence

$$r_N(2\varepsilon_p) \longrightarrow \varepsilon_p - \mu + \sqrt{(\varepsilon_p - \mu)^2 + \Delta^2} > 0. \quad (22.26)$$

This sign is proved even below μ , without a Richardson-root distribution. Substitution into the Toda response gives

$$D_{N,p}^{\text{per}} = \sqrt{(\varepsilon_p - \mu)^2 + \Delta^2} - (\varepsilon_p - \mu) + O(\Omega M^{-1}). \quad (22.27)$$

It is the known square root of [2], now obtained through the recurrence linked to the one-pair expansion.

22.4.1 All leading coefficients for the blocked response

At fixed blocked energy put

$$\eta_0 = \varepsilon_p - \varepsilon_{F0} + \frac{\Omega\sigma}{1-\sigma} > 0, \quad q_0 = \frac{\Omega(1+\sigma)}{1-\sigma}, \quad d_0 = \frac{4\Omega^2\sigma}{(1-\sigma)^2}.$$

The leading response $D(\nu)$ obeys the closed formal equation

$$D^2 + 2(\eta_0 - q_0\nu)D - d_0\nu(1 - \nu) = 0, \quad D(0) = 0. \quad (22.28)$$

For $D = \sum_{\ell \geq 1} d_\ell \nu^\ell$,

$$\begin{aligned} 2\eta_0 d_1 &= d_0, & 2\eta_0 d_2 &= 2q_0 d_1 - d_0 - d_1^2, \\ 2\eta_0 d_\ell &= 2q_0 d_{\ell-1} - \sum_{j=1}^{\ell-1} d_j d_{\ell-j} \quad (\ell \geq 3). \end{aligned} \quad (22.29)$$

Higher density coefficients generally survive and sum to the square root. At the two edge levels the exact finite responses are

$$D_{N,0}^{\text{per}} = \frac{2\Omega\nu}{1 - \sigma}, \quad D_{N,M}^{\text{per}} = \frac{2\Omega\sigma\nu}{1 - \sigma},$$

so the higher coefficients vanish at these special positions.

22.5 Odd, broken-pair, and pair-preserving excitations

Use $\xi = \varepsilon_p - \varepsilon_{F0}$, $\tilde{\mu} = \mu - \varepsilon_{F0}$, and $\mathcal{E}(\xi) = \sqrt{(\xi - \tilde{\mu})^2 + \Delta^2}$. Restoring the energy of the singly occupied orbital gives

$$E_{2N+1,p}^{\text{per}} - E_{2N}^{\text{per}} = \mu + \mathcal{E}(\xi_p) + O(\Omega/M), \quad (22.30)$$

$$E_{2N-1,p}^{\text{per}} - E_{2N}^{\text{per}} = -\mu + \mathcal{E}(\xi_p) + O(\Omega/M). \quad (22.31)$$

The second line has $N - 1$ active pairs; its finite filling shift contributes to the remainder.

For two distinct blocked levels and n active pairs, exactly

$$\frac{Z_n^{[p,q]}}{Z_n} = \frac{P_n(z_p)P_{n+1}(z_q) - P_{n+1}(z_p)P_n(z_q)}{z_q - z_p}, \quad z_j = 2\varepsilon_j. \quad (22.32)$$

For separated limiting energies, factor out $P_n(z_p)P_n(z_q)$. The residual divided difference of r_{n+1} tends to a finite nonzero value because

$$r'(z) = \frac{1}{2} + \frac{z - 2\mu}{2\sqrt{(z - 2\mu)^2 + 4\Delta^2}} > 0.$$

Its coupling derivative contributes only $O(\Omega/M)$. Thus the paired responses add. Taking $n = N - 1$ and restoring both single-electron energies cancels the chemical work:

$$\boxed{E_{B_2;p,q}^{\text{per}} - E_{2N}^{\text{per}} = \mathcal{E}(\xi_p) + \mathcal{E}(\xi_q) + O(\Omega/M)}. \quad (22.33)$$

This short proof is uniform for macroscopic separation. A common limiting energy requires a controlled confluent determinant, as in Chapter 10. Taking the separation to zero *after* the thermodynamic limit gives $2\mathcal{E}(\xi)$; it alone does not prove an error bound for two adjacent finite-lattice indices.

A seniority-zero marked insertion is different: for $a < N$ it uses $P_a^\downarrow = (-1)^{N+a} Q_{N,aa}$, with positive double sum

$$P_a^\downarrow = \sum_{b=0}^a \sum_{\ell=N}^M \frac{(M-b)! \ell! x^{\ell-b}}{(M-\ell)!(\ell-N)!b!(N-1-b)!(a-b)!(\ell-a)!(\ell-b)}. \quad (22.34)$$

Its two saddle coordinates are

$$\beta_\alpha, \tau_\alpha = \frac{\alpha + \nu + x \mp \sqrt{D_\alpha}}{2(1+x)}, \quad D_\alpha = (\alpha + \nu + x)^2 - 4(1+x)\alpha\nu.$$

The rate satisfies $x\partial_x\Phi = \tau_\alpha - \beta_\alpha = \sqrt{D_\alpha}/(1+x)$. Since $\Omega^2 D_\alpha = \mathcal{E}(\Omega\alpha)^2$, the marked-period energy is

$$-\partial_u \log P_a^\downarrow = 2\mathcal{E}(\Omega\alpha) + O(\Omega/M). \quad (22.35)$$

The upper chart follows by exact electron–hole reflection. The particle–hole convolution in Chapter 10 also explains this factor two before taking the limit.

The chemical-work-subtracted odd threshold is

$$\Delta_{\text{qp}} = \begin{cases} \Omega(\nu + x), & 0 < \nu \leq \nu_0, \\ \Delta, & \nu_0 \leq \nu \leq 1 - \nu_0, \\ \Omega(1 - \nu + x), & 1 - \nu_0 \leq \nu < 1, \end{cases} \quad \nu_0 = \frac{\sigma}{1 + \sigma}. \quad (22.36)$$

The candidate neutral even threshold is $2\Delta_{\text{qp}}$. These are constrained minima over $0 \leq \xi \leq \Omega$. The crossings $\tilde{\mu} = 0, \Omega$ are distinct from the Case A–B contour rate boundary; Δ is not always the smallest quasiparticle energy.

22.6 Case B: a finite scalar certificate for the complete cascade

Let $c = N - k > 0$ and choose lower cutoffs $1, \dots, k$, with the remaining c variables on all-pole contours. On an N -element residue subset $S \subset \{0, \dots, M\}$, define

$$W(S) = \prod_{j \in S} (-\sigma)^j \binom{M}{j} \Delta(S)^2, \quad H_r(S) = r - \#\{j \in S : j < r\}, \quad Z_0 = \sum_{|S|=N} W(S).$$

Counting allowable residue assignments gives exactly

$$Z_{B,k}^{\text{red}} = c! \sum_{|S|=N} W(S) \prod_{r=1}^k (c + H_r), \quad R_k = \frac{Z_{B,k}^{\text{red}}}{c! c^k Z_0} = \left\langle \prod_{r=1}^k (1 + H_r/c) \right\rangle_A. \quad (22.37)$$

The signed average is algebraic. At fixed k ,

$$E_{B,k}^{\text{per}} - E_A^{\text{per}} = \frac{2}{\rho(1-\sigma)} D \log R_k, \quad D = x\partial_x.$$

Writing $\xi_a = \mathbf{1}_{a \notin S}$, Boolean expansion gives

$$R_k = 1 + \sum_{\emptyset \neq A \subset \{0, \dots, k-1\}} C(A) \det Q_{N,A}, \quad 0 \leq C(A) \leq \prod_{a \in A} g_a, \quad g_a = \left(1 + \frac{1}{c}\right)^{k-a} - 1. \quad (22.38)$$

Here $C(\{a\}) = g_a$. The majorant allows each selected index to choose factors independently, removing the restriction that one factor select at most one index; it holds for every support size.

For $i, j < k < N$ the sign-gauged kernel has the exact representation

$$K_{ij} := (-1)^{N-j} Q_{N,ij} = \sum_{b=0}^i \sum_{\ell=N}^M \frac{U_i(b) V_j(\ell)}{\ell - b},$$

$$U_i(b) = \frac{(i)_b x^{-b} (M-b)!}{b! (N-1-b)!}, \quad V_j(\ell) = \frac{x^\ell \ell!}{(M-\ell)! (\ell-N)! (\ell-j)! j!}. \quad (22.39)$$

All summands are positive and $c+1 \leq \ell-b \leq M$. Put $P_i = K_{ii}$, $\Lambda = M/(c+1)$. Since $K_{ij} \leq \bar{U}_i \bar{V}_j / (c+1)$ and $P_i \geq \bar{U}_i \bar{V}_i / M$, permutation expansion gives, for $m = |A|$,

$$|\det Q_{N,A}| \leq m! \Lambda^m \prod_{a \in A} P_a, \quad |D \det Q_{N,A}| \leq mM m! \Lambda^m \prod_{a \in A} P_a. \quad (22.40)$$

The derivative bound uses the positive powers $x^{\ell-b}$, all of degree at most M . No confluent asymptotics for a growing minor is required.

Theorem 22.2 (Finite cascade certificate). *For the positive computable scalar*

$$\mathcal{S} = \Lambda \sum_{a=0}^{k-1} g_a P_a, \quad (22.41)$$

if $\mathcal{S} < 1$ then

$$|R_k - 1| \leq \frac{\mathcal{S}}{1 - \mathcal{S}}, \quad |DR_k| \leq \frac{M\mathcal{S}}{(1 - \mathcal{S})^2}.$$

If $\mathcal{S} < 1/2$, $R_k > 0$ and

$$\boxed{|E_{B,k}^{\text{per}} - E_A^{\text{per}}| \leq \frac{2\Omega}{1 - \sigma} \frac{\mathcal{S}}{(1 - \mathcal{S})(1 - 2\mathcal{S})}}. \quad (22.42)$$

Proof. Use $m!e_m(w) \leq (\sum_a w_a)^m$ for $w_a = \Lambda g_a P_a$. Summation over all support sizes gives $\sum_{m \geq 1} \mathcal{S}^m$ and $M \sum_{m \geq 1} m \mathcal{S}^m$. Also $R_k \geq (1 - 2\mathcal{S})/(1 - \mathcal{S})$. Division and (22.1) prove the result. The empty cascade $k = 0$ has $R_0 = 1$ separately. $\square$

Retain the signed singleton sum

$$T = \sum_{a=0}^{k-1} g_a (-1)^{N+a} P_a, \quad \widehat{\delta E} = \frac{2}{\rho(1 - \sigma)} \frac{DT}{1 + T}.$$

Then

$$\boxed{|(E_{B,k}^{\text{per}} - E_A^{\text{per}}) - \widehat{\delta E}| \leq \frac{4\Omega}{1 - \sigma} \frac{\mathcal{S}^2}{(1 - \mathcal{S})(1 - 2\mathcal{S})}}. \quad (22.43)$$

To verify the constant, put $U = R_k - 1 - T$. The omitted supports obey $|U| \leq \mathcal{S}^2/(1 - \mathcal{S})$ and $|DU| \leq M\mathcal{S}^2(2 - \mathcal{S})/(1 - \mathcal{S})^2$. Insert these bounds into $D \log R_k - DT/(1 + T) = DU/R_k - DTU/[R_k(1 + T)]$. The scalar certificate uses only (22.34); DP_a multiplies each summand by $\ell - b$.

22.6.1 Compact endpoint and thermodynamic consequence

For reproducibility let $L(z) = z \log z$, $L(0) = 0$. The rate is

$$\begin{aligned} \Phi(\alpha) &= L(1 - \beta) + L(\tau) - L(1 - \tau) - L(\tau - \nu) - L(\beta) - L(\nu - \beta) \\ &\quad - L(\alpha - \beta) - L(\tau - \alpha) + (\tau - \beta) \log x, \end{aligned} \quad (22.44)$$

where $\beta = \beta_\alpha$, $\tau = \tau_\alpha$. In particular,

$$\Phi(0) = \nu \log \sigma - \log(1 - \sigma) - \nu \log \nu - (1 - \nu) \log(1 - \nu).$$

Choose a compact lower Case B sequence with

$$k = \kappa M + O(1), \quad \Phi(\kappa) = 0, \quad \Phi'(\kappa) > 0, \quad 0 < \kappa < \nu < \frac{1}{2},$$

with $\kappa, \nu - \kappa$, and the slope separated from zero. The one-point saddle gives $P_a = M^{-1}C(a/M)e^{M\Phi(a/M)}[1 + O(M^{-1})]$. For $a = k - r$, $r \geq 1$, one has $g_{k-r} = O(r/M)$ and $P_{k-r} = O(M^{-1}e^{-\lambda_1 r})$, for some $\lambda_1 > 0$. Deeper supports are exponentially small, so $\mathcal{S} = O(M^{-2})$. The finite inequalities yield

$$\boxed{E_{B,k}^{\text{per}} = E_A^{\text{per}} + O(\Omega M^{-2}) = E_A^{\text{per}} + \widehat{\delta E} + O(\Omega M^{-4})}. \quad (22.45)$$

This transfers the extensive one-pair result to the specified cascade. An arbitrary growing cascade need not satisfy this estimate; an $O(\log M)$ endpoint displacement can change the power of M . The explicit scalar certificate remains valid whenever it is small. The rule $k = \kappa(\sigma)M + O(1)$ selects a sequence only and must not be differentiated. Upper Case B follows by exact electron-hole duality $N \mapsto M + 1 - N$; the finite $+1$ affects subleading terms.

22.7 Excitations in Case B: established and conditional transfers

For a marked point macroscopically separated above the lower endpoint, Chapter 12 controls the mixed-minor background factor:

$$\mathcal{M}_{N,k}^\downarrow(a) = 1 + O(M^{-2}), \quad D \log \mathcal{M}_{N,k}^\downarrow(a) = O(M^{-1}).$$

Its marked dispersion is therefore (22.35), with an additional cascade correction $O(\Omega M^{-2})$. The lower-chart minimum satisfies $0 < \kappa < \alpha_* = \tilde{\mu}/\Omega < \nu$ in the compact regime; the controlled separated marked sector reaches the leading threshold. Endpoint collisions and cascade birth require the separate treatment of Chapter 13.

For a finite blocked set, the exact algebra is available but the unblocked certificate does not automatically apply. Let n be the active pair number and $\chi_{\mathbf{p}}(S) = \prod_j \prod_{\ell \in S} (p_j - \ell)$. Then exactly

$$R_k^{[\mathbf{p}]} = \left\langle \prod_{r=1}^k (1 + H_r/(n-k)) \right\rangle_{[\mathbf{p}]}, \quad \langle F \rangle_{[\mathbf{p}]} = \frac{\langle \chi_{\mathbf{p}} F \rangle_A}{\langle \chi_{\mathbf{p}} \rangle_A}. \quad (22.46)$$

Its Boolean expansion uses the complementary kernel $Q_n^{[\mathbf{p}]}$ of the Christoffel-deformed ensemble. Smallness of the unblocked $R_k - 1$ alone bounds neither this kernel nor its normalization.

There is a concrete obstruction to simply copying $R_k = 1 + o(1)$. If a blocked index $a_0 < k$ is a deterministic hole, the insertion already contains the coupling-independent factor $(1 + 1/(n-k))^{k-a_0}$ when all other lower indices are occupied. This factor can have a nonunit finite limit. It should be removed before estimating the remaining deformation. For several blocked indices, the baseline factor is

$$A_{\mathbf{p}} = \prod_{r=1}^k \left(1 + \frac{\#\{p_j < r\}}{n-k} \right),$$

again independent of coupling at fixed finite labels.

Assumption 22.3 (Blocked-cascade control). *The Christoffel normalization does not vanish on the parameter set, and the normalized correction $\widehat{R}_k^{[\mathbf{p}]} = R_k^{[\mathbf{p}]} / A_{\mathbf{p}}$ obeys $D \log \widehat{R}_k^{[\mathbf{p}]} = o(M)$, uniformly for the specified fixed number of blocked levels and contour sequence.*

Under this assumption the blocked cascade energy changes by $o(\Omega)$. Thus the Case B leading period formulas are

$$\begin{aligned} E_{2N\pm 1,p;B}^{\text{per}} - E_{2N;B}^{\text{per}} &= \pm\mu + \mathcal{E}(\xi_p) + o(1), \\ E_{B_2;p,q;B}^{\text{per}} - E_{2N;B}^{\text{per}} &= \mathcal{E}(\xi_p) + \mathcal{E}(\xi_q) + o(1). \end{aligned} \quad (22.47)$$

Here the unblocked reference and the $N - 1$ -pair background must use the corresponding controlled sequences. A deformed analogue of $\mathcal{S} = O(M^{-2})$ would yield the stronger cascade error, but has not been proved in these additions. Positivity of the unblocked kernel does not establish it. Consequently the ground and separated marked Case B periods are controlled, while a complete uniform result for its blocked excitations remains open.

22.8 Physical interpretation and reproducibility

The formal theorem, exact Case A identities, and Case B certificate are complementary results with different scopes. A physical ground energy density requires a contour-to-eigenbranch identification with an $o(M)$ remainder. A physical gap requires the difference of the corresponding corrections to be $o(1)$. Neither condition follows merely from the cancellation theorem or the small cascade correction. The agreement with known μ, Δ , and excitation formulas is an independent consistency test. The continuum-root and Gaussian-branch calculations elsewhere in this manuscript are not silently imported into this derivation.

The directory `code/one_pair_saddle/` preserves the three 17 September 2026 working-source directories and their original algorithms. Its `PROVENANCE.md` distinguishes archived outputs from reruns for this revision. The rerun checks include:

1. exact rational planar loop recursion through δ^{25} , with an independent full finite- N Gaussian Wick expansion through δ^{11} , retaining symbolic σ ;
2. fifteen exact rational blocked logarithmic-derivative/Toda equalities, at both edges and below, at, and above μ , for $M = 100, 400, 1600$;
3. small-system signed subset sums compared with independent positive-diagonal Case B certificates, including the error of the differentiated singleton approximation.

The Gaussian program computes from potential coefficients and Wick rules rather than inserting the answer. Floating evaluations of positive Case B sums support the exact inequalities but are not rigorous interval bounds. These checks neither diagonalize the physical Richardson Hamiltonian nor prove uniform analytic continuation of the local formal series.

Part X

Duality and self-dual Richardson states

Chapter 23

Self-dual finite systems: a real boundary problem and controlled ground energy

The particle-hole construction supplies a second route to physical observables, starting directly from a finite Richardson system. In this chapter it gives a polynomial equation with a provably selected ground branch, an exact real boundary equation, and explicit energy bounds. The large-system BCS energy then follows by taking the limit of these finite bounds. No assumed curve of Richardson roots is used in that argument. The familiar BCS answer itself is not a new physical prediction. The relation to the self-dual states of Ref. [4] is made explicit below.

23.1 Normalization and the finite self-dual identity

Here there are N pairs on $2N$ unblocked orbitals. The physical single-electron energies, in the equally spaced specialization, are

$$\epsilon_{\pm i} = \pm(i - \frac{1}{2})d, \quad G = V = \lambda d > 0, \quad \omega_D = Nd. \quad (23.1)$$

The last equality defines the band edge by the edges of the level cells; the outermost orbital is at $(N - \frac{1}{2})d$. In this chapter and the next two, energies and spectral arguments inside the finite equations are measured in units of d :

$$H = H/d = \sum_{j=1}^{2N} x_j n_j - \lambda \sum_{j,k=1}^{2N} b_j^\dagger b_k, \quad x_{\pm i} = \pm a_i, \quad e = E/d. \quad (23.2)$$

The general finite arguments allow any distinct $a_i > 0$; the equally spaced choice is $a_i = 2i - 1$. The pair number n_j is zero or one. The diagonal interaction is included, and contributes $-\lambda N$ in this sector. For comparison with earlier chapters, their orbital count L is now $2N$. When used below, $q = 1/\lambda$ is a local inverse-coupling variable, not a polynomial or a contour parameter.

Let P_{jk} interchange the two local pair-occupation spaces. The commuting rational charges may be written

$$R_j = n_j + \lambda \sum_{k \neq j} \frac{I - P_{jk}}{x_j - x_k}. \quad (23.3)$$

They commute with H and obey

$$R_j^2 = R_j + \lambda \sum_{k \neq j} \frac{R_j - R_k}{x_j - x_k}, \quad \sum_j x_j R_j = H + \lambda N(N + 1)I. \quad (23.4)$$

These finite identities follow from the swap algebra. In particular, $\sum_{j < k} (I - P_{jk}) = N^2 I - \sum_{j \neq k} b_j^\dagger b_k$ in the N -pair space, which checks the energy constant. Reflection combined with particle-hole interchange sends R_{+i} to $I - R_{-i}$. A self-dual joint eigenvalue therefore satisfies $r_{+i} + r_{-i} = 1$. This is a condition on the joint charge data; self-duality is not synonymous with the absence of broken pairs or with membership in the entire even eigenspace of the reflection-hole operator.

Write the monic Richardson polynomial and its logarithmic derivative as

$$Q(z) = \prod_{\alpha=1}^N (z - r_\alpha), \quad \Lambda(z) = \frac{Q'(z)}{Q(z)}, \quad e = \sum_{\alpha} r_\alpha. \quad (23.5)$$

At a nonsingular Bethe solution, direct action of (23.3) gives $r_j = \lambda \Lambda(x_j)$. Consequently the self-dual relation reads $\lambda[\Lambda(a_i) + \Lambda(-a_i)] = 1$. Multiplication by $Q(a_i)Q(-a_i)$, followed by comparison of degree and leading coefficient, gives the finite polynomial identity

$$\boxed{\begin{aligned} \mathcal{F}_\lambda[Q](z) &:= Q(z)Q(-z) - \lambda\{Q'(z)Q(-z) + Q'(-z)Q(z)\} \\ &= A_N(z) := (-1)^N \prod_{i=1}^N (z^2 - a_i^2). \end{aligned}} \quad (23.6)$$

Here $Q'(-z)$ denotes the derivative polynomial evaluated at $-z$; it does not mean $dQ(-z)/dz$. The identity is polynomial and thus extends across the singular configurations of individual Bethe roots by coefficient continuation.

There is a useful converse check. At a simple root r of Q away from the orbital positions, differentiation of (23.6) and division by its value at r give

$$\frac{A'_N(r)}{A_N(r)} = \frac{Q''(r)}{Q'(r)} - \frac{1}{\lambda}. \quad (23.7)$$

Since $Q''(r_\alpha)/Q'(r_\alpha) = 2 \sum_{\beta \neq \alpha} (r_\alpha - r_\beta)^{-1}$, this is exactly

$$\frac{1}{\lambda} + \sum_{j=1}^{2N} \frac{1}{r_\alpha - x_j} - 2 \sum_{\beta \neq \alpha} \frac{1}{r_\alpha - r_\beta} = 0. \quad (23.8)$$

Thus the polynomial roots are actual Richardson roots wherever the individual-root equations are regular. Singular roots are represented by the regular polynomial branch, rather than by dividing a vanishing root difference.

For subsequent branch and numerical checks it is convenient to use

$$y_i = 2\lambda\Lambda(a_i) - 1 = r_{+i} - r_{-i}. \quad (23.9)$$

Reduction of (23.4) gives

$$y_i^2 = 1 + 2\lambda \sum_j B_{ij} y_j, \quad (23.10)$$

$$\begin{aligned} B_{ii} &= \frac{1}{a_i} + 2a_i \sum_{j \neq i} \frac{1}{a_i^2 - a_j^2}, \\ B_{ij} &= \frac{2a_j}{a_j^2 - a_i^2} \quad (i \neq j), \\ e &= \sum_i a_i y_i - \lambda N(N+1). \end{aligned} \quad (23.11)$$

For example, the energy formula also follows by multiplying (23.8) by r_α , summing, and then using the self-dual relation at each pair of orbitals.

23.2 The exact effective-spin spectrum and its scope

Proposition 23.1 (Finite spectral representation). *The 2^N self-dual joint branches continued from all sign choices $y_i = \pm 1$ at zero coupling have the same energy spectrum, with multiplicities, as the fixed 2^N -dimensional Hermitian matrix*

$$\boxed{H_{\text{SD}} = - \sum_i a_i X_i - \lambda(T^2 + T), \quad T = \sum_i Z_i.} \quad (23.12)$$

The matrices X_i, Y_i, Z_i are Pauli matrices. Its ground energy coincides with the physical ground energy of (23.2).

Proof. One can check the spectral statement using explicit commuting matrices, without postulating a local change of physical variables. Define

$$\begin{aligned} \hat{y}_i = & -X_i + \lambda B_{ii} I - \frac{\lambda}{a_i} Z_i \\ & - 2\lambda \sum_{j \neq i} \frac{a_i Z_i Z_j + a_j (X_i X_j + Y_i Y_j)}{a_i^2 - a_j^2}. \end{aligned} \quad (23.13)$$

Pauli multiplication gives

$$\begin{aligned} [\hat{y}_i, \hat{y}_j] &= 0, \quad \hat{y}_i^2 = I + 2\lambda \sum_j B_{ij} \hat{y}_j, \\ \sum_i a_i \hat{y}_i - \lambda N(N+1)I &= H_{\text{SD}}. \end{aligned} \quad (23.14)$$

At $\lambda = 0$ the simultaneous eigenvalues are all independent signs. The Jacobian of (23.10) at any such point is $2 \text{diag}(y_i)$ and is invertible. Hence each sign choice defines a unique local joint branch. The physical charges and (23.13) generate the same branches there.

For clarity, continuation here concerns the spectrum with algebraic multiplicities, not a claim that distinct energies can never cross. The polynomial equations $y_i^2 - 1 - 2\lambda \sum_j B_{ij} y_j = 0$ have leading monomials y_i^2 ; their quotient algebra has the 2^N square-free monomials as a basis. It therefore provides a finite algebraic continuation of all these branches. Equivalently, the characteristic polynomial of multiplication by $\sum_i a_i y_i - \lambda N(N+1)$ in that algebra agrees near zero with that of (23.12); both are monic polynomial identities in the coupling and so agree everywhere. Matching the physical joint branches near zero gives the stated equality of their continued spectral multiset, including collision limits. In particular, the effective characteristic polynomial divides the full physical characteristic polynomial near zero. Division by this monic polynomial has coefficients polynomial in λ ; its remainder vanishes on an interval and hence identically. The effective energies therefore remain physical energies at every real coupling. This formulation does not require the physical self-dual subspace to remain a fixed subspace as the parameters change.

Finally, the physical configuration matrix has connected nonpositive off-diagonal entries for $\lambda > 0$. Perron–Frobenius gives a unique positive ground vector. Particle–hole reflection preserves it and therefore makes its joint charge data self-dual. Alternatively, the ground branch is identified directly in the next theorem. Since the effective energies are physical self-dual energies and include the physical ground branch, their lowest energy is the same. $\square$

For $N = 1$, (23.12) gives $e = -\lambda \pm \sqrt{\lambda^2 + a_1^2}$, verifying both the normalization and the diagonal interaction. Omitting diagonal pairing would add λN to the entire matrix. That shift is harmless for gaps within one size, but must be restored when comparing different sizes.

The spin Hamiltonian itself belongs to a previously studied integrable family; see Ref. [47]. What is used here is its explicit spectral identification with the self-dual pairing branches. The proposition does not identify an effective local Pauli operator with a physical occupation operator, and it does not identify the first self-dual excitation with the first excitation of the full pairing space.

23.3 A finite ground-branch theorem

Call a real polynomial Hurwitz if all its roots lie in the open left half-plane. This stability condition selects the ground branch without requiring its roots to remain real.

Theorem 23.2 (Existence, uniqueness, and regularity). *For distinct $a_i > 0$ and every finite $\lambda \geq 0$, the identity (23.6) has a unique monic real Hurwitz solution continued from*

$$Q_w(z) = \prod_i (z + a_i). \quad (23.15)$$

Its coefficients are locally analytic in the coupling, its coefficient Jacobian is nonsingular, and it represents the physical ground state. For $\lambda > 0$ its roots and level variables satisfy

$$\Re r_\alpha < -2\lambda, \quad -1 < y_i < 0. \quad (23.16)$$

The coefficient statement remains regular when individual Richardson roots collide.

Proof. Exclusion and localization. On the imaginary axis the identity becomes

$$|Q(it)|^2 - 2\lambda \Re[Q'(it)\overline{Q(it)}] = \prod_i (t^2 + a_i^2) > 0. \quad (23.17)$$

Thus no real-coefficient solution has an imaginary-axis root, and the number of roots in each half-plane is constant on a continuous branch. For a Hurwitz solution put $r_\alpha = -u_\alpha + iv_\alpha$ with $u_\alpha > 0$. Division of (23.17) gives

$$1 - 2\lambda \sum_\alpha \frac{u_\alpha}{u_\alpha^2 + (t - v_\alpha)^2} > 0. \quad (23.18)$$

Every term is positive; setting $t = v_\alpha$ proves $u_\alpha > 2\lambda$. Moreover $\Re \Lambda(z) > 0$ for $\Re z \geq 0$. The boundary inequality in (23.18), decay at infinity, and the harmonic maximum principle imply $0 < \Lambda(a) < 1/(2\lambda)$ for every positive real a . This proves the box for y_i .

Uniqueness and Jacobian. If P, Q are two monic real Hurwitz solutions, then $w(z) = \log |P(z)/Q(z)|$ is harmonic in the closed right half-plane and tends to zero at infinity. With $x = \Re z$, its boundary condition is

$$w_x(it) = \frac{A_N(it)}{2\lambda |Q(it)|^2} [1 - e^{-2w(it)}]. \quad (23.19)$$

A positive maximum must be on the finite boundary, where its inward derivative is nonpositive, contradicting the strictly positive right side. A negative minimum gives the reverse contradiction. Hence $w = 0$ and the analytic ratio P/Q , normalized at infinity, equals one. For the Jacobian let $Q \mapsto Q + \varepsilon p$ with $\deg p < N$ and assume the linearized identity vanishes. The same calculation for $v = \Re(p/Q)$ yields

$$v_x(it) = \frac{A_N(it)}{\lambda |Q(it)|^2} v(it). \quad (23.20)$$

The maximum principle forces $v = 0$, and then $p = 0$. There are N nontrivial even coefficient equations and N unknown monic coefficients, so the Jacobian is invertible.

Existence and continuation. At zero coupling the linearization is

$$p(z)Q_w(-z) + p(-z)Q_w(z) = 0.$$

The two factors $Q_w(z), Q_w(-z)$ are coprime; hence Q_w divides p , so $p = 0$. The implicit-function theorem gives an analytic Hurwitz branch near zero. To exclude a finite endpoint of this branch, put $S = \sum_i a_i$ and $U = -e = \sum_\alpha u_\alpha$. Equations (23.11) and (23.16) give

$$\lambda N(N+1) < U < S + \lambda N(N+1). \quad (23.21)$$

The first even moment of (23.6) gives

$$\sum_\alpha v_\alpha^2 = \sum_\alpha u_\alpha^2 - 2\lambda U - \sum_i a_i^2 \leq U^2 - 2\lambda U - \sum_i a_i^2. \quad (23.22)$$

All roots and coefficients thus remain bounded on a finite coupling interval. Any limiting polynomial satisfies the identity; its roots cannot reach the imaginary axis by (23.17). Uniqueness and the nonsingular Jacobian extend it through the supposed endpoint.

Finally (23.7) identifies the branch near zero with the filled-negative-level ground eigenstate of the physical Hamiltonian. Its nondegenerate ground eigenvalue remains isolated at every positive coupling by Perron–Frobenius. Analytic continuation therefore keeps the polynomial branch on that ground eigenvalue. At exceptional root collisions the same conclusion follows by continuity of the coefficients. $\square$

The theorem gives a branch certificate at an exact polynomial solution and local quadratic convergence of coefficient Newton iteration. It does not prove convergence from an arbitrary initial polynomial or a condition number uniform in N . The box $-1 < y_i < 0$ is a necessary ground-state property; a stand-alone uniqueness criterion based only on that box has not been used.

23.4 Exact reduction to a real boundary equation

Let Q be the ground polynomial from Theorem 23.2 and set

$$u(y) = \log \left| \frac{Q(iy)}{Q_w(iy)} \right|, \quad \rho(y) = \sum_i \frac{a_i}{a_i^2 + y^2}, \quad D = |\partial_y|. \quad (23.23)$$

Our Fourier multiplier convention is $\widehat{Df}(k) = |k|\widehat{f}(k)$; equivalently

$$Df(y) = \frac{1}{\pi} \text{PV} \int_{\mathbb{R}} \frac{f(y) - f(t)}{(y-t)^2} dt. \quad (23.24)$$

The analytic logarithm $f(z) = \log[Q(z)/Q_w(z)]$ exists in the closed right half-plane and vanishes at infinity. Its real part is the Poisson extension of u , so $\Re f'(iy) = -Du(y)$. Dividing (23.17) by $|Q(iy)|^2$ now gives the exact equation

$$\boxed{2\lambda Du + 1 - e^{-2u} = 2\lambda \rho.} \quad (23.25)$$

Every original finite level remains in ρ . Integration along the auxiliary real coordinate is not a replacement of the orbital list by a continuum density.

The coefficient at infinity and the Schwarz representation give

$$f(z) = \frac{-e_0 - \sum_i a_i}{z} + O(z^{-2}), \quad \boxed{e_0 = -\sum_i a_i - \frac{1}{\pi} \int_{\mathbb{R}} u(y) dy.} \quad (23.26)$$

The exact profile is smooth, real, even, and $O(y^{-2})$ at infinity. It is strictly positive: a negative or zero global minimum would have $Du \leq 0$ and $1 - e^{-2u} \leq 0$, contradicting the positive source. In particular it is integrable. Since $\int Du = 0$, integration of (23.25) gives

$$\int_{\mathbb{R}} (1 - e^{-2u}) dy = 2\pi\lambda N. \quad (23.27)$$

The same Hamiltonian has filled-Fermi-sea expectation $e_{\text{FS}} = -\sum_i a_i - \lambda N$. Therefore its positive correlation-energy gain $C = E_{\text{FS}} - E_0$ has the exact representation

$$\boxed{\frac{C}{d} = \frac{1}{2\pi} \int_{\mathbb{R}} [2u - 1 + e^{-2u}] dy > 0.} \quad (23.28)$$

The diagonal interaction cancels in this definition. Counting its constant shift $-GN$ as a correlation energy would change the finite-size criterion artificially. At fixed finite spectrum, expansion of (23.25) gives

$$\begin{aligned} u &= \lambda\rho + \lambda^2(\rho^2 - D\rho) + O(\lambda^3), \\ e_0 &= -\sum_i a_i - \lambda N - \lambda^2 \sum_{i,j} \frac{1}{a_i + a_j} + O(\lambda^3). \end{aligned} \quad (23.29)$$

Here the integral identity $\pi^{-1} \int a_i a_j / [(a_i^2 + y^2)(a_j^2 + y^2)] dy = (a_i + a_j)^{-1}$ recovers the usual virtual transfer of a pair from a negative level to a positive one. This is a fixed-system expansion, not a uniform weak-coupling theorem as N grows.

23.4.1 Recovering the Richardson polynomial

The real profile retains the polynomial, not only its energy. The same Schwarz representation gives, for $\Re z > 0$,

$$Q(z) = Q_w(z) \exp \left[\frac{1}{\pi} \int_{\mathbb{R}} \frac{u(y)}{z - iy} dy \right]. \quad (23.30)$$

The branch is fixed by $Q/Q_w \rightarrow 1$ at infinity. Because the exact solution is known to be a monic polynomial of degree N , its values at N distinct positive real arguments determine its remaining coefficients by a linear interpolation problem. Its zeros are then the Richardson rapidities. One can alternatively solve the finite coefficient identity directly or reconstruct the polynomial from the level variables in (1.4). These are equivalent exact routes; numerical conditioning of interpolation and root extraction must be assessed separately. No accuracy uniform in large degree is inferred from the formal reconstruction formula.

23.5 Comparison and convergent refinement

Denote the residual of (23.25) by $F(v) = 2\lambda Dv + 1 - e^{-2v} - 2\lambda\rho$. For bounded decaying admissible functions ℓ, S ,

$$F(\ell) \leq 0, \quad F(S) \geq 0 \quad \implies \quad \ell \leq u \leq S. \quad (23.31)$$

Indeed, at a positive maximum of $\ell - u$, the nonlocal difference $D(\ell - u)$ is nonnegative and the nonlinear difference is strictly positive. The supersolution proof is the same. The

trial functions need not come from polynomials. Thus comparison immediately supplies two energy bounds through their areas.

The equation is also the Euler equation of the strictly convex functional

$$\mathcal{J}[v] = \lambda \langle v, Dv \rangle + \int_{\mathbb{R}} \left[v + \frac{e^{-2v} - 1}{2} - 2\lambda \rho v \right] dy. \quad (23.32)$$

On bounded admissible functions in $H^{1/2} \cap L^2$, its second variation is strictly positive. Existence here is already supplied by the polynomial theorem, and comparison proves uniqueness in the decaying bounded class; no separate coercivity assumption is needed.

A useful exact refinement starts at $u_0 = 0$ and solves

$$(2\lambda D + 2e^{-2u_n})h_n = -F(u_n), \quad u_{n+1} = u_n + h_n. \quad (23.33)$$

For completeness, an explicit global supersolution is

$$S_0(y) = \frac{1}{2} \sum_i \log \frac{y^2 + (a_i + 2\lambda N)^2}{y^2 + a_i^2}. \quad (23.34)$$

To check it, put $b_i = a_i + 2\lambda N$ and $d_i = (b_i^2 - a_i^2)/(y^2 + b_i^2)$. Then $F(S_0) = 1 - \prod_i (1 - d_i) - 2\lambda \sum_i b_i/(y^2 + b_i^2) \geq 0$, because $b_i^2 - a_i^2 \geq 2\lambda N b_i$ and $N^{-1} \sum_i d_i \leq \max_i d_i \leq 1 - \prod_i (1 - d_i)$.

Proposition 23.3 (Finite-parameter monotone Newton construction). *For each finite positive spectrum and $\lambda > 0$, the sequence (23.33) satisfies*

$$0 \leq u_n \leq u_{n+1} \leq u \leq S_0, \quad F(u_n) \leq 0, \quad u_n \longrightarrow u \quad (23.35)$$

in L^1 and uniformly. Convergence is eventually quadratic. Consequently the associated energies converge monotonically from above.

Proof. Write $M = \|S_0\|_{\infty}$, $m = 2e^{-2M} > 0$. Every linearized operator in the interval $[0, S_0]$ has a positive inverse of L^p norm at most $1/m$. This follows from the positive Poisson semigroup of D and the resolvent of a bounded multiplication potential bounded below by m . Concavity of $1 - e^{-2v}$ shows both that $F(u_n + h_n) \leq 0$ and that $0 \leq h_n \leq u - u_n$. If $s_n = -F(u_n)$, the exact new residual is

$$0 \leq s_{n+1} = e^{-2u_n}(2h_n - 1 + e^{-2h_n}) \leq 2h_n^2. \quad (23.36)$$

Since $\sum_n h_n \leq u$, dominated convergence makes h_n and then s_n tend to zero in each finite L^p . The error $e_n = u - u_n$ satisfies $(2\lambda D + V_n)e_n = s_n$ with $m \leq V_n \leq 2$; hence $e_n \rightarrow 0$ in those spaces. For the uniform norm, the kernel G_m of $(2\lambda D + m)^{-1}$ obeys $\|G_m\|_2^2 = (2\pi\lambda m)^{-1}$, so $\|e_n\|_{\infty} \leq \|s_n\|_2 / \sqrt{2\pi\lambda m} \rightarrow 0$. Finally Taylor's theorem gives $\|e_{n+1}\|_p \leq e^{2M} \|e_n\|_{\infty} \|e_n\|_p$. $\square$

The first step is already physical:

$$u_1 = \lambda(1 + \lambda D)^{-1}\rho, \quad \frac{1}{\pi} \int u_1 = \lambda N, \quad e_1 = e_{\text{FS}}. \quad (23.37)$$

One can maintain a simultaneous upper sequence. Given a subsolution ℓ and supersolution S , put $f(v) = 1 - e^{-2v}$, $s = -F(\ell)$, and $c(y) = [f(S) - f(\ell)]/(S - \ell)$, with the derivative used when they coincide. Solve

$$(2\lambda D + f'(\ell))h = s, \quad (2\lambda D + c)w = s, \quad \ell_{\text{new}} = \ell + h, \quad S_{\text{new}} = \ell + w. \quad (23.38)$$

Positivity and the chord inequality for a concave function give $\ell \leq \ell_{\text{new}} \leq u \leq S_{\text{new}} \leq S$; the residual signs are retained. The area interval shrinks to the exact energy. These are continuous analytic bounds. A Fourier or quadrature discretization needs its own error enclosure before its computed endpoints can be called certified. Neither a uniformly small iteration count nor numerical superiority over finite level-variable solvers has been proved.

23.6 A collective subsolution retaining the discrete spectrum

The saturated nonlinear equation suggests the auxiliary obstacle problem

$$w \geq 0, \quad k := 2\lambda Dw + 1 - 2\lambda\rho \geq 0, \quad wk = 0. \quad (23.39)$$

If $2\lambda \sum_i a_i^{-1} \leq 1$, take $\Delta_c = 0$ and $w = 0$. Otherwise there is exactly one $\Delta_c > 0$ such that

$$\sum_i \frac{1}{\sqrt{a_i^2 + \Delta_c^2}} = \frac{1}{2\lambda}. \quad (23.40)$$

This follows from strict monotonicity of its left side. With $b_i = \sqrt{a_i^2 + \Delta_c^2}$ and $r = \sqrt{\Delta_c^2 - y^2}$, the explicit solution is

$$w(y) = \begin{cases} \sum_i [\operatorname{atanh}(r/b_i) - r/b_i], & |y| < \Delta_c, \\ 0, & |y| \geq \Delta_c. \end{cases} \quad (23.41)$$

It is nonnegative and vanishes as the $3/2$ power of the distance to the endpoints. Its first derivative is continuous, and its fractional regularity suffices for comparison.

Here is a direct verification that fixes the normalization. In the right half-plane take $S(z) = \sqrt{z^2 + \Delta_c^2}$, positive on the positive real axis, and define the derivative of a decaying analytic function by

$$f'_c(z) = \frac{1}{2\lambda} + \sum_i \frac{a_i}{z^2 - a_i^2} - zS(z) \sum_i \frac{1}{b_i(z^2 - a_i^2)}. \quad (23.42)$$

The poles at $z = a_i$ are removable. The constant at infinity vanishes by (23.40); choose $f_c(\infty) = 0$. Inside the cut $\Re f'_c(iy) = 1/(2\lambda) - \rho(y)$ and

$$w'(y) = -y\sqrt{\Delta_c^2 - y^2} \sum_i \frac{1}{b_i(a_i^2 + y^2)}. \quad (23.43)$$

Integration to the endpoint gives (23.41). Outside the cut its real primitive is zero, and evaluation of (23.42) gives

$$k(y) = 2\lambda|y|\sqrt{y^2 - \Delta_c^2} \sum_i \frac{1}{b_i(a_i^2 + y^2)} \geq 0 \quad (|y| \geq \Delta_c). \quad (23.44)$$

Thus (23.39) holds everywhere. Inside its support $F(w) = -e^{-2w} < 0$; outside, $Dw \leq 0$ because $w = 0$ is a global minimum, so $F(w) = 2\lambda Dw - 2\lambda\rho < 0$. Comparison therefore proves $0 \leq w \leq u$.

Expansion of (23.42) at infinity gives its area and the collective energy:

$$\begin{aligned} \frac{1}{\pi} \int w &= \sum_i (b_i - a_i) - \frac{\Delta_c^2}{4\lambda}, \\ e_0 \leq e_{\text{coll}} &:= - \sum_i b_i + \frac{\Delta_c^2}{4\lambda}. \end{aligned} \quad (23.45)$$

This is the familiar discrete BCS expression arising as a rigorous upper bound from a finite auxiliary problem. The dimensional quantity $d\Delta_c$ equals twice the usual BCS order parameter in that approximation; it has not been identified with an exact finite spectral gap. The appearance of its positive solution is a threshold of the auxiliary obstacle problem, not a nonanalyticity of the exact finite ground state. In particular the separate inequality $e_0 \leq -\sum_i a_i - \lambda N$ does not justify subtracting λN from (23.45) by hand.

23.7 An explicit finite error bound and the leading BCS limit

Theorem 23.4 (Collective energy enclosure). *Let Δ_c be defined above, including its zero branch, and put*

$$K = \Delta_c^2 + a_{\max}^2, \quad \mathcal{L} = \max\{16\lambda, 4\sqrt{K}\}, \quad c = \max\left\{2, \frac{5}{8} \log \frac{\mathcal{L}}{\lambda}\right\}. \quad (23.46)$$

Then for every finite system

$$w(y) \leq u(y) \leq w(y) + \frac{c\mathcal{L}^2}{\mathcal{L}^2 + y^2}, \quad \boxed{e_{\text{coll}} - c\mathcal{L} \leq e_0 \leq e_{\text{coll}}}. \quad (23.47)$$

In physical units the enclosure width is $d c \mathcal{L}$.

Proof. For $|y| \geq \Delta_c$, (23.44) and the normalized weights $2\lambda/b_i$ give

$$k(y) \geq 1 - \frac{K}{y^2}. \quad (23.48)$$

Indeed $k = \sqrt{1 - \Delta_c^2/y^2} \sum_i (2\lambda/b_i)/(1 + a_i^2/y^2)$, and $\sqrt{1-s}/(1+t) \geq 1-s-t$ for $0 \leq s \leq 1$, $t \geq 0$. For the zero branch the same bound follows from $2\lambda \sum_i a_i \leq a_{\max}^2$. Set $v = c\mathcal{L}^2/(\mathcal{L}^2 + y^2)$. The exact derivative is

$$Dv = \frac{c\mathcal{L}(\mathcal{L}^2 - y^2)}{(\mathcal{L}^2 + y^2)^2}, \quad F(w+v) = k + 2\lambda Dv - e^{-2(w+v)}. \quad (23.49)$$

Writing $r = \mathcal{L}/\lambda \geq 16$ gives $2 \leq c \leq r/8$ and $e^{-8c/5} \leq 1/r$. On $|y| \leq \mathcal{L}/2$, $k \geq 0$, $Dv \geq 12c/(25\mathcal{L})$, and $v \geq 4c/5$, so $F(w+v) \geq (\lambda/\mathcal{L})(24c/25 - 1) > 0$. On $\mathcal{L}/2 \leq |y| \leq \mathcal{L}$, the support of w is absent, $k \geq 3/4$, $Dv \geq 0$, and $v \geq c/2$; hence the residual is at least $3/4 - e^{-c} > 0$. Finally, on $|y| \geq \mathcal{L}$, use $Dv \geq -c\mathcal{L}/y^2$ and $1 - e^{-2v} \geq 2v/(1+2v)$ to obtain

$$y^2 F(w+v) \geq \frac{c\mathcal{L}^2}{1+c} - K - 2\lambda c\mathcal{L} \geq \left(\frac{2}{3} - \frac{1}{16} - \frac{1}{4}\right) \mathcal{L}^2 > 0. \quad (23.50)$$

Thus $w+v$ is a global supersolution. Comparison and $\pi^{-1} \int v = c\mathcal{L}$ prove the enclosure. $\square$

For $a_i = 2i - 1$ at fixed $\lambda > 0$, the finite equation (23.40) has a positive root for all sufficiently large N . Ordinary Riemann sums, taken only at this stage, give

$$\frac{\Delta_c}{N} \rightarrow \frac{2}{\sinh(1/\lambda)}, \quad \frac{e_{\text{coll}}}{N^2} \rightarrow -\coth(1/\lambda). \quad (23.51)$$

Indeed the first limit solves $\frac{1}{2} \int_0^2 (x^2 + \delta^2)^{-1/2} dx = 1/(2\lambda)$, and insertion into the collective energy cancels the logarithmic integral term. Meanwhile $\mathcal{L} = O(N)$ and $c = O(\log N)$. The finite bound therefore proves

$$0 \leq e_{\text{coll}} - e_0 = O(N \log N), \quad \boxed{\frac{E_0}{dN^2} \rightarrow -\coth(1/\lambda)}. \quad (23.52)$$

At fixed $\omega_D = Nd$ this is an $O(\omega_D \log N)$ bound on the total energy remainder and an error per pair tending to zero. It recovers the established thermodynamic answer [11, 13, 23] from the finite self-dual equation. The constants can deteriorate when λ varies with N ; a fixed-coupling theorem cannot be used as a uniform crossover estimate.

With the same Fermi-sea reference, the leading correlation energy and the conventional bulk order parameter are

$$C_{\text{BCS}} = dN^2[\coth(1/\lambda) - 1], \quad \Delta_0 = \frac{\omega_D}{\sinh(1/\lambda)}. \quad (23.53)$$

The absolute-energy enclosure is deliberately coarse: it does not establish the true next coefficient of order dN , an $O(d)$ remainder, or relative accuracy for a very small C . The archived first-correction boundary calculation reproduces the known Yuzbashyan–Baytin–Altshuler coefficient, but its logarithmic edge layer has not been given a uniform remainder proof in this deposit. That formal calculation is kept separate from Theorem 23.4. The next chapters treat finite excitation differences and crossover statements with their own error estimates and sector restrictions.

23.8 Verification and limits of the construction

The finite algebra, half-plane maximum principles, and the explicit supersolution are the proofs used in this chapter. The source archive retains the derivations `background_ground_branch.md`, `boundary_newton_proof.md`, `discrete_obstacle_proof.md`, and the independent supersolution audit. The recovered `selfdual_iteration5` archive contains the explicit effective charges and the separate proof about the first self-dual level used in the next chapter. As an additional check during this integration, direct exact rational matrix multiplication verified (23.14) at $N = 2$, $a = (1, 3)$ and at $N = 3$, $a = (1, 2, 5)$, both with $\lambda = 2/5$. These small checks verify conventions and signs; they do not replace the arbitrary- N operator identities.

The method supplies an exact finite reformulation and a controlled leading limit. Its boundary coordinate is auxiliary, its collective threshold is not a finite-system phase transition, and its spin representation is spectral rather than a supplied map of all observables. Those distinctions are essential when using the results for excitation energies or for claims of computational improvement.

Chapter 24

Blocked defects, the first self-dual gap, and controlled crossover

This chapter turns the exact ground-state boundary equation into estimates for differences of energies. It treats the cost of blocking a reflected pair of orbitals and the first excitation within the self-dual (SD) spectrum. A finite coalescence argument relates them with an error of at most one level spacing. Convexity permits differentiation of the collective approximation without differentiating an uncontrolled remainder. Together with a variational estimate for the condensation energy, this gives joint weak-coupling limits in which the gap has its BCS value while the condensation energy does not.

The finite SD spectral correspondence, the Hurwitz ground branch, and the boundary comparison estimates are the dependencies supplied by Chapter 23. The excitation-ordering argument below is also retained in `selfdual_iteration5.zip`. These are mathematical arguments in the working manuscript; numerical checks do not replace their proofs or independent external review. In particular, the results do not identify the minimum over all pair-preserving states with the first SD excitation.

24.1 Observables and a common normalization

Let the one-electron orbitals and coupling be

$$\varepsilon_{\pm i} = \pm \frac{da_i}{2}, \quad 0 < a_1 < \cdots < a_N, \quad V = \lambda d > 0. \quad (24.1)$$

There are N pairs at half filling and the diagonal interaction is included. Write $e = E/d$, $q = \lambda^{-1}$, and $D = |\partial_y|$. For equally spaced levels, $a_i = 2i - 1$ and $\omega = Nd$ is the cell-edge half bandwidth; the outermost orbital is at $(N - \frac{1}{2})d$.

For $A = \{a_1, \dots, a_N\}$ let $e_0(A)$ be the exact ground energy at the same λ in every system being compared. Removing $\pm a_j$ leaves $N - 1$ pairs on $2N - 2$ orbitals. Placing one unpaired electron on each removed orbital adds zero total kinetic energy. Thus

$$B_j = e_0(A \setminus \{a_j\}) - e_0(A) \quad (24.2)$$

is the dimensionless energy of a physical mirrored blocked excitation; its physical energy is dB_j . This concerns a specified blocked sector, not a minimization over asymmetric blocking patterns. The second observable is the first SD pair-preserving excitation,

$$\Gamma_{\text{SD}} = d(e_{1,\text{SD}} - e_0). \quad (24.3)$$

24.2 An exact response formula for a blocked defect

Set $p_a(y) = a/(a^2 + y^2)$ and introduce the positive source interpolation

$$\rho_\eta = \sum_{i \neq j} p_{a_i} + \eta p_{a_j}, \quad 2Du_\eta + q(1 - e^{-2u_\eta}) = 2\rho_\eta, \quad 0 \leq \eta \leq 1, \quad (24.4)$$

with interpolated energy

$$\mathcal{E}_\eta = - \sum_{i \neq j} a_i - \eta a_j - \frac{1}{\pi} \int_{\mathbb{R}} u_\eta(y) dy.$$

The endpoints are the exact physical ground energies by (23.26). Fractional η is an auxiliary source interpolation, not a fractional number of physical pairs. The comparison construction in Chapter 23 applies to nonnegative weights and bounds the solution throughout the interpolation. Its positive linearized inverse supplies smooth parameter dependence. The profiles and derivatives used below have integrable decay.

Put $f_\eta = e^{-2u_\eta}$ and $\mathcal{L}_\eta = 2D + 2qf_\eta$. Distinguish the profile u_η from its derivative $v_\eta = \partial_\eta u_\eta$. Differentiation gives

$$\mathcal{L}_\eta v_\eta = 2p_{a_j}, \quad \mathcal{L}_\eta \partial_q u_\eta = -(1 - f_\eta). \quad (24.5)$$

Hence $v_\eta \geq 0$ and $\partial_q u_\eta \leq 0$. Integration and self-adjoint reciprocity give

$$q \int f_\eta v_\eta dy = \pi, \quad (24.6)$$

$$2 \int p_{a_j} \partial_q u_\eta dy = - \int (1 - f_\eta) v_\eta dy. \quad (24.7)$$

Adding them yields

$$\frac{1}{\pi} \int v_\eta dy = \lambda - \frac{2}{\pi} \int p_{a_j} \partial_q u_\eta dy. \quad (24.8)$$

No constant needs to be treated as an L^2 test vector. Define the Poisson average $F_\eta(q) = \pi^{-1} \int p_{a_j} u_\eta dy$. Integrating $-\partial_\eta \mathcal{E}_\eta$ proves the finite identity

$$\boxed{B_j = a_j + \lambda - 2 \int_0^1 F'_\eta(q) d\eta, \quad B_j \geq a_j + \lambda.} \quad (24.9)$$

All the response of the remaining system is included in this integral. It is not a first-order expansion in the defect. The factors of two are specific to $q = 1/\lambda$; part of the retained source material uses $1/(2\lambda)$.

24.3 Convexity, the collective profile, and finite bounds

A small function error cannot generally be differentiated. Here a second q derivative gives the additional control:

$$\mathcal{L}_\eta \partial_q^2 u_\eta = -4f_\eta \partial_q u_\eta + 4qf_\eta (\partial_q u_\eta)^2 \geq 0. \quad (24.10)$$

Thus $F''_\eta \geq 0$. Set $w_i = 1$ except $w_j = \eta$. The collective parameter $\Delta_{c,\eta} \geq 0$ solves

$$\sum_i \frac{w_i}{\sqrt{a_i^2 + \Delta_{c,\eta}^2}} = \frac{q}{2} \quad (24.11)$$

when a positive solution exists, and is zero otherwise. This auxiliary parameter is not an exact finite spectral gap. With $R_y = \sqrt{\Delta_{c,\eta}^2 - y^2}$ and $b_i = \sqrt{a_i^2 + \Delta_{c,\eta}^2}$, the collective profile is

$$w_\eta(y) = \begin{cases} \sum_i w_i [\operatorname{atanh}(R_y/b_i) - R_y/b_i], & |y| < \Delta_{c,\eta}, \\ 0, & |y| \geq \Delta_{c,\eta}. \end{cases} \quad (24.12)$$

The weighted comparison bound is

$$0 \leq u_\eta - w_\eta \leq \frac{cL^2}{L^2 + y^2}, \quad L = \max\{16\lambda, 4\sqrt{\Delta_{c,\eta}^2 + a_{\max}^2}\}, \quad (24.13)$$

$$c = \max\{2, \frac{5}{8} \log(L/\lambda)\}. \quad (24.14)$$

For $G_\eta = \pi^{-1} \int p_{a_j} w_\eta$ this gives the smaller bound

$$0 \leq F_\eta - G_\eta \leq \frac{cL}{a_j + L}. \quad (24.15)$$

It follows by integrating the product of two Lorentzian kernels. Differentiating the profile and (24.11) cancels the common sum $\sum_i w_i/b_i^3$, leaving

$$\partial_q w_\eta(y) = -\frac{1}{2} \sqrt{\Delta_{c,\eta}^2 - y^2}_+.$$

There is no moving-endpoint contribution; the profile vanishes there. The identity extends continuously through the zero-parameter threshold. The Poisson average of a semicircle is

$$\frac{1}{\pi} \int_{-\delta}^{\delta} \frac{a \sqrt{\delta^2 - y^2}}{a^2 + y^2} dy = \sqrt{a^2 + \delta^2} - a,$$

so

$$G'_\eta(q) = \frac{a_j - \sqrt{a_j^2 + \Delta_{c,\eta}^2}}{2}. \quad (24.16)$$

The quasiparticle square root has arisen from an area response of the real boundary equation, without an assumed arc of Richardson roots.

For a finite bound choose $0 < h < q$. Let δ_s be the collective parameter of the full list at $q - h$, and δ_w that of the deleted list at $q + h$, allowing zero values. Set

$$\lambda_s = (q - h)^{-1}, \quad \lambda_w = (q + h)^{-1}, \quad (24.17)$$

$$L_* = \max\{16\lambda_s, 4\sqrt{\delta_s^2 + a_{\max}^2}\}, \quad c_* = \max\{2, \frac{5}{8} \log(L_*/\lambda_w)\}, \quad (24.18)$$

$$\epsilon_* = c_* L_*/(a_j + L_*). \quad (24.19)$$

Convexity bounds $F'_\eta(q)$ by its adjacent secants. Replacing F_η by G_η in those secants costs at most ϵ_*/h , since the error in (24.15) is one sided. Monotonicity of the collective root in the weights and q then proves

$$\lambda + \sqrt{a_j^2 + \delta_w^2} - \frac{2\epsilon_*}{h} \leq B_j \leq \lambda + \sqrt{a_j^2 + \delta_s^2} + \frac{2\epsilon_*}{h}. \quad (24.20)$$

Only two monotone scalar roots are needed to evaluate the bracket. Its width prevents interpreting the displayed λ as the first finite-size correction to the square-root spectrum.

24.4 Why the continued branch is the first SD excitation

Proposition 23.1 realizes the finite SD energies as the spectrum of

$$H_N = - \sum_i a_i X_i - \lambda(T^2 + T), \quad T = \sum_i Z_i. \quad (24.21)$$

The constant $-\lambda N$ is retained; it matters when comparing different system sizes. The following finite ordering argument concerns this spectral realization. No local-operator identification between the effective spins and the physical pairing operators is used.

More generally consider $H = - \sum_i a_i X_i - \sum_{i < j} J_{ij} Z_i Z_j - \sum_i b_i Z_i$, with $a_i, J_{ij} > 0$ and real b_i . Perron–Frobenius gives a positive ground vector ψ_0 . Its logarithm is strictly supermodular:

$$\Delta_{ij} \log \psi_0 = \log \psi_0(++) + \log \psi_0(--) - \log \psi_0(+-) - \log \psi_0(-+) > 0. \quad (24.22)$$

To prove this, evolve $\psi_t = e^{-tH} \mathbf{1}$ and put $\ell_t = \log \psi_t$. The componentwise equation is $\dot{\ell}_t(s) = \sum_k a_k \exp[\ell_t(s^k) - \ell_t(s)] - V(s)$. On a boundary face $\Delta_{ij} \ell = 0$ of the supermodular cone, the $k = i, j$ mixed differences vanish. For $k \neq i, j$, the two increments x, y of $v(s) = \ell(s^k) - \ell(s)$ have the same sign, and $\Delta_{ij} v \geq 0$. Consequently $\Delta_{ij} e^v \geq e^{v--} (e^x - 1)(e^y - 1) \geq 0$. The potential adds $-\Delta_{ij} V = 4J_{ij} > 0$. The flow therefore enters the cone strictly. A zero mixed difference in the stationary ground-state limit would give the same strict inequality against a constant stationary right-hand side, a contradiction.

The ground transform defines the reversible generator

$$(\mathcal{L}f)(s) = \sum_i c_i(s) [f(s^i) - f(s)], \quad c_i(s) = a_i \frac{\psi_0(s^i)}{\psi_0(s)}.$$

For the oriented difference $\nabla_i f = f(s_i = +1) - f(s_i = -1)$, (24.22) makes these rates attractive. Writing $v = \nabla f$, direct subtraction gives

$$\begin{aligned} \nabla_i \mathcal{L}f = & - [c_i(s^+) + c_i(s^-)] v_i(\sigma) + \sum_{j \neq i} c_j(s^+) [v_i(\sigma^j) - v_i(\sigma)] \\ & + \sum_{j \neq i} \kappa_{ij}(\sigma) v_j(\eta_{ij}), \end{aligned} \quad (24.23)$$

$$\kappa_{ij}(\sigma) = -\sigma_j [c_j(s^+) - c_j(s^-)] > 0. \quad (24.24)$$

Here η_{ij} fixes spin i at -1 . Thus $\nabla \mathcal{L} = \mathcal{A} \nabla$ with an irreducible matrix $\mathcal{A}$ having nonnegative off-diagonal entries on oriented edges. Positive c_j connect edges of the same direction and positive κ_{ij} connect directions.

Its leading eigenvalue β is simple. The corresponding positive vector is a gradient: the invariant space $\text{im } \nabla$ contains $\nabla T = 2\mathbf{1}$, whose nonzero Perron projection remains in that space. Every nonconstant eigenfunction of $\mathcal{L}$ also lifts to an eigenvector of $\mathcal{A}$. Moreover $\beta \neq 0$: otherwise $\mathcal{L}f$ would be constant; averaging with ψ_0^2 would make it zero, and irreducibility would force f constant. Since $\mathcal{L}$ is similar to $-(H - E_0)$, this proves $\beta = -(E_1 - E_0)$, simplicity of E_1 , and, after choosing its sign, $\nabla(\psi_1/\psi_0) > 0$.

For (24.21), $J_{ij} = 2\lambda$, $b_i = \lambda$. At $\lambda = 0$ the unique first branch for an ordered list is $(+, -, \dots, -)$. Simplicity prevents it from changing SD spectral order at positive λ . The finite polynomial continuation in the retained branch proof identifies its one right-half-plane zero as a simple real root $0 < r < a_1$, all other roots lying to the left. Imaginary-axis crossings are excluded by the positive polynomial identity on that axis, with the endpoint singularities handled by the finite branch argument. This orders the SD spectrum. Agreement with the first full paired level in small-system diagonalizations remains a check, not a theorem excluding a lower non-SD state.

24.5 Coalescence gives an error of at most one spacing

Fix $a_2, \dots, a_N$ and λ , and let $a_1 = \epsilon \downarrow 0$. Then $r \rightarrow 0$ on the first SD branch. The finite polynomial coefficients remain bounded. In the Richardson equations, a purported divergent group of roots would leave a nonzero multiple of $1/\lambda$ upon summing its equations, whereas the mutual terms cancel. For the polynomial functional of Chapter 23,

$$\mathcal{F}_\lambda[Q] = Q(z)Q(-z) - \lambda[Q'(z)Q(-z) + Q'(-z)Q(z)],$$

one has $\mathcal{F}_\lambda[z^m R] = (-1)^m z^{2m} \mathcal{F}_\lambda[R]$. The double zero of its prescribed right-hand side at the origin forces a single zero of Q_1 there. The remaining Hurwitz factor is the deleted-system ground polynomial:

$$Q_1(z; 0) = zR_0(z), \quad e_{1,\text{SD}}(0) = e_b, \quad e_b = e_0(\{a_2, \dots, a_N\}). \quad (24.25)$$

Physically, the two coincident central pairing pseudospins may form $(|1, 0\rangle - |0, 1\rangle)/\sqrt{2}$, annihilated by their total pair creation and annihilation operators. This is a singlet of pair occupations, distinct from two real unpaired electrons.

On a fixed effective Hilbert space, $\|H_N(\epsilon) - H_N(0)\| = \epsilon$. The min-max principle for the ordered first level therefore gives $|e_{1,\text{SD}}(\epsilon) - e_b| \leq \epsilon$. Subtracting the same full ground energy yields

$$\left| \frac{\Gamma_{\text{SD}}}{d} - B_1 \right| \leq a_1. \quad (24.26)$$

For $a_1 = 1$, $|\Gamma_{\text{SD}} - dB_1| \leq d$, not $2d$: no second ground-energy estimate was needed. Together with (24.9), it also gives $\Gamma_{\text{SD}}/d \geq \lambda$.

24.6 Controlled BCS limits for the excitation energies

For $a_i = 2i - 1$ at fixed $\lambda > 0$, let $\Delta_{c,N}$ denote the full-list collective root. Sum-integral comparison gives $\Delta_{c,N}/N \rightarrow 2/\sinh(1/\lambda)$. The Poisson error is $O_\lambda(\log N)$, deletion of one weight changes the collective square root by $O_\lambda(1)$, and its q derivative is $O_\lambda(N)$. Choosing $h \asymp \sqrt{\log N/N}$ in (24.20) gives, uniformly in the blocked index,

$$B_j = \sqrt{a_j^2 + \Delta_{c,N}^2} + O_\lambda(\sqrt{N \log N}), \quad (24.27)$$

$$\Gamma_{\text{SD}}/d = \Delta_{c,N} + O_\lambda(\sqrt{N \log N}). \quad (24.28)$$

At fixed $\omega = Nd$, the conventional one-quasiparticle parameter is

$$\Delta_0 = \frac{\omega}{\sinh(1/\lambda)}. \quad (24.29)$$

Thus dB_j has leading value $2\sqrt{\varepsilon_j^2 + \Delta_0^2}$ and $\Gamma_{\text{SD}} \rightarrow 2\Delta_0$.

The constants in (24.28) alone do not permit $\lambda \rightarrow 0$. For a uniform version set

$$m = \frac{2N}{\sinh(1/\lambda)}, \quad \ell = 1 + \log(N/\lambda).$$

There exist absolute C, C_0 such that for $0 < \lambda \leq 1/2$ and $m \geq C_0\ell$,

$$\left| \frac{\Gamma_{\text{SD}}}{d} - m \right| \leq C[1 + \sqrt{m\ell}]. \quad (24.30)$$

These constants are not optimized; the explicit finite bracket is preferable for a numerical certificate. Indeed, for $S_N(\delta) = \sum_i [(2i-1)^2 + \delta^2]^{-1/2}$,

$$\left| S_N(\delta) - \frac{1}{2} \operatorname{arsinh}(2N/\delta) \right| \leq \delta^{-1}.$$

Deleting or weighting the central term adds at most another δ^{-1} . The corresponding roots are $m + O(1)$. On a fixed small q interval their variation is $O(mh + 1)$, since $\partial_q \log m = -\coth q$, while $\epsilon_* = O(\ell)$. Equations (24.20) and (24.26) give the error $C[1 + mh + \ell/h]$. Taking $h \asymp \sqrt{\ell/m}$ proves (24.30). A sufficient physical criterion is consequently

$$\frac{d}{\Delta_0} \left[1 + \log \frac{\omega}{\lambda d} \right] \rightarrow 0. \quad (24.31)$$

The additional logarithm measures this proof's accuracy; it is not a new physical threshold replacing $d \sim \Delta_0$.

24.7 Distinct criteria for condensation and excitation energies

In the same Hamiltonian,

$$E_{\text{FS}} = -dN^2 - \lambda dN, \quad C_N = E_{\text{FS}} - E_0, \quad C_{\text{BCS}} = dN^2 [\coth(1/\lambda) - 1]. \quad (24.32)$$

The first-order diagonal interaction cancels in C_N . Using a noninteracting reference with an interacting E_0 would define a different energy difference.

A finite variational bound needs no thermodynamic assumption. Let $|ij\rangle$ move a whole pair from the i th filled negative orbital to the j th positive orbital, and take

$$|\chi\rangle = |\text{FS}\rangle + \sum_{i,j=1}^N \frac{\lambda}{2(i+j-1)} |ij\rangle.$$

Its norm is $1 + \lambda^2 T_N/4$, where

$$A_{2,N} = \frac{1}{2} \sum_{i,j} \frac{1}{i+j-1} = N(H_{2N} - H_N), \quad (24.33)$$

$$T_N = \sum_{i,j} \frac{1}{(i+j-1)^2} = 2H_N - H_{2N} + 2N(H_{2N}^{(2)} - H_N^{(2)}). \quad (24.34)$$

The kinetic cost is $2d(i+j-1)$. Combining it with the Fermi-sea off-diagonal matrix elements gives $-d\lambda^2 A_{2,N}$ relative to E_{FS} . The remaining off-diagonal terms between positive trial amplitudes are nonpositive, proving

$$\boxed{\frac{C_N}{d} \geq \frac{\lambda^2 N(H_{2N} - H_N)}{1 + \lambda^2 T_N/4}}. \quad (24.35)$$

Keeping those terms improves the numerator to $\lambda^2 A_{2,N} + \lambda^3 A_{3,N}$, where $A_{3,N} = (\sum_i h_i^2 - T_N)/2$ and $h_i = H_{N+i-1} - H_{i-1}$. The trial state need not itself be SD; the physical ground is the branch selected in Chapter 23.

As $A_{2,N} = N \log 2 - \frac{1}{4} + O(N^{-1})$ and $T_N = \log N + \gamma - \log 2 + 1 + O(N^{-2})$,

$$C_N \geq (\log 2 + o(1))\lambda^2 \omega \quad \text{if } \lambda \rightarrow 0, \quad \lambda \log N = O(1). \quad (24.36)$$

This lower bound is not itself an additive energy expansion. If its finite right-hand side exceeds $(1 + \eta)C_{\text{BCS}}$, it certifies failure of that relative tolerance. Failure of this test does not certify BCS accuracy.

The distinction between cumulative energy and spectral criteria is established physics [20, 48]. Historically, those works identified the earlier mesoscopic energy scale and explained it through the contribution of levels far from the Fermi energy. The explicit interpretation of that smaller scale as the binding energy of a single Cooper pair was emphasized in the later analysis of Ref. [26]; it should not be attributed to the earlier scale comparison as though the two arguments were identical. For the specified sharp-cutoff reduced model the weak-coupling finite-size expansion supplies the contribution $(\log 2)\lambda^2\omega$ to C_N and a leading correction of order d to the gap [23]. Balancing with the bulk answers gives the scales

$$d_E \asymp \frac{\Delta_0^2}{2(\log 2)\lambda^2\omega}, \quad d_\Gamma \asymp \Delta_0. \quad (24.37)$$

The first expression uses the leading weak-coupling coefficient, not an exact threshold. The cited finite-size analysis emphasizes cutoff sensitivity of the energy correction. Translating it to a real material requires an interaction model outside the specified layer; the finite reduced Hamiltonian itself remains unambiguous.

The Cooper scale describes the exponential separation [26]. For a single continuum Cooper pair in a layer of width 2ω , with this coupling convention,

$$1 = \frac{\lambda}{2} \log \frac{4\omega + \epsilon_C}{\epsilon_C},$$

and therefore

$$\epsilon_C = \frac{4\omega}{e^{2/\lambda} - 1} = \frac{\Delta_0^2}{\omega} (1 - e^{-2/\lambda}). \quad (24.38)$$

The continuum Cooper binding, Δ_0 , and the two-particle gap $2\Delta_0$ are different quantities. In particular, $d_E \asymp \epsilon_C/[2(\log 2)\lambda^2(1 - e^{-2/\lambda})]$. The exponential scales coincide, but this algebraic factor matters in joint weak coupling. The approximate energy interpolation discussed in Ref. [26] is not used as an exact identity here.

24.8 Closed joint limits and their boundaries

Keep $\omega = Nd$ fixed, let $\lambda \downarrow 0$, and take $N = \lceil \exp(\alpha/\lambda) \rceil$ with fixed $\alpha > 0$. Then $m \sim 4e^{(\alpha-1)/\lambda}$ and $C_{\text{BCS}} \sim 2\omega e^{(\alpha-2)/\lambda}$. For $\alpha > 1$, $m/\ell \rightarrow \infty$, so (24.30) proves $\Gamma_{\text{SD}}/(2\Delta_0) \rightarrow 1$. For $\alpha < 2$, (24.36) proves $C_N/C_{\text{BCS}} \rightarrow \infty$. For $\alpha < 1$, even $\Gamma_{\text{SD}}/d \geq \lambda$ gives divergence of its BCS ratio. Finally, integration of the ground-profile envelope and the discrete-to-integral comparison bound the energy error by $O(\omega\ell)$. For $\alpha > 2$ this is negligible relative to C_{BCS} and proves the remaining limit.

Fixed exponent	C_N/C_{BCS}	$\Gamma_{\text{SD}}/(2\Delta_0)$
$\alpha > 2$	1	1
$1 < \alpha < 2$	$+\infty$	1
$0 < \alpha < 1$	$+\infty$	$+\infty$

Table 24.1: Proved limits using the finite ground-state and comparison results. Boundary exponents $\alpha = 1, 2$ are not covered.

These estimates resolve specified joint weak-coupling limits. Unresolved by these estimates are the boundary crossover $q - \log N = O(1)$, removal of the logarithmic loss, and the

comparison with the lowest non-SD paired states. Chapter 25 provides exact finite equations and a conditional limiting system at that gap crossover.

24.9 Finite coefficients and numerical evidence

At fixed N , perturbation of the two isolated SD branches gives

$$\frac{\Gamma_{\text{SD}}}{d} = 2 + \frac{\lambda^2}{N} + A_N \lambda^3 + O_N(\lambda^4), \quad A_N = 2 + \frac{2}{N}(H_{2N} - H_N - 1) - \frac{1}{N^2}. \quad (24.39)$$

The second coefficient is the telescoping sum $1 + \sum_{i=2}^N (1/i - 1/(i-1)) = 1/N$. After removing the common diagonal shift, the third coefficient is the sum over oriented triangles of one-pair transfers. For a reflected self-dual occupation set it is

$$-2 \sum_{i \in \text{occ}} \left[S_i^2 - \sum_{a \in \text{empty}} (x_a - x_i)^{-2} \right], \quad S_i = \sum_{a \in \text{empty}} (x_a - x_i)^{-1},$$

with dimensionless pair energies x . For the ground state $S_i = (H_{N+i-1} - H_{i-1})/2$. For the first state $S'_1 = (H_{N-1} - 1)/2$ and $S'_i = S_i + [2i(i-1)]^{-1}$ for $i \geq 2$. Subtracting and summing by parts yields A_N . The archived recurrence checks these coefficients with exact rational arithmetic for several sizes, including $A_1 = 0$. The remainder is not uniform in N . Substituting $\lambda = (\log N + \tau)^{-1}$ into this truncated series does not prove an asymptotic expansion of a universal crossover function.

N	d/Δ_0	C_N/C_{BCS}	$\Gamma_{\text{SD}}/(2\Delta_0)$
64	1.1594	8.76193	1.35501
128	0.5797	4.99202	0.87860
256	0.2899	3.02312	0.81176
512	0.1449	2.01808	0.91457
1024	0.0725	1.51065	0.96143

Table 24.2: Finite quadratic-SD results at $\lambda = 0.2$, retained from crossover iteration 3. Floating-point calculations, not interval bounds.

The nonmonotone gap ratio also shows why an isolated equality between two curves does not define recovery of BCS accuracy.

The source derivations and original checks are retained in `code/duality_selfdual/checkpoint_v0_1/previous_notes/` and its `input_archives/`. The finite ordering proof is in `code/duality_selfdual/selfdual_iteration5.zip`. Current reproduction and independent small-matrix comparisons are documented in `revision/DUALITY_NUMERICS_v1_3.md`. Full physical paired-Hamiltonian diagonalization was performed only at small N . Larger computations solve the SD equations and do not alone prove absence of a lower non-SD excitation.

Chapter 25

The finite self-dual gap from one root and a smooth real response

The first self-dual excitation can be computed without locating the remaining complex Richardson roots. After removing its single positive root, the remainder is described by a smooth real boundary response to the exact ground state of the deleted system. A scalar equation selects the positive root, and an area identity returns the energy. These equations are exact at finite N on the branch selected in Chapters 23 and 24.

The finite identities are exact. The small response and rapid Newton improvement are numerical observations at the reported parameters. The N -independent crossover equations below are a conditional limiting candidate: global uniqueness of their scalar root and convergence of the full gap are not proved by these calculations.

25.1 Factor out the positive root

Use $a_i = 2i - 1$, $q = \lambda^{-1}$, $D = |\partial_y|$, and $e = E/d$. The first SD polynomial has the factorization

$$Q_1(z) = (z - r)R(z), \quad 0 < r < 1, \quad \deg R = N - 1, \quad (25.1)$$

with every zero of R in the left half-plane. Let R_0 be the ground polynomial on the remaining levels $\pm 3, \pm 5, \dots, \pm(2N - 1)$ at the same λ ; its energy is e_b . The full ground energy is e_0 , and $B_N = e_b - e_0$ is the central blocked energy of (24.2). Define

$$u_B(y) = \log \left| \frac{R_0(iy)}{\prod_{i=2}^N (iy + a_i)} \right|, \quad A_B(y) = qe^{-2u_B(y)} > 0, \quad (25.2)$$

$$s(y) = \log \left| \frac{R(iy)}{R_0(iy)} \right|, \quad b_r(y) = \frac{y^2 + r^2}{y^2 + 1}. \quad (25.3)$$

The response s can have either sign; it is not a density. The narrow root contribution has been removed analytically. For a finite physical polynomial, s is smooth, even, and integrable, with $s(y) = O(y^{-2})$.

For

$$\mathcal{F}_\lambda[P](z) = P(z)P(-z) - \lambda[P'(z)P(-z) + P'(-z)P(z)],$$

direct multiplication gives

$$\boxed{\mathcal{F}_\lambda[(z - r)R](z) = (r^2 - z^2)\mathcal{F}_\lambda[R](z) + 2\lambda r R(z)R(-z).} \quad (25.4)$$

Here $P'(-z)$ means the derivative evaluated at $-z$, not the derivative of the composite function. This fixes the sign of the last term. The finite SD identity is $\mathcal{F}_\lambda[Q_1] = (-1)^N \prod_i (z^2 - a_i^2)$.

25.2 The exact smooth response equation

On the imaginary axis, (25.4) yields

$$\boxed{2b_r Ds + A_B(1 - e^{-2s}) = \frac{A_B(1 - r^2) - 2r}{1 + y^2}.} \quad (25.5)$$

Derivation through the full excited modulus also checks the contribution of the right-half-plane zero. Write

$$u_1 = u_B + s + \frac{1}{2} \log \frac{y^2 + r^2}{y^2 + 1}.$$

Dividing the analytic function by its Blaschke factor $(z - r)/(z + r)$ removes the positive zero. The boundary equations are

$$2Du_1 + q(1 - e^{-2u_1}) = 2\rho_N - \frac{4r}{r^2 + y^2}, \quad (25.6)$$

$$2Du_B + q(1 - e^{-2u_B}) = 2\rho_B, \quad \rho_B = \sum_{i=2}^N \frac{a_i}{a_i^2 + y^2}. \quad (25.7)$$

The elementary identity

$$D \left[\frac{1}{2} \log \frac{y^2 + r^2}{y^2 + 1} \right] = \frac{1}{1 + y^2} - \frac{r}{r^2 + y^2}$$

then proves (25.5) by subtraction and multiplication by b_r . The Lorentzian of width r has been eliminated explicitly; the response mesh need not resolve that width just to recover the singular root contribution.

For fixed $0 < r < 1$, divide by b_r and set

$$c_r = A_B/b_r, \quad F_r = \frac{A_B(1 - r^2) - 2r}{y^2 + r^2}.$$

The equation becomes $2Ds + c_r(1 - e^{-2s}) = F_r$. The difference of two bounded decaying solutions obeys $(2D + m(y))v = 0$ with $m(y) > 0$. At a positive maximum $Dv \geq 0$, contradicting the equation; a negative minimum is excluded similarly. The decaying response is therefore unique for fixed r .

For finite N , $A_{\min} = \inf_y A_B(y) > 0$, and constant barriers give

$$-\frac{1}{2} \log(1 + 2r/A_{\min}) \leq s(y) \leq -\log r. \quad (25.8)$$

For example, writing $h(y) = (1 + y^2)^{-1}$, the upper-barrier residual is $A_B(1 - r^2)(1 - h) + 2rh \geq 0$; the lower residual is nonpositive. Comparison and a bounded-domain limit supply the finite decaying solution. These coarse bounds do not establish uniformly small s . The linearized operator

$$\mathcal{M}_{r,s} = 2b_r D + 2A_B e^{-2s} \quad (25.9)$$

has a positivity-preserving inverse after division by b_r . The divided equation is the Euler equation of the strictly convex functional

$$\int s Ds \, dy + \int \left\{ c_r \left[s + \frac{e^{-2s} - 1}{2} \right] - F_r s \right\} dy.$$

Fixed- r uniqueness and convexity do not establish uniqueness of the scalar root to be determined next.

25.3 A scalar condition selects the positive root

The scalar condition follows directly from the finite polynomial identity. At $z = r$ its value is $\mathcal{F}_\lambda[Q_1](r) = -\lambda Q_1'(r)Q_1(-r)$. Differentiating cancels the two cross-derivative terms, giving

$$\frac{\partial_z \mathcal{F}_\lambda[Q_1](r)}{\mathcal{F}_\lambda[Q_1](r)} = -q + \frac{Q_1''(r)}{Q_1'(r)} = -q + 2\frac{R'(r)}{R(r)}.$$

The prescribed right-hand side has logarithmic derivative $2r \sum_i (r^2 - a_i^2)^{-1}$, hence

$$q = 2r \sum_{i=1}^N \frac{1}{a_i^2 - r^2} + 2\frac{R'(r)}{R(r)}. \quad (25.10)$$

Because R/R_0 is analytic and zero-free in the right half-plane, its logarithm follows by Poisson extension from its boundary modulus. Define the derivative of that extension by

$$\mathcal{P}'_r[f] = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y^2 - r^2}{(y^2 + r^2)^2} f(y) dy. \quad (25.11)$$

This derivative moves the evaluation point while holding the boundary function f fixed. It is not a total derivative of s_r . The scalar equation is consequently

$$0 = q - 2r \sum_{i=1}^N \frac{1}{a_i^2 - r^2} - 2 \sum_{i=2}^N \frac{1}{a_i + r} - 2\mathcal{P}'_r[u_B + s_r]. \quad (25.12)$$

For each trial r , solve the unique response and then impose this real scalar condition. The identities hold on the physical branch. A functional solution does not automatically prove polynomial reconstruction, spectral completeness, or global uniqueness of all possible scalar roots. Physical continuation and finite SD checks remain the branch-selection tools in the reported computations.

25.4 The energy is an area, not the positive root

Let e_R be the sum of the zeros of R . At large positive z , $\log(R(z)/R_0(z)) = (e_b - e_R)/z + O(z^{-2})$. The Poisson formula identifies that coefficient with $\pi^{-1} \int s$. Since $e_{1,\text{SD}} = r + e_R$,

$$\frac{\Gamma_{\text{SD}}}{d} = B_N + r - \frac{1}{\pi} \int_{\mathbb{R}} s(y) dy. \quad (25.13)$$

The deleted-system background and response both contribute; r is not the gap. For $N = 256$, $\lambda = (\log 256)^{-1}$, the retained calculation gives $r = 0.1899981289$ while $\Gamma_{\text{SD}}/d = 3.3402889813$. For $N = 1$, $s = 0$ and $r = \sqrt{1 + \lambda^2} - \lambda$. The formula then returns $2\sqrt{1 + \lambda^2}$, checking the sign and diagonal convention.

25.5 Linear response and one Newton correction

Replacing $1 - e^{-2s}$ by $2s$ gives

$$(2b_r D + 2A_B) s_{\text{lin}} = \frac{A_B(1 - r^2) - 2r}{1 + y^2}. \quad (25.14)$$

For the residual $\mathcal{R}_r(s)$ of (25.5), Newton's equation at the same trial r is

$$(2b_r D + 2A_B e^{-2s_k}) \delta s_k = -\mathcal{R}_r(s_k), \quad s_{k+1} = s_k + \delta s_k. \quad (25.15)$$

Each approximation requires its own scalar-root search using that response. One Newton correction means one response solve per trial root, not one solve for the entire scalar search. Since $1 - e^{-2s} \leq 2s$, the linear response is a subsolution. Concavity and the positive inverse give monotone Newton improvement at fixed r , bounded above by the exact response. This does not fix the sign of the final gap error, because the selected root changes with the response approximation.

τ	Finite Γ_{SD}/d	$ \delta\Gamma_{\text{lin}} /d$	$ \delta\Gamma_{\text{Newton}} /d$
-1	9.6517801392	$2.5141 \cdot 10^{-4}$	$1.6150 \cdot 10^{-9}$
0	3.3460145603	$1.2311 \cdot 10^{-3}$	$2.9609 \cdot 10^{-7}$
1	2.2692879347	$4.7888 \cdot 10^{-4}$	$6.4663 \cdot 10^{-8}$
2	2.0829575160	$1.6520 \cdot 10^{-4}$	$8.0865 \cdot 10^{-9}$
4	2.0175707498	$2.9947 \cdot 10^{-5}$	$2.7336 \cdot 10^{-10}$

Table 25.1: Retained response benchmark at $N = 64$, $\lambda = (\log 64 + \tau)^{-1}$, on a 512-point auxiliary real-coordinate grid. The last column uses one Newton correction. The reference is the finite SD energy, not a limiting function.

For these five examples $\max |s|$ ranges from approximately 0.0041 to 0.0274. This supports the method's efficiency but is not a uniform theorem that one Newton step suffices. At $N = 256$, $\tau = 0$, increasing the nonlinear grid from 512 to 1024 reduces the retained energy error from $1.61 \cdot 10^{-8}d$ to $6.08 \cdot 10^{-12}d$.

The code maps the real line by $y = L \tan(\theta/2)$. Conformal covariance gives $D_y u = k D_\theta u$ with $k = (1 + \cos \theta)/L$, while $s = kv$ gives $\pi^{-1} \int_{\mathbb{R}} s \, dy = 2 \text{mean}(v)$. This grid discretizes an auxiliary real coordinate, not the physical energy levels. The positive-root factor is removed analytically.

25.6 An exact digamma separation of the cutoff

Introduce

$$\tau = q - \log N, \quad U_{B,N} = u_B - \frac{1}{2} \log q, \quad A_B = e^{-2U_{B,N}}. \quad (25.16)$$

The constant subtraction is annihilated by both D and $\mathcal{P}'_r$. The finite digamma sum identity yields

$$2DU_{B,N} - e^{-2U_{B,N}} = -\tau - \text{Re} \psi \left(\frac{3 + iy}{2} \right) + \mathfrak{r}_N(y), \quad (25.17)$$

where

$$\mathfrak{r}_N(y) = \text{Re} \psi \left(N + \frac{1 + iy}{2} \right) - \log N.$$

This is an exact rewrite of a discrete sum. The scalar sums also combine:

$$\begin{aligned} 2r \sum_{i=1}^N \frac{1}{a_i^2 - r^2} + 2 \sum_{i=2}^N \frac{1}{a_i + r} \\ = \frac{2r}{1 - r^2} + \sum_{i=2}^N \left(\frac{1}{a_i - r} + \frac{1}{a_i + r} \right). \end{aligned} \quad (25.18)$$

The digamma recurrence and reflection formulas consequently transform (25.12) into

$$\boxed{0 = \tau + \psi \left(\frac{3 + r}{2} \right) - \frac{\pi}{2} \tan \frac{\pi r}{2} - 2\mathcal{P}'_r[U_{B,N} + s_r] + \epsilon_N(r),} \quad (25.19)$$

where

$$\epsilon_N(r) = \log N - \frac{1}{2} \left[\psi \left(N + \frac{1+r}{2} \right) + \psi \left(N + \frac{1-r}{2} \right) \right]. \quad (25.20)$$

The lower endpoints in (25.18) give $\psi((3+r)/2) - (\pi/2) \tan(\pi r/2)$; the upper endpoints give the symmetric average in (25.20). The cancellation of an apparent N^{-1} term is algebraic. Expanding that average gives, uniformly for $0 \leq r \leq 1$,

$$\epsilon_N(r) = \frac{3r^2 - 1}{24N^2} + O(N^{-4}). \quad (25.21)$$

The independent analytic audit also gives the next coefficient $(15r^4 - 30r^2 + 7)/(960N^4)$. From $0 < \psi(z + \frac{1}{2}) - \log z \leq (24z^2)^{-1}$ for $z > 0$, one obtains the finite enclosure

$$-\frac{1}{48} \left[(N + r/2)^{-2} + (N - r/2)^{-2} \right] \leq \epsilon_N(r) \leq -\frac{1}{2} \log \left(1 - \frac{r^2}{4N^2} \right). \quad (25.22)$$

This proves an $O(N^{-2})$ cutoff error in one scalar term. It is not an $O(N^{-2})$ convergence rate for the root, background, or full gap.

25.7 The conditional universal crossover system

At fixed τ , $\mathfrak{r}_N(y) \rightarrow 0$ locally. If the appropriately matched ground backgrounds converge, the deleted-system candidate V_B obeys

$$2DV_B - e^{-2V_B} = -\tau - \operatorname{Re} \psi \left(\frac{3+iy}{2} \right). \quad (25.23)$$

The candidate response and scalar equations are

$$2b_r Ds + e^{-2V_B} (1 - e^{-2s}) = \frac{e^{-2V_B} (1 - r^2) - 2r}{1 + y^2}, \quad (25.24)$$

$$0 = \tau + \psi \left(\frac{3+r}{2} \right) - \frac{\pi}{2} \tan \frac{\pi r}{2} - 2\mathcal{P}'_r[V_B + s]. \quad (25.25)$$

The matching class comes from the finite ground problem; arbitrary solutions with unrelated behavior at infinity are not asserted to be physical limits.

An energy limit also requires control of the blocked difference. If $B_N \rightarrow b(\tau)$, $r_N \rightarrow r$, and the response integrals converge, (25.13) gives

$$g_{\text{SD}}(\tau) = b(\tau) + r - \frac{1}{\pi} \int s(y) dy. \quad (25.26)$$

Equivalently, let V_0 be the matched full-background limit, using $\psi((1+iy)/2)$ on the right-hand side. If the difference integrals converge, then $b(\tau) = 1 + \pi^{-1} \int (V_0 - V_B)$. This identifies the specific ground-state area difference that must be controlled.

Local background convergence alone is insufficient. The additional requirements are compactness or endpoint control of the selected roots, stability and physical selection of the limiting scalar zero, and uniform integrability of the response and ground differences. Fixed- r uniqueness does not establish these global statements. The system is a concrete candidate for completing the crossover analysis, not a proved universal spectral theorem.

The loss of a positive solution of the auxiliary collective gap equation is likewise not an exact spectral threshold. In the retained finite tests the SD gap remains positive and smooth near $\tau = \log 4 + \gamma$. This is compatible with the finite theory; it does not prove a universal limit.

τ	$N = 64$	$N = 256$	$N = 1024$	$4e^{-\tau}$
-1	9.65178014	9.60665247	9.60082829	10.87312731
0	3.34601456	3.34028898	3.33937076	4.00000000
1	2.26928793	2.26818727	2.26800569	1.47151776
2	2.08295752	2.08249553	2.08241784	0.54134113

Table 25.2: Retained finite Γ_{SD}/d along $\lambda = (\log N + \tau)^{-1}$. The last column is the limiting bulk BCS expression, not an exact crossover curve. Clustering is evidence, not proof, of a limiting function.

25.8 Reproduction, provenance, and scope of the checks

The source note, code, and historical outputs are preserved under `code/duality_selfdual/checkpoint_v0_1/current_iteration4/`. The programs `scalar_excited.py`, `response_benchmark.py`, `cutoff_identity.py`, and `finite_universal_scan.py` implement the response calculation, approximations, cutoff identity, and scan. The SD solver and boundary operators are `sd_core.py` and `boundary_support.py`. Current reruns are separate, under `code/duality_selfdual/revision_validation/`, and documented in `revision/DUALITY_NUMERICS_v1_3.md`.

The supplemental audit in `revision/scalar_independent_audit.md` checks the factorization, root equation, energy sign, and analytic cutoff coefficients. At $N = 8, \tau = 0$ on a 256-point grid, its scalar energy $e_{1,\text{SD}} = -67.49828754002039$ agrees with finite level equations within $4.27 \cdot 10^{-13}$ and a separate diagonalization of the 256×256 effective matrix within $4.27 \cdot 10^{-14}$. All 30 bounded cutoff-identity checks in that run passed. This supplement did not rerun its own symbolic suite; the broader revision reproduction is documented separately. Extracting an N^{-4} coefficient by subtracting float64 values at large N loses precision, so its analytic derivation supports the coefficient, not that numerical subtraction.

Small-system polynomial and matrix comparisons test branch selection. Large- N agreement of SD equations does not supply a missing intersector theorem. Grid convergence does not establish a universal limit, and accurate one-step Newton results at five parameters do not constitute a uniform error theorem. With these distinctions, the finite reduction and regularized response are exact usable results.

Part XI

Technical appendices

Appendix A

Binomial moments and all-pole normalization

This appendix records the finite algebra behind [Chapter 4](#) in a form suitable for independent checking. No asymptotic limit is taken anywhere in the derivations below.

A.1 Residue normalization in the scaled variable

Starting from

$$R = 2\varepsilon_{F0} + \frac{2z}{\rho}, \quad (\text{A.1})$$

one has

$$2\varepsilon_m - R = \frac{2}{\rho}(m - z). \quad (\text{A.2})$$

At $z = n$,

$$\prod_{\substack{m=0 \\ m \neq n}}^M (m - n) = (-1)^n n! (M - n)!, \quad (\text{A.3})$$

$$\prod_{m=0}^M (m - z) = -(-1)^n n! (M - n)! (z - n) + O((z - n)^2). \quad (\text{A.4})$$

Consequently,

$$\begin{aligned} \text{Res}_{z=n} \frac{\exp[-2z/(\rho V)]}{\prod_{m=0}^M (m - z)} &= -\frac{(-1)^n \sigma^n}{n! (M - n)!} \\ &= -\frac{1}{M!} (-1)^n \sigma^n \binom{M}{n}. \end{aligned} \quad (\text{A.5})$$

For N variables the product of these residues gives $(-1)^N (M!)^{-N}$. Combining it with the scale factors from the Vandermonde, denominator, and measure yields

$$\mathcal{C}_{M,N,\rho} = (-1)^N (M!)^{-N} \left(\frac{2}{\rho}\right)^{N(N-M-1)}, \quad (\text{A.6})$$

which is [\(4.9\)](#).

As a one-variable sign check, take $M = 1$ and $\varepsilon_{F0} = 0$. Direct residues give

$$\int_{\gamma_{\text{all}}} \frac{e^{-R/V}}{(-R)(2/\rho - R)} \frac{dR}{2\pi i} = -\frac{\rho}{2}(1 - \sigma), \quad (\text{A.7})$$

in agreement with

$$\mathcal{C}_{1,1,\rho} z_{1,1} = -\frac{\rho}{2}(1 - \sigma). \quad (\text{A.8})$$

A.2 Coefficient extraction for the mixed moment

For completeness, expand the generating function in two ways. First,

$$\begin{aligned} \mathcal{B}(u, v) &:= \sum_{n=0}^M (-1)^n \sigma^n \binom{M}{n} (1+u)^n (1+v)^{M-n} \\ &= \sum_{a,b \geq 0} \left[\sum_{n=0}^M (-1)^n \sigma^n \binom{M}{n} (n)_{\underline{a}} (M-n)_{\underline{b}} \right] \frac{u^a}{a!} \frac{v^b}{b!}. \end{aligned} \quad (\text{A.9})$$

Second, the binomial theorem gives

$$\begin{aligned} \mathcal{B}(u, v) &= [(1+v) - \sigma(1+u)]^M \\ &= [(1-\sigma) + v - \sigma u]^M. \end{aligned} \quad (\text{A.10})$$

The coefficient of $u^a v^b$ in the last line is zero if $a+b > M$. Otherwise it is

$$\frac{M!}{a!b!(M-a-b)!} (-\sigma)^a (1-\sigma)^{M-a-b}. \quad (\text{A.11})$$

Multiplication by $a!b!$ proves

$$\sum_{n=0}^M (-1)^n \sigma^n \binom{M}{n} (n)_{\underline{a}} (M-n)_{\underline{b}} = (-1)^a \sigma^a (1-\sigma)^{M-a-b} (M)_{\underline{a+b}}. \quad (\text{A.12})$$

A.3 Falling-factorial Vandermonde identities

Let $p_a(y) = (y)_{\underline{a}}$. Since p_a is monic of degree a , the determinant

$$\det[p_a(y_i)]_{i=1, \dots, N}^{a=0, \dots, N-1} \quad (\text{A.13})$$

is an alternating polynomial of total degree $N(N-1)/2$ with the same leading coefficient as the standard Vandermonde. Hence

$$\det[(y_i)_{\underline{a}}] = \prod_{i < j} (y_j - y_i). \quad (\text{A.14})$$

Setting $y_i = n_i$ gives (4.20). Setting $y_i = M - n_i$ gives

$$\det[(M - n_i)_{\underline{b}}] = \prod_{i < j} (n_i - n_j) = (-1)^{s_N} \prod_{i < j} (n_j - n_i), \quad (\text{A.15})$$

which gives (4.21).

A.4 Discrete Andréief identity

The determinant identity used below is the finite discrete form of Andréief's integration formula [49, 50].

Let $\mathcal{X}$ be a finite set, let $w(x)$ be any scalar weight, and let f_a, g_b be functions on $\mathcal{X}$. Then

$$\sum_{x_1, \dots, x_N \in \mathcal{X}} \left[\prod_{i=1}^N w(x_i) \right] \det[f_a(x_i)]_{i,a=1}^N \det[g_b(x_i)]_{i,b=1}^N = N! \det \left[\sum_{x \in \mathcal{X}} w(x) f_a(x) g_b(x) \right]_{a,b=1}^N. \quad (\text{A.16})$$

Proof. Expand both determinants:

$$\begin{aligned} & \sum_{\pi, \tau \in S_N} \text{sgn}(\pi) \text{sgn}(\tau) \sum_{x_1, \dots, x_N} \prod_{i=1}^N w(x_i) f_{\pi(i)}(x_i) g_{\tau(i)}(x_i) \\ &= \sum_{\pi, \tau \in S_N} \text{sgn}(\pi) \text{sgn}(\tau) \prod_{i=1}^N \left[\sum_x w(x) f_{\pi(i)}(x) g_{\tau(i)}(x) \right]. \end{aligned}$$

Put $\omega = \tau \circ \pi^{-1}$. For each fixed ω there are $N!$ choices of π , and

$$\text{sgn}(\pi) \text{sgn}(\tau) = \text{sgn}(\omega).$$

The remaining sum over ω is the determinant on the right-hand side of (A.16). $\square$

Apply (A.16) with

$$\mathcal{X} = \{0, \dots, M\}, \quad w(n) = (-1)^n \sigma^n \binom{M}{n}, \quad (\text{A.17})$$

$$f_a(n) = (n)_{\underline{a-1}}, \quad g_b(n) = (M-n)_{\underline{b-1}}. \quad (\text{A.18})$$

Together with the sign in the Vandermonde split, this gives

$$z_{M,N} = (-1)^{s_N} N! \det[\mu_{ab}]_{a,b=0}^{N-1}. \quad (\text{A.19})$$

A.5 Evaluation of the remaining Hankel determinant

Define

$$H_{ab}^{(M,N)} = (M)_{\underline{a+b}}, \quad 0 \leq a, b < N. \quad (\text{A.20})$$

The identity

$$(M)_{\underline{a+b}} = (M)_{\underline{a}} (M-a)_{\underline{b}} \quad (\text{A.21})$$

implies

$$\det H^{(M,N)} = \left[\prod_{a=0}^{N-1} (M)_{\underline{a}} \right] \det[(M-a)_{\underline{b}}]_{a,b=0}^{N-1}. \quad (\text{A.22})$$

The second determinant is the Vandermonde at the nodes

$$M, M-1, \dots, M-N+1. \quad (\text{A.23})$$

Therefore

$$\begin{aligned} \det[(M-a)_{\underline{b}}] &= \prod_{0 \leq i < j \leq N-1} [(M-j) - (M-i)] \\ &= (-1)^{s_N} \prod_{0 \leq i < j \leq N-1} (j-i) \\ &= (-1)^{s_N} \prod_{j=1}^{N-1} j! = (-1)^{s_N} \prod_{a=0}^{N-1} a!. \end{aligned} \quad (\text{A.24})$$

This proves (4.25).

A.6 Product and superfactorial normalizations

Starting from

$$A_{M,N} = (-1)^{s_N} N! \prod_{a=0}^{N-1} a! (M)_{\underline{a}}, \quad (\text{A.25})$$

we separately simplify the two products. First,

$$N! \prod_{a=0}^{N-1} a! = \prod_{a=0}^N a! = \text{sf}(N+1). \quad (\text{A.26})$$

Second,

$$\begin{aligned} \prod_{a=0}^{N-1} (M)_{\underline{a}} &= \prod_{a=0}^{N-1} \frac{M!}{(M-a)!} \\ &= (M!)^N \left[\prod_{j=M-N+1}^M j! \right]^{-1} \\ &= (M!)^N \frac{\text{sf}(M-N+1)}{\text{sf}(M+1)}. \end{aligned} \quad (\text{A.27})$$

Combining these expressions gives (4.31).

The convention

$$\text{sf}(K) = 0!1! \cdots (K-1)! \quad (\text{A.28})$$

is fixed throughout the manuscript. In particular,

$$\text{sf}(0) = \text{sf}(1) = 1. \quad (\text{A.29})$$

This permits the exact endpoint $N = M + 1$ without a special case.

A.7 Ordered and unordered sums

Because the Vandermonde square vanishes whenever two indices coincide, every nonzero ordered tuple $(n_1, \dots, n_N)$ consists of N distinct indices. For each subset $S \subset \{0, \dots, M\}$ with $|S| = N$, all $N!$ permutations have the same weight. Therefore

$$z_{M,N} = N! Z_0 \quad (\text{A.30})$$

exactly. The factor $N!$ is part of $A_{M,N}$ and explains why a superfactorial formula written for the unordered subset sum differs by one factorial step from the formula for the ordered contour period.

A.8 Independent exact-rational verification

The file

`code/verify_all_pole.py`

implements six independent checks:

1. direct finite summation of the mixed moment and comparison with (4.17);

2. direct determinant verification of the two falling-factorial Vandermonde matrices;
3. direct verification of (4.25);
4. independent computation of the unordered subset sum and the ordered factor $N!$;
5. comparison of the ordered sum with (4.28);
6. direct comparison of the original- R residue sum with the geometric prefactor (4.9);
7. symbolic differentiation of the closed period and comparison with (4.33).

The archived default report is

`code/all_pole_verification_report.txt.`

Its current contents are:

```
All-pole exact-rational verification: PASS
SymPy version: 1.14.0
Mixed-moment identities checked: 852
Vandermonde splits checked: 456
Hankel determinants checked: 30
Closed all-pole sums checked: 90
Original-R contour prefactors checked: 66
M range: 0..7
N range: 1..min(5, M+1)
Sigma values: 1/5, 2/5, 3/7
Symbolic logarithmic-derivative energy check: PASS
Negative second-cumulant check: PASS
Exact finite-size decomposition check: PASS
All comparisons used exact arithmetic.
```

Appendix B

Krawtchouk-type polynomials and moment diagonalization

B.1 Generating function and coefficient formulas

For fixed M and $0 < \sigma < 1$, define

$$\sum_{s=0}^M K_s(n) z^s = (1+z)^{M-n} \left(1 + \frac{z}{\sigma}\right)^n. \quad (\text{B.1})$$

The generating function is a polynomial of degree M in z for every lattice point $n = 0, \dots, M$. Using

$$\frac{1+z/\sigma}{1+z} = 1 + \frac{z}{x(1+z)}, \quad x = \frac{\sigma}{1-\sigma}, \quad (\text{B.2})$$

we obtain

$$\begin{aligned} (1+z)^{M-n} \left(1 + \frac{z}{\sigma}\right)^n &= (1+z)^M \left[1 + \frac{z}{x(1+z)}\right]^n \\ &= \sum_{b=0}^n \binom{n}{b} x^{-b} z^b (1+z)^{M-b}. \end{aligned} \quad (\text{B.3})$$

Extracting z^s gives two equivalent coefficient formulas:

$$K_s(n) = \sum_{b=0}^{\min(n,s)} \binom{n}{b} \binom{M-b}{s-b} x^{-b}, \quad (\text{B.4})$$

$$K_s(n) = \sum_{b=0}^s \frac{x^{-b}}{b!} \binom{M-b}{s-b} (n)_{\underline{b}}. \quad (\text{B.5})$$

The second formula makes the degree and the leading coefficient manifest:

$$\deg_n K_s = s, \quad [(n)_{\underline{s}}] K_s = \frac{x^{-s}}{s!}. \quad (\text{B.6})$$

Thus the square evaluation matrix

$$P_{ns} = K_s(n), \quad 0 \leq n, s \leq M, \quad (\text{B.7})$$

is invertible.

B.2 Signed orthogonality from a bivariate generating function

Let

$$w(n) = (-\sigma)^n \binom{M}{n}. \quad (\text{B.8})$$

Consider

$$\mathcal{B}(z, t) := \sum_{n=0}^M w(n)(1+z)^{M-n} \left(1 + \frac{z}{\sigma}\right)^n (1+t)^{M-n} \left(1 + \frac{t}{\sigma}\right)^n. \quad (\text{B.9})$$

The binomial theorem gives

$$\begin{aligned} \mathcal{B}(z, t) &= [(1+z)(1+t)]^M \sum_{n=0}^M \binom{M}{n} \left[-\sigma \frac{(1+z/\sigma)(1+t/\sigma)}{(1+z)(1+t)} \right]^n \\ &= [(1+z)(1+t) - \sigma(1+z/\sigma)(1+t/\sigma)]^M \\ &= (1-\sigma)^M \left(1 - \frac{zt}{\sigma}\right)^M. \end{aligned} \quad (\text{B.10})$$

On the other hand, expansion of the two generating functions gives

$$\mathcal{B}(z, t) = \sum_{r,s=0}^M z^r t^s \sum_{n=0}^M w(n) \mathbf{K}_r(n) \mathbf{K}_s(n). \quad (\text{B.11})$$

Comparing coefficients proves

$$\sum_{n=0}^M w(n) \mathbf{K}_r(n) \mathbf{K}_s(n) = (1-\sigma)^M \left(-\frac{1}{\sigma}\right)^s \binom{M}{s} \delta_{rs}. \quad (\text{B.12})$$

This is [Exact identity 5.2](#).

Remark B.1 (Relation to conventional Krawtchouk polynomials). The family (B.1) is a reparametrized Krawtchouk family continued to the signed binomial weight $(-\sigma)^n \binom{M}{n}$. We retain the generating-function normalization because it makes the transition to falling factorials and the complementary-kernel tail particularly transparent. No positivity property of the conventional binomial measure is used.

B.3 Transition matrix and diagonalization of the moment matrix

Let

$$p_b(n) = (n)_{\underline{b}}, \quad 0 \leq b \leq M. \quad (\text{B.13})$$

Equation (B.5) defines the upper-triangular matrix

$$T_{bs} = \mathbf{1}_{b \leq s} \frac{x^{-b}}{b!} \binom{M-b}{s-b}. \quad (\text{B.14})$$

Writing row vectors of functions,

$$(\mathbf{K}_0, \dots, \mathbf{K}_M) = (p_0, \dots, p_M) T. \quad (\text{B.15})$$

The diagonal entries are

$$T_{ss} = \frac{x^{-s}}{s!}. \quad (\text{B.16})$$

For the leading N basis functions, let

$$G_{ab} = \sum_{n=0}^M w(n) p_a(n) p_b(n), \quad 0 \leq a, b < N, \quad (\text{B.17})$$

and let T_N be the leading $N \times N$ block of T . Orthogonality gives

$$T_N^\top G T_N = H_N, \quad H_N = \text{diag}(h_0, \dots, h_{N-1}). \quad (\text{B.18})$$

Hence

$$G = T_N^{-\top} H_N T_N^{-1}. \quad (\text{B.19})$$

In particular,

$$\det G = \frac{\prod_{s=0}^{N-1} h_s}{(\det T_N)^2} = \prod_{s=0}^{N-1} h_s (s! x^s)^2. \quad (\text{B.20})$$

B.4 Independent recovery of the all-pole determinant

Substituting

$$h_s = (1 - \sigma)^M (-1)^s \sigma^{-s} \binom{M}{s} \quad (\text{B.21})$$

and

$$x^s = \sigma^s (1 - \sigma)^{-s} \quad (\text{B.22})$$

into (B.20), we find

$$\begin{aligned} \det G &= (1 - \sigma)^{NM-2} \sum_{s=0}^{N-1} s (-1)^{\sum_{s=0}^{N-1} s} \sigma^{-\sum_{s=0}^{N-1} s} \\ &\quad \times \sigma^{2 \sum_{s=0}^{N-1} s} \prod_{s=0}^{N-1} \binom{M}{s} (s!)^2 \\ &= (-1)^{N(N-1)/2} \sigma^{N(N-1)/2} (1 - \sigma)^{NM-N(N-1)} \prod_{s=0}^{N-1} \frac{M! s!}{(M-s)!}. \end{aligned} \quad (\text{B.23})$$

Since

$$z_{M,N} = N! Z_0 = N! \det G, \quad (\text{B.24})$$

this is exactly the product form (4.29) of the all-pole factorization. The calculation is independent of the mixed-moment Hankel determinant used in Chapter 4 and Appendix A.

B.5 Full completeness and the spectral split

Let

$$W = \text{diag}(w(0), \dots, w(M)), \quad H = \text{diag}(h_0, \dots, h_M). \quad (\text{B.25})$$

Equation (B.12) is

$$P^\top W P = H. \quad (\text{B.26})$$

Because both P and H are invertible,

$$P H^{-1} P^\top W = I_{M+1}. \quad (\text{B.27})$$

In entries,

$$\boxed{\delta_{ij} = w(j) \sum_{s=0}^M \frac{K_s(i) K_s(j)}{h_s}.} \quad (\text{B.28})$$

Splitting the sum into $s < N$ and $s \geq N$ gives

$$K_{N,ij}^{\text{occ}} = w(j) \sum_{s=0}^{N-1} \frac{\mathsf{K}_s(i)\mathsf{K}_s(j)}{h_s}, \quad (\text{B.29})$$

$$Q_{N,ij} = w(j) \sum_{s=N}^M \frac{\mathsf{K}_s(i)\mathsf{K}_s(j)}{h_s}. \quad (\text{B.30})$$

The first kernel is the projection onto polynomial degrees $0, \dots, N-1$ in the signed bilinear form, and the second is the complementary projection onto degrees $N, \dots, M$.

The projection identities follow directly. For example,

$$\begin{aligned} (K_N^{\text{occ}})_{ij}^2 &= w(j) \sum_{r,s=0}^{N-1} \frac{\mathsf{K}_r(i)}{h_r} \left[\sum_{n=0}^M w(n) \mathsf{K}_r(n) \mathsf{K}_s(n) \right] \frac{\mathsf{K}_s(j)}{h_s} \\ &= K_{N,ij}^{\text{occ}}. \end{aligned} \quad (\text{B.31})$$

The same calculation applies to Q_N .

B.6 Exact verification targets

The Krawtchouk part of `code/verify_kernel.py` checks independently:

1. the coefficient formula (B.4) against direct coefficient extraction from (B.1);
2. all signed orthogonality relations (B.12) in the tested range;
3. the matrix diagonalization (B.18);
4. the full completeness identity (B.28);
5. equality of the spectral kernels (B.29)–(B.30) with the inverse-moment construction.

All comparisons use rational values of σ and exact arithmetic.

Appendix C

Derivation of the complementary-kernel factorization

C.1 Mixed moments and their triangular factorization

Use the left and right polynomial bases

$$\phi_b(n) = (n)_{\underline{b}}, \quad \psi_c(n) = (M - n)_{\underline{c}}, \quad 0 \leq b, c < N. \quad (\text{C.1})$$

Their mixed moment matrix is

$$\mathcal{M}_{bc} = \sum_{n=0}^M w(n) \phi_b(n) \psi_c(n). \quad (\text{C.2})$$

By (4.17),

$$\mathcal{M}_{bc} = (-1)^b \sigma^b (1 - \sigma)^{M-b-c} (M)_{\underline{b+c}}. \quad (\text{C.3})$$

The expression remains correct when $b + c > M$ if the falling factorial is taken to vanish.

Use

$$(M)_{\underline{b+c}} = (M)_{\underline{b}} (M - b)_{\underline{c}}. \quad (\text{C.4})$$

Then

$$\boxed{\mathcal{M} = DTE}, \quad (\text{C.5})$$

where

$$D_{bb} = (-1)^b \sigma^b (1 - \sigma)^{M-b} (M)_{\underline{b}}, \quad (\text{C.6})$$

$$T_{bc} = (M - b)_{\underline{c}}, \quad (\text{C.7})$$

$$E_{cc} = (1 - \sigma)^{-c}. \quad (\text{C.8})$$

The matrix T is the evaluation matrix of the falling-factorial basis at the distinct nodes

$$M, M - 1, \dots, M - N + 1. \quad (\text{C.9})$$

It is therefore invertible. Hence

$$\mathcal{M}^{-1} = E^{-1} T^{-1} D^{-1}. \quad (\text{C.10})$$

C.2 Biorthogonal form of the occupation kernel

The split Vandermonde identity (4.22) rewrites the signed ensemble as a finite biorthogonal ensemble with left functions ϕ_b and right functions ψ_c . Its occupation kernel is

$$K_{N,ij}^{\text{occ}} = w(j) \sum_{b,c=0}^{N-1} \phi_b(i) (\mathcal{M}^{-1})_{cb} \psi_c(j). \quad (\text{C.11})$$

The order $(\mathcal{M}^{-1})_{cb}$ is fixed by the convention

$$\mathcal{M}_{bc} = \sum_n w(n) \phi_b(n) \psi_c(n). \quad (\text{C.12})$$

Substituting (C.10),

$$K_{N,ij}^{\text{occ}} = w(j) \sum_{b=0}^{N-1} \phi_b(i) D_{bb}^{-1} \sum_{c=0}^{N-1} (T^{-1})_{cb} (1-\sigma)^c (M-j)_{\underline{c}}. \quad (\text{C.13})$$

The remaining inner sum is evaluated by interpolation.

C.3 Lagrange basis on descending lattice nodes

Define

$$\ell_b(y) = \sum_{c=0}^{N-1} (T^{-1})_{cb} (y)_{\underline{c}}. \quad (\text{C.14})$$

Because $T_{rc} = (M-r)_{\underline{c}}$,

$$\ell_b(M-r) = \delta_{br}, \quad 0 \leq r < N. \quad (\text{C.15})$$

Thus ℓ_b is the Lagrange basis polynomial associated with the node $M-b$. Its explicit form is

$$\ell_b(y) = \frac{(-1)^b}{b!(N-1-b)!} \prod_{\substack{r=0 \\ r \neq b}}^{N-1} (y - M + r). \quad (\text{C.16})$$

Indeed, the denominator of the ordinary Lagrange formula is

$$\begin{aligned} \prod_{\substack{r=0 \\ r \neq b}}^{N-1} [(M-b) - (M-r)] &= \prod_{r \neq b} (r-b) \\ &= (-1)^b b!(N-1-b)!. \end{aligned} \quad (\text{C.17})$$

C.4 The binomial-thinning operator

Define a linear operator $\mathcal{R}_q$ on polynomials by

$$\mathcal{R}_q(y)_{\underline{c}} = q^c (y)_{\underline{c}}. \quad (\text{C.18})$$

For an integer $L \geq 0$ it has the binomial representation

$$(\mathcal{R}_q f)(L) = \sum_{r=0}^L \binom{L}{r} q^r (1-q)^{L-r} f(r). \quad (\text{C.19})$$

It is enough to verify this on the basis $(y)_{\underline{c}}$. The right-hand side is the c th falling-factorial moment of a binomial variable and equals

$$(L)_{\underline{c}} q^c. \quad (\text{C.20})$$

The inner sum of (C.13) is therefore

$$\sum_{c=0}^{N-1} (T^{-1})_{cb} (1-\sigma)^c (M-j)_{\underline{c}} = (\mathcal{R}_{1-\sigma} \ell_b)(M-j). \quad (\text{C.21})$$

Apply (C.19) with $L = M-j$ and set $r = M-s$. Then

$$(\mathcal{R}_{1-\sigma} \ell_b)(M-j) = \sum_{s=j}^M \binom{M-j}{s-j} (1-\sigma)^{M-s} \sigma^{s-j} \ell_b(M-s). \quad (\text{C.22})$$

C.5 Values inside and beyond the interpolation set

For $0 \leq s < N$, the interpolation property gives

$$\ell_b(M-s) = \delta_{bs}. \quad (\text{C.23})$$

For $s \geq N$, use (C.16):

$$\ell_b(M-s) = \frac{(-1)^b}{b!(N-1-b)!} \prod_{\substack{r=0 \\ r \neq b}}^{N-1} (r-s). \quad (\text{C.24})$$

Since

$$\prod_{r=0}^{N-1} (r-s) = (-1)^N \frac{s!}{(s-N)!}, \quad s \geq N, \quad (\text{C.25})$$

and the omitted factor is $b-s = -(s-b)$,

$$\ell_b(M-s) = (-1)^{N+b+1} \frac{s!}{b!(N-1-b)!(s-N)!(s-b)}, \quad s \geq N. \quad (\text{C.26})$$

This continuation of a Lagrange polynomial beyond its interpolation nodes is the source of the Cauchy denominator $(s-b)^{-1}$.

C.6 Exact cancellation of the low sector

Substitute the low value (C.23) into (C.13)–(C.22). For $s = b < N$ the coefficient multiplying $(i)_{\underline{b}}$ is

$$\begin{aligned} w(j) D_{bb}^{-1} \binom{M-j}{b-j} (1-\sigma)^{M-b} \sigma^{b-j} \\ = \frac{(-1)^{b-j}}{j!(b-j)!}, \quad b \geq j, \end{aligned} \quad (\text{C.27})$$

where the expression is zero for $b < j$. On the lattice $0 \leq i, j < N$, Newton interpolation gives

$$\delta_{ij} = \sum_{b=j}^{N-1} \frac{(-1)^{b-j}}{j!(b-j)!} (i)_{\underline{b}}. \quad (\text{C.28})$$

For completeness, the coefficient of $(i)_{\underline{b}}$ is the b th forward difference at zero of the function $i \mapsto \delta_{ij}$ divided by $b!$:

$$\frac{\Delta^b \delta_{0j}}{b!} = \frac{(-1)^{b-j}}{j!(b-j)!} \mathbf{1}_{b \geq j}. \quad (\text{C.29})$$

Therefore the entire low part of K_N^{occ} is exactly δ_{ij} for $i, j < N$. When

$$Q_N = I - K_N^{\text{occ}} \quad (\text{C.30})$$

is formed, the low sector cancels identically. No limiting argument is involved.

C.7 Assembly of the tail

Only $s \geq N$ remains. Substitute (C.26). The required elementary factors are

$$w(j) = (-1)^j \sigma^j \frac{M!}{j!(M-j)!}, \quad (\text{C.31})$$

$$D_{bb}^{-1} = (-1)^b \sigma^{-b} (1 - \sigma)^{-M+b} \frac{(M-b)!}{M!}, \quad (\text{C.32})$$

$$\binom{M-j}{s-j} = \frac{(M-j)!}{(s-j)!(M-s)!}. \quad (\text{C.33})$$

The tail of K_N^{occ} has sign $(-1)^{N+j+1}$; hence the tail of $Q_N = I - K_N^{\text{occ}}$ has sign

$$(-1)^{N+j} = (-1)^{N-j}. \quad (\text{C.34})$$

The powers of the coupling combine as

$$\sigma^{s-b} (1 - \sigma)^{b-s} = x^{s-b}. \quad (\text{C.35})$$

After cancellation of $M!$ and $(M-j)!$, one obtains

$$\begin{aligned} Q_{N,ij} &= (-1)^{N-j} \sum_{b=0}^{N-1} (i)_{\underline{b}} x^{-b} \frac{(M-b)!}{b!(N-1-b)!} \\ &\quad \times \sum_{s=N}^M \frac{1}{s-b} x^s \frac{s!}{(M-s)!(s-N)!} \frac{1}{(s-j)!j!}. \end{aligned} \quad (\text{C.36})$$

If $i < k \leq N$, then $(i)_{\underline{b}} = 0$ for $b \geq k$, so the outer sum may be restricted to $0 \leq b < k$. This is exactly

$$Q_N^{(k)} J = F D_1 C D_2 B D_3 \quad (\text{C.37})$$

with the factors (5.42)–(5.47).

C.8 Boundary cases

Saturated lattice. If $N = M + 1$, every site is occupied and $Q_N = 0$. The spectral-tail sum is empty. The factorization theorem is stated for $N \leq M$, where at least one hole degree is present.

Restriction $k = N$. The algebraic factorization remains valid for $k = N$. Later Case B applications use $k < N$ because $N - k$ unrestricted contours remain, but this physical restriction is not needed here.

One pair. For $N = 1$, the index b is forced to zero and the formula reduces to a single positive tail sum. This provides a useful sign check on the general expression.

C.9 Independent checks

The factorization part of `code/verify_kernel.py` constructs the following objects independently:

1. $Q_N = I - \Phi G^{-1} \Phi^T W$ from the symmetric falling-factorial moment matrix;
2. Q_N from the Krawtchouk tail (B.30);
3. $Q_N^{(k)} J$ from the six-factor product $FD_1CD_2BD_3$;
4. the hole insertion minors by direct enumeration of all residue subsets in the tested range.

The first construction does not use mixed moments or interpolation; the second does not invert G ; the third does not use the Krawtchouk polynomials. Exact equality of all three is therefore a sensitive audit of signs, transposes, and factorial shifts.

Appendix D

Cauchy signs and positive Cauchy–Binet expansions

D.1 Total nonnegativity of the Pascal factors

The two non-diagonal factors adjacent to the Cauchy matrix are

$$F_{ib} = (i)_{\underline{b}} = b! \binom{i}{b}, \quad (\text{D.1})$$

$$(BD_3)_{sj} = \frac{1}{j!(s-j)!} = \frac{1}{s!} \binom{s}{j}. \quad (\text{D.2})$$

Thus both are positive diagonal scalings of submatrices of the infinite Pascal matrix

$$\mathbf{P}_{rc}^{\text{Pas}} = \begin{pmatrix} r \\ c \end{pmatrix}, \quad r, c \geq 0. \quad (\text{D.3})$$

Lemma D.1 (Total nonnegativity of the Pascal matrix). *For increasing nonnegative integer sequences*

$$r_1 < \cdots < r_m, \quad c_1 < \cdots < c_m, \quad (\text{D.4})$$

one has

$$\det \left[\begin{pmatrix} r_i \\ c_j \end{pmatrix} \right]_{i,j=1}^m \geq 0. \quad (\text{D.5})$$

Proof. The entry $\binom{r}{c}$ counts directed lattice paths from

$$A_c = (-c, c) \quad \text{to} \quad B_r = (0, r) \quad (\text{D.6})$$

using east and north steps: such a path contains c east steps and $r - c$ north steps. By the Lindström–Gessel–Viennot determinant formula, the determinant (D.5) is the signed count of vertex-disjoint path families connecting the ordered sources A_{c_j} to the ordered sinks B_{r_i} . Planarity and the displayed order exclude every nonidentity matching; the surviving families have positive sign. Hence the determinant is nonnegative. $\square$

It follows immediately that every minor of F and every minor of BD_3 is nonnegative. The same is true of B because D_3 is positive diagonal.

For the principal choice of row and column indices $A = \{a_1 < \cdots < a_m\}$,

$$\det F_{A,A} > 0. \quad (\text{D.7})$$

Indeed, the matrix is lower triangular after the common ordering, with diagonal entries $(a_i)_{\underline{a_i}} = a_i!$.

D.2 The sign of the ordered Cauchy determinant

Let

$$b_1 < \cdots < b_m, \quad s_1 < \cdots < s_m, \quad b_i < s_j \text{ for all } i, j. \quad (\text{D.8})$$

The standard Cauchy determinant formula for $x_i = -b_i$ and $y_j = s_j$ gives

$$\begin{aligned} \det \left[\frac{1}{s_j - b_i} \right]_{i,j=1}^m &= \frac{\prod_{i < j} (x_j - x_i) \prod_{i < j} (y_j - y_i)}{\prod_{i,j} (x_i + y_j)} \\ &= (-1)^{\binom{m}{2}} \frac{\prod_{i < j} (b_j - b_i) \prod_{i < j} (s_j - s_i)}{\prod_{i,j} (s_j - b_i)}. \end{aligned} \quad (\text{D.9})$$

Every factor in the final quotient is positive. Hence

$$\operatorname{sgn} \det C_{B_*, S_*} = (-1)^{\binom{m}{2}}. \quad (\text{D.10})$$

The matrix $C_{bs} = (s - b)^{-1}$ is therefore strictly sign-regular in the displayed increasing order, not totally positive.

Warning D.2 (The indispensable extra sign). If one defines a corrected minor using only the column sign matrix J , the residual factor $(-1)^{\binom{m}{2}}$ remains. The definition

$$\mathfrak{P}(A) = (-1)^{mN + \sum_{a \in A} a + \binom{m}{2}} \det Q_{N,A} \quad (\text{D.11})$$

is precisely the sign that removes both effects.

D.3 Full Cauchy–Binet expansion

Fix

$$A = \{a_1 < \cdots < a_m\} \subset \{0, \dots, k-1\}. \quad (\text{D.12})$$

Apply Cauchy–Binet to

$$\mathbf{K}^{(k)} = F D_1 C D_2 B D_3. \quad (\text{D.13})$$

Let

$$B_* = \{b_1 < \cdots < b_m\} \subset \{0, \dots, k-1\}, \quad (\text{D.14})$$

$$S_* = \{s_1 < \cdots < s_m\} \subset \{N, \dots, M\}. \quad (\text{D.15})$$

Then

$$\begin{aligned} \det \mathbf{K}_{A,A}^{(k)} &= \sum_{B_*, S_*} \det F_{A, B_*} \left(\prod_{b \in B_*} (D_1)_{bb} \right) \det C_{B_*, S_*} \\ &\quad \times \left(\prod_{s \in S_*} (D_2)_{ss} \right) \det B_{S_*, A} \left(\prod_{a \in A} (D_3)_{aa} \right). \end{aligned} \quad (\text{D.16})$$

Multiplication by $(-1)^{\binom{m}{2}}$ and substitution of the diagonal factors gives

$$\mathfrak{P}(A) = \sum_{B_*, S_*} \frac{\det F_{A, B_*} \left[(-1)^{\binom{m}{2}} \det C_{B_*, S_*} \right] \det B_{S_*, A}}{\prod_{a \in A} a!}$$

$$\begin{aligned}
& \times \prod_{b \in B_*} \frac{(M-b)!}{b!(N-1-b)!} \prod_{s \in S_*} \frac{s!}{(M-s)!(s-N)!} \\
& \times x^{\sum_{s \in S_*} s - \sum_{b \in B_*} b}.
\end{aligned} \tag{D.17}$$

Every displayed factor is nonnegative. Therefore each coefficient of the resulting polynomial in x is nonnegative.

D.4 Strict positivity

Assume first that $m \leq M + 1 - N$. To show that the full sum is not zero, choose

$$B_* = A \tag{D.18}$$

and, for example,

$$S_* = \{N, N+1, \dots, N+m-1\}. \tag{D.19}$$

The first determinant is positive by (D.7). The sign-corrected Cauchy determinant is strictly positive by (D.9). Finally,

$$\det \left[\begin{pmatrix} s_i \\ a_j \end{pmatrix} \right]_{i,j=1}^m > 0 \tag{D.20}$$

for these strictly ordered sources and sinks, because there exists at least one nonintersecting path family. Since B differs from this Pascal minor only by positive row and column scalings,

$$\det B_{S_*, A} > 0. \tag{D.21}$$

Thus the chosen term is strictly positive and

$$\mathfrak{P}(A) > 0, \quad m \leq M + 1 - N. \tag{D.22}$$

If $m > M + 1 - N$, no m -element set $S_* \subset \{N, \dots, M\}$ exists. The Cauchy–Binet sum is empty and

$$\mathfrak{P}(A) = 0, \tag{D.23}$$

which is also forced by $\text{rank } Q_N = M + 1 - N$.

D.5 Polynomial degrees and logarithmic derivatives

Each term in (D.17) has degree

$$d(B_*, S_*) = \sum_{s \in S_*} s - \sum_{b \in B_*} b. \tag{D.24}$$

Since $s \geq N$ and $b < k \leq N$, every individual difference $s - b$ is positive. Also

$$0 < d(B_*, S_*) \leq mM. \tag{D.25}$$

Write

$$\mathfrak{P}(A) = \sum_d c_d x^d, \quad c_d \geq 0. \tag{D.26}$$

Then

$$x \partial_x \log \mathfrak{P}(A) = \frac{\sum_d d c_d x^d}{\sum_d c_d x^d} \tag{D.27}$$

is an exact average of the degree in a positive finite measure. More generally,

$$|(x\partial_x)^r \mathfrak{P}(A)| \leq (mM)^r \mathfrak{P}(A). \quad (\text{D.28})$$

Because $\det Q_{N,A}$ differs from $\mathfrak{P}(A)$ by a coupling-independent sign, the same absolute bound holds for $\det Q_{N,A}$.

This degree positivity is the algebraic reason why later WKB derivatives can be read as expectations after the sign correction. It is not a statement about the original residue ensemble.

D.6 One-point specialization

For $A = \{a\}$, the Cauchy–Binet expansion has one intermediate index b and one tail index s . From

$$F_{ab} = (a)_{\underline{b}}, \quad C_{bs} = \frac{1}{s-b}, \quad B_{sa} = \frac{1}{(s-a)!}, \quad (\text{D.29})$$

we obtain

$$\begin{aligned} P_a &= \sum_{b=0}^a \sum_{s=N}^M (a)_{\underline{b}} x^{-b} \frac{(M-b)!}{b!(N-1-b)!} \frac{1}{s-b} \\ &\quad \times x^s \frac{s!}{(M-s)!(s-N)!} \frac{1}{(s-a)!a!} \\ &= \sum_{b=0}^a \sum_{s=N}^M \frac{(M-b)!s!x^{s-b}}{(M-s)!(s-N)!b!(N-1-b)!(a-b)!(s-a)!(s-b)}. \end{aligned} \quad (\text{D.30})$$

This recovers (5.75) directly from the positive expansion.

D.7 Exact-rational verification

The positivity section of `code/verify_kernel.py` performs the following independent checks in the archived parameter range:

1. direct evaluation of the Cauchy determinant and comparison with (D.9);
2. exact equality of the Cauchy–Binet sum (D.17) and the sign-corrected principal minor of Q_N ;
3. positivity of every nonzero Cauchy–Binet summand;
4. positivity of every tested $\mathfrak{P}(A)$;
5. equality of the one-point specialization with (D.30).

The test computes the moment kernel and the factorized expansion separately, so an incorrect global sign cannot cancel between the two constructions.

Appendix E

Projected periods, excluded poles, and Möbius inversion

E.1 Common residue normalization for pole-supported cycles

Consider a product cycle

$$\Gamma = \gamma_{B_1} \times \cdots \times \gamma_{B_N}, \quad B_j \subset \mathcal{L}_M. \quad (\text{E.1})$$

After the scaling (4.3), every residue at $z_j = n_j$ contributes the same one-variable prefactor in (4.8). Therefore

$$Z_\Gamma = \mathcal{C}_{M,N,\rho} e^{-2N\varepsilon_{F0}/V} \sum_{n_1 \in B_1} \cdots \sum_{n_N \in B_N} \left[\prod_{j=1}^N w(n_j) \right] \Delta(n_1, \dots, n_N)^2, \quad (\text{E.2})$$

where $w(n) = (-\sigma)^n \binom{M}{n}$. Repeated indices vanish. Equation (E.2) proves that the reduced normalization (6.4) is common to every pole-supported cycle used in the manuscript.

For the excluded product cycle $B_j = \mathcal{L}_M \setminus A$, every surviving unlabelled subset S is disjoint from A and has $N!$ labelings. Hence

$$\mathcal{Z}_{\text{all}}[A] = N! \sum_{|S|=N} W(S) \xi_A(S), \quad (\text{E.3})$$

which, together with (5.33), proves (6.8) without any contour deformation beyond linearity in one-variable homology.

E.2 Detailed one-contour projection

For the cycle (6.11), fix S . The distinguished variable can be assigned to any member of

$$S \cap \{a, a+1, \dots, M\}, \quad (\text{E.4})$$

whose cardinality is $N - X_a(S)$. The remaining residues have $(N-1)!$ assignments. This proves (6.13).

Using

$$N - X_a = N - a + H_a, \quad H_a = \sum_{j < a} \xi_j, \quad (\text{E.5})$$

we obtain

$$\mathcal{Z}_a^{(1)} = (N-1)! Z_0(N-a) + (N-1)! Z_0 \sum_{j < a} \langle \xi_j \rangle_0$$

$$= \frac{N-a}{N} \mathcal{Z}_{\text{all}} + (N-1)! Z_0 \sum_{j < a} Q_{N,jj}. \quad (\text{E.6})$$

The first term is the exact all-pole contamination. Taking the adjacent difference of the second term gives $Q_{N,aa}$.

There is also a direct cycle interpretation of the combination in (6.17). Since

$$\gamma_{\geq a+1} - \gamma_{\geq a} = -\gamma_a, \quad (\text{E.7})$$

we have

$$\mathcal{Z}_{a+1}^{(1)} - \mathcal{Z}_a^{(1)} = -(N-1)! Z_0 \langle \eta_a \rangle_0, \quad (\text{E.8})$$

where $\eta_a = 1 - \xi_a$ is the occupation indicator. Since

$$\frac{1}{N} \mathcal{Z}_{\text{all}} = (N-1)! Z_0, \quad (\text{E.9})$$

adding $\mathcal{Z}_{\text{all}}/N$ converts $-\eta_a$ into ξ_a . This proves the local projection directly at the insertion level.

E.2.1 Reconciliation with the legacy anomalous determinants

In the notation of AnomalousGap-1, the exact early decomposition had the form

$$\mathcal{Z}_a^{(1)} = (N-1)! \left[(N-a) Z_0 + \sum_{b=0}^{a-1} D_b \right]. \quad (\text{E.10})$$

Comparison with (E.6) gives the exact identification

$$\boxed{\sum_{b=0}^{a-1} D_b = Z_0 \sum_{j=0}^{a-1} Q_{N,jj}.} \quad (\text{E.11})$$

Thus the triple finite-difference sum that remained after the conjectural reductions in the legacy text is the cumulative diagonal of the complementary kernel. No approximation is needed to pass from that object to the local period: the adjacent projected difference is exactly $Q_{N,aa}$.

E.3 Complete assignment proof for arbitrary cutoffs

Let $\mathbf{r} = (r_1, \dots, r_k)$ satisfy (6.23), and fix a residue subset S . We count bijections from the N labelled integration variables to the elements of S subject to the k cutoff constraints.

Order the trapped variables as $k, k-1, \dots, 1$. Suppose that variables $t+1, \dots, k$ have been assigned. Their residues all lie above r_t because

$$r_t < r_{t+1} < \dots < r_k. \quad (\text{E.12})$$

The set S contains $N - X_{r_t}(S)$ residues at or above r_t , and $k-t$ of these have already been used. Hence the conditional number of choices is

$$N - X_{r_t}(S) - (k-t) = c + t - X_{r_t}(S). \quad (\text{E.13})$$

If no admissible choice exists for a higher contour, the product count has already acquired a zero. Conditional on reaching a given step, (E.13) is a nonnegative integer. After all trapped variables have been assigned, the c remaining residues have $c!$ assignments to the unrestricted variables. Therefore the total count is

$$m_{\mathbf{r}}(S) = c! \prod_{t=1}^k [c + t - X_{r_t}(S)]. \quad (\text{E.14})$$

This establishes (6.26). The same argument with $S \cap E = \emptyset$ establishes (6.31).

E.4 Boolean Möbius inversion and coefficient recursion

Let $[R] = \{0, \dots, R-1\}$. Every function $f : \{0, 1\}^R \rightarrow \mathbb{C}$ has a unique multilinear representative

$$f(\boldsymbol{\xi}) = \sum_{A \subseteq [R]} c(A) \xi_A. \quad (\text{E.15})$$

Evaluating at the indicator vector $\mathbf{1}_B$ gives the zeta transform

$$f(\mathbf{1}_B) = \sum_{A \subseteq B} c(A). \quad (\text{E.16})$$

Möbius inversion on the subset lattice yields

$$c(A) = \sum_{B \subseteq A} (-1)^{|A|-|B|} f(\mathbf{1}_B). \quad (\text{E.17})$$

Taking $f = F_r$ gives (6.35).

For completeness, multiply a canonical polynomial

$$F_{t-1} = \sum_A C_{t-1}(A) \xi_A \quad (\text{E.18})$$

by

$$d_t + \sum_{j \in I_t} \xi_j. \quad (\text{E.19})$$

The coefficient of ξ_A receives:

1. $d_t C_{t-1}(A)$ from the constant;
2. one copy of $C_{t-1}(A)$ for every $j \in A \cap I_t$, since $\xi_A \xi_j = \xi_A$;
3. $C_{t-1}(A \setminus \{j\})$ for every $j \in A \cap I_t$.

Their sum is precisely (6.39).

As a minimal example, the consecutive cascade with $k = 2$ has

$$\begin{aligned} (c + H_1)(c + H_2) &= (c + \xi_0)(c + \xi_0 + \xi_1) \\ &= c^2 + (2c + 1)\xi_0 + c\xi_1 + \xi_0\xi_1. \end{aligned} \quad (\text{E.20})$$

The singleton coefficients agree with (6.43).

E.5 Determinant insertion formula

Let $\mathcal{I}$ be any finite set of lattice nodes and let D be diagonal on $\mathcal{I}$. Expanding the determinant in principal minors gives

$$\det(I + DQ_{N,\mathcal{I}}) = \sum_{A \subseteq \mathcal{I}} \left(\prod_{a \in A} D_{aa} \right) \det Q_{N,A}. \quad (\text{E.21})$$

Using (5.33),

$$\det(I + DQ_{N,\mathcal{I}}) = \left\langle \prod_{j \in \mathcal{I}} (1 + D_{jj} \xi_j) \right\rangle_0. \quad (\text{E.22})$$

For D in (6.46), Booleanity gives

$$\begin{aligned} 1 + D_{jj}\xi_j &= (1 - \xi_j) + \xi_j e^{U_j} \prod_{e \in E} (1 + z_e \mathbf{1}_{j=e}) \\ &= e^{U_j \xi_j} \prod_{e \in E} (1 + z_e \xi_j)^{\mathbf{1}_{j=e}}. \end{aligned} \quad (\text{E.23})$$

Multiplication over j proves (6.47).

Finally, for a singleton insertion a , differentiate

$$A(z) = I + D(\mathbf{u}, z)Q_N. \quad (\text{E.24})$$

At $z = 0$,

$$\partial_z A = e^{U_a} E_{aa} Q_N, \quad (\text{E.25})$$

where E_{aa} is the matrix unit. Jacobi's formula yields

$$\begin{aligned} \partial_z \det A &= \det A \operatorname{tr}(A^{-1} \partial_z A) \\ &= \det A e^{U_a} \operatorname{tr}(Q_N A^{-1} E_{aa}) \\ &= \det A e^{U_a} [Q_N A^{-1}]_{aa}, \end{aligned} \quad (\text{E.26})$$

which is (6.50).

E.6 Scope of the exact generator

The determinant in (6.48) generates every insertion built from cumulative holes and exclusions. A blocked one-particle level is a different finite deformation: it inserts the characteristic polynomial $\prod_{n \in S} (p - n)$ rather than the Boolean hole ξ_p . Its Heine–Christoffel representation will be introduced in Chapter 9 and Appendix I; it should not be conflated with simple pole exclusion.

Appendix F

Positive lattice Laplace method and the interior prefactor

F.1 Uniform Stirling form of the exact summand

Start from (7.4) and use the exact ratios

$$N = \nu M, \quad a = \alpha M, \quad b = \beta M, \quad s = \tau M. \quad (\text{F.1})$$

Writing every factorial as a gamma function and applying

$$\Gamma(My + 1) = \sqrt{2\pi My} \left(\frac{My}{e} \right)^{My} \left[1 + \frac{1}{12My} + O(M^{-2}) \right] \quad (\text{F.2})$$

uniformly for y separated from zero gives

$$T_{b,s}^{(a)} = \frac{e^{M\Psi(\beta,\tau)}}{(2\pi)^2 M^2} \mathcal{A}(\beta, \tau) [1 + O(M^{-1})]. \quad (\text{F.3})$$

All powers M^{My} cancel because the total numerator and denominator factorial arguments have the same sum. The residual factor $(s - b)^{-1}$ supplies one power M^{-1} ; the shifted factorial $(N - 1 - b)!$ supplies the factor $\nu - \beta$ in the numerator of $\mathcal{A}$. Direct collection yields exactly (7.15) and (7.16).

For later reference, the action separates:

$$\Psi_{\alpha,\nu}(\beta, \tau) = U_{\alpha,\nu}(\beta) + V_{\alpha,\nu}(\tau), \quad (\text{F.4})$$

where

$$U_{\alpha,\nu}(\beta) = L(1 - \beta) - L(\beta) - L(\nu - \beta) - L(\alpha - \beta) - \beta \log x, \quad (\text{F.5})$$

$$V_{\alpha,\nu}(\tau) = L(\tau) - L(1 - \tau) - L(\tau - \nu) - L(\tau - \alpha) + \tau \log x. \quad (\text{F.6})$$

Consequently the Hessian is diagonal.

F.2 The unique interior maximum

Differentiating (F.4) gives

$$\partial_\beta U = \log \frac{(\alpha - \beta)(\nu - \beta)}{x\beta(1 - \beta)}, \quad \partial_\tau V = \log \frac{x\tau(1 - \tau)}{(\tau - \nu)(\tau - \alpha)}. \quad (\text{F.7})$$

Thus the critical equations are (7.17). The two equations reduce to the common quadratic (7.18). Since

$$f(0) = \alpha\nu > 0, \quad f(\alpha) = -x\alpha(1 - \alpha) < 0, \quad (\text{F.8})$$

$$f(\nu) = -x\nu(1 - \nu) < 0, \quad f(1) = (1 - \alpha)(1 - \nu) > 0, \quad (\text{F.9})$$

the roots have the ordering (7.21).

The second derivatives at the saddle are

$$-U''(\beta_\alpha) = \frac{1}{\nu - \beta_\alpha} + \frac{1}{\alpha - \beta_\alpha} + \frac{1}{\beta_\alpha} - \frac{1}{1 - \beta_\alpha}, \quad (\text{F.10})$$

$$-V''(\tau_\alpha) = -\frac{1}{\tau_\alpha} + \frac{1}{1 - \tau_\alpha} + \frac{1}{\tau_\alpha - \nu} + \frac{1}{\tau_\alpha - \alpha}. \quad (\text{F.11})$$

Using the saddle equations and

$$f'(\beta_\alpha) = -(1 + x)q, \quad f'(\tau_\alpha) = (1 + x)q \quad (\text{F.12})$$

reduces these expressions to (7.24). They are positive and uniformly bounded away from zero and infinity on compact subsets of (7.12).

F.3 The two-dimensional lattice Gaussian

Let $\mathcal{K}$ be a compact subset of the interior domain. Choose a fixed small neighborhood $\mathcal{U}$ of the saddle, uniform on $\mathcal{K}$. Strict concavity gives

$$\sup_{(\beta, \tau) \notin \mathcal{U}} \Psi(\beta, \tau) \leq \Phi - c_{\mathcal{K}} \quad (\text{F.13})$$

for some $c_{\mathcal{K}} > 0$. The contribution outside $\mathcal{U}$ is therefore exponentially small.

Inside $\mathcal{U}$, put

$$b = M\beta_\alpha + u\sqrt{M}, \quad s = M\tau_\alpha + v\sqrt{M}. \quad (\text{F.14})$$

Taylor expansion gives

$$M\Psi = M\Phi - \frac{1}{2}h_\beta u^2 - \frac{1}{2}h_\tau v^2 + O(M^{-1/2}(|u| + |v|)^3). \quad (\text{F.15})$$

The lattice spacing in (u, v) is $M^{-1/2}$, hence the Gaussian lattice sum contributes

$$M \int_{\mathbb{R}^2} e^{-(h_\beta u^2 + h_\tau v^2)/2} du dv = \frac{2\pi M}{\sqrt{h_\beta h_\tau}}. \quad (\text{F.16})$$

Multiplication by the local factor $(2\pi)^{-2}M^{-2}\mathcal{A}$ in (F.3) yields

$$P_a = \frac{e^{M\Phi}}{2\pi M} \frac{\mathcal{A}(\beta_\alpha, \tau_\alpha)}{\sqrt{h_\beta h_\tau}} [1 + O(M^{-1})]. \quad (\text{F.17})$$

The standard odd Gaussian terms vanish; the first surviving relative correction is $O(M^{-1})$.

F.4 Simplification of the prefactor

Insert (7.16) and (7.24) into (F.17). Before using the saddle equations, the coefficient is

$$\frac{x}{2\pi(1+x)q^2} \sqrt{\frac{(1 - \beta_\alpha)^2 \tau_\alpha^2 (\nu - \beta_\alpha)}{(\tau_\alpha - \nu)(\alpha - \beta_\alpha)(\tau_\alpha - \alpha)}}. \quad (\text{F.18})$$

The two saddle identities imply

$$\frac{x^2 \tau_\alpha (1 - \beta_\alpha)(\nu - \beta_\alpha)}{(\tau_\alpha - \nu)(\alpha - \beta_\alpha)(\tau_\alpha - \alpha)} = \frac{(\nu - \beta_\alpha)^2}{\beta_\alpha(1 - \tau_\alpha)}. \quad (\text{F.19})$$

Substitution gives (7.26).

F.5 Envelope derivatives and the constant term in the degree

Since $\partial_\beta \Psi = \partial_\tau \Psi = 0$ at the saddle,

$$x\partial_x \Phi = x\partial_x \Psi|_* = \tau_\alpha - \beta_\alpha = q. \quad (\text{F.20})$$

Similarly, explicit differentiation with respect to α gives the second identity in (7.23).

Implicit differentiation of the common quadratic gives

$$x\partial_x \beta_\alpha = -\frac{x\beta_\alpha(1-\beta_\alpha)}{(1+x)q}, \quad x\partial_x \tau_\alpha = \frac{x\tau_\alpha(1-\tau_\alpha)}{(1+x)q}. \quad (\text{F.21})$$

Applying these identities to (7.26) gives

$$\begin{aligned} x\partial_x \log C &= \frac{\alpha - \beta_\alpha + x}{(1+x)q} - \frac{x}{1+x} \\ &\quad - \frac{2x[\beta_\alpha(1-\beta_\alpha) + \tau_\alpha(1-\tau_\alpha)]}{(1+x)q^2}, \end{aligned} \quad (\text{F.22})$$

which is (7.28). Uniform differentiability of the local expansion on compact positive x -intervals proves (7.27).

F.6 Why the cumulative and local objects have the same exponent

The summand of $\tilde{\mathcal{H}}_a$ in (7.6) differs from the corresponding summand of P_a by

$$\frac{T_{b,s}^{(\tilde{\mathcal{H}})}}{T_{b,s}^{(P)}} = \frac{a-b}{s-b}. \quad (\text{F.23})$$

For an interior saddle this ratio tends to

$$\frac{\alpha - \beta_\alpha}{\tau_\alpha - \beta_\alpha} > 0. \quad (\text{F.24})$$

It changes only the algebraic prefactor, not the rate Φ or the leading logarithmic derivative. This explains retrospectively why the older cumulative construction found the correct square-root exponent even though it had not isolated the pure local period.

Appendix G

Discrete lower-edge layer, matching, and the finite staircase

G.1 Endpoint representation of P_0

From (7.35), set $s = N + r$ and use

$$\frac{1}{N+r} = \int_0^1 t^{N+r-1} dt. \quad (\text{G.1})$$

The binomial sum over r gives (7.36):

$$P_0 = N \binom{M}{N} x^N \int_0^1 t^{N-1} (1+xt)^{M-N} dt. \quad (\text{G.2})$$

Write

$$h(t) = \nu \log t + (1-\nu) \log(1+xt), \quad g(t) = t^{-1}. \quad (\text{G.3})$$

The endpoint derivative is

$$h'(1) = \nu + \frac{x(1-\nu)}{1+x} = \frac{\nu+x}{1+x} = \tau_0 > 0. \quad (\text{G.4})$$

Watson's endpoint expansion gives

$$\int_0^1 g(t) e^{Mh(t)} dt = \frac{e^{Mh(1)}}{M\tau_0} \left[1 + \frac{x(1-\nu)}{M(\nu+x)^2} + O(M^{-2}) \right]. \quad (\text{G.5})$$

On the other hand,

$$\begin{aligned} N \binom{M}{N} x^N &= \sqrt{\frac{M\nu}{2\pi(1-\nu)}} \exp\{M[-\nu \log \nu - (1-\nu) \log(1-\nu) + \nu \log x]\} \\ &\times \left[1 + \frac{1-\nu^{-1} - (1-\nu)^{-1}}{12M} + O(M^{-2}) \right]. \end{aligned} \quad (\text{G.6})$$

Multiplication of (G.5) and (G.6) proves (7.40) and (7.39).

Differentiating the uniform expansion gives

$$x\partial_x \Phi_0 = \tau_0, \quad x\partial_x \log C_0 = c_0, \quad x\partial_x \delta_0^{\log} = d_0, \quad (\text{G.7})$$

with c_0, d_0 in (7.42). This proves (7.41).

G.2 The fixed-index expansion

Let $T_s^{(0)}$ be the summand of P_0 , and normalize it to a positive distribution

$$\pi_0(s) = \frac{T_s^{(0)}}{P_0}. \quad (\text{G.8})$$

The ratio of a general summand of P_a to $T_s^{(0)}$ is exactly

$$R_{a,b}(s) = \frac{x^{-b}}{b!(a-b)!} \frac{(N-1)!}{(N-1-b)!} \frac{(M-b)!}{M!} \frac{s!}{(s-a)!} \frac{s}{s-b}. \quad (\text{G.9})$$

Thus

$$\frac{P_a}{P_0} = \left\langle \sum_{b=0}^a R_{a,b}(s) \right\rangle_{\pi_0}. \quad (\text{G.10})$$

For fixed a, b , with $s = M\tau$,

$$\frac{(N-1)!}{(N-1-b)!} \frac{(M-b)!}{M!} = \nu^b \left[1 + \frac{1}{M} \left(\frac{b(b-1)}{2} - \frac{b(b+1)}{2\nu} \right) + O(M^{-2}) \right], \quad (\text{G.11})$$

$$\frac{s!}{(s-a)!} \frac{s}{s-b} = M^a \tau^a \left[1 + \frac{b-a(a-1)/2}{M\tau} + O(M^{-2}) \right]. \quad (\text{G.12})$$

The edge distribution has

$$\langle s \rangle = M\tau_0 + c_0 + O(M^{-1}), \quad \text{Var}(s) = M \frac{x(1-\nu)}{(1+x)^2} + O(1). \quad (\text{G.13})$$

Expanding τ^a about τ_0 , averaging with (G.13), and performing the finite binomial sums over b yields

$$\sum_{b=0}^a \frac{(\nu/x)^b}{b!(a-b)!} = \frac{1}{a!} \left(1 + \frac{\nu}{x} \right)^a \quad (\text{G.14})$$

and finally (7.46) with Λ and B_a given by (7.44) and (7.45). The calculation uses only a finite number of moments for every fixed a .

G.3 Small- α expansion and overlap

The roots of (7.18) have the expansions

$$\beta_\alpha = \frac{\nu}{\nu+x} \alpha - \frac{\nu x(1-\nu)}{(\nu+x)^3} \alpha^2 + O(\alpha^3), \quad (\text{G.15})$$

$$\tau_\alpha = \frac{\nu+x}{1+x} + \frac{x(1-\nu)}{(1+x)(\nu+x)} \alpha + \frac{\nu x(1-\nu)}{(\nu+x)^3} \alpha^2 + O(\alpha^3). \quad (\text{G.16})$$

Substitution in the envelope derivative (7.23) gives

$$\partial_\alpha \Phi = \log \frac{\Lambda}{\alpha} + \Xi \alpha + O(\alpha^2), \quad (\text{G.17})$$

which integrates to (7.49). Direct expansion of (7.26) gives (7.51).

Now put $\alpha = a/M$ in the interior formula. From (7.49),

$$M\Phi(a/M) = M\Phi_0 + a \log \frac{M\Lambda}{a} + a + \frac{\Xi a^2}{2M} + O(a^3/M^2). \quad (\text{G.18})$$

Together with (7.51) and

$$a! = \sqrt{2\pi a} \left(\frac{a}{e}\right)^a [1 + O(a^{-1})], \quad (\text{G.19})$$

this reduces the interior result to (7.53) whenever $a \rightarrow \infty$ and $a^2/M \rightarrow 0$. This proves the existence of the overlap domain stated in (7.52).

G.4 Double scaling at $\tilde{\mu} = 0$

The critical filling is $\nu_0 = x/(1 + 2x)$. Take

$$\nu = \nu_0 + \frac{\eta}{M}, \quad a = O(1). \quad (\text{G.20})$$

To obtain the energy through M^{-2} , retain

$$\log P_a = M\Phi_0 - \frac{1}{2} \log M + \log C_0 + a \log(M\Lambda) - \log a! + \frac{\delta_0^{\log} + B_a}{M} + O(M^{-2}). \quad (\text{G.21})$$

The dependence of every coefficient on $\nu = \nu_0 + \eta/M$ must be expanded consistently before applying

$$\Delta E_a^{\text{per}} = \frac{2\Omega(1+x)}{M} x \partial_x \log P_a. \quad (\text{G.22})$$

Direct algebra gives (7.57) and (7.58). Subtracting the results at $a + 1$ and a cancels the common terms and yields (7.59). The transition values (7.60) follow immediately.

The key scale separation is now explicit. The common critical gap changes by $O(M^{-1})$ across the double-scaled window, while the differences between neighboring marked indices are only $O(M^{-2})$. Thus the staircase affects the location of the minimizer but not the leading thermodynamic threshold.

Appendix H

Finite electron–hole duality

H.1 Complement Vandermonde identity

Let $\mathcal{L}_M = \{0, \dots, M\}$, let $S \subset \mathcal{L}_M$, and write $T = \mathcal{L}_M \setminus S$. Define

$$p_M(z) = \prod_{m=0}^M (z - m). \quad (\text{H.1})$$

The full Vandermonde splits as

$$\Delta(\mathcal{L}_M)^2 = \Delta(S)^2 \Delta(T)^2 \prod_{\substack{s \in S \\ t \in T}} (t - s)^2. \quad (\text{H.2})$$

For fixed $t \in T$,

$$p'_M(t)^2 = \prod_{s \in S} (t - s)^2 \prod_{\substack{t' \in T \\ t' \neq t}} (t - t')^2. \quad (\text{H.3})$$

Multiplying over $t \in T$ gives

$$\prod_{t \in T} p'_M(t)^2 = \left[\prod_{s \in S, t \in T} (t - s)^2 \right] \Delta(T)^4. \quad (\text{H.4})$$

Elimination of the cross product between (H.2) and (H.4) yields

$$\boxed{\Delta(S)^2 = \Delta(\mathcal{L}_M)^2 \frac{\Delta(T)^2}{\prod_{t \in T} p'_M(t)^2}.} \quad (\text{H.5})$$

On the consecutive lattice,

$$p'_M(t) = (-1)^{M-t} t! (M - t)!, \quad \Delta(\mathcal{L}_M) = \prod_{j=1}^M j!. \quad (\text{H.6})$$

H.2 Derivation of the weight factor

Let

$$w(t) = (-\sigma)^t \binom{M}{t}. \quad (\text{H.7})$$

Equation (H.6) gives

$$\frac{1}{w(t)p'_M(t)^2} = \frac{(-\sigma)^{-t}}{(M!)^2} \binom{M}{t}. \quad (\text{H.8})$$

Using (H.5),

$$\begin{aligned} W(S) &= \left[\prod_{n=0}^M w(n) \right] \Delta(\mathcal{L}_M)^2 \Delta(T)^2 \prod_{t \in T} \frac{1}{w(t)p'_M(t)^2} \\ &= \left[\prod_{n=0}^M w(n) \right] \Delta(\mathcal{L}_M)^2 (M!)^{-2\bar{N}} \Delta(T)^2 \prod_{t \in T} (-\sigma)^{-t} \binom{M}{t}. \end{aligned} \quad (\text{H.9})$$

The full-lattice product simplifies because

$$\prod_{n=0}^M \binom{M}{n} = \frac{(M!)^{M+1}}{\left(\prod_{n=0}^M n! \right)^2}, \quad (\text{H.10})$$

$$\Delta(\mathcal{L}_M)^2 = \left(\prod_{n=0}^M n! \right)^2. \quad (\text{H.11})$$

Hence

$$\left[\prod_{n=0}^M \binom{M}{n} \right] \Delta(\mathcal{L}_M)^2 = (M!)^{M+1}. \quad (\text{H.12})$$

Now reflect $t = M - a$. The Vandermonde square is invariant, $\binom{M}{M-a} = \binom{M}{a}$, and

$$\prod_{t \in T} (-\sigma)^{-t} = (-\sigma)^{-M\bar{N}} \prod_{a \in \mathcal{D}(S)} (-\sigma)^a. \quad (\text{H.13})$$

The total power of $M!$ is

$$M + 1 - 2\bar{N} = N - \bar{N} = \delta_N, \quad (\text{H.14})$$

and the total power of $-\sigma$ is

$$\frac{M(M+1)}{2} - M\bar{N} = \frac{M(N - \bar{N})}{2} = \chi_{M,N}. \quad (\text{H.15})$$

This proves (8.9).

H.3 All-pole normalization and sign

From (4.9),

$$\frac{\mathcal{C}_{M,N,\rho}}{\mathcal{C}_{M,\bar{N},\rho}} = (-1)^{\delta_N} (M!)^{-\delta_N} \left(\frac{2}{\rho} \right)^{N(N-L) - \bar{N}(\bar{N}-L)}. \quad (\text{H.16})$$

Since

$$N(N-L) = -N\bar{N} = \bar{N}(\bar{N}-L), \quad (\text{H.17})$$

the scale factor $2/\rho$ cancels. Multiplication by the reduced relation (8.16) cancels the remaining power of $M!$ and leaves

$$(-1)^{\delta_N} (-\sigma)^{\chi_{M,N}} = \varsigma_{M,N} \sigma^{\chi_{M,N}}. \quad (\text{H.18})$$

Restoring $\exp(-2N\varepsilon_{F0}/V)$ proves (8.20) including its exact sign.

The energy shift follows immediately:

$$-\partial_{1/V} \left[\chi_{M,N} \log \sigma - \frac{2\delta_N \varepsilon_{F0}}{V} \right] = \frac{2\chi_{M,N}}{\rho} + 2\delta_N \varepsilon_{F0} \quad (\text{H.19})$$

$$= \delta_N (2\varepsilon_{F0} + \Omega). \quad (\text{H.20})$$

H.4 Reciprocity of the Krawtchouk basis

Let

$$G_n(z) = (1+z)^{M-n} \left(1 + \frac{z}{\sigma}\right)^n = \sum_{s=0}^M K_s(n) z^s. \quad (\text{H.21})$$

Then

$$z^M G_n(z^{-1}) = (1+z)^{M-n} (z + \sigma^{-1})^n \quad (\text{H.22})$$

$$= \sigma^{-n} (1+z)^{M-n} (1 + \sigma z)^n = \sigma^{-n} G_{M-n}(\sigma z). \quad (\text{H.23})$$

The coefficient of z^r gives

$$K_{M-r}(n) = \sigma^{r-n} K_r(M-n). \quad (\text{H.24})$$

To obtain the kernel map, start from

$$Q_{N,ij} = w(j) \sum_{s=N}^M \frac{K_s(i) K_s(j)}{h_s}. \quad (\text{H.25})$$

After setting $s = M - r$. The coefficient identity

$$\boxed{\frac{w(j) \sigma^{2r-i-j}}{h_{M-r}} = \sigma^{j-i} \frac{w(M-j)}{h_r}} \quad (\text{H.26})$$

follows directly from

$$w(j) = (-\sigma)^j \binom{M}{j}, \quad h_s = (1-\sigma)^M \left(-\frac{1}{\sigma}\right)^s \binom{M}{s}. \quad (\text{H.27})$$

Thus

$$Q_{N,ij} = \sigma^{j-i} w(M-j) \sum_{r=0}^{\overline{N}-1} \frac{K_r(M-i) K_r(M-j)}{h_r}, \quad (\text{H.28})$$

which is (8.29).

H.5 Upper sign correction

Let $A \subset \{N, \dots, M\}$ and $m = |A|$. The reflected set $\mathcal{R}(A)$ lies in $\{0, \dots, \overline{N} - 1\}$. The lower-hole sign exponent is

$$m\overline{N} + \sum_{b \in \mathcal{R}(A)} b + \binom{m}{2} = m(M+1-N) + mM - \sum_{a \in A} a + \binom{m}{2}. \quad (\text{H.29})$$

Its difference from

$$m(N+1) + \sum_{a \in A} a + \binom{m}{2} \quad (\text{H.30})$$

is

$$2m(M-N) - 2 \sum_{a \in A} a, \quad (\text{H.31})$$

an even integer. Hence the sign in (8.42) is precisely the reflection of the lower sign. Together with (8.30), this proves (8.43).

For $m = 1$ the sign becomes $(-1)^{N+a+1}$, giving the upper one-point factor (8.37). If $N \leq a \leq M$, set

$$\bar{a} = M - a, \quad \bar{N} = M + 1 - N. \quad (\text{H.32})$$

Substitution into the lower double sum (5.75) gives the explicit positive upper formula

$$P_a^{\uparrow, (N)} = \sum_{r=0}^{\bar{a}} \sum_{s=\bar{N}}^M \frac{(M-r)! s! x^{s-r}}{(M-s)!(s-\bar{N})! r! (\bar{N}-1-r)! (\bar{a}-r)!(s-\bar{a})!(s-r)}. \quad (\text{H.33})$$

Every summand is positive for $x > 0$.

H.6 Saddle reflection

Let $\beta_\alpha, \tau_\alpha$ be the roots of

$$(1+x)z^2 - (\alpha + \nu + x)z + \alpha\nu = 0. \quad (\text{H.34})$$

Set

$$\bar{\alpha} = 1 - \alpha, \quad \bar{\nu} = 1 - \nu, \quad \bar{\beta} = 1 - \tau_\alpha, \quad \bar{\tau} = 1 - \beta_\alpha. \quad (\text{H.35})$$

Their sum and product are those of the transformed quadratic, so they are its two ordered roots. Direct substitution into the action (7.15) merely permutes the eight entropy terms:

$$\Psi_{1-\alpha, 1-\nu}(1-\tau, 1-\beta) = \Psi_{\alpha, \nu}(\beta, \tau). \quad (\text{H.36})$$

This proves (8.50). The prefactor relation (8.51) follows by substituting the transformed roots into (7.26).

H.7 Operator derivation of the physical shift

In the paired subspace the hard-core operators obey

$$[b_n, b_m^\dagger] = \delta_{nm}(1 - 2n_n), \quad n_n = b_n^\dagger b_n. \quad (\text{H.37})$$

Define reflected holes by (8.61). Then

$$n_{M-m} = 1 - \bar{n}_m. \quad (\text{H.38})$$

Using

$$2\varepsilon_{M-m} = 2(2\varepsilon_{F0} + \Omega) - 2\varepsilon_m, \quad (\text{H.39})$$

one obtains

$$\sum_{n=0}^M 2\varepsilon_n n_n = \sum_{m=0}^M 2\varepsilon_m \bar{n}_m + (2N - L)(2\varepsilon_{F0} + \Omega). \quad (\text{H.40})$$

Furthermore,

$$B^\dagger = \bar{B}, \quad B = \bar{B}^\dagger, \quad (\text{H.41})$$

and therefore

$$B^\dagger B = \bar{B} \bar{B}^\dagger = \bar{B}^\dagger \bar{B} + \sum_{m=0}^M (1 - 2\bar{n}_m) \quad (\text{H.42})$$

$$= \bar{B}^\dagger \bar{B} + L - 2\bar{N} = \bar{B}^\dagger \bar{B} + \delta_N. \quad (\text{H.43})$$

Equations (H.40) and (H.43) prove the Hamiltonian relation (8.64).

In a residue-subset basis the same statement is a literal matrix similarity. The reflected-complement permutation maps an electron subset S to $\mathcal{D}(S)$; pair hops are mapped bijectively to hole-pair hops, while every diagonal element acquires the same constant $\delta_N(2\varepsilon_{F0} + \Omega - V)$. This finite matrix form is checked exactly by the verification script of Chapter 8.

Appendix I

Heine–Christoffel identities and finite seniority

I.1 Monic normalization and signed norms

Recall the signed lattice weight

$$w(n) = (-\sigma)^n \binom{M}{n}, \quad n = 0, \dots, M, \quad (\text{I.1})$$

and the Krawtchouk-type polynomials of Appendix B. The monic normalization is

$$\pi_s^{(M)}(z; x) = s!x^s \mathsf{K}_s^{(M)}(z; \sigma). \quad (\text{I.2})$$

Its signed squared norm is

$$\mathfrak{h}_s = \sum_{n=0}^M w(n) \pi_s^{(M)}(n; x)^2 = (s!x^s)^2 h_s, \quad (\text{I.3})$$

where h_s is given in (5.16). Thus

$$\mathfrak{h}_s = (s!x^s)^2 (1 - \sigma)^M \left(-\frac{1}{\sigma}\right)^s \binom{M}{s}. \quad (\text{I.4})$$

The sign alternates with s ; none of the determinant identities below requires a positive measure.

For an n -element residue subset S , write

$$W_n(S) = \prod_{j \in S} w(j) \Delta(S)^2, \quad Z_0^{(n)} = \sum_{|S|=n} W_n(S). \quad (\text{I.5})$$

The subscript on W_n only records the subset size; the one-body weight is unchanged.

I.2 One-characteristic-polynomial Heine identity

Let

$$\mu_k = \sum_{m=0}^M w(m) m^k \quad (\text{I.6})$$

be the power moments of the signed lattice weight, and define the Hankel determinant

$$D_n = \det[\mu_{i+j}]_{i,j=0}^{n-1}. \quad (\text{I.7})$$

It is nonzero for $0 \leq n \leq M + 1$ because it is related by a unit triangular basis change to the moment determinant of Chapter 5.

Define

$$H_n(z) = \frac{1}{Z_0^{(n)}} \sum_{|S|=n} W_n(S) \chi_S(z). \quad (\text{I.8})$$

Expanding one Vandermonde determinant in the ordered form of the subset sum and applying finite Andreief gives the standard Heine determinant

$$H_n(z) = \frac{1}{D_n} \det \begin{pmatrix} \mu_0 & \mu_1 & \cdots & \mu_n \\ \mu_1 & \mu_2 & \cdots & \mu_{n+1} \\ \vdots & \vdots & & \vdots \\ \mu_{n-1} & \mu_n & \cdots & \mu_{2n-1} \\ 1 & z & \cdots & z^n \end{pmatrix}. \quad (\text{I.9})$$

The coefficient of z^n is $D_n/D_n = 1$, so H_n is monic. If $0 \leq k < n$, multiply (I.9) by $w(z)z^k$ and sum over the lattice. The last row becomes

$$(\mu_k, \mu_{k+1}, \dots, \mu_{k+n}), \quad (\text{I.10})$$

which duplicates one of the first n rows. Hence

$$\sum_{z=0}^M w(z) z^k H_n(z) = 0, \quad 0 \leq k < n. \quad (\text{I.11})$$

Thus H_n is the unique monic orthogonal polynomial of degree n . Comparing with (I.2) proves

$$H_n(z) = \pi_n^{(M)}(z; x). \quad (\text{I.12})$$

This proves [Exact identity 9.1](#). The argument is purely determinant algebra and is unchanged by the alternating signs of the norms.

I.3 Multiple characteristic polynomials

Let $z_1, \dots, z_r$ be distinct complex numbers and assume $n + r \leq M + 1$. The combined Vandermonde identity is

$$\Delta(S) \Delta(z_1, \dots, z_r) \prod_{i=1}^r \chi_S(z_i) = \Delta(S, z_1, \dots, z_r), \quad (\text{I.13})$$

with the variables ordered as displayed. Applying finite Andreief to one copy of the enlarged determinant gives

$$\frac{1}{Z_0^{(n)}} \sum_{|S|=n} W_n(S) \prod_{i=1}^r \chi_S(z_i) = \frac{\det[\pi_{n+j-1}^{(M)}(z_i; x)]_{i,j=1}^r}{\Delta(z_1, \dots, z_r)}. \quad (\text{I.14})$$

The quotient is symmetric and polynomial in the z_i , because the determinant in the numerator is antisymmetric and divisible by the Vandermonde. This proves [Exact identity 10.1](#).

For $r = 2$,

$$\frac{1}{Z_0^{(n)}} \sum_{|S|=n} W_n(S) \chi_S(p) \chi_S(q) = \frac{\pi_n(p) \pi_{n+1}(q) - \pi_{n+1}(p) \pi_n(q)}{q - p}. \quad (\text{I.15})$$

When $q \rightarrow p$, the right-hand side tends to the Wronskian

$$\pi_n(p)\pi'_{n+1}(p) - \pi_{n+1}(p)\pi'_n(p). \quad (\text{I.16})$$

Higher confluent limits replace repeated rows by derivatives. Since the number of rows is fixed, these derivatives modify only algebraic prefactors in the large- M asymptotics.

I.4 Christoffel transformation at one blocked level

Fix a lattice point p and define

$$w_{[p]}(n) = (p - n)w(n). \quad (\text{I.17})$$

Because $\pi_j(p) > 0$ for integer p and $x > 0$, the ratios

$$c_j(p) = \frac{\pi_{j+1}(p)}{\pi_j(p)} \quad (\text{I.18})$$

are finite. Define

$$\tilde{\pi}_j^{[p]}(z) = \frac{\pi_{j+1}(z) - c_j(p)\pi_j(z)}{z - p}. \quad (\text{I.19})$$

The numerator vanishes at $z = p$, and its leading terms are $z^{j+1} - c_j z^j$, so the quotient is monic of degree j .

Let q have degree below j . Then

$$\sum_{n=0}^M w_{[p]}(n) \tilde{\pi}_j^{[p]}(n) q(n) = - \sum_n w(n) [\pi_{j+1}(n) - c_j \pi_j(n)] q(n) = 0. \quad (\text{I.20})$$

Thus (I.19) is the monic orthogonal polynomial for the blocked weight. Its norm is

$$\begin{aligned} \tilde{\mathfrak{h}}_j^{[p]} &= - \sum_n w(n) \tilde{\pi}_j^{[p]}(n) [\pi_{j+1}(n) - c_j \pi_j(n)] \\ &= c_j(p) \mathfrak{h}_j, \end{aligned} \quad (\text{I.21})$$

which proves (9.47).

The occupation kernel for n active pairs is

$$K_{n,ij}^{[p],\text{occ}} = (p - j)w(j) \sum_{s=0}^{n-1} \frac{\tilde{\pi}_s^{[p]}(i) \tilde{\pi}_s^{[p]}(j)}{\tilde{\mathfrak{h}}_s^{[p]}}. \quad (\text{I.22})$$

Finite Andreief then gives

$$\frac{\sum_{|S|=n} W_n(S) \chi_S(p) \xi_A(S)}{\sum_{|S|=n} W_n(S) \chi_S(p)} = \det[I - K_n^{[p],\text{occ}}]_A. \quad (\text{I.23})$$

Repeated Christoffel transformations or the multi-determinant (I.14) produce the finite-seniority kernel $Q_n^{[p]}$ used in (10.9).

I.5 Exact derivation of the marked convolution

We prove (10.15) term by term. Let $y = xt$. The first blocked polynomial is

$$\pi_{N-1}^{(M)}(a; y) = (N-1)! \sum_{b=0}^a \binom{a}{b} \binom{M-b}{N-1-b} y^{N-1-b}. \quad (\text{I.24})$$

The reflected hole polynomial is

$$\pi_{M-N}^{(M)}(M-a; y) = (M-N)! \sum_{c=0}^{M-N} \binom{M-a}{c} \binom{M-c}{M-N-c} y^{M-N-c}. \quad (\text{I.25})$$

Set

$$c = M - s. \quad (\text{I.26})$$

Then s runs from $\max(N, a)$ to M , and the product of the two monomials is

$$y^{N-1-b+s-N} = x^{s-b-1} t^{s-b-1}. \quad (\text{I.27})$$

Therefore

$$\int_0^1 t^{s-b-1} dt = \frac{1}{s-b}. \quad (\text{I.28})$$

The coefficient from the first polynomial is

$$\frac{(N-1)! a! (M-b)!}{b!(a-b)!(N-1-b)!(M-N+1)!}. \quad (\text{I.29})$$

The coefficient from the second is

$$\frac{(M-N)! (M-a)! s!}{(M-s)!(s-a)! N! (s-N)!}. \quad (\text{I.30})$$

Multiplying them by

$$\frac{N(M-N+1)x}{a!(M-a)!} \quad (\text{I.31})$$

and by the integral (I.28) leaves

$$\frac{(M-b)! s! x^{s-b}}{(M-s)!(s-N)! b!(N-1-b)!(a-b)!(s-a)!(s-b)}, \quad (\text{I.32})$$

which is exactly the summand of (7.3). This proves the convolution identity.

I.6 Endpoint asymptotics of the convolution

For $0 < \alpha < \nu < 1$, the electron polynomial in (10.15) has blocked saddle β_α . The hole polynomial has reflected parameters

$$\bar{\alpha} = 1 - \alpha, \quad \bar{\nu} = 1 - \nu. \quad (\text{I.33})$$

By the root transformation of Chapter 8, its lower saddle is

$$\bar{\beta}_{\bar{\alpha}} = 1 - \tau_\alpha. \quad (\text{I.34})$$

Hence the degree densities are

$$\nu - \beta_\alpha, \quad (1 - \nu) - (1 - \tau_\alpha) = \tau_\alpha - \nu. \quad (\text{I.35})$$

Their sum is $q(\alpha) = \tau_\alpha - \beta_\alpha$.

Let

$$G_M(t) = \pi_{N-1}^{(M)}(a; xt) \pi_{M-N}^{(M)}(M-a; xt). \quad (\text{I.36})$$

All coefficients of G_M are positive. Uniformly on compact interior parameter sets,

$$t \partial_t \log G_M(t)|_{t=1} = Mq(\alpha) + O(1), \quad (\text{I.37})$$

and the derivative remains bounded below by a positive multiple of M in a fixed one-sided neighborhood of $t = 1$. Watson's endpoint lemma gives

$$\int_0^1 G_M(t) dt = \frac{G_M(1)}{Mq(\alpha)} [1 + O(M^{-1})]. \quad (\text{I.38})$$

Taking the logarithmic derivative and applying (9.35) to the electron and reflected hole factors proves (10.24).

I.7 Special lower-edge values

The first two lattice values of the monic polynomial are elementary:

$$\boxed{\pi_n^{(M)}(0; x) = (M)_{\underline{n}} x^n,} \quad (\text{I.39})$$

$$\boxed{\pi_n^{(M)}(1; x) = \pi_n^{(M)}(0; x) \left(1 + \frac{n}{Mx}\right).} \quad (\text{I.40})$$

The second formula follows because only $m = 0, 1$ survive in (9.8). Therefore

$$\begin{aligned} & \pi_n(0) \pi_{n+1}(1) - \pi_{n+1}(0) \pi_n(1) \\ &= \pi_n(0) \pi_{n+1}(0) \left[\frac{n+1}{Mx} - \frac{n}{Mx} \right] \\ &= \boxed{\frac{\pi_n(0) \pi_{n+1}(0)}{Mx}.} \end{aligned} \quad (\text{I.41})$$

This is (10.29). It will be used in Chapter 17 to isolate the exact period of two adjacent blocked levels at the lower Debye edge.

I.8 Verification map

The exact part of `code/verify_seniority.py` constructs the signed subset ensemble directly and compares it with:

1. the monic Krawtchouk polynomial for one characteristic polynomial;
2. the determinant (I.14) for up to three characteristic polynomials;
3. the one-step Christoffel polynomial, norm, and hole kernel;
4. the two-blocked-level determinant and its confluent limits where applicable;
5. the marked convolution (10.15);
6. the edge values (I.39)–(I.41).

The asymptotic part independently tests (9.24), (9.26), the electron/hole degree split, and the endpoint slope (I.37).

Appendix J

Endpoint estimates for the Case B cascade

J.1 Off-diagonal Stirling expansion

Start from the exact positive entry obtained from (5.49):

$$\mathbf{K}_{ij} = \sum_{b=0}^i \sum_{s=N}^M \frac{(i)_b}{j!} \frac{(M-b)!}{b!(N-1-b)!} \frac{x^{s-b}}{s-b} \times \frac{s!}{(M-s)!(s-N)!(s-j)!}. \quad (\text{J.1})$$

Set

$$i = \alpha M, \quad j = \gamma M, \quad b = \beta M, \quad s = \tau M, \quad N = \nu M. \quad (\text{J.2})$$

Gamma-function Stirling expansion gives

$$T_{b,s}^{(i,j)} = \frac{e^{M[U_{\alpha,\nu}(\beta) + V_{\gamma,\nu}(\tau)]}}{(2\pi)^2 M^2} \mathcal{A}_{\alpha,\gamma}(\beta, \tau) [1 + O(M^{-1})], \quad (\text{J.3})$$

with U, V from (12.12) and (12.13) and

$$\mathcal{A}_{\alpha,\gamma}(\beta, \tau) = \frac{1}{\tau - \beta} \sqrt{\frac{\alpha(1-\beta)(\nu-\beta)\tau}{\gamma(\alpha-\beta)\beta(1-\tau)(\tau-\nu)(\tau-\gamma)}}. \quad (\text{J.4})$$

The power M^{-2} deserves explicit notice. The ratio $(N-1-b)!$ is $\Gamma[M(\nu-\beta)]$, not $\Gamma[M(\nu-\beta)+1]$; together with the explicit factor $(s-b)^{-1}$, this produces exactly the displayed power.

The saddle equations are

$$(\alpha - \beta)(\nu - \beta) = x\beta(1 - \beta), \quad (\text{J.5})$$

$$(\tau - \nu)(\tau - \gamma) = x\tau(1 - \tau). \quad (\text{J.6})$$

Thus the maximizers are β_α and τ_γ . The curvatures are

$$h_\beta(\alpha) = \frac{(1+x)q(\alpha)}{x\beta_\alpha(1-\beta_\alpha)}, \quad (\text{J.7})$$

$$h_\tau(\gamma) = \frac{(1+x)q(\gamma)}{x\tau_\gamma(1-\tau_\gamma)}. \quad (\text{J.8})$$

The two-dimensional lattice Gaussian therefore gives

$$\mathbf{K}_{ij} = \frac{e^{M[u(\alpha)+v(\gamma)]}}{2\pi M} \frac{\mathcal{A}_{\alpha,\gamma}(\beta_\alpha, \tau_\gamma)}{\sqrt{h_\beta(\alpha)h_\tau(\gamma)}} [1 + O(M^{-1})]. \quad (\text{J.9})$$

Direct simplification yields (12.17).

For later use, write the prefactor in the separated Cauchy form

$$\mathcal{G}(\alpha, \gamma) = \frac{\ell(\alpha)r(\gamma)}{\tau_\gamma - \beta_\alpha}, \quad (\text{J.10})$$

where

$$\ell(\alpha) = \sqrt{\frac{x}{2\pi(1+x)}} \frac{1 - \beta_\alpha}{\sqrt{q(\alpha)}} \sqrt{\frac{\alpha(\nu - \beta_\alpha)}{\alpha - \beta_\alpha}}, \quad (\text{J.11})$$

$$r(\gamma) = \sqrt{\frac{x}{2\pi(1+x)}} \frac{\tau_\gamma}{\sqrt{q(\gamma)}} \frac{1}{\sqrt{\gamma(\tau_\gamma - \nu)(\tau_\gamma - \gamma)}}. \quad (\text{J.12})$$

On the diagonal,

$$\ell(\alpha)r(\alpha) = C(\alpha)q(\alpha). \quad (\text{J.13})$$

J.2 Derivative-uniform Laplace expansion and confluent stability

The determinant in the endpoint problem is parametrically smaller than the product of its individual entries. Consequently, the entrywise relative estimate in (12.16) is not by itself sufficient: the remainder must retain enough smoothness in both source indices for the confluent row and column cancellations to be controlled.

An unrestricted gamma-function continuation of the complete finite sum is *not* used. At integer i the falling factorial $(i)_b$ vanishes for $b > i$, whereas its continuation to a noninteger source generally does not. The additional alternating tail need not be exponentially small. Smoothness of such an exact continuation therefore cannot be inferred from the positive integer-source Laplace problem.

We instead construct a smooth *local approximation* and transfer its confluent determinant back to the exact lattice. Throughout this argument, $\nu = N/M$ is the exact finite-system ratio. Uniformity refers to admissible integer sequences with (ν, x) in a fixed compact parameter set.

Let $\mathcal{K} \subseteq (0, \nu)^2$ be a compact source set and enlarge it slightly without reaching a boundary. Compactness gives a number $h > 0$ such that the rectangles

$$|\beta - \beta_\alpha| < 2h, \quad |\tau - \tau_\gamma| < 2h \quad (\text{J.14})$$

remain strictly inside $0 < \beta < \alpha < \nu < \tau < 1$, with all displayed gaps bounded below, uniformly on the enlarged set. Choose a fixed smooth function ϑ with $0 \leq \vartheta \leq 1$, equal to 1 on $[-1, 1]$ and zero outside $(-2, 2)$, and put

$$\chi_{\alpha,\gamma}(\beta, \tau) = \vartheta\left(\frac{\beta - \beta_\alpha}{h}\right) \vartheta\left(\frac{\tau - \tau_\gamma}{h}\right). \quad (\text{J.15})$$

The cutoff also depends smoothly on the compact parameters (ν, x) . Let $T_{M;b,s}(\alpha, \gamma)$ denote the gamma-function expression for the summand in (J.1), evaluated at $i = M\alpha$ and $j = M\gamma$.

It is used only on the support of χ , where every gamma argument is positive and of order M . The localized kernel is

$$\tilde{\mathbf{K}}_M^{\text{loc}}(\alpha, \gamma) = \sum_{b=0}^{N-1} \sum_{s=N}^M \chi_{\alpha, \gamma}(b/M, s/M) T_{M; b, s}(\alpha, \gamma). \quad (\text{J.16})$$

A summand outside the cutoff support is defined to be zero; no gamma expression beyond this positive-argument region is needed. The summation ranges

$$0 \leq b \leq N-1, \quad N \leq s \leq M \quad (\text{J.17})$$

are fixed, but (J.16) is not asserted to interpolate the exact kernel. Its error at lattice sources is estimated separately.

Lemma J.1 (Exponentially accurate lattice localization). *There are constants $c, C > 0$ and a finite exponent B , uniform on the compact sets just specified, such that at integer sources $i = M\alpha, j = M\gamma$,*

$$\begin{aligned} & \left| Me^{-M[u(\alpha)+v(\gamma)]} (\mathbf{K}_{ij} - \tilde{\mathbf{K}}_M^{\text{loc}}(\alpha, \gamma)) \right| \\ & + \left| x\partial_x \left\{ Me^{-M[u(\alpha)+v(\gamma)]} (\mathbf{K}_{ij} - \tilde{\mathbf{K}}_M^{\text{loc}}(\alpha, \gamma)) \right\} \right| \leq CM^B e^{-cM}. \end{aligned} \quad (\text{J.18})$$

The derivative holds M, N, i, j fixed. No derivative with respect to an interpolated source is claimed for this error.

Proof. For integer sources, the exact summands are nonnegative and terminate at $b = i$. Their scaled domain is $0 \leq \beta \leq \alpha, \nu \leq \tau \leq 1$. The separated action $U_{\alpha, \nu}(\beta) + V_{\gamma, \nu}(\tau)$ has the unique maximum $u(\alpha) + v(\gamma)$ there. Its two factors are strictly concave in the open domains, and the saddles remain separated from the boundaries. Consequently, outside the smaller rectangle where $\chi = 1$, compactness gives a fixed action gap $c_1 > 0$.

Uniform factorial bounds of the form $\log(n!) = n \log n - n + O(\log(M+1))$ for $0 \leq n \leq M$ bound every exact summand by a fixed power of M times the exponential of this action, including at $\beta = 0, \alpha$ and $\tau = \nu, 1$. The explicit denominator obeys $s - b \geq N - i \geq c_2 M$. There are at most $(M+1)^2$ summands. Positivity then bounds the discarded sum by $CM^B e^{M(u+v)-c_1 M}$.

The exact logarithmic derivative of an uncut summand is $s - b$, between 1 and M . Differentiating the cutoff adds only bounded smooth factors supported away from the smaller saddle rectangle; differentiating the normalization adds another factor of order M . These polynomial losses are absorbed by the same exponential estimate, with slightly smaller c . This proves (J.18). $\square$

Lemma J.2 (Derivative-uniform localized Laplace expansion). *For every fixed integer $m \geq 1$ there are smooth functions $\mathcal{G}_0, \mathcal{G}_1$ on $\mathcal{K}$, with $\mathcal{G}_0 = \mathcal{G}$ from (12.17), such that*

$$\begin{aligned} \widehat{K}_M^{\text{loc}}(\alpha, \gamma) &:= Me^{-M[u(\alpha)+v(\gamma)]} \tilde{\mathbf{K}}_M^{\text{loc}}(\alpha, \gamma) \\ &= \mathcal{G}_0(\alpha, \gamma) + \frac{\mathcal{G}_1(\alpha, \gamma)}{M} + O_{C^{2m}}(M^{-2}). \end{aligned} \quad (\text{J.19})$$

All mixed source derivatives of total order at most $2m$ obey the displayed bound, and the same bound holds after one application of $x\partial_x$.

Proof. The cutoff keeps every factorial argument a fixed positive multiple of M . Stirling expansion, with differentiated remainders to any fixed required order, is therefore uniform throughout its support. The leading amplitude is (J.4). Contributions from the cutoff

transition have a uniform action gap and remain exponentially small after any fixed number of source or coupling derivatives.

In the region where the cutoff equals one, set $y = (b - M\beta_\alpha)/\sqrt{M}$ and $z = (s - M\tau_\gamma)/\sqrt{M}$. The Hessian is uniformly positive after reversing the sign of the action. Taylor expansion of the normalized summand gives a Gaussian in (y, z) times polynomials and successive powers of $M^{-1/2}$. This expansion can be taken to an arbitrarily high fixed order before differentiation. For a Gaussian times a polynomial, Poisson summation shows that its shifted lattice sum equals its Gaussian integral up to $O(M^{-L})$ for any prescribed fixed L ; the estimate remains true for the finite set of parameter derivatives in the lemma. The Gaussian tails outside a slowly growing neighborhood are smaller than the required remainder. Odd terms integrate to zero, leaving integer powers of M^{-1} .

To control the derivatives, first take enough Taylor and Stirling terms that the polynomial losses from the at most $2m$ source derivatives and one coupling derivative are dominated by the discarded order. The normalized action, saddle, Hessian, and amplitudes are smooth with bounded derivatives on the compact sets. The resulting integrated coefficients can thus be differentiated term by term with an $O(M^{-2})$ remainder in the stated norms. The leading Gaussian integral is precisely (J.9); it gives $\mathcal{G}_0 = \mathcal{G}$. $\square$

Lemma J.3 (Confluent determinant stability). *Let*

$$H_M = H_0 + M^{-1}H_1 + O_{C^{2m}}(M^{-2}) \quad (\text{J.20})$$

near (κ, κ) , and let $\delta_1, \dots, \delta_m$ be distinct bounded numbers. Then

$$\begin{aligned} & \det \left[H_M \left(\kappa + \frac{\delta_i}{M}, \kappa + \frac{\delta_j}{M} \right) \right]_{i,j=1}^m \\ &= M^{-m(m-1)} \mathcal{V}(\boldsymbol{\delta})^2 \left\{ \det \left[\frac{\partial_\alpha^{p-1} \partial_\gamma^{q-1} H_0(\kappa, \kappa)}{(p-1)!(q-1)!} \right]_{p,q=1}^m + O(M^{-1}) \right\}. \end{aligned} \quad (\text{J.21})$$

Proof. Replace the rows successively by divided differences in their first argument. This factors

$$M^{-\binom{m}{2}} \mathcal{V}(\boldsymbol{\delta}). \quad (\text{J.22})$$

Apply the same operation to the columns in the second argument, producing a second identical factor. The remaining matrix converges to the mixed-derivative matrix in (J.21). The C^{2m} remainder controls every divided difference uniformly. In particular, the term $M^{-1}H_1$ inherits both Vandermonde factors and perturbs the coefficient in braces by $O(M^{-1})$. When its leading determinant is nonzero, as in the Cauchy application below, this is a relative $O(M^{-1})$ correction; an entrywise remainder cannot bypass the Vandermonde factors. If the expansion is controlled after one coupling derivative, the same conclusion holds for that derivative: divided differences at fixed source nodes commute with $x\partial_x$. The nodes, rather than their offsets from a coupling-dependent center, must be held fixed. $\square$

Apply the smooth confluent lemma to

$$H_M^{\text{loc}}(\alpha, \gamma) = \frac{\widehat{K}_M^{\text{loc}}(\alpha, \gamma)}{\ell(\alpha)r(\gamma)}. \quad (\text{J.23})$$

Its leading term is $H_0(\alpha, \gamma) = (\tau_\gamma - \beta_\alpha)^{-1}$. The factors ℓ, r are smooth, positive, and bounded away from zero on these compact sets. For integer sources define the exact normalized matrix

$$H_{M,ij}^{\text{lat}} = \frac{M e^{-M[u(\alpha_i) + v(\alpha_j)]} \mathbf{K}_{a_i, a_j}}{\ell(\alpha_i)r(\alpha_j)}, \quad \alpha_i = a_i/M. \quad (\text{J.24})$$

The localization lemma gives an entrywise difference $O(M^B e^{-cM})$ between this matrix and $[H_M^{\text{loc}}(\alpha_i, \alpha_j)]$, also after one fixed-label coupling derivative. Both matrices have uniformly bounded entries and coupling derivatives. Multilinearity of a determinant of the fixed size m therefore gives

$$\det H_M^{\text{lat}} = \det[H_M^{\text{loc}}(\alpha_i, \alpha_j)] + O(M^B e^{-cM}), \quad (\text{J.25})$$

with the same type of derivative estimate. In the confluent regime the smooth determinant has size $M^{-m(m-1)}$ with nonzero Cauchy coefficient. Since $M^B e^{-cM} = o(M^{-L})$ for every fixed L , the transfer error remains negligible even relative to this small determinant. Thus the Vandermonde cancellations are proved on the localized smooth kernel and transferred to the exact lattice; no uncontrolled off-lattice tail enters the argument.

J.3 Cauchy determinant and confluent endpoint scaling

For fixed m and $A = \{a_1 < \dots < a_m\}$ in the bounded-offset endpoint regime, put $\alpha_i = a_i/M$. The localized confluent expansion and (J.25) justify the following formula after extracting the row and column factors in (J.10):

$$\begin{aligned} \mathfrak{P}(A) &= M^{-m} e^{M \sum_i \Phi(\alpha_i)} \left[\prod_i \ell(\alpha_i) r(\alpha_i) \right] \\ &\times \frac{\Delta(\beta_{\alpha_1}, \dots, \beta_{\alpha_m}) \Delta(\tau_{\alpha_1}, \dots, \tau_{\alpha_m})}{\prod_{i,j} (\tau_{\alpha_j} - \beta_{\alpha_i})} [1 + O(M^{-1})]. \end{aligned} \quad (\text{J.26})$$

The fixed Cauchy sign has already been removed in the definition of $\mathfrak{P}$.

Let $\alpha_i = \kappa + \delta_i/M$, with bounded distinct offsets. Taylor expansion gives

$$\Delta(\beta_{\alpha_1}, \dots, \beta_{\alpha_m}) = M^{-\binom{m}{2}} (\beta'_\kappa)^{\binom{m}{2}} \mathcal{V}(\boldsymbol{\delta}) [1 + O(M^{-1})], \quad (\text{J.27})$$

$$\Delta(\tau_{\alpha_1}, \dots, \tau_{\alpha_m}) = M^{-\binom{m}{2}} (\tau'_\kappa)^{\binom{m}{2}} \mathcal{V}(\boldsymbol{\delta}) [1 + O(M^{-1})]. \quad (\text{J.28})$$

Also

$$M \sum_i \Phi(\alpha_i) = d_\kappa \sum_i \delta_i + O(M^{-1}), \quad (\text{J.29})$$

$$\prod_i \ell(\alpha_i) r(\alpha_i) = [C_\kappa q_\kappa]^m [1 + O(M^{-1})], \quad (\text{J.30})$$

$$\prod_{i,j} (\tau_{\alpha_j} - \beta_{\alpha_i}) = q_\kappa^{m^2} [1 + O(M^{-1})]. \quad (\text{J.31})$$

Substitution yields (12.25).

When differentiating with respect to x , the lattice indices a_i are held fixed. One may not differentiate the centering relation $a_i = \kappa(x)M + \delta_i$ as though the integer labels moved with the coupling. At fixed labels, the leading derivative is

$$x \partial_x \log \mathfrak{P}(A) = M \sum_i q(\alpha_i) + O(1) = mMq_\kappa + O(1). \quad (\text{J.32})$$

J.4 Uniform envelopes for arbitrary supports

Choose $\epsilon > 0$ so that the endpoint strip remains in a compact subset of $(0, \nu)$. By [Period asymptotics 12.2](#), there is a constant C such that

$$\mathbf{K}_{ij} \leq \frac{C}{M} e^{M[u(i/M) + v(j/M)]} \quad (\text{J.33})$$

when both indices lie in the strip. Expanding a principal determinant by permutations gives

$$\begin{aligned} \mathfrak{P}(A) &\leq \sum_{\pi \in S_m} \prod_{r=1}^m \mathbf{K}_{a_r, a_{\pi(r)}} \\ &\leq m! \prod_{a \in A} \left[\frac{C}{M} e^{M\Phi(a/M)} \right]. \end{aligned} \quad (\text{J.34})$$

The identity $\sum_r v(a_{\pi(r)}/M) = \sum_r v(a_r/M)$ is what makes the bound multiplicative in the diagonal rate Φ .

If at least one index lies in the deep region, positivity of the exact double sum and the uniform factorial bounds give the coarser entry estimate

$$\mathbf{K}_{ij} \leq CM^{B_0} e^{M[u(i/M)+v(j/M)]}, \quad B_0 \geq 0, \quad (\text{J.35})$$

with fixed constants. To convert this into a support-wise bound, let a support of size m contain r deep indices. In any permutation product, at most $2r$ entries have a deep row or a deep column. If their number is $t \leq 2r$, its polynomial prefactor is bounded by

$$C^m M^{-m+(B_0+1)t} \leq C^m M^{-(m-r)+(2B_0+1)r}. \quad (\text{J.36})$$

Thus one may take the exponent in the deep-node weight (12.32) to be $B = 2B_0 + 1$, retaining C/M for every endpoint-strip node. The permutation of column actions again leaves $\sum_a \Phi(a/M)$ unchanged. Summing the at most $m!$ permutation terms proves the claimed product envelope for arbitrary supports, with constants independent of their size. This estimate uses entry bounds only; it does not extend the fixed- m confluent asymptotic to growing m . Since $\Phi \leq -c_1 < 0$ in the deep region, the polynomial loss is exponentially dominated.

For $a = k - r$ in the strip, use $c \geq c_0 M$ and

$$\left(1 + \frac{1}{c}\right)^r - 1 \leq e^{r/c} - 1 \leq C \frac{r}{M} \quad (\text{J.37})$$

for $r \leq \epsilon M + O(1)$. The mean-value theorem and $\Phi'(\alpha) \geq d_1 > 0$ in the strip imply

$$M\Phi((k-r)/M) \leq C_0 - d_1 r. \quad (\text{J.38})$$

Therefore

$$S_M \leq \frac{C}{M^2} \sum_{r \geq 1} r e^{-d_1 r} + CM^{B+1} e^{-c_1 M} = O(M^{-2}). \quad (\text{J.39})$$

For a support of size m , the Boolean product bound and the minor envelope give

$$\widehat{C}_{k,c}(A) \mathfrak{P}(A) \leq m! \prod_{a \in A} w_a, \quad w_a = g_{M,k}(a) \omega_M(a). \quad (\text{J.40})$$

Since $m! e_m(\mathbf{w}) \leq (\sum_a w_a)^m = S_M^m$, summing over all support sizes proves (12.41). The exact degree estimate (5.73) adds at most mM , so

$$|x \partial_x \mathcal{R}| \leq M \sum_{m \geq 1} m S_M^m = O(M^{-1}). \quad (\text{J.41})$$

This completes the full support summation without selecting any endpoint component.

J.5 The leading singleton correction

Here $\nu_M = N/M$ is exact, and all endpoint quantities are evaluated at $\kappa_M = \kappa(\nu_M, x)$; suppress their subscript M in the following display. Assume $k = \kappa_M M + \delta_M$, $|\delta_M| = O(1)$, and put $a = k - r$ with fixed $r \geq 1$. The singleton coefficient and one-point asymptotics give

$$\hat{C}_{k,c}(\{k-r\}) = \frac{r}{(\nu - \kappa)M} + O(M^{-2}), \quad (\text{J.42})$$

$$Q_{N,k-r,k-r} = (-1)^{N+k-r} \frac{C_\kappa}{M} e^{d_\kappa(\delta_M-r)} [1 + O(M^{-1})]. \quad (\text{J.43})$$

All supports of size at least two contribute $o(M^{-2})$, and the exponentially decaying tail permits summation over all fixed r . Hence

$$\begin{aligned} \mathcal{R} - 1 &= \frac{(-1)^{N+k} C_\kappa e^{d_\kappa \delta_M}}{(\nu - \kappa)M^2} \sum_{r \geq 1} r (-e^{-d_\kappa})^r + o(M^{-2}) \\ &= (-1)^{N+k+1} \frac{C_\kappa e^{d_\kappa \delta_M}}{(\nu - \kappa)M^2} \frac{e^{-d_\kappa}}{(1 + e^{-d_\kappa})^2} + o(M^{-2}), \end{aligned} \quad (\text{J.44})$$

which is (12.48).

J.6 Mixed minors and the separated marked ratio

Let a/M remain a fixed distance above κ and below ν . Include this separated compact source range together with the endpoint strip in the regular-index region. The C/M off-diagonal bound applies when both indices are regular; the preceding at-most- $2r$ argument bounds entries incident on deep indices. For $A = \{a_1, \dots, a_m\} \subset \{0, \dots, k-1\}$, expanding the $(m+1)$ -point determinant therefore gives

$$|\det Q_{N,A \cup \{a\}}| \leq (m+1)! \left[\frac{C}{M} e^{M\Phi(a/M)} \right] \prod_{b \in A} \omega_M(b). \quad (\text{J.45})$$

The one-point asymptotics supply the matching lower bound

$$|Q_{N,aa}| = P_a \geq \frac{c_a}{M} e^{M\Phi(a/M)} \quad (\text{J.46})$$

with $c_a > 0$ uniform in the separated compact range. Division initially leaves a fixed factor K_0 , bounded by the uniform ratio of the marked-entry upper constant to this lower constant. For $m \geq 1$, replace every $\omega_M(b)$ by $\max(1, K_0)\omega_M(b)$; their m -fold product absorbs that factor. The common constant in (12.32) may be chosen this large from the start, without changing $S_M = O(M^{-2})$. The $m = 0$ quotient equals one exactly. This proves (12.55) with all prefactors accounted for.

Summing supports as before,

$$|\mathcal{M}(a) - 1| \leq \sum_{m \geq 1} (m+1) S_M^m = O(M^{-2}). \quad (\text{J.47})$$

For the derivative, the sign-corrected numerator and denominator are positive polynomials. Hence

$$\left| x \partial_x \frac{\det Q_{A \cup \{a\}}}{Q_{aa}} \right| \leq (m+2)M \left| \frac{\det Q_{A \cup \{a\}}}{Q_{aa}} \right|. \quad (\text{J.48})$$

The same support sum gives $x \partial_x \mathcal{M} = O(M^{-1})$. Since $\mathcal{M} = 1 + O(M^{-2})$, this is also the logarithmic-derivative estimate used in Period asymptotics 12.6.

Appendix K

Gaudin determinant and spectral-shift representation

K.1 Hessian normalization and saddle expansion

After the change of variables (15.1), the reduced period has the form

$$Z_\Gamma = \mathcal{N}_{M,N,\rho} e^{-2N\varepsilon_{F0}/V} \int_\Gamma e^{\mathcal{S}(\mathbf{r};u)} \prod_{j=1}^N dr_j, \quad (\text{K.1})$$

where $\mathcal{N}_{M,N,\rho}$ is independent of V , and $\mathcal{S}$ is given by (15.2). Let $\mathcal{J}$ denote the real middle-dimensional steepest-descent coordinates on the selected thimble. In these coordinates the quadratic term is

$$\mathcal{S}(\mathbf{r}_* + \boldsymbol{\eta}; u) = \mathcal{S}(\mathbf{r}_*; u) - \frac{1}{2} \boldsymbol{\eta}^\top \mathbf{H} \boldsymbol{\eta} + O(\eta^3), \quad (\text{K.2})$$

with the holomorphic measure and orientation included consistently in the Gaussian determinant. At a regular saddle this gives the usual local Gaussian expression below. Its uniform validity when N grows with M , and across any collisions of root coordinates, is part of $\mathbf{B}_0$ rather than a consequence of this local formula.

$$Z_\Gamma^{\text{th}} = \mathcal{N}_{M,N,\rho} e^{-2N\varepsilon_{F0}/V} e^{\mathcal{S}(\mathbf{r}_*;u)} \frac{(2\pi)^{N/2} \mathbf{n}_\Gamma}{\sqrt{\det(-\mathcal{G})}} \exp \mathcal{R}_M(u), \quad \partial_u \mathcal{R}_M = O(M^{-1}). \quad (\text{K.3})$$

The action has $O(M^2)$ contributions and the number of integration variables also grows, $N \sim M$. These counts alone do not determine a uniform fluctuation scale or the order of connected diagrams. A proof would require bounds on the inverse Hessian, higher derivatives and index sums, together with control after $\partial_{1/V}$ differentiation. Root collisions also require regular coordinates or a separate uniform argument. The stated remainder is retained here only under $\mathbf{B}_0$. A constant rescaling of every integration coordinate contributes a coupling-independent Jacobian and does not affect the energy derivative.

Under the differentiated remainder and contour-selection clauses of $\mathbf{B}_0$, differentiation of (K.3) gives

$$E_0^{\text{per}} = E_0^{\text{RG}} + \frac{1}{2} \partial_{1/V} \log \det(-\mathcal{G}) + o(1), \quad (\text{K.4})$$

which is (15.5). The $o(1)$ also includes the full-period selection error in $\mathbf{B}_{0,\text{sel}}$; the stronger $O(M^{-1})$ local derivative bound does not improve that separate selection error.

K.2 Leading spectral curve and the finite auxiliary function

The leading Richardson resolvent may be written

$$\mathcal{R}_0(z) = -\frac{1}{2}\mathcal{E}_M(z)\mathcal{F}_M(z), \quad (\text{K.5})$$

where $\mathcal{E}_M$ and $\mathcal{F}_M$ are defined in [Equations \(15.11\)](#) and [\(15.16\)](#). The conditions at infinity are exactly the two equations [\(15.8\)](#). In particular,

$$\mathcal{F}_M(z) = \frac{2}{Vz} + \frac{1}{z^2} \sum_{n=0}^M \frac{\xi_n}{\mathcal{E}_{n,M}} + O(z^{-3}). \quad (\text{K.6})$$

Comparison with the product representation [\(15.14\)](#) gives the useful sum rule

$$\boxed{\sum_{n=0}^M \xi_n - \sum_{\ell=1}^M y_\ell = \tilde{\mu}_M + V \left(\frac{M+1}{2} - N \right)}. \quad (\text{K.7})$$

Indeed,

$$\frac{V}{2} \sum_n \frac{\xi_n}{\mathcal{E}_{n,M}} = \frac{V}{2} (M+1-2N) + \tilde{\mu}_M \quad (\text{K.8})$$

by the finite number equation.

K.3 Known Richardson response and the missing Hessian identification

The following expression is the first correction in the standard Richardson resolvent expansion; it is equation (2.14) of [\[23\]](#), written in the present notation. We use it as an attributed input for the auxiliary spectral calculation, rather than claiming a new derivation from $\mathcal{G}$. It reads

$$\boxed{\mathcal{R}_1(z) = \frac{1}{2\mathcal{E}_M(z)} \left[\sum_{n=0}^M \frac{z + \xi_n - 2\tilde{\mu}_M}{\mathcal{E}_M(z) + \mathcal{E}_{n,M}} - \sum_{\ell=1}^M \frac{z + y_\ell - 2\tilde{\mu}_M}{\mathcal{E}_M(z) + \mathcal{E}_M(y_\ell)} - \frac{z - \tilde{\mu}_M}{\mathcal{E}_M(z)} \right]}. \quad (\text{K.9})$$

The elementary identity

$$\frac{z + w - 2\tilde{\mu}_M}{\mathcal{E}_M(z) + \mathcal{E}_M(w)} = \frac{\mathcal{E}_M(z) - \mathcal{E}_M(w)}{z - w} \quad (\text{K.10})$$

shows that each displayed difference quotient is regular at $z = w$ whenever the chosen square-root branch is analytic and nonzero there. The two sums in [\(K.9\)](#) run over fixed poles and auxiliary zeros; they do not constitute a list of the N eigenvalues of the Gaudin matrix.

Remark K.1 (Unproved connection to the period Hessian). The earlier version described Cauchy interpolation and a Schur complement as if they provided a determinant factorization. No interpolation matrix, full set of interpolation data, or factorization of $\det \mathcal{G}$ was supplied. To close that argument one must derive the response equation from the finite Hessian, specify the map to the auxiliary pole/zero representation, and control its differentiated remainder uniformly in M . In particular, the M auxiliary zeros cannot be identified with the eigenvalues of an $N \times N$ Hessian without an explicit augmented construction. This connection is the independent hypothesis G in [Assumption 15.2](#).

Expand (K.9) at infinity. The constant terms in the bracket cancel because there are $M + 1$ poles, M finite zeros, and one subtraction at infinity. The coefficient of z^{-2} is

$$\frac{1}{2} \left[\sum_n \xi_n - \sum_\ell y_\ell - \tilde{\mu}_M + \sum_\ell \mathcal{E}_M(y_\ell) - \sum_n \mathcal{E}_{n,M} \right]. \quad (\text{K.11})$$

Using (K.7), the first three terms become

$$V \left(\frac{M+1}{2} - N \right). \quad (\text{K.12})$$

Thus twice the z^{-2} coefficient of the known response field is the auxiliary spectral-shift expression below. Its identification with a logarithmic Hessian derivative is precisely G:

$$-\frac{1}{2} \partial_{1/V} \log \det(-\mathcal{G}) = V \left(\frac{M+1}{2} - N \right) + \sum_\ell \mathcal{E}_M(y_\ell) - \sum_n \mathcal{E}_{n,M} + O(M^{-1}), \quad (\text{K.13})$$

The coefficient algebra recovers the standard Richardson correction (equation (3.1) of [23]). It does not prove G or its finite-defect extension; [Period asymptotics 15.3](#) remains conditional on that hypothesis.

K.4 Interlacing and the argument-principle sum

The positivity of the residues of $\mathcal{F}_M$ implies strict monotonicity between poles, as in (15.12). Moreover,

$$\mathcal{F}_M(z) \rightarrow +\infty \quad (z \downarrow \xi_{\ell-1}), \quad \mathcal{F}_M(z) \rightarrow -\infty \quad (z \uparrow \xi_\ell), \quad (\text{K.14})$$

so there is exactly one simple zero in every interval.

For a function g analytic near the window,

$$\boxed{\sum_{\ell=1}^M g(y_\ell) - \sum_{n=0}^M g(\xi_n) = \frac{1}{2\pi i} \oint_{\mathcal{C}} g(z) d \log \mathcal{F}_M(z),} \quad (\text{K.15})$$

where $\mathcal{C}$ encloses all finite poles and zeros but not infinity. This identity is exact. In the continuum limit the contour is collapsed onto the two sides of $[0, \Omega]$. The discontinuity of $\log \mathcal{F}$ is $-2\pi i a(\epsilon)$, with the branch convention of (15.20). Integration by parts gives

$$-g(0) - \int_0^\Omega g'(\epsilon) a(\epsilon) d\epsilon, \quad (\text{K.16})$$

proving (15.23).

The same shift follows locally. Write

$$y_\ell = \xi_\ell - da_\ell, \quad 0 < a_\ell < 1. \quad (\text{K.17})$$

Separating the locally singular lattice sum from its smooth principal value and using

$$\text{PV} \sum_{m \in \mathbb{Z}} \frac{1}{m+a} = \pi \cot(\pi a) \quad (\text{K.18})$$

gives

$$\cot(\pi a_\ell) = \frac{\mathcal{E}(\xi_\ell)}{\pi} \text{PV} \int_0^\Omega \frac{d\epsilon'}{(\xi_\ell - \epsilon') \mathcal{E}(\epsilon')} + o(1). \quad (\text{K.19})$$

This is equivalent to

$$a_\ell \rightarrow -\frac{1}{\pi} \arg \mathcal{F}(\xi_\ell + i0). \quad (\text{K.20})$$

K.5 Explicit continuum Cauchy transform

With $\epsilon = \tilde{\mu} + \Delta \sinh u$,

$$\mathcal{F}(\epsilon(u) + i0) = \frac{1}{\Delta} \text{PV} \int_{u_-}^{u_+} \frac{du'}{\sinh u - \sinh u'} - \frac{i\pi}{\Delta \cosh u}. \quad (\text{K.21})$$

The principal value is elementary:

$$\text{PV} \int_{u_-}^{u_+} \frac{du'}{\sinh u - \sinh u'} = \frac{\mathcal{L}(u)}{\cosh u}, \quad (\text{K.22})$$

where $\mathcal{L}$ is (15.29). One way to verify (K.22) is to put $t = e^{u'}$, perform the partial-fraction decomposition, and take symmetric limits around $u' = u$. Equations (15.28) and (15.30) follow immediately.

The phase form also gives an alternative expression for the fluctuation energy. Integrating (15.24) by parts and using $a(0) = 1$, $a(\Omega) = 0$,

$$\boxed{\delta E_{\text{fl}} = v\Omega \left(\frac{1}{2} - \nu \right) + \int_0^\Omega \mathcal{E}(\epsilon) a'(\epsilon) d\epsilon.} \quad (\text{K.23})$$

The derivative form is useful analytically; (15.33) is numerically more stable because it avoids differentiating the endpoint logarithms.

K.6 Half-filled reduction

At half filling, $u_{\pm} = \pm t$, $t = 1/v$, and

$$\mathcal{L}(u) = \log \frac{\sinh(t+u)}{\sinh(t-u)}. \quad (\text{K.24})$$

The symmetry is

$$a(-u) = 1 - a(u). \quad (\text{K.25})$$

Using (K.23),

$$-\frac{\delta E_{\text{fl}}}{\Delta} = \int_{-t}^t \cosh u [-a'(u)] du. \quad (\text{K.26})$$

The Fourier representation of the Poisson kernel for the strip gives

$$\int_{-t}^t \cosh u [-a'(u)] du = \cosh t \, 2 \int_0^\infty \frac{ds}{\pi(1+s^2)} \frac{\cosh(\pi s/2)}{\sqrt{\cosh^2(\pi s/2) + \sinh^2 t}}. \quad (\text{K.27})$$

Since $\Delta \cosh t = D \coth t$, this is exactly Equations (15.46) and (15.47).

K.7 Smoothness and electron–hole covariance

Under the thermodynamic electron–hole map,

$$\nu \mapsto 1 - \nu, \quad \tilde{\mu} \mapsto \Omega - \tilde{\mu}, \quad \Delta \mapsto \Delta. \quad (\text{K.28})$$

Therefore

$$\mathcal{E}_{1-\nu}(\epsilon) = \mathcal{E}_\nu(\Omega - \epsilon). \quad (\text{K.29})$$

Changing variables in the Cauchy transform shows

$$a_{1-\nu}(\epsilon) = 1 - a_\nu(\Omega - \epsilon). \quad (\text{K.30})$$

Substitution into (15.32) yields

$$\delta E_{\text{fl}}(\nu, v) - \delta E_{\text{fl}}(1 - \nu, v) = v\Omega(1 - 2\nu). \quad (\text{K.31})$$

For smoothness, fix a compact parameter set with $0 < \nu < 1$ and $v > 0$. Then Δ is bounded away from zero. Near either endpoint,

$$\mathcal{L}(u) = \pm \log |u - u_\pm| + O(1), \quad (\text{K.32})$$

so $a(u)$ is bounded and all parameter derivatives needed at fixed order are dominated after subtracting this universal logarithm. Hence δE_{fl} is smooth. The equation $\Phi_0 = 0$ defining the A–B topology change introduces no singularity into $\tilde{\mu}$, Δ , or the phase integral.

Appendix L

Derivation and real-axis form of the finite-defect response

L.1 Linear response of the gap and number constraints

The continuum spectral-curve equations are

$$\frac{2}{v} = \int_0^\Omega \frac{d\epsilon}{\mathcal{E}(\epsilon)}, \quad \Omega(1 - 2\nu) = \int_0^\Omega \frac{\epsilon - \tilde{\mu}}{\mathcal{E}(\epsilon)} d\epsilon. \quad (\text{L.1})$$

To represent two blocked lattice levels at a macroscopic point ξ , introduce a bookkeeping parameter h with dimensions of energy and write

$$\frac{2}{v} = \int_0^\Omega \frac{d\epsilon}{\sqrt{(\epsilon - \tilde{\mu}_h)^2 + \Delta_h^2}} - \frac{2h}{\sqrt{(\xi - \tilde{\mu}_h)^2 + \Delta_h^2}}, \quad (\text{L.2})$$

$$\begin{aligned} \Omega(1 - 2\nu) &= \int_0^\Omega \frac{\epsilon - \tilde{\mu}_h}{\sqrt{(\epsilon - \tilde{\mu}_h)^2 + \Delta_h^2}} d\epsilon \\ &\quad - 2h \frac{\xi - \tilde{\mu}_h}{\sqrt{(\xi - \tilde{\mu}_h)^2 + \Delta_h^2}}. \end{aligned} \quad (\text{L.3})$$

The physical defect corresponds to $h = d$. Put

$$\tilde{\mu}_h = \tilde{\mu} + h\dot{\mu}_\xi + O(h^2), \quad \Delta_h = \Delta + h\dot{\Delta}_\xi + O(h^2). \quad (\text{L.4})$$

Differentiation at $h = 0$ gives

$$I_1\dot{\mu}_\xi - \Delta I_3\dot{\Delta}_\xi = \frac{2}{\mathcal{E}_\xi}, \quad (\text{L.5})$$

$$-\Delta^2 I_3\dot{\mu}_\xi - \Delta I_1\dot{\Delta}_\xi = \frac{2\eta_\xi}{\mathcal{E}_\xi}, \quad (\text{L.6})$$

where

$$I_1 = \int_0^\Omega \frac{\epsilon - \tilde{\mu}}{\mathcal{E}(\epsilon)^3} d\epsilon, \quad I_3 = \int_0^\Omega \frac{d\epsilon}{\mathcal{E}(\epsilon)^3}. \quad (\text{L.7})$$

The integrals are elementary:

$$I_1 = \frac{1}{\mathcal{E}(0)} - \frac{1}{\mathcal{E}(\Omega)} = A, \quad (\text{L.8})$$

$$-\Delta I_3 = -\frac{1}{\Delta} \left[\frac{\Omega - \tilde{\mu}}{\mathcal{E}(\Omega)} + \frac{\tilde{\mu}}{\mathcal{E}(0)} \right] = B. \quad (\text{L.9})$$

Equations (L.5)–(L.6) are therefore exactly (16.16).

At $\xi = \tilde{\mu}$, use

$$\mathcal{E}(0) = \Omega(\nu + x), \quad \mathcal{E}(\Omega) = \Omega(1 - \nu + x), \quad \Delta = 2\Omega\sqrt{x(1+x)\nu(1-\nu)}. \quad (\text{L.10})$$

Direct substitution yields (16.20) and (16.21).

L.2 Variation of the auxiliary spectral functional

The one-loop functional before the uniform mode is

$$\mathcal{I}[\mathcal{E}, \mathcal{F}] = \frac{1}{2\pi i} \oint_{\mathcal{C}} \mathcal{E}(z) \partial_z \log \mathcal{F}(z) dz. \quad (\text{L.11})$$

The perturbed Cauchy transform is

$$\mathcal{F}_h(z) = \int_0^\Omega \frac{d\epsilon}{(z - \epsilon) \sqrt{(\epsilon - \tilde{\mu}_h)^2 + \Delta_h^2}} - \frac{2h}{(z - \xi) \sqrt{(\xi - \tilde{\mu}_h)^2 + \Delta_h^2}}. \quad (\text{L.12})$$

Therefore

$$\partial_h \mathcal{F}_h(z)|_{h=0} = \dot{\mathcal{F}}_\xi(z) \quad (\text{L.13})$$

with (16.27). Varying $\mathcal{I}$ gives

$$\partial_h \mathcal{I}[\mathcal{E}_h, \mathcal{F}_h]|_{h=0} = \frac{1}{2\pi i} \oint_{\mathcal{C}} \left[\dot{\mathcal{E}}_\xi \partial_z \log \mathcal{F} + \mathcal{E} \partial_z \left(\frac{\dot{\mathcal{F}}_\xi}{\mathcal{F}} \right) \right] dz. \quad (\text{L.14})$$

The second term is integrated by parts on the closed contour. No endpoint term is produced, and the result is (16.29).

The contour may be chosen as any narrow positively oriented loop around $[0, \Omega]$ whose vertical extent is smaller than Δ . It then encloses the direct defect pole $z = \xi$ but excludes the branch points $\tilde{\mu} \pm i\Delta$. The result is invariant under continuous deformations within that class.

L.3 Closed form of the Cauchy transform

Set

$$u_- = \operatorname{arsinh} \frac{-\tilde{\mu}}{\Delta}, \quad u_+ = \operatorname{arsinh} \frac{\Omega - \tilde{\mu}}{\Delta}, \quad (\text{L.15})$$

$$t_- = e^{u_-}, \quad t_+ = e^{u_+}, \quad (\text{L.16})$$

and

$$\zeta_\pm(z) = \frac{z - \tilde{\mu} \pm \mathcal{E}(z)}{\Delta}. \quad (\text{L.17})$$

The substitution $\epsilon = \tilde{\mu} + \Delta \sinh u$ gives

$$\boxed{\mathcal{F}(z) = \frac{1}{\mathcal{E}(z)} \log \frac{(t_+ - \zeta_-)(t_- - \zeta_+)}{(t_+ - \zeta_+)(t_- - \zeta_-)}}. \quad (\text{L.18})$$

The logarithm is fixed by $\mathcal{F}(z) > 0$ for real $z > \Omega$. On the upper side of the physical interval this becomes

$$\mathcal{F}(\epsilon + i0) = \frac{\mathcal{L}(\epsilon) - i\pi}{\mathcal{E}(\epsilon)}. \quad (\text{L.19})$$

Equation (L.18) makes the alternative form

$$\dot{\mathcal{F}}_\xi = (\dot{\mu}_\xi \partial_\mu + \dot{\Delta}_\xi \partial_\Delta) \mathcal{F} - \frac{2}{(z - \xi) \mathcal{E}_\xi} \quad (\text{L.20})$$

fully explicit and is the expression used by the contour verification code.

L.4 Real-axis principal-value representation

The ground one-loop phase functional is

$$\delta E_{\text{fl}} = v\Omega \left(\frac{1}{2} - \nu \right) - \mathcal{E}(0) - \int_0^\Omega \mathcal{E}'(\epsilon) a(\epsilon) d\epsilon. \quad (\text{L.21})$$

The explicit first term does not vary under $L \mapsto L - 2$, $N \mapsto N - 1$. Differentiating the remaining terms directly gives

$$\mathcal{H}(\xi) = -\dot{\mathcal{E}}_\xi(0) - \int_0^\Omega \left[\partial_\epsilon \dot{\mathcal{E}}_\xi a + \mathcal{E}' \dot{a}_\xi \right] d\epsilon, \quad (\text{L.22})$$

which is (16.35). This form is preferable to integrating the first term by parts: $a'(\epsilon)$ is integrable but approaches the endpoints only as $[\epsilon \log^2 \epsilon]^{-1}$.

To obtain the explicit distribution, use

$$\frac{1}{\epsilon - \xi + i0} = \text{PV} \frac{1}{\epsilon - \xi} - i\pi \delta(\epsilon - \xi). \quad (\text{L.23})$$

The smooth parameter variation gives

$$\dot{a}_{\text{bulk}} = -\frac{\dot{\mathcal{L}}_\xi}{\mathcal{L}^2 + \pi^2}. \quad (\text{L.24})$$

The direct removed-level source gives

$$\dot{a}_{\text{src}} = \frac{2\mathcal{E}}{\mathcal{E}_\xi(\mathcal{L}^2 + \pi^2)} \text{PV} \frac{1}{\epsilon - \xi} - \frac{2\mathcal{L}(\xi)}{\mathcal{L}(\xi)^2 + \pi^2} \delta(\epsilon - \xi). \quad (\text{L.25})$$

Combining the two terms gives (16.36).

For numerical work one may remove the principal value explicitly. Define

$$f_\xi(\epsilon) = \frac{\mathcal{E}'(\epsilon)\mathcal{E}(\epsilon)}{\mathcal{L}(\epsilon)^2 + \pi^2}. \quad (\text{L.26})$$

Then

$$\begin{aligned} \text{PV} \int_0^\Omega \frac{f_\xi(\epsilon)}{\epsilon - \xi} d\epsilon &= \int_0^\Omega \frac{f_\xi(\epsilon) - f_\xi(\xi)}{\epsilon - \xi} d\epsilon \\ &\quad + f_\xi(\xi) \log \frac{\Omega - \xi}{\xi}. \end{aligned} \quad (\text{L.27})$$

Thus

$$\begin{aligned} \mathcal{H}(\xi) &= -\dot{\mathcal{E}}_\xi(0) - \int_0^\Omega \left[\partial_\epsilon \dot{\mathcal{E}}_\xi a - \mathcal{E}' \frac{\dot{\mathcal{L}}_\xi}{\mathcal{L}^2 + \pi^2} \right] d\epsilon \\ &\quad - \frac{2}{\mathcal{E}_\xi} \text{PV} \int_0^\Omega \frac{\mathcal{E}'\mathcal{E}}{(\mathcal{L}^2 + \pi^2)(\epsilon - \xi)} d\epsilon \\ &\quad + 2\mathcal{E}'(\xi) \frac{\mathcal{L}(\xi)}{\mathcal{L}(\xi)^2 + \pi^2}. \end{aligned} \quad (\text{L.28})$$

At the interior minimum, $\mathcal{E}'(\xi) = 0$, so the local delta contribution vanishes, although the principal-value source remains finite.

L.5 Pending confluent period and physical-determinant comparisons

Take the lower chart and let two blocked levels be

$$p = \alpha_* M + O(1), \quad q = p + 1. \quad (\text{L.29})$$

The exact seniority-two period contains

$$\frac{\pi_{N-1}(p)\pi_N(q) - \pi_N(p)\pi_{N-1}(q)}{q - p}. \quad (\text{L.30})$$

The intended route is to substitute the one-point saddle expansions of Chapter 9 into all four factors and retain the first surviving term after the leading cancellation at $q - p = 1$. The full expansion and a uniform bound on its coupling derivative remain to be written. The candidate exponential response of the $(N - 1)$ -pair sea is

$$-d(1 + 2x), \quad (\text{L.31})$$

which is equivalently the shift $d\dot{\Delta}_*$ of (16.20). The candidate contribution from the discrete Wronskian of the two prefactors is

$$d \frac{\Delta}{\Omega(1 - \nu)} \quad (\text{L.32})$$

in the lower chart. Under the pending confluent estimate C_{per} , their sum gives

$$\Delta_{B_2}^{\text{per}} = 2\Delta + d \left[-(1 + 2x) + \frac{\Delta}{\Omega(1 - \nu)} \right] + O(d^2). \quad (\text{L.33})$$

For the upper chart the exact dual filling is $\bar{\nu}_M = 1 - \nu + M^{-1}$. The raw lower-hole prefactor contains $\Delta/(\Omega\nu)$, but the M^{-1} displacement of the leading term $2\Delta(\bar{\nu}_M)$ contributes at the same order. Their sum is again

$$-(1 + 2x) + \frac{\Delta}{\Omega(1 - \nu)}. \quad (\text{L.34})$$

This checks the algebra of the proposed coefficient and its finite-duality convention, but does not prove the confluent asymptotic estimate (16.22). That estimate is kept as [Assumption 16.4](#).

The marked convolution (10.15) offers a second possible route: its endpoint $t = 1$ couples the electron and hole one-blocked saddles. Showing that its endpoint correction reproduces the Wronskian coefficient with an $O(d^2)$ differentiated remainder would close C_{per} . The exact convolution alone does not supply that error estimate.

For the physical determinants, the block determinant identity yields the Schur complement (16.39). No explicit matrix identity equating the remaining sea block to the two Christoffel defects is presently available in this appendix. The assertion that the residual logarithmic energy derivative is $O(d^2)$ is the matching clause of B_{def} , not a consequence established by the exact scalar convolution. Under that clause, and G , both sectors acquire the same $d\mathcal{H}(\tilde{\mu})$ term.

L.6 Half-filled reduction of the defect response

At half filling, write

$$t = \frac{1}{v}, \quad D = \frac{\Omega}{2}, \quad \Delta = \frac{D}{\sinh t}. \quad (\text{L.35})$$

Symmetry gives

$$\dot{\mu}_* = 0, \quad \dot{\Delta}_* = -\coth t. \quad (\text{L.36})$$

The complete residue-period coefficient also contains the confluent prefactor contribution $1/\sinh t$; hence

$$-\coth t + \frac{1}{\sinh t} = -\tanh \frac{t}{2}. \quad (\text{L.37})$$

Evaluating (16.29) by the same strip Fourier transform used in Appendix K.6 yields

$$\mathcal{H}(D) = \frac{\Delta}{D} \varphi(v), \quad (\text{L.38})$$

with $\varphi(v)$ from (15.47). Combining it with

$$\frac{2c_*^{\text{eff}}}{1-\sigma} = -\tanh \frac{t}{2} \quad (\text{L.39})$$

proves (16.46). The equivalent form (16.47) follows from

$$\coth t - \frac{1}{\sinh t} = \tanh \frac{t}{2}. \quad (\text{L.40})$$

Appendix M

Soft edge roots and logarithmic asymptotics

M.1 Exact lower auxiliary-zero equation

Use the finite spectral curve of [Section 15.2](#) and write

$$\xi_n = nd, \quad d = \frac{\Omega}{M}, \quad \mathcal{E}_{n,M} = \sqrt{(nd - \tilde{\mu}_M)^2 + \Delta_M^2}. \quad (\text{M.1})$$

The first zero $y_1^{(F)} = d\lambda$ of $\mathcal{F}_M$ satisfies

$$\frac{1}{d\lambda\mathcal{E}_{0,M}} + \sum_{n=1}^M \frac{1}{d(\lambda - n)\mathcal{E}_{n,M}} = 0. \quad (\text{M.2})$$

Multiplication by $d\mathcal{E}_{0,M}$ gives [\(17.18\)](#).

Define

$$w_{n,M} = \frac{\mathcal{E}_{0,M}}{\mathcal{E}_{n,M}}, \quad (\text{M.3})$$

and the harmonic moments

$$A_M = \sum_{n=1}^M \frac{w_{n,M}}{n}, \quad B_M = \sum_{n=1}^M \frac{w_{n,M}}{n^2}. \quad (\text{M.4})$$

For $0 < \lambda < 1/2$,

$$\frac{1}{n - \lambda} = \frac{1}{n} + \frac{\lambda}{n^2} + \frac{\lambda^2}{n^2(n - \lambda)}. \quad (\text{M.5})$$

Hence

$$\frac{1}{\lambda} = A_M + \lambda B_M + \lambda^2 C_M(\lambda), \quad (\text{M.6})$$

where $C_M(\lambda)$ is uniformly bounded on compact parameter sets.

M.2 Harmonic–Euler–Maclaurin decomposition

Split

$$A_M = H_M + \sum_{n=1}^M \frac{w_{n,M} - 1}{n}. \quad (\text{M.7})$$

Since

$$H_M = \log M + \gamma + \frac{1}{2M} + O(M^{-2}), \quad (\text{M.8})$$

and

$$\sum_{n=1}^M \frac{w_{n,M} - 1}{n} = d \sum_{n=1}^M \frac{\mathcal{E}_-/\mathcal{E}(nd) - 1}{nd} + O(M^{-1}), \quad (\text{M.9})$$

the Riemann sum gives

$$\boxed{A_M = \log M + \gamma + \mathcal{C}_- + O(M^{-1})}. \quad (\text{M.10})$$

The apparent singularity in (17.19) is removable because

$$\frac{\mathcal{E}_-/\mathcal{E}(\epsilon) - 1}{\epsilon} \longrightarrow \frac{\tilde{\mu}}{\mathcal{E}_-^2} \quad (\epsilon \downarrow 0). \quad (\text{M.11})$$

For the second moment, dominated convergence gives

$$\boxed{B_M = \frac{\pi^2}{6} + O\left(\frac{\log M}{M}\right)}. \quad (\text{M.12})$$

Indeed, for every fixed n , $w_{n,M} \rightarrow 1$, while the tail is dominated by a constant multiple of n^{-2} .

M.3 Solution of the auxiliary equation

Let $L_- = \log M + \gamma + \mathcal{C}_-$. Solving (M.6) by reversion gives

$$\lambda_M^{(F)} = \frac{1}{A_M} - \frac{B_M}{A_M^3} + O(A_M^{-4}). \quad (\text{M.13})$$

Using (M.10)–(M.12), all explicit M^{-1} remainders are smaller than any fixed inverse power of $\log M$, and

$$\boxed{\lambda_M^{(F)} = \frac{1}{L_-} - \frac{\pi^2}{6L_-^3} + O(L_-^{-4})}. \quad (\text{M.14})$$

Multiplication by d proves (17.21).

The upper formula follows after reflection $\epsilon \mapsto \Omega - \epsilon$. The removable endpoint limit in (17.25) is

$$\frac{\mathcal{E}_+/\mathcal{E}(\epsilon) - 1}{\Omega - \epsilon} \longrightarrow \frac{\Omega - \tilde{\mu}}{\mathcal{E}_+^2} \quad (\epsilon \uparrow \Omega). \quad (\text{M.15})$$

This proves (17.28).

M.4 Physical root equation and conservative remainder

Separate the real root $r_1^{\text{RG}} = d\lambda_M^{\text{RG}}$ from the remaining $N - 1$ roots. A complete proof would write the exact sea-resolvent contribution to its Richardson equation and bound the remainder after extracting the singular fixed-level sum. That derivation is not supplied here. The compact-edge clause of $\mathbf{B}_{\text{def}}$ is the following assumed dressed representation:

$$\frac{1}{\lambda_M^{\text{RG}}} = A_M + \mathfrak{b}_M(\lambda_M^{\text{RG}}), \quad (\text{M.16})$$

where

$$\sup_{0 < \lambda \leq 1/2} |\mathbf{b}_M(\lambda)| \leq C \quad (\text{M.17})$$

for compact lower-edge parameter sets. This bound is an assumption, not a consequence proved from the interior response formula in Chapter 16. In particular, one must still show that the physical sea introduces no additional harmonic divergence.

Equations (M.10) and (M.16) imply

$$\lambda_M^{\text{RG}} = \frac{1}{L_-} + O(L_-^{-2}), \quad (\text{M.18})$$

This proves (17.30) only conditional on the dressed representation and uniform bound above. It does not prove those inputs. The bounded remainder also leaves the L_-^{-2} coefficient undetermined, so the auxiliary $-\pi^2/(6L_-^3)$ term must not be copied to the physical root.

The branch calculations in `code/verify_edge_soft_scale.py` show that

$$\lambda_M^{\text{RG}} - \lambda_M^{(F)} \longrightarrow 0 \quad (\text{M.19})$$

along the tested sequence and much faster than the conservative estimate requires. This stronger behavior is evidence, not an additional theorem.

M.5 Exact edge energy bookkeeping

For the lower broken-pair sector, insert $n = N - 1$, $p = 0$, $q = 1$ into the exact period formula. Using (17.7),

$$\begin{aligned} \Delta_{B_2,-}^{\text{per}} &= E_{N-1}^{\text{per}} - E_N^{\text{per}} + d + \frac{2(1+x)d}{1} 2(N-1) \\ &= \frac{\Omega}{M} (2Mx + 2N - 1). \end{aligned} \quad (\text{M.20})$$

This proves (17.8).

For the upper edge, the reflected lower sector has $\bar{N} = M + 1 - N$. Thus

$$2\Omega \left(\frac{\bar{N}}{M} + x \right) - d = 2\Omega(1 - \nu + x) + d, \quad (\text{M.21})$$

which proves (17.13). This identity is the simplest place where dropping the M^{-1} term in the exact dual filling changes an $O(d)$ coefficient.

The upper real-root coefficient follows similarly from

$$P_M^{\uparrow,(N)} = P_0^{\downarrow,(\bar{N})}. \quad (\text{M.22})$$

Expanding both the leading term $2\Delta(\bar{N}/M)$ and the lower-hole endpoint coefficient gives (17.14).

M.6 Soft-coordinate contribution to the physical gaps

At the lower edge,

$$\mathcal{E}(r_1^{\text{RG}}) = \mathcal{E}_- + v_- r_1^{\text{RG}} + O\left(\frac{(r_1^{\text{RG}})^2}{\Delta}\right). \quad (\text{M.23})$$

In the conditional dressed-energy expansion, the real-root excitation has leading energy $2\mathcal{E}(r_1^{\text{RG}})$, producing the term $2v_-r_1^{\text{RG}}$. This energy identification and its uniform remainder are part of the edge finite-defect input. For two blocked levels, the corresponding dispersion expansion is

$$\mathcal{E}(0) + \mathcal{E}(d) = 2\mathcal{E}_- + dv_- + O(d^2/\Delta). \quad (\text{M.24})$$

Assuming the edge matching and Hessian-response clauses, the remaining collective-sea correction is common and equals $d\mathcal{H}_-$. Subtraction then gives (17.36). Neither the common sea term nor its physical remainder follows from the Taylor expansions alone.

The upper calculation is identical after replacing the distance from the lower edge by the distance from the upper edge. The slope of $\mathcal{E}$ at Ω is negative, but moving the real root inward by s_1^{RG} raises its energy by $v_+s_1^{\text{RG}}$, producing (17.40).

M.7 Numerical extraction and diagnostics

The archived verification uses the branch conventions of Appendix N. For the lower real-root branch the weak label is

$$S_{a=0} = \{1, 2, \dots, N\}. \quad (\text{M.25})$$

The eigenvalue-based variables are continued to $V = v\Omega/M$, refined with the exact sector sum, and converted to a Chebyshev Baxter polynomial. The unique real zero in the first pair interval is divided by two to obtain the coordinate r_1^{RG} of (15.1).

The verification report records separately:

1. the auxiliary root $\lambda_M^{(F)}$ and both logarithmic approximations;
2. the physical root λ_M^{RG} and Baxter interpolation residual;
3. the physical gap splitting divided by d ;
4. the leading coefficient v_- and the refined value $v_-(1 - 2\lambda_M^{(F)})$;
5. the broken-pair and real-root one-loop remainders after the soft coordinate is separated.

Appendix N

Numerical continuation and reproducibility

N.1 Purpose and separation of roles

The computations in this appendix share the LLM provenance and incomplete author verification stated at the beginning of the manuscript. The numerical archive has three distinct functions.

1. It checks exact finite identities through separate implementations for small integer data.
2. It evaluates positive finite sums in regimes where direct factorial arithmetic would overflow.
3. It follows named physical Richardson branches and reconstructs their root topology.

The third task is the subject of this appendix. The continuation code does not replace the analytic weak-coupling identification of [Branch identification 14.4](#); it provides numerical diagnostics that the labelled branches used in the finite sequences reach the stated target parameters. These diagnostics do not constitute a rigorous exclusion of every possible undetected branch exchange.

All archived branch tables use

$$\Omega = 1, \quad \varepsilon_{F0} = 0, \quad E_n = \frac{2n}{M}, \quad V = \frac{v}{M}. \quad (\text{N.1})$$

Restoring Ω and ε_{F0} is immediate from the formulas in the main text.

N.2 Vectorized eigenvalue-based system

For an active array of L_{act} pair energies E_n , define

$$F_n(\boldsymbol{\lambda}, V) = \lambda_n^2 - \lambda_n - V \sum_{m \neq n} \frac{\lambda_n - \lambda_m}{E_n - E_m}. \quad (\text{N.2})$$

The analytic Jacobian used by Newton correction is

$$\frac{\partial F_n}{\partial \lambda_m} = \frac{V}{E_n - E_m}, \quad m \neq n, \quad (\text{N.3})$$

while

$$\frac{\partial F_n}{\partial \lambda_n} = 2\lambda_n - 1 - V \sum_{m \neq n} \frac{1}{E_n - E_m}. \quad (\text{N.4})$$

The inverse level-difference matrix is precomputed once. Both F and the off-diagonal part of its Jacobian are then evaluated by vectorized array operations.

The zero-coupling initial data are exactly binary:

$$\lambda_n(0) = \mathbf{1}_{n \in S}. \quad (\text{N.5})$$

For a blocked sector, the active-level array excludes the blocked nodes, and the subset positions refer to this reduced array.

N.3 Continuation schedule and branch guards

The archived calculations use a dense monotone schedule

$$V_s = V_{\text{target}} \left(\frac{s}{S} \right)^{1.6}, \quad s = 1, \dots, S. \quad (\text{N.6})$$

The power 1.6 places additional points in the weak-coupling region, where roots first collide and direct root variables are least convenient. At each step the previous λ is the initial Newton vector and (N.3)–(N.4) supply the exact Jacobian.

The production report records the following diagnostics:

$$r_F = \max_n |F_n|, \quad (\text{N.7})$$

$$r_N = \left| \sum_n \lambda_n - N \right|, \quad (\text{N.8})$$

$$r_{\text{step}} = \max_{s,n} |\lambda_n(V_s) - \lambda_n(V_{s-1})|. \quad (\text{N.9})$$

The first is the nonlinear residual. The second is an important branch guard: the polynomial system by itself contains solutions belonging to all pair-number sectors, whereas an N -root Richardson solution must satisfy (14.22). A small equation residual with a large sector error is therefore rejected as an accidental solution exchange.

For the convergence sequence of Table 14.3,

$$S = 20M. \quad (\text{N.10})$$

The $M = 36$ marked branch was additionally compared on the schedules $S = 20M, 22M$, and $24M$; the accepted energies differ by less than 3×10^{-6} and all have sector errors below 2×10^{-6} . Coarser paths that solve (N.2) but violate (N.8) are deliberately excluded.

A pseudo-arclength implementation is not required for the parameter paths archived in this release. The code is organized so that such a corrector can be added without changing the definitions of the branch labels, residuals, or energy formulas.

N.4 Energy without root reconstruction

For L_{act} active levels and N active pairs, the code evaluates

$$E^{\text{RG}} = \sum_{n=1}^{L_{\text{act}}} E_n \lambda_n - V N (L_{\text{act}} - N + 1). \quad (\text{N.11})$$

This is the primary energy estimator. It avoids the conditioning loss associated with finding the zeros of a degree- N polynomial. The sector sum (N.8) and the exact Hamiltonian checks below make the estimator independently auditable.

For one blocked level p , $L_{\text{act}} = M$ and the total odd energy is

$$E_{\text{odd}} = E_{\text{pair}}^{[p]} + \varepsilon_p. \quad (\text{N.12})$$

The active zero-coupling subset is the set of the lowest N unblocked original levels.

N.5 Direct Hamiltonian diagonalization

For small M , the paired Hamiltonian is constructed directly in the subset basis. Its dimension is

$$\binom{M+1}{N}. \quad (\text{N.13})$$

The diagonal and off-diagonal entries are exactly (14.6)–(14.7). Dense Hermitian diagonalization is used only for the small systems listed in Table 14.2; it is an independent check of the eigenvalue-based energy and not the method used for thermodynamic sequences.

The same construction on the reduced active-level array verifies a blocked branch. In the archived test

$$M = 7, \quad N = 2, \quad p = 1, \quad v = 1, \quad (\text{N.14})$$

the pair energy from (N.11) and the lowest eigenvalue of the reduced Hamiltonian both equal

$$-0.153533188585 \quad (\text{N.15})$$

to the digits shown.

N.6 Chebyshev reconstruction of the Baxter polynomial

Let

$$c = \frac{E_0 + E_M}{2}, \quad s = \frac{E_M - E_0}{2}, \quad t_n = \frac{E_n - c}{s}. \quad (\text{N.16})$$

Expand

$$\mathcal{Q}(z) = \sum_{r=0}^N q_r T_r \left(\frac{z - c}{s} \right). \quad (\text{N.17})$$

The interpolation equations (14.38) become

$$\sum_{r=0}^N \left[\frac{1}{s} T'_r(t_n) - \frac{\lambda_n}{V} T_r(t_n) \right] q_r = 0. \quad (\text{N.18})$$

The coefficient vector is the right singular vector associated with the smallest singular value of this $(M+1) \times (N+1)$ matrix. The roots of the resulting Chebyshev polynomial are then mapped back to the physical pair-energy plane.

Four diagnostics are printed:

$$r_{\text{svd}} = s_{\min}, \quad (\text{N.19})$$

$$r_{\text{int}} = \max_n \left| \mathcal{Q}'(E_n) - \frac{\lambda_n}{V} \mathcal{Q}(E_n) \right|, \quad (\text{N.20})$$

$$r_E = \left| \Re \sum_j R_j - E_\lambda \right|, \quad (\text{N.21})$$

$$r_{\text{RG}} = \max_j \left| \frac{1}{V} - \sum_n \frac{1}{E_n - R_j} - 2 \sum_{\ell \neq j} \frac{1}{R_j - R_\ell} \right|. \quad (\text{N.22})$$

For the degree-four examples of Table 14.1, all four residuals are between 10^{-15} and 10^{-9} . At larger degree the energy is never inferred from numerically delicate zeros; (N.11) remains the authoritative estimator.

N.7 Root classification

A reconstructed root is classified as real when

$$|\Im R_j| < 2 \times 10^{-6} \quad (\text{N.23})$$

for the double-precision tables. A real root is then classified as:

1. below the pair window;
2. above the pair window;
3. inside the interval (E_n, E_{n+1}) identified by binary search.

The tolerance is much larger than the interpolation residual but much smaller than the level spacing in all displayed examples. The report prints the actual intervals, not only the total number of real roots.

N.8 Stable evaluation of period predictions

The exact marked polynomial P_a is positive but its factorial summands span many orders of magnitude. The code evaluates

$$\log P_a = \text{logsumexp}_{b,s} \log T_{b,s} \quad (\text{N.24})$$

and obtains

$$x \partial_x \log P_a = \frac{\sum_{b,s} (s-b) T_{b,s}}{\sum_{b,s} T_{b,s}} \quad (\text{N.25})$$

from normalized log-weights. The exact finite period gap is then

$$\Delta_{\text{mark}}^{\text{per}} = \frac{2\Omega(1+x)}{M} x \partial_x \log P_a. \quad (\text{N.26})$$

The blocked polynomial is treated by an analogous one-dimensional log-sum-exp. Its degree expectation gives the exact finite pair response, to which the kinetic energy of the unpaired electron is added as in (N.12).

N.9 Reproduction commands and archive map

From the project root, the branch audit is reproduced by

```
cd code
python verify_branch_identification.py
```

The default report is

```
code/branch_identification_report.txt.
```

Its final line must be

```
Branch-identification verification: PASS.
```

The branch script imports only the already archived stable evaluators

```
code/verify_caseA.py,    code/verify_seniority.py.
```

The earlier transition script

```
code/verify_AB_transition.py
```

contains the independent half-filled real-root count sequence used in [Chapter 13](#).

The soft-edge audit is reproduced by

```
cd code
python verify_edge_soft_scale.py
```

Its default report is

```
code/edge_soft_scale_verification_report.txt.
```

The script prints the auxiliary zero, physical Richardson root, Baxter residual, physical gap splitting, leading and refined asymptotic coefficients, and the sea-response remainder separately; it therefore never identifies $y_1^{(F)}$ with r_1^{RG} .

The numerical archive currently has the following division of responsibility. Earlier reports are retained in `code/archive_module18_reports/`. Only the Revision 01 lattice report has been regenerated for this release; the other rows describe the retained programs and their historical tests:

Script	Primary task
<code>verify_all_pole.py</code>	Exact all-pole identities and energy derivative.
<code>verify_kernel.py</code>	Moment, kernel, projection, and positive-minor identities.
<code>verify_pure_contours.py</code>	Excluded-pole and arbitrary nested-contour algebra.
<code>verify_caseA.py</code>	Positive marked sums, WKB prefactors, and edge matching.
<code>verify_eh_duality.py</code>	Exact finite electron–hole transformations.
<code>verify_seniority.py</code>	Blocked Heine–Christoffel sectors and marked convolution.
<code>verify_caseB_exact.py</code>	Exact consecutive-cascade ground, marked, and blocked periods.
<code>verify_caseB_asymptotics.py</code>	Complete endpoint support resummation.
<code>verify_AB_transition.py</code>	Cascade birth law and representative ground root counts.
<code>verify_branch_identification.py</code>	Weak labels, physical continuation, Hamiltonian spectrum, Baxter roots, and convergence tables.
<code>verify_gaudin_ground.py</code>	Ground-state Gaudin determinant and spectral-shift formulas.
<code>verify_finite_defect.py</code>	Interior finite-defect response and exact-dual upper coefficient.
<code>verify_edge_soft_scale.py</code>	Auxiliary and physical soft roots and edge gap splitting.
<code>verify_physical_synthesis.py</code>	Algebraic and numerical comparisons of the physical summary formulas.
<code>verify_revision01.py</code>	Separately implemented exact lattice-minor and fixed-label derivative checks; regression of the rejected continuation.

N.10 Numerical scope

The continuation data in this appendix establish reproducibility for the paths and sizes quoted in the manuscript. They do not constitute a global theorem that every excited branch can be continued through every positive coupling without a singular Jacobian. The exact statements are:

1. local branch existence follows analytically from [Branch identification 14.4](#);
2. global ground uniqueness follows analytically from [Branch identification 14.9](#);
3. the archived excited paths satisfy all diagnostics (N.7)–(N.22);
4. later asymptotic physical formulas use these named paths together with the explicit unresolved assumptions B_0 , G , B_{def} , and, where required, C_{per} .

Open proof obligations and scope boundaries

1. **Ground contour selection.** Prove an estimate for the full period relative to the selected saddle cycle, not merely the existence of the physical ground branch. The corrected B_0 is a sufficient assumption, not a theorem established in this revision.
2. **Growing-dimensional Gaussian analysis.** Bound the inverse Hessian, higher derivatives, summed loop terms, and parameter derivatives of the remainder. The order of the action alone is insufficient.
3. **Hessian versus Richardson spectral shift.** Supply an explicit operator or determinant identity proving G . The known Richardson resolvent correction is retained as a comparison and not relabeled as a new derivation.
4. **Finite defects.** Complete the confluent blocked-period prefactor calculation, then derive the physical marked/blocked determinant comparison and bounded physical edge-sea response. The auxiliary Gaudin zero has a different and more direct proof.
5. **Physical root density.** Establish, or correct, the identification of the period endpoint κ with the asymptotic real-root count. The Lambert inversion alone proves a law for the rate-defined endpoint.
6. **Joint weak-coupling limits.** At fixed filling, $M\sigma = O(1)$ implies $\varepsilon_c \sim d \ll \Delta$; a simultaneous dilute limit is needed to make all three scales comparable. The earlier period route supplies no uniform theorem in either joint limit. The selected half-filled self-dual joint limits in Chapter 24 are now established separately; boundary crossover limits and simultaneous dilute scaling remain distinct.
7. **Extensions.** The new Case A Pfaffian and positive compact identities allow arbitrary real spectra at finite size, but controlled nonuniform-spectrum thermodynamics remains separate. Nonconstant or multiband interactions, higher finite-size orders and finite temperature also remain separate problems. A Debye interaction cutoff must not be identified with a physical band bottom without specifying the excluded states.
8. **Higher inverse temperature.** Nonvanishing of the full fourth-power Case A period is proved in Chapter 20. It does not control the ratio to a selected physical thimble. The fixed-system large-multiplier ground-state limit is now established by the vector argument of Chapter 21. Uniform thermodynamic errors of this limit or of the two-temperature extrapolation, and an extension to mixed Case B contours, remain open.
9. **Formal one-pair summation.** Chapter 22 proves cancellation of every higher extensive coefficient. Uniform analytic summation and continuation of that formal expansion are not consequences of this coefficientwise theorem. The blocked Case B cascade also needs its own Christoffel-kernel and derivative estimates.

The invalid source-index interpolation in the previous Appendix J is addressed in this iteration by saddle localization and transfer back to the integer lattice. It is no longer a pending assumption. The other items above remain open.

Duality-specific extensions

The finite self-dual ground construction does not require the contour bridges above. Its remaining questions have a different form: prove the ordering against non-self-dual pair-preserving levels; control the physical scalar root and energy integral in the limiting excited crossover system; derive uniform numerical error or convergence estimates for a short response iteration; and extend the construction beyond the present half-filled symmetric setting. The explicit $O(N^{-2})$ cutoff remainder in the scalar equation is not an $O(N^{-2})$ error bound for the full gap. The energy and gap criteria already known in the mesoscopic literature are recovered, not proposed as new physical scales.

Reproducibility and contents of the deposit package

The deposit package provides this PDF as a separately accessible file and a source archive containing the full LaTeX manuscript, bibliography, the retained fifteen scientific verification programs, a separate Case A Pfaffian library with two verification drivers, their dependency instructions, the relevant historical reports, and the Revision 01 checker and report. The main source file is `rg_master.tex`; build it with `python3 build_pdf.py` from the source root. Detailed build instructions are in `README.md`.

Run `python code/verify_revision01.py` from the source root to reproduce the current lattice checks. This program uses the Python standard library only. The other scientific programs require the packages listed in `requirements_optional.txt`. Their purposes and default report paths are documented in `code/README.md`. The older reports are retained in `code/archive_module18_reports/`; they are historical outputs, not a full rerun of this release. When earlier verification notes give a filename under `code/`, that is the script's default new-output location; the archived copy resides in the historical-report subdirectory.

The Case A addition is self-contained in `code/pfaffian_caseA/`. Its README gives the exact commands and its JSON reports record the checks performed for version 1.1. The scope is the full common-contour period, its positive integral, and fixed-spectrum coefficients; no new Case B Pfaffian code or numerical data are bundled.

All these programs belong to the LLM-assisted computational work described at the beginning of the manuscript. Independence of the current checker means a separate implementation without imports from the original verification modules. It does not mean independent human validation. Neither a successful finite test nor a historical PASS report proves the open contour-selection, uniform Gaussian, or finite-defect assumptions.

The source archive also contains a change record, a citation file, and checksums. The deposit description repeats the authorship and incomplete-verification statement. No DOI for this working manuscript is assumed in the source package; Zenodo can assign one when the record is created.

Third-edition additions

The arbitrary-even-exponent checks and their instructions are in `code/even_beta_caseA/`. The formal one-pair and scalar cascade checks are in `code/one_pair_saddle/`. Their release reports distinguish current executions from archived exploratory data. The report `revision/BUILD_QA_v1_2.json` records the historical build and checks run for the third edition. The two new chapters and their supporting materials are included in the same source archive; no previous source archive is needed to build the complete manuscript.

Fourth-edition additions

The complete duality checkpoint v0.1 is retained in

`code/duality_selfdual/checkpoint_v0_1/`.

An additional recovered archive, `code/duality_selfdual/selfdual_iteration5.zip`, supplies its finite spectral representation and first-level ordering dependency. The included source proofs are cited by exact filename in the new chapters. Fresh validation has its own directory, `code/duality_selfdual/revision_validation/`, and its scope is recorded in `revision/DUALITY_NUMERICS_v1_3.md`.

The new runs include small-system polynomial reconstruction, scalar identities, direct paired-Hamiltonian comparisons, and selected finite crossover and response calculations. Mathematical audits distinguish algebraic proof from a finite symbolic test and numerical agreement. The final build is documented in `revision/BUILD_QA_v1_3.json`. All files needed to compile the fourth-edition manuscript and inspect its scientific dependencies are included in the current source archive.

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